

Self-Embeddings of the Surreal Numbers

Rigidity, Hahn lifts, topology,
and size-safe foundations

with order and simplicity, algebra, analysis, exponential and model-theoretic structure, large cardinals, and omega-series

Built from six manuscripts dated 3 October 2026

(batch 84: base manuscript 03; members 02, 04, 05, 06, 07)

Central separation [03]. For every set $A \subseteq \mathbf{No}$, there is a *proper, cofinal, continuous ordered-field self-embedding* fixing $A \cup \mathbb{R} \cup \mathbf{On}$ pointwise. It preserves all admissible Hahn sums, normal-form truncations, omnific integrality, and omnific floors. Its image is a closed nowhere-dense subfield. These constraints therefore do not force surjectivity or rigidity. In sharp contrast, preserving birthdays without increase forces an order embedding to be the identity; an order embedding with simplicity-initial image must be onto; and an exponential field self-embedding fixing the natural valuation pointwise must be the identity.

Status. AI-assisted research manuscripts, merged and edited for the ProveIt Surreal collection; unrefereed. No statement of this report is formalized in Lean as part of it (section 1). Each source is a research synthesis with proofs supplied in the text, not a claim of publication priority, and roughly half of each re-proves material already in the collection, which is credited label by label.

Repository snapshots read by the sources: af04de3fa (02), b0d4af4f3 (03 and 07), a006a77a0 (04), 51b0a69d7 (05), 79049d58c (06). Placement commit e1395c66c; provenance in Appendix C.

Abstract

A self-embedding of the surreal numbers can mean an injection of sign sequences, an order embedding, a simplicity-tree embedding, a field embedding, an elementary embedding in a specified language, or a map respecting valuation, normal forms, exponentiation, a derivation, or particular distinguished subclasses. These notions have markedly different rigidity and size behavior. We organize them into a common framework. Our main constructive device is a two-stage Hahn lift that faithfully converts any increasing class embedding of **No** into a strongly $\mathbb{R}$ -linear ordered-field self-embedding. It gives explicit proper maps fixing every real, every ordinal, and any prescribed set of additional surreals; it also gives bounded elementary subfield copies fixing any prescribed set. We classify monomial strong embeddings by an additive exponent embedding and a positive real character, compute their fixed fields and images, and analyze their iterated-image cores. For the full surreal order topology we prove a field-embedding dichotomy: cofinal images give continuous embeddings, whereas noncofinal images are discrete and the maps are nowhere continuous. We separate this topology from Hahn summability, simplicity limits, and differentiation. Building on the repository's displacement argument, we prove exponential valuation rigidity without surjectivity and a uniqueness theorem for cofinal exponential embeddings with the same value-group action. Finally we give class-theoretic, universe-indexed, and Lean-oriented formulations that avoid powerclass, truth-predicate, and improper-cut paradoxes, and formulate a focused research program.

[write] The paragraph above is the abstract of the base, source 03 (Part I). Five independent texts answering the same question are merged as Parts II–VI. They add: a description of every order-and-simplicity embedding by local padding data, tree embeddings fixing any set, all reals and all ordinals, and a characterization of convex tree images; a large-cardinal map $\mathcal{J}_j$ induced by an elementary $j : V \rightarrow M$ (conditional); an exact criterion for the vanishing derivative of a weighted monomial lift; a continuous non-strong automorphism; under GBC with elementary transfinite recursion and a class satisfaction relation, bounded and cofinal proper elementary self-embeddings of o-minimal expansions, including the exponential field; abstract cofinal pair rigidity and an exponent-lift criterion for commuting with $\exp$; the exponential defect cocycle of a derivation-compatible embedding; a classification of strong exponential self-embeddings of the omega-series field resting on imported literature; the relative real closure as the class of unavoidable points; and club localization of class maps to birthday stages. About half of every source re-proves results already in the collection (double Hahn lifts and parameter copies, closed images and the continuity dichotomy, the field lift, Taylor flows and dilation derivatives, exponential rigidity, a conditional omega-map embedding). These are printed once, with the existing labels credited, never as new.

Contents

I	The base text (source 03)	7
1	The merged report: sources, conventions, and status	7
1.1	Six sources, one question	7
1.2	Notation across the sources	8
1.3	Status against the collection	9
1.4	What is not claimed	11
2	The question is a family of questions	11
2.1	Relation to existing work and to the repository	12
3	A foundation that does not manufacture paradoxes	12
3.1	Set-length signs and set-supported normal forms	12
3.2	Three legitimate size registers	13
3.3	What is forbidden, and what needs extra axioms	14
4	Order and simplicity: flexibility and rigidity	14
4.1	Three explicit order embeddings	14
4.2	Simplicity without order	15

4.3	Rigidity statements with different hypotheses	16
4.4	Closed substructures and their fixed points	17
5	Initial images and the distinction between two truncations	18
6	Hahn relabeling and a double-lift construction	18
6.1	The additive lift of an order embedding	19
6.2	The field lift of an additive embedding	19
6.3	Classification with real weights	21
7	Proper embeddings fixing every real and ordinal	21
7.1	A completely explicit example	21
7.2	Fixing an arbitrary prescribed set	22
7.3	Bounded field copies fixing prescribed data	23
8	Omnific floors, fixed fields, and iteration	24
8.1	Omnific integrality is preserved and reflected	24
8.2	Fixed fields do not determine surjectivity	24
8.3	The largest reversible core	25
9	Topology: cofinality, discrete copies, and closed images	26
9.1	Why set-indexed limits are misleading	26
9.2	The field-embedding dichotomy	26
9.3	An everywhere-zero derivative of a nonconstant automorphism	27
10	Valuation, leading terms, and the ω-map	28
10.1	Three different valuation constraints	28
10.2	A controlled unit twist fixes even the leading term	29
10.3	Fixing residues need not fix real elements	30
10.4	Truncation thresholds and the ω -map	30
11	Exponential rigidity from finite displacement	31
11.1	The finite engine	31
11.2	Profiles in an ordered exponential field	32
11.3	The actual surreal exponential	33
11.4	A uniqueness theorem beyond automorphisms	34
11.5	What remains flexible in the exponential category	35
12	Elementarity, definability, and other signatures	35
12.1	Field elementary does not mean set-theoretically elementary	35
12.2	O-minimal definability is a strong rigidity condition	35
12.3	Algebraic variants and surcomplex extensions	36
13	Differential constraints and composition	36
14	Cutoffs, universes, and a concrete formalization plan	37
14.1	Set-sized cutoff models require a smaller quantifier	37
14.2	Coherent restrictions rather than a fictitious last stage	38
14.3	What was actually inspected in ProveIt	38
14.4	Proposed modules and proof obligations	39
15	A research agenda with precise targets	40
15.1	Exponential and ω -map embeddings	40
15.2	Combining tree structure with arithmetic	40
15.3	Hahn images, fixed fields, and dynamics	41
15.4	Topology, differentiation, and foundations	41
16	Synthesis and an assumption checklist	42

II Source 02: Order, simplicity, algebra, analysis, and paradox-free foundations	42
17 Source 02 at a glance	42
18 Order and simplicity in source 02	46
19 Hahn lifts and explicit field embeddings in source 02	48
19.1 The compression double lift	48
20 Named constants, omnific integers, and limits in source 02	50
21 Exponential, omega-map and differential constraints in source 02	50
22 Elementarity and definability in source 02	51
23 Birthday fragments and universe changes in source 02	52
24 Embeddings induced by large cardinals (conditional)	53
25 Lexicographic coding and iteration in source 02	56
26 The formalization plan of source 02	56
27 Research questions of source 02	57
28 Source audit and regression checks of source 02	60
28.1 Repository boundary	60
28.2 External literature boundary	61
28.3 Finite regression checks	61
29 Status and non-claims of source 02	62
III Source 04: A Structural and Foundational Atlas	62
30 Source 04 at a glance	62
30.1 Renames	63
30.2 Correspondence table	64
31 Foundational setting and small examples of source 04	66
32 Hahn lifts: a bounded copy, fixed fields, and cores	66
32.1 An explicit bounded field copy	66
32.2 A proper cofinal embedding with fixed field exactly the reals	67
32.3 The iterated-image core of a double lift	67
33 Automatic preservation, the Taylor twist, and value actions	68
34 Elementary self-embeddings of o-minimal surreal structures (under GBC+ETR)	69
34.1 The class-model setting	69
34.2 Bounded elementary images	70
34.3 Proper cofinal elementary images	71
34.4 Internal definability, by a calculus argument	72
35 Exponential growth-scale rigidity for embeddings	72
35.1 A criterion for monomial lifts to preserve exponentiation	74
36 The omega-map, other languages, omnific integers, and complexification	74
36.1 Differential self-embeddings	75
36.2 Omnific integers	76
36.3 Complexification	76

37 Cutoffs and universe-polymorphic formalization	76
38 Integration with the ProveIt repository	77
38.1 What source 04 inspected	77
38.2 Existing interfaces	78
38.3 A staged implementation plan [04]	78
38.4 The finite regression script	79
39 Proof-obligation ledger and audit of invalid arguments of source 04	80
39.1 A compact audit of common invalid arguments [04]	80
40 Research questions of source 04	81
40.1 Classification and explicit constructions	81
40.2 Omega and differential constraints	82
40.3 Images, fixed fields, and dynamics	82
40.4 Foundations and verified implementation	83
41 Status and non-claims of source 04	83
 IV Source 05: Order, simplicity, algebra, topology, exponential structure, and foundations	 84
42 Source 05 at a glance	85
43 Scope and foundations in source 05	89
44 Order and tree self-embeddings	90
45 Hahn lifts, explicit copies, and the two truncations	92
46 Topology and ordinary differentiability	94
47 Strong weighted maps: fixed fields and an exact differentiability criterion	95
48 Exponential comparison, the omega-map and omnific integers	97
49 Elementary embeddings and internal definability	98
50 Differential, hyperserial and higher-dimensional variants	103
51 Formalization architecture and finite checks of source 05	104
52 Research questions of source 05	106
52.1 Near-term formalization projects	106
52.2 Structural classification projects	107
52.3 Foundational and computational projects	109
53 Status and non-claims of source 05	110
 V Source 06: Structure, Rigidity, and Foundations	 111
54 Source 06 at a glance	111
55 Order, simplicity, and birthdays (source 06, Section 2)	115
56 Field embeddings and normal forms (source 06, Section 3)	117
57 Fixed parameters, ordinals, and logic (source 06, Section 4)	120

58 The omnific predicate (source 06, Section 5)	121
59 Exponential, differential, and hyperserial constraints (source 06, Section 6)	125
60 Foundations, localization, and large cardinals (source 06, Section 7)	129
61 Formalization status recorded by source 06 (its Section 8)	132
62 Research questions of source 06	134
63 Status and non-claims of source 06	136
 VI Source 07: Order, simplicity, normal forms, analytic structure, and foundations without paradoxes	 137
64 Source 07 at a glance	137
65 Order, simplicity and the sign tree [07]	141
65.1 Convex images	141
65.2 Local coding data and ordinal lifts	142
65.3 Rigidity: initial images and prefix growth	143
65.4 Fixed points	143
66 Normal-form lifts [07]	144
66.1 Truncation compatibility	145
66.2 The Taylor flow and omnific integers	145
67 Continuity and set nets [07]	146
68 Exponentiation, the omega-map and the derivation [07]	148
68.1 The omega-series field	149
69 Foundations: objects and quantifiers [07]	150
70 Elementarity and homogeneity in the field language [07]	151
71 Set-sized fragments and universes [07]	153
72 Set-theoretic elementary embeddings [07]	154
73 A staged formalization programme [07]	154
74 Source audit and corrections of source 07	155
74.1 A correction concerning Taylor supports [07]	156
74.2 Two sign conventions in the KKS preprint [07]	156
75 Research questions of source 07	157
75.1 The omega-map and analytic intersections	157
75.2 Series structure and integer parts	158
75.3 Simplicity and controlled deformation	159
75.4 Foundations and proof-assistant targets	159
76 Status and non-claims of source 07	160
A Result and verification ledger	161
B Source audit and reproducibility	162
C Provenance of the merged report	162
C.1 Archives, pins and placement	162

C.2	The superseded edition 01	163
C.3	What each source contributed	163
C.4	Where the merge had to choose	164
References		165

Part I

The base text (source 03)

1 The merged report: sources, conventions, and status

[write] This section was written when the report joined the collection. It says what each source contributed, fixes one notation for all six texts, and records which results the collection already had.

1.1 Six sources, one question

On 3 October 2026 seven archives arrived, all titled *Self-Embeddings of the Surreal Numbers* and all answering one research prompt. They were numbered 01–07 by archive time (batch 84). Manuscript 01 is an earlier edition of 05: 05 says that it “incorporates and extends earlier working notes”, about nine tenths of 01’s distinct eight-word sequences recur in 05, and all 52 labels of 01 reappear there. 01 is therefore *not* a source; [Appendix C](#) records what of it survives only in its archive. The other six are independent texts (between 0.4% and 1.7% of eight-word sequences shared by any two) proving a largely common spine, each with different additions.

Source	Archive and subtitle	Pages	Pin	Printed as
03	Surreal_Self_Embeddings(2): <i>Rigidity, Hahn lifts, topology, and size-safe foundations</i>	33	b0d4af4f3	Part I, Sections 2–15 and Appendices A, B (the base; its section n is Section $n + 1$ here)
02	surreal_self_embeddings(1): <i>Order, simplicity, algebra, analysis, and paradox-free foundations</i>	36	af04de3fa	Part II
04	Surreal_Self_Embeddings_Atlas: <i>A Structural and Foundational Atlas</i> (title page: “prepared with ChatGPT”)	35	a006a77a0	Part III
05	surreal_self_embeddings_comprehensive: <i>Order, simplicity, algebra, topology, exponential structure, and foundations</i>	39	51b0a69d7	Part IV
06	surreal_self_embeddings(3): <i>Structure, Rigidity, and Foundations</i>	47	79049d58c	Part V
07	surreal_self_embeddings(4): <i>Order, simplicity, normal forms, analytic structure, and foundations without paradoxes</i>	42	b0d4af4f3	Part VI

Why 03 is the base. Where the sources prove the same theorem, 03's version has the weakest hypotheses or the greatest generality. Its stabilizer theorem (theorem 7.4) gives, for every set A , a proper cofinal continuous field embedding fixing $A \cup \mathbb{R} \cup \mathbf{On}$, and a nonidentity automorphism doing the same; the ordinal-fixing examples of sources 02, 04 and 05 are its case $A = \emptyset$, and the automorphisms of 06 and 07 fixing all ordinals are its second clause. Its initial-image theorem (theorem 5.2) does not assume that simplicity is preserved (05 and 07 assume it). Its profile rigidity (theorem 11.3) is proved for any ordered exponential field with a convex valuation. It uses no elementary transfinite recursion, satisfaction class or large cardinal. Source 05 was the other candidate: it works with embeddings throughout and has the cofinal two-map uniqueness, but its model theory needs GBC with elementary transfinite recursion and its initial-image theorem assumes simplicity preservation. Everything the other sources add is printed in their Parts.

How duplicates are printed. A result proved by several sources is printed once, in the base when the base has it, and otherwise in the Part of its *home* source; each Part opens with a correspondence table that sends *every* numbered item of its source to the place where it is printed. A genuinely different proof of a printed result is kept as a marked second route in its source's Part. Notes beginning [write] were added when the report was written; the base's own text is otherwise unchanged except for label prefixes and the notes listed in Appendix C.

1.2 Notation across the sources

The base's symbols are used throughout. The member Parts rename their sources' symbols to them (each Part lists its renames); no normalization changes. The table records the letters that carry several meanings across the six sources and the neighbouring reports.

Notion	Symbol in this report	Other sources and reports
birthday	$b(x)$	bd (02, 04), ℓ (06, 07)
sign-prefix (simplicity) order	$x \preccurlyeq_s y$, strict $x \prec_s y$	
prefix map and its image	$P_p(x) = p \frown x$, image $P_p[\mathbf{No}]$	cone C_p (06, 07)
sign repetition	D_m (m an integer), D_δ (δ an ordinal)	block map B_δ (02), B (07)
interval compression	$c(x) = x/(1 + x)$	$C(x)$ (05)
displacement of σ	$D(a) = \sigma(a) - a$	D_σ (exponential-automorphism-rigidity)
additive (first) lift of an increasing f	A_f	L_f (02), T_h (06); inner lift ι_μ (omnific-preserving-automorphisms)
field lift of an additive embedding τ	H_τ	H_φ (05), F_τ (07), $M_{1,\tau}$ (opa), Φ_τ (isc)
double lift	$J_f = H_{A_f}$	E_f (02, 05); M_{1,ι_f} (opa)
weighted monomial embedding	$H_{\tau,\chi}$	$H_{\varphi,c}$ (05), $J_{g,\chi}$ (06), $J_{\tau,\rho}$ (07, opa), M (saut)
exponent dilation	$S_c = H_{a \mapsto ca}$	T_q (05), D_c (07), D_q (02), T_r (exp.-rigidity), S_a (saut)
iterated-image core	$K_\infty(j)$	Core (04)
Taylor maps	T (equation (10.3)), in general $\mathcal{T}_{d,\delta}$	T_D (04), V_d (06); Φ_d (saut); $\mathcal{T}_{d,\delta}$ (opa)
truncation	$T_{>c}$	
omega-map	$\Omega(x) = \omega^x$	W (07)

Notion	Symbol in this report	Other sources and reports
footprint; abstract exponential	$E(A)$; (F, E, w) with $v = w \circ E$	E for a double lift (02, 05)
natural valuation; value action	$v(x) = -(\text{leading exponent})$; τ_j	
large-cardinal map from $j : V \rightarrow M$	$\mathcal{J}_j$	J_j (02), J (lce)
multiplicative conjugate $\exp \circ \varphi \circ \log$	$\mathcal{M}_\varphi$	M_φ (05)
diagonal derivations	∂_λ	D_λ (05)
birthday stages	$\mathbf{No}_{<\lambda}$	N_λ (06)

Tempting false readings, each excluded in the text:

- A_f is an additive embedding, never a field embedding (unless f is additive with $f(0) = 0$, when $A_f = H_f$); even $A_f(1) = \omega^{f(0)}$.
- The letter D means three things in the base, distinguished by their arguments: D_m (sign repetition), $D(a)$ (displacement), and the derivation Δ inside the Taylor shift. $E(A)$ (a set of exponents) is not the abstract exponential E . T (the Taylor shift) is not the truncation $T_{>c}$, and S_c (a dilation) is not the Taylor map.
- J_f (double lift), J_- and J_c are field embeddings built from normal forms; $\mathcal{J}_j$ is induced by an elementary embedding of the set-theoretic universe and is not a double lift.
- $\Omega(x) = \omega^x$ is the Conway omega-map onto the monomials, not Gonshor’s exponential $\exp$; preserving one is unrelated to preserving the other.
- “Continuous” always means continuous in the full surreal order topology; “strong” means preserving set-indexed Hahn sums. Neither implies the other (corollary 9.5).
- “Initial” refers to the sign-prefix order unless “normal-form truncation” is said.

1.3 Status against the collection

None of the six sources inspected docs/surreal/omnific-preserving-automorphisms beyond its README, docs/surreal/independent-surreal-copies or docs/surcomplex/surcomplex-field-automorphisms; all six cite docs/surreal/exponential-automorphism-rigidity. About half of each re-proves collection results. The table lists the results of Part I that the collection already had; the Parts credit their own.

Here	Already in the collection
proposition 4.6 (order-and-simplicity automorphisms are trivial)	the subsection “Simplicity restores the canonical surreal numbers” of <code>saut:sec:exponential</code> (<code>surcomplex-field-automorphisms</code>); its cutoff form <code>hset:lem:rigidity</code> ; the case $E = \mathbf{No}$ of <code>isg:cf:prop:coherence</code> ; KKS Prop. 5.1
theorem 6.1 (additive lift)	<code>opa:as:lem:inner</code> with <code>opa:as:eq:innerlift</code> (the inner lift ι_μ)

Here	Already in the collection
theorem 6.2 (field lift)	<code>isc:thm:lift</code> (any ordered additive embedding), <code>opa:as:thm:classification</code> ; for automorphisms <code>prop:hahn-lift</code> of <code>exponential-automorphism-rigidity</code>
theorem 6.3 (double lift, functor, faithfulness)	the double lifts M_{1,ι_μ} of <code>omnific-preserving-automorphisms</code> (paragraph after <code>opa:as:lem:inner</code>); functor and the witness equation (6.11): <code>opa:es:prop:lifts</code> , <code>opa:es:eq:witness</code>
theorem 6.4 (weighted classification)	<code>isc:prop:classification</code> ($\chi = 1$), <code>saut:thm:monomial</code> (automorphisms), <code>opa:as:thm:classification</code> (coefficient-map version); fixed fields <code>opa:es:thm:fixedsupport</code>
theorem 7.4, theorem 7.6 (parameter copies)	<code>opa:as:thm:parameters</code> : for every set A , a bounded and a cofinal proper copy fixing $A \cup \mathbb{R}$, the bounded one built with the same compression as f_γ^+ ; the fixing of all ordinals is new here
proposition 8.2 (fixed fields real closed)	<code>opa:es:lem:realclosed</code> (for groups of automorphisms)
proposition 9.1	<code>saut:prop:discrete</code>
theorem 9.3	<code>opa:as:thm:topology</code> (valuation form)
theorem 9.4	<code>opa:as:thm:closed</code>
proposition 9.6	<code>saut:thm:zeroderiv</code> , with the dichotomy <code>saut:thm:derivative</code>
proposition 10.1	<code>opa:as:lem:natural</code> , <code>saut:lem:order</code>
Taylor shift equation (10.3)	<code>saut:thm:coeffflow</code> ; <code>opa:as:eq:taylor</code>
theorem 10.3	the subsection “The omega-map detects exact value preservation” of <code>saut:sec:exponential</code> ; KKS Prop. 5.5
lemma 11.1, corollary 11.2, theorem 11.3, theorem 11.4, theorem 11.5	<code>exponential-automorphism-rigidity</code> : its <code>lem:amplify</code> , <code>cor:bounded-displacement</code> , <code>thm:profile</code> , <code>cor:embedding</code> , <code>thm:no</code> , <code>thm:cofinal</code> (credited by the base)
proposition 12.1	<code>opa:as:thm:elementary</code>
surcomplex extensions (section 12)	<code>saut:thm:axis</code> , <code>opa:as:thm:complexembed</code> , <code>hset:thm:two-lifts</code>
homogeneity remarks	<code>saut:thm:orderedhomogeneity</code>

New to the collection, as far as a search of its reports finds: birthday rigidity and canonical-cut rigidity (theorem 4.5, proposition 4.7), surjectivity of order embeddings with simplicity-initial image (theorem 5.2), prefix and repetition maps and closed sign images (propositions 4.2 and 4.3 and theorem 4.9, the last credited to Bagayoko–van der Hoeven), proper field embeddings fixing all ordinals (theorems 7.1 and 7.4), iterated-image cores (proposition 8.4), the leading-term-fixing unit twists (proposition 10.2), the cofinal two-map uniqueness (lemma 11.6 and theorem 11.7; it strengthens `cor:two-profiles` of `exponential-automorphism-rigidity` and the Lean lemma `eq_of_profile_eq`, which need one map to be surjective), o-minimal definability rigidity (theorem 12.2) and the necessary differential conditions (proposition 13.1). The new results of the members are listed at the head of their Parts.

1.4 What is not claimed

The six sources are unrefereed AI-assisted texts. Each states that “proved here” marks where a proof is supplied, not historical priority; none supplies new Lean code or a repository build; their finite checks (sources 02–05) are regression tests on finite sign words and finite Hahn polynomials, not verification of class-sized statements. Results resting on GBC with elementary transfinite recursion and a class satisfaction relation (sources 04, 05), on an elementary embedding $j : V \rightarrow M$ (source 02), or on imported literature (source 07’s omega-series classification) are stated with those hypotheses. Each Part ends with its source’s status and non-claims, and [Appendix A](#) is the base’s ledger. Placement beside the Lean development of `Algebra/SurrealNumbers` confers no formal status; the README of this report names the declarations that formalize parts of the material.

2 The question is a family of questions

An embedding is always relative to a signature and to a category of allowable maps. In this article, an *order embedding* preserves and reflects the numerical order; a *field embedding* is unital; and a *proper embedding* is injective but not surjective. Unless a cutoff or universe is explicitly named, the domain and codomain are the entire proper class **No** in one fixed ambient set/class universe.

We use $\mathbb{N} = \{0, 1, 2, \dots\}$. The two fundamental orders must not be conflated. Numerical order is linear. Simplicity is the tree order of being an initial segment of a sign sequence. Being born earlier is weaker than being a proper sign prefix. Likewise, an image that is closed under deleting late *normal-form terms* need not be closed under deleting late *signs*.

Structure retained	Meaning	Main outcome here
Bare class carrier	Injective class map	Numerous proper maps; little arithmetic content.
Numerical order	$x < y \iff jx < jy$	Copies in any open interval; proper cofinal copies too.
Simplicity tree	Prefix preservation/reflection	Rich tree maps; branch swaps are allowed without order.
Order and simplicity	Both relations	Proper substructures exist; automorphisms are rigid.
Birthdays or ambient cuts	Nonincreasing birthdays, or canonical cut preservation	Identity.
Additive group	Additive, ordered	Hahn relabeling of exponents is sufficient.
Ordered field	$0, 1, +, \cdot, <$	Proper cofinal and bounded copies exist.
Hahn structure	Strong sums and monomial data	Explicit classification within the monomial class.
Natural valuation	Comparisons, or pointwise values	These are different requirements; neither alone gives general field rigidity.
ω -map / exponential	$j(\omega^x) = \omega^{jx} / j(\exp x) = \exp(jx)$	Distinct operations; valuation-pointwise compatibility gives rigidity.
Elementary structure	All first-order formulas of a fixed language	Automatic for ordered-field embeddings in the field language, not in arbitrary expansions.
Definability	Definable graph in a named language	O-minimal definable field self-embeddings are the identity.

Structure retained	Meaning	Main outcome here
Omnific / differential data	Oz , its floor, a chosen derivation	Extra structures require separate compatibility statements.

No finite list can exhaust all expansions of **No** by arbitrarily chosen predicates. The meaningful aim is to cover the standard structures and provide a method for analyzing additional constraints.

2.1 Relation to existing work and to the repository

The language of *surreal substructures*, developed systematically by Bagayoko and van der Hoeven, studies images isomorphic to $(\mathbf{No}, <, \prec_s)$ and provides much more than the elementary prefix examples below [4]. Our constructions in the field category use the classical normal-form and ordered-field theory of Conway and Gonshor, with the set-sized field analysis of van den Dries and Ehrlich [1, 2, 3].

Kaplan, Krapp, and Serra’s version 3 of *Decomposing the automorphism group of the surreal numbers* is the relevant recent comparison point [5]. It separates valuation, monomial, strong-linearity, exponential, and ω -map constraints. Its Question 5.4 asks about exponential 1-automorphisms; Question 5.6 concerns ω -map-preserving field automorphisms. The repository’s exponential-rigidity report supplies a finite displacement argument directed at the former question [13]. We reproduce the applicable proof and extend the comparison argument to cofinal self-embeddings, while making no priority claim and not treating the latter question as solved.

Evidence boundary. The pinned repository contains actual Lean source for universe-indexed sign sequences and generic exponential-profile lemmas. The inspected generic exponential module explicitly lists the instantiation with the actual surreal exponential as pending. This article uses conventional mathematical proofs for that bridge. The source inspection is not an independent kernel build. The new Hahn-lift, stabilizer, topology, and iteration arguments below are proved in this article, not certified by the included finite regression tests.

3 A foundation that does not manufacture paradoxes

3.1 Set-length signs and set-supported normal forms

Represent a surreal by a function $x : \alpha \rightarrow \{-, +\}$, where α is an ordinal and hence a set. Its birthday is $b(x) = \alpha$. For comparison, extend the sign function by the termination symbol 0, ordered by $- < 0 < +$. The first disagreement gives numerical order. Write $x \preceq_s y$ when x is a sign prefix of y , and $x \prec_s y$ when it is a proper prefix. The empty sequence is 0.

The canonical option sets are

$$L_x = \{x \upharpoonright \beta : \beta < b(x), x(\beta) = +\}, \quad (3.1)$$

$$R_x = \{x \upharpoonright \beta : \beta < b(x), x(\beta) = -\}. \quad (3.2)$$

They are sets, and $x = \{L_x \mid R_x\}$. More precisely, the numbers between these canonical options are exactly the sign extensions of x . For arbitrary sets $L < R$, the notation $\{L \mid R\}$ denotes the unique simplest surreal strictly between them. These standard sign and cut facts are the foundational input used below [1, 2].

Every surreal has a unique normal form

$$x = \sum_{a \in A} r_a \omega^a, \quad r_a \in \mathbb{R} \setminus \{0\}, \quad (3.3)$$

where A is a *set* reverse well-ordered by numerical order. Equivalently, its exponents form a strictly decreasing ordinal-indexed sequence. The notation $\mathbb{R}((\omega^E))$ will mean the class of such series with support contained in E . It is only an additive subgroup when E is an arbitrary subclass; if E is an ordered additive subgroup, it is a Hahn subfield.

For $x \neq 0$, let $a_0 = \max \text{supp}(x)$ and set

$$v(x) = -a_0, \quad v(0) = \infty. \quad (3.4)$$

Thus $v(\omega^a) = -a$. The valuation group is identified with $(\mathbf{No}, +, <)$, but the occurrence of $\mathbf{No}$ as value group should remain logically distinct from its occurrence as field.

Definition 3.1 (Strong summability). A *set-indexed* family $(x_i)_{i \in I}$ is Hahn-summable if the union of its normal-form supports is reverse well-ordered and every exponent occurs in only finitely many supports. The sum is then formed coefficient by coefficient. A map is strongly additive if it preserves these families and their sums; it is strongly $\mathbb{R}$ -linear if it also respects real scalar multiplication.

This is a support condition, not a limit in the full surreal order topology. In particular, an arbitrary proper-class family is not a legal Hahn sum.

3.2 Three legitimate size registers

Definable classes in ZFC. Each surreal is a set code, and a specified embedding is given by a formula, possibly with set parameters. Assertions about every definable embedding are metatheoretic schemes, not quantification over a set of all formulas with a universally available truth predicate. All explicit maps in our Hahn constructions are definable from their stated data.

Two-sorted class theory. In GBC, or in a suitable NGB presentation, an embedding may be a class graph. Class replacement ensures that its image of a set is a set. This suffices for mapping supports, taking preimages of a set inside an injective image, and performing the explicit constructions below. Global choice is useful for a global back-and-forth using a set-like enumeration; it is not needed for the displayed rational formulas and Hahn relabelings.

One may quantify over an individual class embedding, or finitely many class embeddings, in class theory. One may *not* form a class whose members are all those proper-class graphs: classes are not elements of classes. Thus “the group of all automorphisms of $\mathbf{No}$ ” is an external collection with operations, or an abbreviation for finite class-quantified statements. A genuine carrier of all such maps requires a higher sort, a larger universe, or a restricted set-coded family. KKS explicitly highlights this point in its foundational conventions [5].

Universe-indexed types. Fix a small universe U and work externally in a larger universe V . The surreals with U -small ordinal lengths form a V -set, denoted $\mathbf{No}_U$. They are not a U -set. Maps $\mathbf{No}_U \rightarrow \mathbf{No}_U$, maps $\mathbf{No}_U \rightarrow \mathbf{No}_V$, and restrictions of maps on a still larger surreal universe are different objects. An external set of signs need not be an internally available support. An external cut need not be an admissible internal cut.

This is directly reflected by the inspected repository declaration

```
structure SignSequence : Type (u + 1) where
  birthday : Ordinal.{u}
  signAt : Ordinal.{u} -> SignType
  signAt_ne_zero_iff : forall i, signAt i != 0 <-> i < birthday
```

The display uses ASCII logical notation for readability; the exact Lean source uses its usual Unicode syntax. The file proves that bounded birthday fragments are small and that the entire carrier is not small at level u [12].

3.3 What is forbidden, and what needs extra axioms

The expression $\{\mathbf{No} \mid \emptyset\}$ is not a surreal cut: its left side is a proper class. Nor is there a single surreal sign sequence of length “all ordinals.” These errors would falsely produce an element above every surreal and immediately contradict ordinary field arithmetic.

Similarly, a universal satisfaction predicate for an arbitrary proper-class structure is not supplied merely by writing the word “elementary.” One can state preservation for each fixed formula. In the ordered-field language, quantifier elimination gives an especially clean reduction to finite algebraic/order data.

Set-valued recursion along $\mathbf{On}$, with a set of earlier values at each set stage, is different from recursion that assigns an entire class at each stage. The latter leads to elementary transfinite recursion principles stronger than bare GBC in the usual class-theoretic setting [10]. We do not silently use such principles to construct the maps in this article. In particular, a set-like global back-and-forth has set-sized partial maps at every ordinal stage; its final union is a class, not a set.

Finally, a field self-embedding of $\mathbf{No}$ is not an elementary embedding of the set-theoretic universe $(V, \in)$. The underlying membership relation on set codes is not part of the ordered-field signature. Large-cardinal and Kunen-style questions about universe embeddings therefore cannot be transferred to the present field maps without a separate interpretation theorem. Coding $\mathbf{No}$ by sets does not provide such a theorem by itself.

4 Order and simplicity: flexibility and rigidity

4.1 Three explicit order embeddings

Proposition 4.1 (Interval compression). *The map*

$$c(x) = \frac{x}{1 + |x|}$$

is an increasing bijection $\mathbf{No} \rightarrow (-1, 1)$, with inverse $c^{-1}(y) = y/(1 - |y|)$. Consequently every nonempty open interval contains an order-isomorphic copy of the entire surreal class.

Proof. For $x \geq 0$, the formula is $1 - (1 + x)^{-1}$, which is strictly increasing with image $[0, 1)$. The negative half is obtained by oddness. The inverse formula is verified separately on the two signs. An affine increasing bijection carries $(-1, 1)$ to any prescribed bounded open interval; the half-line variants follow by equally elementary fractional formulas. $\square$

[write] Also in sources 02 (Proposition 3.1), 04 (Proposition 3.1), 05 (Section 3.1) and 06 (Section 2.6), with the same proof.

No cardinal paradox occurs: an interval of a proper-class field is itself a proper class. This map is not additive and is not a field embedding.

Proposition 4.2 (Prefix copies). *For a fixed sign sequence p , put $P_p(x) = p \smallfrown x$. Then P_p preserves and reflects order and simplicity. Its image is the convex class of all extensions of p , and*

$$b(P_p(x)) = b(p) + b(x),$$

where addition is ordinal addition. If $p \neq 0$, the embedding is proper.

Proof. The common prefix p postpones every first disagreement by its length. It also preserves exactly the initial-segment relations among the suffixes. The characterization of the image is immediate. If two extensions bracket another sign sequence numerically, that sequence cannot disagree with p at an earlier position, so the image is convex. The empty sequence is not in the image when p is nonempty. $\square$

[write] Also in sources 02 (Proposition 3.2), 04 (Section 3.2), 05 (Section 3.2), 06 (Proposition 2.1) and 07 (Proposition 3.1), with the same proof.

Ordinal addition matters: a nonempty prefix need not strictly increase an infinite birthday. For example $1 + \omega = \omega$.

Proposition 4.3 (Sign repetition). *For an integer $m \geq 2$, let $D_m(x)$ replace every sign of x by m consecutive copies of that sign. Then D_m is a proper, cofinal and coinital order-and-simplicity embedding fixing 0, with*

$$b(D_m(x)) = m \cdot b(x).$$

Its image is discrete in the ambient order topology and D_m is nowhere continuous.

Proof. At a first differing sign, repeated minus, termination, and repeated plus occur in the same order as the original three symbols. Prefix relations are preserved and reflected. The one-sign sequence $+$ is omitted, so the map is proper. Images of sufficiently long all-plus sequences lie above any given surreal; all-minus sequences give coinitality.

For $z = D_m(x)$, let $z^- = z \frown (-)$ and $z^+ = z \frown (+)$. The interval (z^-, z^+) contains no other point of the image: any image extending z has at least two copies of its next sign and lies beyond the corresponding endpoint. All other image points disagree earlier or terminate earlier. Thus each image point is isolated. The source has no isolated points, so continuity at any point is impossible. $\square$

[write] Source 02 extends the repetition maps to blocks of any ordinal length δ (proposition 18.4). Also in sources 04 (Section 3.2), 05 (Section 3.2), 06 (Sections 2.2, 2.8) and 07 (Proposition 3.2). Source 02 extends the repetition maps to blocks of any ordinal length δ (proposition 18.4); source 05 describes every order-and-simplicity embedding by local padding data (theorem 44.1).

For example $D_2(1) = 2$, but $D_2(1/2)$ has signs $++--$ and equals $5/4$; therefore D_2 is not additive. Also $D_m(\omega) = \omega$, since $m \cdot \omega = \omega$. These examples already show that cofinality, continuity, and birthday growth are different issues for order embeddings.

4.2 Simplicity without order

The rooted binary sign tree has many automorphisms if the order of the two children is forgotten. At each node one may decide whether to interchange the two outgoing branches, recursively. These automorphisms preserve birthdays and prefix relations but need not preserve numerical order. Global sign reversal $x \mapsto -x$ is the simplest example: it reverses numerical order and is not a unital field homomorphism. Every order-reversing self-embedding is the composite of negation with an order-preserving one.

Lemma 4.4 (Reflection follows from order). *An order embedding that preserves sign-prefix relations automatically reflects them.*

Proof. If $y \prec_s x$, preservation and injectivity preclude $jx \preccurlyeq_s jy$. If x, y are incomparable, their longest common prefix p lies strictly between them numerically, after interchanging x, y if needed. The images

extend jp and lie on opposite sides of it. They therefore depart along opposite branches and are incomparable. These exhaust the alternatives to $x \preccurlyeq_s y$. $\square$

[write] Source 05 derives the same statement from its padding description (theorem 44.1).

4.3 Rigidity statements with different hypotheses

Theorem 4.5 (Birthday rigidity). *If $j : \mathbf{No} \rightarrow \mathbf{No}$ is an order embedding and $b(jx) \leq b(x)$ for every x , then $j = \text{id}$.*

Proof. Induct on $\alpha = b(x)$. Every canonical option of x has smaller birthday and is fixed by induction. Therefore jx lies between L_x and R_x , so it extends x . Its birthday cannot exceed α by hypothesis; hence it is exactly x . $\square$

[write] Not in the collection before this report. Proved independently, by the same induction, in sources 02 (Theorem 4.2), 04 (Theorem 3.2), 05 (Theorem 4.1), 06 (Theorem 2.4) and 07 (Theorem 3.7).

Proposition 4.6. *Every simplicity embedding satisfies $b(jx) \geq b(x)$. Every order-and-simplicity automorphism of $\mathbf{No}$ is the identity.*

Proof. The proper prefixes of x form a chain of order type $b(x)$; their images form a strictly increasing chain of proper prefixes of jx . Consequently $b(x) \leq b(jx)$. For an automorphism, apply the same argument to its inverse to obtain equality of birthdays, then use theorem 4.5. $\square$

[write] The automorphism clause is already in the collection: the subsection “Simplicity restores the canonical surreal numbers” of `saut:sec:exponential` in `docs/surcomplex/surcomplex-field-automorphisms` proves it by the same induction, `hset:lem:rigidity` is its form for a birthday cutoff, and the uniqueness clause of `isg:cf:prop:coherence` gives it as the case $E = \mathbf{No}$. It is KKS Proposition 5.1. Also in sources 02 (Lemma 4.1, Corollary 4.3), 04 (Theorem 3.4), 05 (Corollaries 3.2, 4.3), 06 (Lemma 2.3, Corollary 2.5) and 07 (Corollary 3.8).

This is compatible with the proper maps P_p and D_m : an embedding into an image is not an automorphism of the whole tree.

Proposition 4.7 (Canonical construction rigidity). *An order embedding preserving ambient canonical cuts,*

$$j\{L \mid R\} = \{j[L] \mid j[R]\}$$

for every admissible set cut, is the identity. It is enough to require this for the canonical cuts of individual surreals.

Proof. Transfinite induction fixes all options and then fixes the simplest number between them. The empty cut fixes 0 at the first step. $\square$

[write] Also in sources 02 (Theorem 4.4), 04 (Corollary 3.3), 05 (Proposition 4.4), 06 (Section 2.6) and 07 (Section 3.5), with the same proof.

Equivalently, preserving the root, the two labeled immediate successors, and unions of all set-sized prefix chains forces the identity. Omitting limit-chain preservation leaves room for extra signs at limit birthdays. Preservation only for countable chains does not justify a conclusion at arbitrary uncountable birthdays.

Remark 4.8 (Relative cuts are not ambient cuts). If S is the image of an order-and-simplicity embedding, the image of a cut is the simplest suitable point *inside* S . It need not be the simplest suitable point in all of **No**. This relative notion is central to surreal substructures [4]; replacing it with the ambient notion would incorrectly rule out the elementary prefix examples.

4.4 Closed substructures and their fixed points

The rigidity of automorphisms implies uniqueness of the parameterization of any given surreal substructure: compose two proposed parameterizations with one another's inverses. Monomials and purely infinite numbers supply further standard substructure examples beyond the prefix cones; their substructure parameterizations are not thereby field embeddings [2, 4].

There is also a useful fixed-point theorem under an explicit closure hypothesis. The following gives a direct proof of this basic closed- substructure phenomenon, part of the broader theory in [4].

Theorem 4.9 (Closed sign images have a full fixed-point substructure). *Let j be an order-and-simplicity embedding. Suppose its image is closed under unions of nonempty set-sized sign-prefix chains. Then j commutes with those unions, and $\text{Fix}(j)$ is itself a surreal substructure.*

Proof. Let C be a nonempty set-sized prefix chain and $x = \bigcup C$. By the image-closure assumption, $y = \bigcup j[C]$ lies in the image. Its preimage is an upper bound of C , while jx is an upper bound of $j[C]$. The two least-upper-bound properties give $jx = y$.

Whenever $u \preccurlyeq_s ju$, put

$$c_j(u) = \bigcup_{n < \omega} j^n(u).$$

This is fixed by union preservation. It is the least fixed sign extension of u : if $u \preccurlyeq_s z = jz$, induction gives $j^n u \preccurlyeq_s z$ for every n . In particular $c_j(0)$ is the least fixed point in simplicity. If x is fixed, then $u = x \smallfrown (\pm)$ satisfies $u \preccurlyeq_s ju$. Indeed ju extends x and lies on the same numerical side of x as u . Thus $c_j(x \smallfrown (-))$ and $c_j(x \smallfrown (+))$ are the least fixed extensions on the two respective branches.

Construct a map Φ from sign sequences by sending the empty sequence to $c_j(0)$, adjoining a sign by the corresponding least-fixed-extension operation, and taking unions at limit lengths. All these unions are sets; union preservation by j keeps every output fixed. At each successor step the image strictly extends its predecessor on the specified side. Hence Φ preserves and reflects numerical order and simplicity.

To see that its image is all of $\text{Fix}(j)$, take a fixed z . Its least fixed prefix is $c_j(0)$. As long as the constructed prefix is not z , follow the next sign of z and then take the least fixed extension on that branch. This remains a prefix of z . At limits take unions. The process must reach z : otherwise it would produce more than $b(z)$ strict prefix extensions inside the set-length sequence z . Thus Φ is onto the fixed-point class. $\square$

[write] Source 05's least fixed point of a chain-continuous tree map (proposition 44.3) is the fixed-point part of this theorem, and the fixed-point class is the intersection of the iterated images (theorem 55.8, after Bagayoko–van der Hoeven).

For example, D_m commutes with sign-chain unions and its image is closed under them, so its fixed points form a surreal substructure. Yet D_m is nowhere continuous in the numerical order topology by proposition 4.3. The two kinds of continuity are not interchangeable.

5 Initial images and the distinction between two truncations

Definition 5.1. A subclass $S \subseteq \mathbf{No}$ is *simplicity-initial* if $x \in S$ and $y \preccurlyeq_s x$ imply $y \in S$. A normal-form truncation discards a final segment of the decreasing list of exponents in (3.3). These two operations are unrelated definitions and must be tracked separately.

Theorem 5.2 (An initial full-sized order copy is the whole class). *If an order embedding $j : \mathbf{No} \rightarrow \mathbf{No}$ has simplicity-initial image S , then $S = \mathbf{No}$. Simplicity preservation by j is not required.*

Proof. The order S inherits interpolation for every set cut: take preimages under j , interpolate in $\mathbf{No}$, then apply j . Suppose $S \neq \mathbf{No}$, and choose an omitted x of least birthday. Every proper prefix of x belongs to S . In particular $L_x, R_x \subseteq S$. Interpolation in S gives $y \in S$ with $L_x < y < R_x$. Such a y extends x . Initiality then gives $x \in S$, a contradiction. $\square$

[write] With simplicity preservation added, the conclusion “identity” is also the uniqueness clause of `isg:cf:prop:coherence` for $E = \mathbf{No}$. Sources 05 and 07 prove only that version. Also in sources 02 (Theorem 4.6), 04 (Theorem 3.5, Corollary 3.6) and 06 (Theorem 2.6), with the same proof. Sources 05 and 07 prove the version with simplicity preserved, concluding identity, by other routes (corollary 44.6; section 65).

The conclusion is surjectivity, not identity. A nonidentity ordered-field automorphism has image $\mathbf{No}$, which is certainly initial. Adding simplicity preservation gives identity by [proposition 4.6](#).

This theorem does not conflict with initial embeddings of set-sized ordered fields into $\mathbf{No}$ [7]. A set-sized field need not interpolate every set cut of its underlying order. The full class $\mathbf{No}$ does, and that extra property drives the proof.

Proposition 5.3 (A proper subfield is not convex). *A convex subfield of an ordered field is the whole field. More generally, a subfield containing a nonempty open interval is the whole field.*

Proof. For the first statement, if $a > 0$ lies above the convex subfield, then $0 < a^{-1} < 1$ puts a^{-1} in the subfield, hence also a . For the second, subtract two points of the interval to obtain an interval $(-\delta, \delta)$ inside the subfield. For arbitrary x in the ambient field choose $0 < t < \delta/(1 + |x|)$. Both t and tx lie in that interval, so $x = (tx)/t$ belongs to the subfield. $\square$

[write] Also in sources 02 (Proposition 4.7) and 04 (Proposition 3.7).

Thus the convex prefix copies and interval copies are inherently different from proper field copies. The field images constructed next are closed under normal-form truncation, but by [theorem 5.2](#) they cannot be simplicity-initial.

6 Hahn relabeling and a double-lift construction

The principal construction is deliberately modular. Relabeling the exponents of a surreal needs only an order embedding and produces an *additive* embedding. Relabeling exponents by an *additive* embedding then produces a field embedding. Keeping these two levels separate prevents a common false claim that every monotone change of exponent preserves multiplication.

6.1 The additive lift of an order embedding

Let $f : \mathbf{No} \rightarrow \mathbf{No}$ be an increasing class embedding. Define

$$A_f \left(\sum_a r_a \omega^a \right) = \sum_a r_a \omega^{f(a)}. \quad (6.1)$$

The sum on the right is legal: the image of its support is a set and an order-isomorphic copy of a reverse well-order.

Theorem 6.1 (Additive lift). *The map A_f is a strongly $\mathbb{R}$ -linear ordered additive-group embedding. It preserves and reflects Hahn summability, and*

$$\text{im } A_f = \mathbb{R}((\omega^{f(\mathbf{No})})), \quad (6.2)$$

$$A_f \circ A_g = A_{f \circ g}, \quad (6.3)$$

$$\text{Fix}(A_f) = \mathbb{R}((\omega^{\text{Fix}(f)})). \quad (6.4)$$

It is surjective exactly when f is surjective, and its image is cofinal exactly when $f(\mathbf{No})$ is cofinal.

Proof. Addition and real scalar multiplication are coefficientwise. An increasing injection preserves the largest differing exponent and its coefficient, so it preserves order and is injective. The two summability conditions are invariant under an injective order relabeling: support order type is unchanged, as is the number of indices contributing to a coefficient. Equation (6.2) follows by pulling back a support contained in $f(\mathbf{No})$ under the inverse class map on that image. Class replacement makes the pulled-back support a set. Composition is immediate termwise.

If $A_f(x) = x$, its support B satisfies $f[B] = B$. The restriction of f is an order automorphism of the reverse well-order B , hence is the identity on B . Conversely, support in $\text{Fix}(f)$ gives a fixed series. This proves (6.4) without assuming f is onto on the whole class.

Surjectivity follows from the image formula; an omitted exponent omits its monomial. If $f(\mathbf{No})$ is cofinal, images of monomials dominate any given normal form. If it is not cofinal, choose a strict upper bound b for it. Every nonzero $A_f(x)$ has largest exponent below b , so $|A_f(x)| < \omega^b$. Its image is bounded. $\square$

[write] This is the inner lift of docs/surreal/omnific-preserving-automorphisms: its opa:as:lem:inner, with $A_f = \iota_f$ of opa:as:eq:innerlift (image, composition, preservation of the leading coefficient). Its fixed-point formula is opa:es:thm:fixedsupport, stated there for order automorphisms and groups of them; the proof above for an injective f is the same support argument. Also in sources 02 (Theorem 5.2), 04 (Theorem 4.1), 05 (Proposition 5.1), 06 (Theorem 3.4) and 07 (Theorem 4.3); source 02 adds the converse that A_f is multiplicative only when f is additive (proposition 19.1).

Even the image of 1 changes in general: $A_f(1) = \omega^{f(0)}$. The symbol A_f should never be called a field embedding unless the additional multiplicative conditions have been verified.

6.2 The field lift of an additive embedding

Let $\tau : (\mathbf{No}, +, <) \rightarrow (\mathbf{No}, +, <)$ be an ordered additive-group embedding. Define

$$H_\tau \left(\sum_a r_a \omega^a \right) = \sum_a r_a \omega^{\tau(a)}. \quad (6.5)$$

Theorem 6.2 (Field lift). *The map H_τ is a unital ordered-field embedding, fixes $\mathbb{R}$ pointwise, and is strongly $\mathbb{R}$ -linear. It preserves and reflects Hahn summability. Moreover,*

$$\text{im } H_\tau = \mathbb{R}((\omega^{\tau(\mathbf{No})})), \quad (6.6)$$

$$H_\tau H_\rho = H_{\tau \circ \rho}, \quad (6.7)$$

$$v(H_\tau x) = \tau(vx) \quad (x \neq 0), \quad (6.8)$$

$$\text{Fix}(H_\tau) = \mathbb{R}((\omega^{\text{Fix}(\tau)})). \quad (6.9)$$

Surjectivity and cofinality are respectively equivalent to surjectivity and cofinality of τ .

Proof. The additive-lift proof applies to addition and supports. Since $\tau(0) = 0$, constants are fixed. For multiplication, the identity $\tau(a + b) = \tau(a) + \tau(b)$ carries every convolution term $r_a s_b \omega^{a+b}$ to the corresponding term of the product of the images. The usual finiteness of contributions to a Hahn product coefficient is unchanged. Thus multiplication is preserved. Injectivity then implies preservation of inverses, because $H_\tau(x)H_\tau(x^{-1}) = 1$.

The image, composition, fixed-point, and cofinality statements follow by the same support arguments as above. Finally, if a is the leading exponent of x , then $v(H_\tau x) = -\tau(a) = \tau(-a) = \tau(vx)$. $\square$

[write] This is `isc:thm:lift` of `docs/surreal/independent-surreal-copies` (the monomial lift Φ_h of any ordered additive embedding, with the same image), and the outer lift $M_{1,\tau}$ of `opa:as:thm:classification`; for automorphisms it is `prop:hahn-lift` of `docs/surreal/exponential-automorphism-rigidity`. In Lean the lift is formalized only for ordered additive *equivalences* (`exponentAutomorphism`, see the README). Also in sources 05 (Theorem 5.2), 06 (Theorem 3.5) and 07 (Theorem 4.3).

The image is a real-closed field, because it is isomorphic to $\mathbf{No}$. This uses the real closedness of the surreal field, not an unsupported assertion that arbitrary birthday cutoffs are fields. Each image is also closed under all normal-form truncations.

Theorem 6.3 (Double lift). *For an increasing class embedding $f : \mathbf{No} \rightarrow \mathbf{No}$, put*

$$J_f = H_{A_f}. \quad (6.10)$$

Then J_f is a strongly $\mathbb{R}$ -linear ordered-field self-embedding, and

$$J_f J_g = J_{f \circ g}, \quad J_{\text{id}} = \text{id}.$$

The assignment is faithful: $J_f = J_g$ implies $f = g$. It preserves and reflects surjectivity and cofinality.

Proof. The first lift is an ordered additive embedding, so the second is a field embedding. Both composition laws give the displayed identity. Faithfulness can be read off from a single family of test elements:

$$J_f(\omega^{\omega^a}) = \omega^{\omega^{f(a)}}. \quad (6.11)$$

Injectivity of the ω -map twice recovers $f(a)$. The remaining assertions follow by applying the image criteria for the two lifts. $\square$

[write] The double lift is M_{1,ι_f} of `docs/surreal/omnific-preserving-automorphisms` (paragraph after `opa:as:lem:inner`); functoriality, faithfulness and the test element equation (6.11) are `opa:es:prop:lifts` and `opa:es:eq:witness` there, for order automorphisms. Also in sources 02 (Theorem 5.4), 04 (Theorem 4.4) and 05 (Theorem 5.3), with the same witness.

The construction converts an order-embedding problem into a field-embedding problem without pretending that the original f was additive. Statements about the “monoid” of all such maps are understood in the external or finite-class-quantified sense of section 3. The formulas also give ordinary monoid homomorphisms on any legitimate set-coded family closed under composition.

6.3 Classification with real weights

There is a useful exact classification within a specified strong monomial class. It should not be mistaken for a classification of every field embedding of $\mathbf{No}$.

Theorem 6.4 (Weighted monomial classification). *Let τ be an ordered additive embedding of $\mathbf{No}$, and let $\chi : (\mathbf{No}, +) \rightarrow (\mathbb{R}_{>0}, \cdot)$ be a group homomorphism. Then*

$$H_{\tau, \chi} \left(\sum_a r_a \omega^a \right) = \sum_a r_a \chi(a) \omega^{\tau(a)} \quad (6.12)$$

is a strongly $\mathbb{R}$ -linear ordered-field embedding. Conversely, every strongly $\mathbb{R}$ -linear field embedding sending every monomial ω^a to a positive real multiple of a monomial has this unique form. Its image is $\mathbb{R}((\omega^{\tau(\mathbf{No})}))$, and

$$\text{Fix}(H_{\tau, \chi}) = \mathbb{R}((\omega^{\text{Fix}(\tau) \cap \ker \chi})). \quad (6.13)$$

Here $\ker \chi = \{a : \chi(a) = 1\}$. Composition is given by

$$(\tau_1, \chi_1)(\tau_2, \chi_2) = (\tau_1 \tau_2, a \mapsto \chi_2(a) \chi_1(\tau_2(a))).$$

Proof. The character law and additivity of τ prove multiplicativity on monomials; strong sums extend it to all products. Positive coefficients preserve the sign of the leading term. Conversely, unique normal forms of the images of monomials determine $\tau(a)$ and $\chi(a)$. Multiplication of monomials forces additivity and the character law. The strict order of Archimedean classes of monomials forces τ to be an order embedding. Strong real linearity determines the map on every series.

Every series supported on $\tau(\mathbf{No})$ is in the image: pull back each exponent and divide its real coefficient by the corresponding nonzero $\chi(a)$. For fixed points, equality of supports again makes τ the identity on the support. Equality of nonzero coefficients then requires $\chi(a) = 1$ at each exponent. The composition formula follows by applying the two maps to ω^a . $\square$

[write] The case $\chi = 1$ is `isc:prop:classification`; for automorphisms the weighted maps are the real-axis case of `saut:thm:monomial`, with the same composition law; `opa:as:thm:classification` classifies the coefficient-map version, and `opa:es:thm:fixedsupport` computes the fixed fields for automorphisms. Also in sources 02 (Theorems 5.7, 7.1), 04 (Theorems 4.7, 5.2), 05 (Theorem 9.1, Lemma 9.2), 06 (Theorem 3.7) and 07 (Section 4.2, Proposition 4.5).

General strong embeddings can send a monomial to a monomial times a nonconstant unit, and general field embeddings need not be strong at all. The theorem makes no claim about those larger classes. The full surreal-field and Hahn-field automorphism literature includes such distinctions [5, 6].

7 Proper embeddings fixing every real and ordinal

7.1 A completely explicit example

Define the increasing map

$$f_-(a) = \begin{cases} \frac{a}{1-a}, & a < 0, \\ a, & a \geq 0. \end{cases} \quad (7.1)$$

It maps the negative half-line bijectively onto $(-1, 0)$ and fixes the nonnegative half-line. Therefore it is an order isomorphism $\mathbf{No} \rightarrow (-1, \infty)$, proper and cofinal. Let

$$\tau_- = A_{f_-}, \quad J_- = H_{\tau_-}.$$

Theorem 7.1 (An ordinal-fixing proper field copy). *The field embedding J_- is proper and cofinal, fixes every real and every ordinal, and preserves all Hahn-summable families and all normal-form truncations. It moves the explicit omnific integer*

$$J_-(\omega^{\omega^{-1}}) = \omega^{\omega^{-1/2}}.$$

The element $\omega^{\omega^{-1}}$ does not belong to its image.

Proof. Properness and cofinality follow from theorem 6.3. Real constants are fixed by every field lift. The Cantor normal form of an ordinal has ordinal exponents and is its Conway normal form. Since f_- fixes every nonnegative exponent, A_{f_-} fixes every ordinal. Applying the outer lift now fixes the Cantor normal form of every ordinal as well.

The displayed motion follows from $f_-(-1) = -1/2$. The exponent -1 is omitted by f_- , so ω^{-1} is omitted by A_{f_-} . Consequently the outer monomial $\omega^{\omega^{-1}}$ is omitted by J_- . $\square$

[write] Fixing every ordinal is new to the collection; the closest earlier statement, `opa:as:thm:parameters`, fixes a set and all reals. The same map is in sources 04 (Theorem 4.6) and 05 (Theorem 6.2); source 02 uses a shift map instead (example 19.5).

There is no contradiction between “omitted” and “moved”: an element of the domain need not belong to the range, although it certainly has an image. Also, $J_-(\omega^{-1}) = \omega^{-1}$, because A_{f_-} fixes the constant -1 . The map does not merely replace every visible occurrence of ω by a new base; the level of each normal-form exponent matters.

7.2 Fixing an arbitrary prescribed set

Definition 7.2 (Two-level exponent footprint). For a set $A \subseteq \mathbf{No}$, define the set

$$E(A) = \bigcup_{x \in A} \bigcup_{a \in \text{supp}(x)} \text{supp}(a). \quad (7.2)$$

This consists of the exponents occurring in the normal forms of the exponents occurring in members of A .

The expression is a set by replacement and union. That elementary size observation makes the following stabilizer theorem possible.

Lemma 7.3 (Footprint criterion). *If f fixes $E(A)$ pointwise, then J_f fixes A pointwise.*

Proof. For every outer exponent a in a member of A , the support of a is fixed by f . Thus $A_f(a) = a$. The outer field lift then leaves every normal form in A unchanged. $\square$

[write] This is `opa:as:lem:parameters` of `docs/surreal/omnific-preserving-automorphisms` (two-level support control).

Theorem 7.4 (Large stabilizers, including proper maps). *For every set $A \subseteq \mathbf{No}$, there exists a proper cofinal strongly $\mathbb{R}$ -linear field embedding fixing $A \cup \mathbb{R} \cup \mathbf{On}$ pointwise. There also exists a nonidentity strongly $\mathbb{R}$ -linear field automorphism fixing that same class pointwise.*

Proof. Choose $\beta < 0$ below every member of the set $E(A)$, using set-cut interpolation. Define

$$f_\beta(a) = \begin{cases} \beta + \frac{a - \beta}{1 - (a - \beta)}, & a < \beta, \\ a, & a \geq \beta. \end{cases} \quad (7.3)$$

This is an increasing bijection from **No** onto $(\beta - 1, \infty)$. It fixes $E(A)$ and every nonnegative exponent. The footprint criterion fixes A ; the ordinal argument of theorem 7.1 fixes **On**; and theorem 6.3 supplies properness and cofinality of J_{f_β} .

For the automorphism, replace the lower branch by an affine dilation:

$$g_\beta(a) = \begin{cases} \beta + 2(a - \beta), & a < \beta, \\ a, & a \geq \beta. \end{cases} \quad (7.4)$$

This is a nonidentity order automorphism of **No** with the same fixed upper tail. Its double lift is a nonidentity field automorphism by faithfulness, with all the required fixed points. $\square$

[write] The embedding clause extends the cofinal copy g_A of `opa:as:thm:parameters` (which fixes $A \cup \mathbb{R}$ but moves large ordinals) by fixing **On** as well; the automorphism clause has no earlier counterpart in the collection, whose parameter-fixing automorphisms (`isc:cor:parameters` gives embeddings) do not fix all ordinals. Sources 06 (Proposition 4.1, with its own witness, example 57.1) and 07 (Theorem 4.7) prove automorphisms fixing all reals and all ordinals, 07 also any prescribed set: these are the second clause here, with proposition 8.1.

Corollary 7.5 (No set of extra parameters is a rigidity base). *Even after all reals and all ordinals are fixed, no set of additional surreal parameters forces every strongly $\mathbb{R}$ -linear monomial field self-embedding to be the identity, or even to be surjective.*

The fixed class **On** is itself proper. The statement is therefore stronger than simply observing that a saturated object has nontrivial automorphisms over a small set.

7.3 Bounded field copies fixing prescribed data

Theorem 7.6 (Bounded copies). *For every set $A \subseteq \mathbf{No}$, there is a strongly $\mathbb{R}$ -linear field self-embedding fixing $A \cup \mathbb{R}$ whose entire image is bounded in **No**. It can also be chosen to fix ω .*

Proof. Choose $\gamma > 0$ above every member of $E(A)$, and put

$$f_\gamma^+(a) = \begin{cases} a, & a \leq \gamma, \\ \gamma + \frac{a - \gamma}{1 + (a - \gamma)}, & a > \gamma. \end{cases} \quad (7.5)$$

The range is $(-\infty, \gamma + 1)$. Therefore for every x ,

$$|A_{f_\gamma^+}(x)| < \omega^{\gamma+1}.$$

The exponents of every image under $J_{f_\gamma^+}$ are consequently below $\omega^{\gamma+1}$, giving the common strict bound

$$|J_{f_\gamma^+}(y)| < \omega^{\omega^{\gamma+1}} \quad (y \in \mathbf{No}). \quad (7.6)$$

The footprint criterion fixes A . Constants are fixed, and $f_\gamma^+(0) = 0$ gives $A_{f_\gamma^+}(1) = 1$, hence $J_{f_\gamma^+}(\omega) = \omega$. $\square$

[write] This is the bounded copy f_A of `opa:as:thm:parameters`: the same compression of the upper tail (μ^{cmp} there is f_γ^+ here), with the annulus bound $\Omega^{-1} < |y| < \Omega$ there. Fixing ω is the extra clause here. Bounded copies are also in sources 02 (J_c , theorem 19.3), 04 (the same J_c with the bound ω^ω , theorem 32.1), 05 (the annulus $\omega^{-1} < |j(x)| < \omega$, theorem 45.3), 06 (Theorem 4.3) and 07 (Theorem 5.2).

A bounded subfield is not an Archimedean subfield. It can contain infinite elements, all their inverses, and arbitrarily complicated prescribed set families, while a still larger ambient surreal bounds the entire class image. Such an embedding cannot fix every ordinal, since the ordinals are cofinal in **No**.

8 Omnific floors, fixed fields, and iteration

8.1 Omnific integrality is preserved and reflected

Conway's omnific integers have the normal-form description

$$\mathbf{Oz} = \{x : \text{supp}(x) \subseteq \mathbf{No}_{\geq 0}, \quad [x]_0 \in \mathbb{Z}\}, \quad (8.1)$$

where $[x]_0$ is the coefficient of ω^0 , taken as zero when absent. Every surreal has a unique omnific floor a with $a \leq x < a + 1$. These are the usual omnific conventions, also used in the repository's constructed omnific ring [1, 11].

Proposition 8.1 (Omnific compatibility). *Every weighted monomial embedding $H_{\tau, \chi}$ satisfies*

$$x \in \mathbf{Oz} \iff H_{\tau, \chi}(x) \in \mathbf{Oz}, \quad H_{\tau, \chi}([x]_{\mathbf{Oz}}) = [H_{\tau, \chi}(x)]_{\mathbf{Oz}}.$$

It preserves the constant coefficient and commutes with normal-form truncations at corresponding support positions.

Proof. An additive ordered embedding fixes zero and preserves and reflects the sign of every exponent. Since $\chi(0) = 1$, the constant coefficient is unchanged. This proves the membership equivalence. If a is the omnific floor of x , apply the ordered field embedding to $a \leq x < a + 1$. The image of a is omnific, and uniqueness of the floor gives the formula. Termwise relabeling plainly commutes with deleting a final support segment. $\square$

[write] For automorphisms this is in `opa:es:prop:lifts` (preservation of $\mathbf{Oz}$, its constant term and the floor), and for the monomial lifts Φ_h in `isc:prop:arithmeticlift`. Also in sources 02 (Proposition 8.1), 04 (Proposition 11.1), 05 (Theorem 11.2) and 07 (Section 4.3).

This includes the negative-infinitesimal boundary: the floor of a negative infinitesimal is -1 , not zero. The proof does not rely on ignoring that boundary. Preserving $\mathbf{Oz}$ as a predicate is much weaker than fixing it pointwise: the moved element in [theorem 7.1](#) belongs to $\mathbf{Oz}$.

8.2 Fixed fields do not determine surjectivity

Proposition 8.2 (General fixed-field facts). *The fixed field of any ordered-field self-embedding of $\mathbf{No}$ is real closed. No point has a nontrivial finite orbit, and $\text{Fix}(j^n) = \text{Fix}(j)$ for every positive integer n .*

Proof. The fixed points form a subfield. For a polynomial with fixed coefficients, its finite set of roots in $\mathbf{No}$ is carried injectively into itself. The induced map is a strictly increasing permutation of a finite ordered set, so fixes every root. In particular positive square roots and a root of any odd-degree polynomial lie in the fixed field, proving real closedness. If $jx > x$, its forward iterates are strictly increasing; if $jx < x$, they are strictly decreasing. Neither case gives a periodic point. $\square$

[write] Real closedness of fixed fields of groups of automorphisms is `opa:es:lem:realclosed`; the embedding case above is the same root-permutation argument. Also in sources 02 (Proposition 5.8), 04 (Proposition 5.7), 05 (Proposition 11.3) and 07 (Proposition 4.1).

Let

$$B = \mathbb{R}((\omega^{\mathbf{No}_{\geq 0}})). \quad (8.2)$$

This is the additive subgroup of purely infinite parts plus real constants. It need not be confused with $\mathbf{Oz}$, whose constant term must be integral. Since $\text{Fix}(f_-) = \mathbf{No}_{\geq 0}$, the two lift formulas give

$$\text{Fix}(\tau_-) = B, \quad \text{Fix}(J_-) = \mathbb{R}((\omega^B)). \quad (8.3)$$

Now take $g(a) = 2a$ for $a < 0$ and $g(a) = a$ for $a \geq 0$. Its double lift is an automorphism, but it has exactly the same fixed field $\mathbb{R}((\omega^B))$ as the proper embedding J_- . Consequently neither this fixed field nor its inclusion of all ordinals can detect whether an embedding is onto.

8.3 The largest reversible core

Definition 8.3. For an injective field endomorphism j , define its iterated-image core by

$$K_\infty(j) = \bigcap_{n < \omega} j^n(\mathbf{No}). \quad (8.4)$$

This is an intersection specified by quantification over natural numbers, not a set containing proper-class fields as elements.

Proposition 8.4. *The restriction of j to $K_\infty(j)$ is an automorphism. This is the largest subfield on which j restricts to an automorphism. The core is real closed and can strictly contain $\text{Fix}(j)$.*

Proof. Forward invariance is immediate. For $x \in K_\infty(j)$ let $j(y) = x$. Since $x \in j^{n+1}(\mathbf{No})$ for every n , injectivity implies $y \in j^n(\mathbf{No})$ for every n . Thus y lies in the core, proving surjectivity of the restriction. A subfield on which j is bijective lies in every iterated image, proving maximality.

Each iterated image is real closed. Their intersection is real closed: for a polynomial over the intersection, all of its real roots in $\mathbf{No}$ belong to each image field, since a real-closed subfield has no proper ordered algebraic extension inside $\mathbf{No}$. Thus the positive square-root and odd-degree-root tests hold in the intersection. Strict containment is shown in the next example. $\square$

[write] Not in the collection before this report. Source 04 proves the same (its Proposition 5.5) and computes the core of every double lift (theorem 32.3).

If j is proper, its finite iterated images are strictly decreasing: $j^n(\mathbf{No}) = j^{n+1}(\mathbf{No})$ would, by injectivity of j^n , imply $\mathbf{No} = j(\mathbf{No})$. Nevertheless the hierarchy stabilizes at the countable intersection because $j(K_\infty) = K_\infty$. No stronger transfinite recursion is needed for this stabilization.

For a pure monomial lift,

$$K_\infty(H_\tau) = \mathbb{R}((\omega^{\bigcap_{n < \omega} \tau^n(\mathbf{No})})). \quad (8.5)$$

Indeed, membership is exactly the requirement that every exponent belong to every iterated exponent image.

For f_- , direct iteration gives

$$f_-^n(a) = \frac{a}{1 - na} \quad (a < 0), \quad f_-^n(\mathbf{No}) = (-1/n, \infty) \quad (n \geq 1).$$

Let

$$C = \bigcap_{n \geq 1} (-1/n, \infty), \quad G = \mathbb{R}((\omega^C)).$$

The class C consists of nonnegative surreals together with negative real-infinitesimals. Although C itself need not be an additive group, G is an additive subgroup: supports in any fixed class are closed under addition and negation of series. The formulas for the lifts give

$$K_\infty(J_-) = \mathbb{R}((\omega^G)). \quad (8.6)$$

This core is larger than $\mathbb{R}((\omega^B)) = \text{Fix}(J_-)$. For example, the exponent $a = \omega^{-1/\omega}$ lies in $G \setminus B$. Hence ω^a lies in the core but is not fixed. A map can therefore act reversibly and nontrivially on its core while being proper on the entire class.

9 Topology: cofinality, discrete copies, and closed images

Throughout this section, the order topology is understood through interval neighborhoods in the *full* surreal class. We do not form a set of all open subclasses. At a universe cutoff, the internally available neighborhoods may be fewer, so an external ambient subspace topology can be strictly finer than an internal order topology.

9.1 Why set-indexed limits are misleading

Proposition 9.1 (Set-sized subsets are closed and discrete). *Every set $A \subseteq \mathbf{No}$ is closed and discrete in the full surreal order topology. Every convergent set-indexed net is eventually constant.*

Proof. Fix $x \in \mathbf{No}$. The nonzero distances $|x - a|$ for $a \in A \setminus \{x\}$ form a set of positive surreals. By interpolating the cut between 0 and that set, choose $\delta > 0$ below every such distance. Then $(x - \delta, x + \delta)$ meets A in at most $\{x\}$. This isolates points of A and gives a neighborhood disjoint from A at every point outside A . For a set-indexed net, apply the same argument to its set of values and its proposed limit. The isolating neighborhood forces eventual constancy. $\square$

[write] This is `saut:prop:discrete` of `docs/surcomplex/surcomplex-field-automorphisms`, proved there for the surcomplex field. Also in sources 02 (Proposition 9.3), 04 (Proposition 7.2), 06 (Proposition 2.8) and 07 (Proposition 5.3); source 05 adds a uniform radius (theorem 46.1).

The proposition is not a statement that every subclass is discrete. The full class $\mathbf{No}$ is densely ordered. It says that a set-sized probe cannot capture every infinitesimal scale of the ambient class.

Corollary 9.2 (Existing set suprema are trivial). *If a nonempty set $A \subseteq \mathbf{No}$ has a supremum in $\mathbf{No}$, then it has a maximum, equal to that supremum. Thus every order embedding preserves suprema of nonempty sets whenever those suprema exist in $\mathbf{No}$.*

Proof. If u is a supremum not attained in A , choose $\delta > 0$ below all positive distances $u - a$. Then $u - \delta$ is still an upper bound, a contradiction. A maximum is preserved by any order embedding. $\square$

Accordingly, “preserves all existing set suprema” is far weaker here than ambient simplest-cut preservation. A simplest surreal between two sets is not generally their numerical supremum or infimum.

9.2 The field-embedding dichotomy

For a general order embedding, continuity at x requires its images of the left and right neighborhoods to approach jx from the corresponding sides. Cofinality of the entire image says nothing by itself about this local condition, as D_m shows. Field structure changes the answer.

Theorem 9.3 (Cofinality is exactly continuity for field embeddings). *For an ordered-field self-embedding j of $\mathbf{No}$, the following are equivalent:*

- (i) $j(\mathbf{No})$ is cofinal in $\mathbf{No}$;
- (ii) j is continuous at 0;
- (iii) j is continuous everywhere;
- (iv) j is a homeomorphism onto its image with the ambient subspace topology.

If these conditions fail, the image is discrete in $\mathbf{No}$ and j is nowhere continuous.

Proof. A cofinal subfield has positive elements arbitrarily close to zero, by inversion. Given $\varepsilon > 0$, choose $\delta > 0$ in the domain with $0 < j\delta < \varepsilon$. Then $|h| < \delta$ implies $|jh| < \varepsilon$, proving continuity at zero. Additivity translates this to every point.

Conversely, continuity at zero provides arbitrarily small positive image elements: for a neighborhood of radius ε , use a positive $h < \delta$ in a source neighborhood mapped into it. Inverses of these positive image elements are cofinal.

The inverse from the image is always continuous: a source interval $(x - \delta, x + \delta)$ is the inverse image, within the range, of the ambient interval $(jx - j\delta, jx + j\delta)$. Thus continuity of j makes it a topological embedding.

If the subfield image K is not cofinal, choose $B > 0$ strictly above all of K . For $0 < k \in K$, its inverse also belongs to K , so $k^{-1} < B$ and $k > B^{-1}$. Distinct points of K are therefore separated by more than B^{-1} . The image is discrete; continuity would isolate a point of the densely ordered domain, which is impossible. $\square$

[write] This is `opa:as:thm:topology of docs/surreal/omnific-preserving-automorphisms`, stated there with the value map: (i) $\tau_f(\mathbf{No})$ cofinal gives continuity and a nondiscrete image, (ii) otherwise the image is closed and uniformly discrete and f is nowhere continuous. Cofinality of $\tau_f(\mathbf{No})$ and of $f(\mathbf{No})$ are equivalent. Also in sources 02 (Theorem 9.1), 04 (Theorem 7.1, with the value-group form [proposition 33.2](#)), 05 (Theorem 7.1) and 06 (Theorem 3.9); source 07 proves it for embeddings between arbitrary ordered fields ([theorem 67.1](#)).

Theorem 9.4 (Monomial images are closed). *For any ordered additive subgroup $G \subseteq \mathbf{No}$, the subfield $\mathbb{R}((\omega^G))$ is closed in the ambient order topology. If it is proper, it is nowhere dense. In particular every proper image in [theorem 6.4](#) is closed and nowhere dense.*

Proof. Take $x \notin \mathbb{R}((\omega^G))$. Its support has a largest exponent a outside G . Let p be the truncation of x to exponents strictly greater than a ; then $p \in \mathbb{R}((\omega^G))$. Write $r = [x]_a \neq 0$. For any $y \in \mathbb{R}((\omega^G))$, either $y - p$ has a nonzero term above a , in which case that term dominates $x - y$, or the leading term of $x - y$ at exponent a is exactly $r\omega^a$. In both cases

$$|x - y| > \frac{1}{2}|r|\omega^a.$$

Hence a neighborhood of x misses the subfield. This proves closedness. A proper subfield has no interior by [proposition 5.3](#), so a closed proper subfield is nowhere dense. $\square$

[write] This is `opa:as:thm:closed of docs/surreal/omnific-preserving-automorphisms`, there for the images of its classification theorem.

Corollary 9.5 (Strengthened stabilizer conclusion). *The proper embeddings in [theorem 7.4](#) are continuous and are homeomorphisms onto closed nowhere-dense subfields. The embeddings of [theorem 7.6](#) have discrete images and are nowhere continuous. Both families preserve all admissible Hahn sums.*

There is therefore no implication from strong Hahn summability to continuity in the ambient order topology. Nor does a continuous cofinal field embedding have to have a dense image.

9.3 An everywhere-zero derivative of a nonconstant automorphism

There are two different uses of “derivation” in surreal mathematics: an algebraic operator on field elements, and the pointwise derivative of a function in the order topology. The latter already exhibits a striking separation.

Proposition 9.6. *Let $c > 1$ be a surreal and let $S_c = H_{a \mapsto ca}$. This is an increasing field automorphism. As a function $\mathbf{No} \rightarrow \mathbf{No}$, its order-topological derivative exists at every point and equals zero. Its inverse $S_{1/c}$ has no surreal-valued derivative at any point.*

Proof. Multiplication by c is an ordered additive automorphism, so the field-lift theorem applies. Additivity makes every difference quotient independent of the base point:

$$\frac{S_c(x+h) - S_c(x)}{h} = \frac{S_c(h)}{h}.$$

For $h \neq 0$,

$$v\left(\frac{S_c(h)}{h}\right) = (c-1)v(h). \quad (9.1)$$

Given $\varepsilon > 0$, choose η with $(c-1)\eta > v(\varepsilon)$, and then take $0 < |h| < \omega^{-\eta}$. The quotient has valuation greater than $v(\varepsilon)$, so its absolute value is below ε . This proves derivative zero by the full ε - δ definition. For $S_{1/c}$ the coefficient $(1/c - 1)$ is negative. As h becomes small through monomials of arbitrarily large valuation, the positive quotient exceeds every prescribed positive surreal. It cannot have a surreal limit. $\square$

[write] This is `saut:thm:zeroderiv` of `docs/surcomplex/surcomplex-field-automorphisms` (with S_a for S_c), and the base-point independence is its dichotomy `saut:thm:derivative`. The same dilations appear in sources 02, 05 (Theorem 8.2) and 07 (Example 4.4); source 05 proves an exact criterion for a vanishing derivative of a weighted lift (theorem 47.3) and the embedding form of the dichotomy (proposition 46.4).

The familiar real mean-value conclusion “derivative zero implies constant” requires hypotheses unavailable here. A set-indexed sequence of h ’s is not sufficient to test the limit, by proposition 9.1. This example is not a statement about the Berarducci–Mantova derivation, considered in section 13.

10 Valuation, leading terms, and the ω -map

10.1 Three different valuation constraints

For a field embedding j , distinguish:

$$\begin{aligned} \text{comparison preservation: } & vx < vy \iff v(jx) < v(jy); \\ \text{pointwise valuation fixation: } & v(jx) = vx; \\ \text{leading-term fixation: } & \text{lt}(jx) = \text{lt}(x). \end{aligned}$$

The last implies the second, which implies the first. The first is automatic for surreal field embeddings.

Proposition 10.1. *Every field self-embedding of $\mathbf{No}$ is order preserving and induces a unique ordered additive-group embedding τ_j of its natural value group satisfying*

$$v(jx) = \tau_j(vx) \quad (x \neq 0). \quad (10.1)$$

The induced map on the residue field $\mathbb{R}$ is the identity. This does not, by itself, say that the distinguished embedded copy of $\mathbb{R}$ is fixed pointwise.

Proof. Positive elements of a real-closed field are exactly nonzero squares, so a field embedding preserves positivity. The Archimedean comparison of two positive elements is expressed using inequalities with ordinary integer multiples. Since j fixes the integers and reflects order, it preserves and

reflects these comparisons. Thus it induces an order embedding on Archimedean classes. Multiplication gives additivity of the induced value map, and surjectivity of v gives uniqueness.

The valuation ring of finite elements and its infinitesimal ideal are preserved and reflected. The induced ordered field embedding of the residue field $\mathbb{R}$ into itself fixes rational cuts, hence is the identity. It determines the standard part of $j(r)$, not necessarily $j(r)$ itself. $\square$

[write] This is `opa:as:lem:natural` of `docs/surreal/omnific-preserving-automorphisms` and `saut:lem:order` of `docs/surcomplex/surcomplex-field-automorphisms`. Also in sources 02 (Section 8.2), 04 (Propositions 6.1, 6.2), 05 (Section 5.4), 06 (Propositions 3.1, 3.2) and 07 (Proposition 4.1).

[write] Added 4 October 2026, batch 90. Lemma 3.4 (`csf:lem:valuation-functor`) of `docs/foundations-and-computation/cantor-families-of-surreal-subfields` (batch-90 manuscript 04; AI-assisted, unrefereed, not formalized) proves the same induced value-group embedding, by the same integer comparisons, for an arbitrary embedding of ordered fields, together with the induced maps of Archimedean rank chains and components. That report applies it to countable real closed subfields of **No**, not to self-embeddings of **No**: there the rank maps classify embeddability in a closed Cantor family of subfields of one countable field, which is a complete analytic quasiorder. It cites this report as background and answers none of its questions.

Within the weighted monomial class, a simple pointwise-valuation example is obtained from $\tau = \text{id}$ and

$$\chi(a) = 2^{\lfloor a \rfloor - 1}.$$

Coefficient extraction is additive, so this is a positive real character. The resulting automorphism fixes $\mathbb{R}$ and every ordinal, preserves **Oz** and its floor, and fixes every valuation, but

$$H_{\text{id}, \chi}(\omega^{\omega^{-1}}) = 2\omega^{\omega^{-1}}.$$

Pointwise value fixation is therefore not field rigidity.

10.2 A controlled unit twist fixes even the leading term

We now give a fully supported example outside the real-weighted monomial class. It demonstrates why leading terms alone do not settle the field problem.

Let $t = \omega^{-1}$, let $\beta < 0$, and let $\ell_\beta(a) = [a]_\beta \in \mathbb{R}$. For $r \in \mathbb{R}$, define the usual formal binomial unit

$$(1 + t)^r = \sum_{n \geq 0} \binom{r}{n} t^n.$$

These are legal Hahn series. Define

$$U_\beta \left(\sum_a r_a \omega^a \right) = \sum_a r_a \omega^a (1 + t)^{\ell_\beta(a)}. \quad (10.2)$$

Proposition 10.2 (Strong leading-term stabilizers). *The map U_β is a strongly $\mathbb{R}$ -linear field automorphism that fixes the leading term of every nonzero surreal. It fixes $\mathbb{R}$ and every ordinal. For any prescribed set A , $\beta < 0$ can be chosen so that it also fixes A pointwise. Nevertheless it is nonidentity.*

Proof. For a reverse well-ordered support B , the support of the proposed image is contained in $B - \mathbb{N}$. This is reverse well-ordered. Each exponent has only finitely many representations $a - n$ with $a \in B$, $n \in \mathbb{N}$: otherwise B would contain an infinite increasing sequence of the form $b + n$. Thus the displayed

family is summable. The same argument, together with the local finiteness of a summable input family, proves strong additivity.

The real binomial identity $(1+t)^{r+s} = (1+t)^r(1+t)^s$ and additivity of ℓ_β prove multiplicativity. The term $n = 0$ preserves the leading term, while all correction terms have strictly smaller exponents. Because $\beta < 0$, we have $\ell_\beta(-1) = 0$, so $U_\beta(t) = t$. Consequently U_β fixes every binomial unit above, by strong real linearity. Replacing ℓ_β by $-\ell_\beta$ in the formula gives an inverse: the two units cancel on every monomial and hence on every series.

Ordinals have no negative exponents in their own normal forms, so ℓ_β vanishes on every ordinal exponent occurring in a Cantor normal form. This fixes all ordinals. Taking β below every member of $E(A)$ similarly fixes every outer exponent of every member of A . Finally,

$$U_\beta(\omega^{\omega^\beta}) = \omega^{\omega^\beta}(1 + \omega^{-1}) \neq \omega^{\omega^\beta},$$

which proves nontriviality. □

[write] Not in the collection before this report.

This example does *not* preserve omnific integrality: for $\beta < 0$, the displayed image has the negative exponent $\omega^\beta - 1$, whereas the original element belongs to **Oz**. It therefore supplies a useful separation from the floor-preserving monomial embeddings. It will also fail exponential compatibility by theorem 11.4.

10.3 Fixing residues need not fix real elements

There is an explicit algebraic mechanism for moving a real by an infinitesimal. In ZFC, choose a transcendental real r and a derivation $d : \mathbb{R} \rightarrow \mathbb{R}$ with $d(r) = 1$, defined first on a transcendence basis and then extended uniquely through the algebraic extension. Let Δ act coefficientwise on surreal normal forms, fixing every monomial. Then

$$T(x) = \sum_{n \geq 0} \frac{t^n \Delta^n(x)}{n!}, \quad t = \omega^{-1}, \quad (10.3)$$

is well-defined by the same support bound $\text{supp}(x) - \mathbb{N}$. The iterated Leibniz rule proves multiplicativity. Since $\Delta(t) = 0$, the series with $-t$ is its inverse, by the binomial cancellation at each coefficient. Thus T is an ordered, strongly additive field automorphism fixing every leading term, but $T(r) = r + t$. It is not $\mathbb{R}$ -linear. This is a formal operator exponential of $t\Delta$, not the Gonshor exponential function on field elements.

[write] This is Φ_d of `saut:thm:coefffflow` of `docs/surcomplex/surcomplex-field-automorphisms` with $\delta = 1$, and the display `opa:as:eq:taylor` of `docs/surreal/omnific-preserving-automorphisms`. A sentence of that report saying that the ordinary Taylor section “is not a map on arbitrary normal forms” was false as stated, as source 07 showed (Part VI); it was corrected in commit `f06e67d10` before this report was placed. Sources 04 (Proposition 6.3), 06 (Theorem 3.8) and 07 (Proposition 4.9) give the same construction.

10.4 Truncation thresholds and the ω -map

Let $T_{>c}$ discard all normal-form terms with exponent at most c . For a weighted monomial embedding the precise covariance formula is

$$H_{\tau, \chi} T_{>c} = T_{>\tau(c)} H_{\tau, \chi}. \quad (10.4)$$

Thus preservation of corresponding truncations does not mean commuting with every *fixed* threshold operator. Within this class, commuting with every $T_{>c}$ forces $\tau = \text{id}$: otherwise choose a threshold strictly between an exponent and its image and test the monomial at that exponent. Real weights can still remain.

The Conway map $\Omega(x) = \omega^x$ is an increasing isomorphism from the additive group of exponents to the multiplicative monomial group. It is not Gonshor's total exponential $\exp$. Its image is only the monomials, whereas $\exp$ is onto all positive surreals.

Theorem 10.3 (ω -map and valuation rigidity). *A field self-embedding satisfying $j(\omega^x) = \omega^{j(x)}$ for every x and $v(jy) = v(y)$ for every $y \neq 0$ is the identity.*

Proof. For every x ,

$$-jx = v(\omega^{jx}) = v(j(\omega^x)) = v(\omega^x) = -x.$$

No surjectivity or strong additivity is used. □

[write] This is the computation of the subsection “The omega-map detects exact value preservation” of `saut:sec:exponential` in `docs/surcomplex/surcomplex-field-automorphisms` (for automorphisms), and KKS Proposition 5.5; the embedding case needs nothing more. Also in sources 02 (Proposition 10.4), 04 (proposition 36.1), 05 (Proposition 11.1), 06 (Proposition 6.6) and 07 (Proposition 6.1).

For a general field embedding commuting with Ω , the same calculation shows $\tau_j(x) = j(x)$. In the pure monomial class $j = H_\tau$, compatibility is therefore exactly the functional equation

$$\tau = H_\tau. \tag{10.5}$$

Solving this equation, and then treating nonmonomial embeddings, is a substantial separate problem. The ordinal-fixing maps J_- do not solve it: at $x = \omega^{-1}$, their exponent action and field action differ.

11 Exponential rigidity from finite displacement

This section separates three claims: a finite identity about arbitrary field endomorphisms, a profile theorem for ordered exponential fields, and its application to the actual surreal exponential. The first two have generic counterparts in the inspected repository source [13, 14]; the third still needs its formal bridge in that development.

11.1 The finite engine

Lemma 11.1 (Displacement amplification). *Let F be an ordered field and $\sigma : F \rightarrow F$ a unital field endomorphism. If $D(a) = \sigma(a) - a \neq 0$, then for every $B > 0$ the element*

$$b = \frac{2B(1 + |\sigma(a)|)}{|D(a)|} \tag{11.1}$$

satisfies

$$\max\{|D(b)|, |D(ab)|\} > B.$$

Consequently every nonidentity field endomorphism has displacement image cofinal and coinital in F .

Proof. The exact identity is the twisted Leibniz rule

$$D(ab) = \sigma(a)D(b) + bD(a), \quad (11.2)$$

not $aD(b) + bD(a)$. If both displayed displacements were at most B in absolute value, then

$$|bD(a)| = |D(ab) - \sigma(a)D(b)| \leq B + |\sigma(a)|B.$$

But the definition of b makes its left side $2B(1 + |\sigma(a)|)$, a contradiction. This proves unbounded absolute displacement. Since D is additive and $D(-x) = -D(x)$, it proves both cofinality and coinitality. $\square$

[write] This is `lem:amplify of docs/surreal/exponential-automorphism-rigidity` (stated there for unital endomorphisms), as the base says at the head of this section. Also in sources 02 (Lemma 10.1), 04 (Lemma 9.1), 05 (Lemma 10.1), 06 (Lemma 6.2) and 07 (Lemma 6.2).

Corollary 11.2 (Bounded displacement is rigid). *If one field element B bounds $|\sigma(x) - x|$ for every x , then $\sigma = \text{id}$. More generally, if all displacements lie in a proper convex additive subgroup of F , then $\sigma = \text{id}$.*

Proof. For the second statement, choose b outside the proper convex subgroup. Every element of the subgroup has absolute value less than $|b|$; otherwise convexity would place b in it. Apply the first statement. $\square$

[write] Its `cor:bounded-displacement`.

The bound may be an infinite field element. Replacing it by an ordinary integer bound would lose the intended surreal application. Notice also that A_{f_-} is a nonidentity *additive* embedding with infinitesimal displacement. Thus the field hypothesis cannot be dropped.

11.2 Profiles in an ordered exponential field

Let $E : (F, +, <) \rightarrow (F_{>0}, \cdot, <)$ be an onto ordered exponential, and let $w : F^\times \rightarrow \Gamma$ be an onto nontrivial convex valuation. Put

$$\nu(x) = w(E(x)).$$

This is an additive, surjective, order-reversing map. Its kernel is a proper convex additive subgroup of F .

Theorem 11.3 (Profile rigidity). *If a field endomorphism σ satisfies $\nu(\sigma x) = \nu(x)$ for every x , then $\sigma = \text{id}$. In particular this holds if σ commutes with E and $w(\sigma y) = w(y)$ for every nonzero y .*

Proof. The hypothesis places every $\sigma x - x$ in $\ker \nu$. This kernel is proper and convex, so corollary 11.2 applies. For the last assertion,

$$\nu(\sigma x) = w(E(\sigma x)) = w(\sigma(E(x))) = w(E(x)) = \nu(x).$$

$\square$

[write] Its `thm:profile`, for a unital endomorphism; Lean `eq_id_of_profile`.

This proof needs no strong summability, no normal forms, no fixed coefficient field, and no surjectivity of σ .

11.3 The actual surreal exponential

Gonshor's exponential is an increasing isomorphism $(\mathbf{No}, +) \rightarrow (\mathbf{No}_{>0}, \cdot)$ extending the real exponential [2, 3, 8]. For its natural-valuation profile,

$$\ker(v \circ \exp) = \mathcal{O}, \quad \mathcal{O} = \{x : |x| < n \text{ for some } n \in \mathbb{N}\}. \quad (11.3)$$

Indeed, a real bound on x gives positive real upper and lower bounds on $\exp x$. If x is positive infinite, monotonicity puts $\exp x$ above every real exponential of an integer, so it is infinite; for negative infinite x , its exponential is infinitesimal. This proves (11.3) without an explicit normal-form formula for $\exp$. Every member of $\mathcal{O}$ has absolute value less than ω .

Theorem 11.4 (Exponential valuation rigidity for self-embeddings). *Suppose $j : \mathbf{No} \rightarrow \mathbf{No}$ is a field embedding, $j \exp = \exp j$, and $v(jy) = v(y)$ for every $y \neq 0$. Then $j = \text{id}$. In particular, every exponential leading-term-preserving self-embedding is the identity.*

Proof. For every x ,

$$v(\exp(jx - x)) = v(j(\exp x)) - v(\exp x) = 0.$$

Thus $jx - x$ is finite by (11.3), and all displacements have the common bound ω . Apply lemma 11.1. $\square$

[write] This is `cor:embedding` with `thm:no of docs/surreal/exponential-automorphism-rigidity` (Lean `eq_id_of_commute`, generic; the surreal instantiation is pending there). Also in sources 02 (Theorem 10.3), 04 (`corollary 35.2`), 05 (Theorem 10.2), 06 (Theorem 6.4) and 07 (Theorem 6.3); all credit the exponential-rigidity report.

The conclusion applies to automorphisms as a special case and therefore addresses the pointwise-leading-term condition in KKS Question 5.4. The repository already presents the generic self-embedding implication; the present contribution is an explicit integration of that argument with the much broader embedding taxonomy. Correctness of the displayed proof is separate from priority, and no exhaustive priority assessment is claimed.

Theorem 11.5 (Nontrivial exponential maps move values cofinally). *If j is a nonidentity exponential field self-embedding and τ_j is its value-group action, then*

$$\{\tau_j(\gamma) - \gamma : \gamma \in \mathbf{No}\}$$

is cofinal and coinital in the value group. Consequently j cannot induce the identity modulo any proper convex subgroup of that group.

Proof. Let $\nu = v \circ \exp$. Compatibility gives

$$\nu(jx - x) = \tau_j(\nu x) - \nu x. \quad (11.4)$$

The image of the displacement $jx - x$ is cofinal and coinital by lemma 11.1. Applying the onto order-reversing map ν preserves these two unboundedness properties, with their directions exchanged. Since ν is onto, the right side ranges over precisely the asserted set. A proper convex subgroup of an ordered group is bounded by any positive element outside it, so it cannot contain this displacement set. $\square$

[write] This is `thm:cofinal` of `docs/surreal/exponential-automorphism-rigidity` (Lean `cofinal_value_displacement`, generic). Also in source 07 (Proposition 6.4); source 04 proves it for an abstract ordered exponential field (`theorem 35.1`).

For J_- and, more generally, $J_{f\beta}$, the exponent action $A_f(x) - x$ has only negative exponents and is therefore infinitesimal. Hence these maps cannot commute with the total exponential. This proves failure of exponential compatibility without guessing a complicated individual value of $\exp$. The unit twists and Taylor shifts of section 10 fail it for the even stronger reason that their value action is exactly the identity.

11.4 A uniqueness theorem beyond automorphisms

A trivial kernel for an automorphism-group action implies faithfulness by composing with an inverse. A self-embedding need not have a global inverse, so the same reasoning is invalid for an embedding monoid. A direct comparison supplies a useful replacement.

Lemma 11.6 (Two-map bounded-displacement comparison). *Let $\sigma_1, \sigma_2 : F \rightarrow F$ be field embeddings of an ordered field. Suppose $\sigma_2(F)$ is cofinal and one $B > 0$ bounds $|\sigma_1(x) - \sigma_2(x)|$ for all x . Then $\sigma_1 = \sigma_2$.*

Proof. Put $d(x) = \sigma_1(x) - \sigma_2(x)$. The correct product identity is

$$d(ab) = \sigma_1(a)d(b) + \sigma_2(b)d(a).$$

If $d(a) \neq 0$, cofinality supplies b such that

$$|\sigma_2(b)d(a)| > B(1 + |\sigma_1(a)|).$$

The product identity and the proposed bound on d give the opposite inequality. Thus there can be no such a . $\square$

Theorem 11.7 (Cofinal exponential embeddings have unique value lifts). *Let $j_1, j_2 : \mathbf{No} \rightarrow \mathbf{No}$ be exponential field embeddings with the same value-group action. If at least one has cofinal image, then $j_1 = j_2$.*

Proof. For every x , equal value actions and exponential compatibility imply

$$v(\exp(j_1x - j_2x)) = v(j_1(\exp x)) - v(j_2(\exp x)) = 0.$$

Thus $j_1x - j_2x$ is finite and has uniform bound ω . Interchange the two maps if necessary and apply lemma 11.6. $\square$

[write] New to the collection: `cor:two-profiles` of `docs/surreal/exponential-automorphism-rigidity` and the Lean lemma `eq_of_profile_eq` require one of the two maps to be surjective (a ring equivalence $F \simeq F$ in Lean), and `saut:thm:expfaithful` treats automorphisms. Here a cofinal image suffices. Proved independently in sources 02 (Proposition 10.6), 05 (lemma 48.1 and theorem 48.2) and 07 (Proposition 6.5); source 04 proves it for an abstract ordered exponential field (theorem 35.3). Source 06 proves only the automorphism case (its Corollary 6.5).

This extends the comparison mechanism from the presence of a surjective map to the presence of a cofinal image. It does *not* prove uniqueness of arbitrary bounded exponential embeddings with the same value action. That is a precise remaining question, rather than an inference from a trivial kernel.

11.5 What remains flexible in the exponential category

The preceding theorems do not say that the whole exponential structure is rigid. The known homogeneity theory for the surreal exponential field allows nontrivial exponential automorphisms; the pointwise value-group kernel is the restrictive condition [5, 7]. Likewise, a nontrivial value action with cofinal displacement is only a necessary condition for lifting to an exponential embedding. Equation (11.4), compatibility with the actual logarithm of monomials, and compatibility with multiplication impose further conditions. The double lift is a field construction, not a universal solution of those conditions.

12 Elementarity, definability, and other signatures

12.1 Field elementary does not mean set-theoretically elementary

Proposition 12.1. *Every field self-embedding of $\mathbf{No}$ is elementary in the language of ordered fields. Every order self-embedding is elementary in the language of dense linear orders without endpoints.*

Proof. Both statements follow from quantifier elimination for the corresponding theories. A field embedding preserves and reflects polynomial equations and inequalities, and an order embedding preserves and reflects the atomic order relations. Quantifier elimination reduces every fixed formula to these data. This argument is a formula-by-formula scheme for the proper-class structures and does not require a powerclass of types. $\square$

[write] The field clause is `opa:as:thm:elementary` of `docs/surreal/omnific-preserving-a-automorphisms`. Also in sources 02 (Section 11.1, with the ordered-group clause `proposition 22.1`), 04 (Proposition 6.1), 05 (Section 12.1), 06 (Proposition 4.4) and 07 (Section 8.1).

In particular, the bounded copies from theorem 7.6 are elementary subfields in the ordered-field language. The discrete image of D_m is an elementary suborder in the pure order language. First-order elementarity is not ambient topological density, continuity, or simplicity preservation.

For the exponential language there is an additional standard input: $(\mathbf{No}, +, \cdot, <, \exp)$ is an elementary extension of the real exponential field, whose theory is model complete [3, 8, 9]. Hence an exponential field self-embedding is elementary in that language as well. This does not make a non-exponential field embedding elementary in the exponential expansion.

Adding simplicity, birthdays, or a predicate for $\mathbf{Oz}$ changes the language again. For example the root and the two labeled immediate successors are first-order definable in $(\mathbf{No}, <, \prec_s)$. A nonempty prefix map fails to fix the root; D_m fails to preserve immediate successors. They are therefore not elementary embeddings of that expanded structure. Their status as relation embeddings is weaker.

12.2 O-minimal definability is a strong rigidity condition

Theorem 12.2 (Definable field self-embeddings are trivial). *Let F be a real-closed field, possibly represented as a class, with an o-minimal expansion. A field self-embedding whose graph is definable in that expansion, with parameters allowed, is the identity.*

Proof. Its fixed field is definable and infinite, since it contains $\mathbb{Q}$. By o-minimality it contains a nonempty interval. A subfield containing an interval is the whole field by proposition 5.3. $\square$

[write] New to the collection; `saut:prop:nondefinability` of `docs/surcomplex/surcomplex-field-automorphisms` is a different statement (nondefinability from parameters of particular maps).

Source 06 (Theorem 4.5) gives the same proof; sources 02, 04 and 05 prove it through o-minimal differentiability and the mean value theorem (second routes in section 22, section 34 and theorem 49.9).

This applies to the pure ordered surreal field and to its standard exponential expansion. In contrast, “definable in set theory from the normal-form construction” is a much larger notion. The map f_- itself is given by a piecewise rational formula in the ordered field, but its double lift J_- cannot have a graph definable in that o-minimal field language. Extracting all normal-form exponents changes the expressive resources.

There is no conflict with the parameter-stabilizer theorem. Fixing a class of named elements does not make a map definable from finitely many of them. A single formula in a language with individual constants still uses only finitely many constants.

12.3 Algebraic variants and surcomplex extensions

A nonzero ring homomorphism from a field to a field is injective and unital: the image of 1 is a nonzero idempotent, hence 1. Dropping the unital condition therefore only adds the zero homomorphism, which is not an embedding. Additive-group embeddings and real-linear embeddings are much less rigid, as A_f shows. The ω -map itself is a useful additive-to-multiplicative embedding, but it is not a field endomorphism.

For the surcomplex field $\mathbf{No}(i)$, every real-surreal field embedding j has two extensions

$$j_+(a + bi) = j(a) + j(b)i, \quad j_-(a + bi) = j(a) - j(b)i.$$

Both are field embeddings, and these are exactly the extensions of j , because i must map to one of the two roots of $X^2 + 1$. Their images are $j(\mathbf{No})(i)$; properness and surjectivity correspond to those of j . These statements concern embeddings respecting the specified real form. Arbitrary embeddings of the algebraically closed field $\mathbf{No}(i)$ need not preserve that real form, so the two-extension classification is not a classification of all surcomplex embeddings.

[write] These two extensions are `saut:thm:axis` of `docs/surcomplex/surcomplex-field-automorphisms` for automorphisms, and `opa:as:thm:complexembed` and `hset:thm:two-lifts` elsewhere in the collection. Sources 04 (with a converse for conjugation-commuting embeddings, proposition 36.6) and 05 (Section 13.4) state the same.

Finally, maps on pre-games or on particular cut presentations must first respect equality of represented surreal values before they descend to $\mathbf{No}$. A graph embedding of presentation trees, a map of sign sequences, and a well-defined map of the quotient field are different constructions. A proof assistant should retain this distinction rather than infer it from similar notation.

13 Differential constraints and composition

Fix a particular surreal derivation ∂ , rather than merely the assertion that some derivation exists. The Berarducci–Mantova construction provides a strong derivation compatible with $\exp$, with constant field $\mathbb{R}$; its distinguished simplest choice satisfies $\partial\omega = 1$ [8]. A differential embedding must satisfy

$$\partial(jx) = j(\partial x).$$

This is not the derivative of the function $x \mapsto jx$ in proposition 9.6.

Proposition 13.1 (Two necessary differential conditions). *If j commutes with a derivation whose constant field is $\mathbb{R}$, then j fixes $\mathbb{R}$ pointwise. If also $\partial\omega = 1$, then $j(\omega) - \omega \in \mathbb{R}$.*

Proof. For $r \in \mathbb{R}$, $\partial(jr) = j(0) = 0$, so $jr \in \mathbb{R}$. The restriction $j|_{\mathbb{R}}$ is an ordered field embedding of $\mathbb{R}$ into itself and fixes every rational cut, hence is the identity. Furthermore $\partial(j\omega - \omega) = j(1) - 1 = 0$. $\square$

[write] Also in sources 02 (Proposition 10.7), 04 (Proposition 10.2) and 05 (Section 13.1); source 06 proves that $c_j = j(\omega) - \omega$ is additive in j (proposition 59.5), and source 07 that $c_j = 0$ under the omega-map (proposition 68.4).

These are necessary conditions, not a classification. In particular, fixing ω alone does not prove differential compatibility for the proper maps constructed above.

On a domain of transseries where substitution by g is defined and the usual chain rule is valid, the substitution map $C_g(f) = f \circ g$ satisfies

$$\partial C_g(f) = C_g(\partial f) \partial g.$$

If it is to commute with the derivation, testing the variable itself forces $\partial g = 1$. A domain closed under this substitution and a support theorem are still needed. One cannot extend this observation to a global composition on all surreals merely by writing a formal symbol $f \circ g$.

The operator exponential in (10.3) illustrates the same point in a controlled setting: a fixed negative support shift made every operator series summable. For an arbitrary surreal derivation, neither $\sum_n \partial^n(x)/n!$ nor a proposed inverse is automatically summable. Differential, exponential, and Hahn-continuity conditions should therefore be formalized independently and combined only after proving their compatibility.

14 Cutoffs, universes, and a concrete formalization plan

14.1 Set-sized cutoff models require a smaller quantifier

Write $\mathbf{No}_{<\kappa}$ for the surreals of birthday below κ . An arbitrary ordinal cutoff need not be closed under field operations. For an uncountable regular cardinal κ , the established bounded-Hahn identification gives

$$\mathbf{No}_{<\kappa} = \mathbb{R}((\omega^{\mathbf{No}_{<\kappa}}))_{<\kappa}, \quad (14.1)$$

where supports have cardinality less than κ ; see van den Dries and Ehrlich and its use in KKS Remark 2.3 [3, 5]. This is the right setting for literal set-sized embedding monoids in ZFC.

Proposition 14.1 (Transfer of the explicit constructions). *At such a cutoff, the Hahn lifts are valid whenever their exponent maps are self-maps of $\mathbf{No}_{<\kappa}$. In particular the rational lower-tail and upper-tail maps above yield the corresponding proper cofinal and bounded field embeddings of $\mathbf{No}_{<\kappa}$. The parameter-stabilizer assertions hold for prescribed sets A of size less than κ , with parameters chosen inside $\mathbf{No}_{<\kappa}$.*

Proof. The rational formulas stay inside the cutoff field. Their inverse formulas on their stated ranges also stay inside it. Relabeling a support does not change its cardinality, so (14.1) supplies both lifts. For $|A| < \kappa$, regularity ensures $|E(A)| < \kappa$, since it is a union of fewer than κ sets, each of size less than κ , twice. The birthdays in this footprint have a bound below κ , providing an internal lower or upper parameter. The earlier proofs then apply. $\square$

[write] Also in sources 02 and 04 (their Sections 12.1).

The hypothesis $|A| < \kappa$ is essential to this transfer. Externally $\mathbf{No}_{<\kappa}$ itself is a set, and an embedding fixing every member of that set is the identity. Thus the full-class theorem “fix any prescribed set” must not be copied verbatim into a larger-universe view of a set-sized cutoff.

Similarly, the topological statement that every set subset is closed and discrete becomes the statement for subsets of size less than κ . Larger nets and larger cuts may behave differently. The field-embedding continuity dichotomy itself needs only ordered-field arguments and continues to hold with the cutoff's own order topology.

14.2 Coherent restrictions rather than a fictitious last stage

A class map j has a set-valued restriction to every bounded birthday fragment. Consequently, for each ordinal α there is an ordinal $F(\alpha)$ bounding the birthdays of all $j(x)$ with $b(x) < \alpha$. There is a closed unbounded class of limit ordinals λ satisfying $F(\alpha) < \lambda$ for every $\alpha < \lambda$: repeatedly close a starting ordinal under F and take a countable supremum. At such λ , $j[\mathbf{No}_{<\lambda}] \subseteq \mathbf{No}_{<\lambda}$.

This does not imply that those closure ordinals are regular cardinals or that their birthday fragments are fields. Closure under a class map and closure under arithmetic are separate obligations. Conversely, a family j_α of stage maps defines a global map only if it is coherent on overlaps. Unrelated embeddings at larger and larger cutoffs do not form a global embedding merely because every stage has some embedding.

[write] Source 06 proves this club localization as a proposition with its corollary (proposition 60.2); source 07 proves the same theorem for finitely many named maps and operations.

An especially safe universe formulation takes U -small supports, builds the carrier in the next universe, and proves lift theorems polymorphically in U . The final equivalence with sign-sequence arithmetic must be an actual theorem, not a type alias silently treating a support representation as the already constructed field.

14.3 What was actually inspected in ProveIt

All paths below are relative to Algebra/SurrealNumbers/ at the snapshot printed on the title page.

Inspected source	Relevant declarations and boundary
Surreal/Foundations/ SignSequence.lean	Universe-indexed signs, first-disagreement order, ordinal embedding, birthdays_bounded, small_bounded, not_small, IsPrefix, and Simpler. This file alone is not the field/Hahn identification.
Surreal/Algebra/ ExponentialProfile.lean	Generic OrderedExp, displacement probes, eq_id_of_commute, value maps, and profile-comparison lemmas. Its header explicitly leaves actual surreal exponential instantiation pending.
docs/surreal/ exponential-automorphism- rigidity/README.md README.md	Research claims, the twisted-product engine, source-version audit, and the distinction between generic Lean results and the actual surreal application. Project-level account of the sign-sequence field, normal forms, omnific integers, and its formalization ledger. This is reported project status, not an independent build performed for this article.

The inspected portions of the two Lean files were their first 200 and 180 lines, respectively; these include definitions, theorem headers, and some actual proofs. Assertions about later declarations were read from the module's detailed header, not independently checked in their complete proof bodies.

The repository also points to `Surreal/Algebra/SigmaDerivation.lean`; that dependency was not separately compiled for this article.

14.4 Proposed modules and proof obligations

A practical development should follow the mathematical dependencies rather than start with a single enormous theorem about “all embeddings.”

Layer 1: the sign tree. Extend the existing sign-sequence API with concatenation and repetition, then prove first-disagreement preservation, prefix reflection, birthday rigidity, and initial-image surjectivity. The properness witnesses are concrete omitted sign sequences. The all-ordinal statements must quantify over `Ordinal.{u}` without assuming it is small at universe u .

Layer 2: a support-relabeling interface. Define a generic map on small-support Hahn series from an ordered embedding of exponent types. Prove support transport, injectivity, strong sum preservation and reflection, and image characterization. Only afterward add an additive hypothesis on the exponent map and prove multiplication. This makes it impossible to confuse A_f with a field homomorphism.

Layer 3: the double lift and stabilizers. Use the proved equivalence between the actual surreal field and small normal forms. Construct J_f , establish functoriality and faithfulness, and implement the two-level footprint. Its smallness theorem is as important as its arithmetic theorem. Prove omitted-monomial witnesses, real and ordinal fixation, and the bounded-image inequality separately.

Layer 4: topology and omnific data. State continuity through the ordinary order topology of the universe-indexed carrier. Keep internal small sets separate from all sets in the larger universe. The cofinality/continuity theorem is generic to ordered fields; closedness of monomial images uses the support API. Omnific-floor compatibility should reuse the actual floor uniqueness theorem, including the negative-infinitesimal correction.

Layer 5: exponential bridges. Instantiate the generic profile lemmas with the actual surreal exponential and natural valuation. Required bridges are: positivity and multiplicative additivity of `exp`, monotonicity, surjectivity onto positives, the valuation conventions, and the finite-kernel statement (11.3). Then add the two-map cofinal comparison lemma, which needs only a finite product identity and an order-cofinality hypothesis.

A schematic, *not compiled*, API boundary is:

```
-- Exponent order only: an additive lift.
additiveLift (f : Exponent -> Exponent) (hf : StrictMono f)

-- Exponent addition as well: a field lift.
fieldLift (tau : AdditiveOrderEmbedding Exponent Exponent)

-- Surreal-specific bridge through the normal-form equivalence.
doubleLift (f : No -> No) (hf : StrictMono f)

-- Quantify only over a small parameter family at universe u.
fixesSmallFamily (A : I -> No) [Small.{u} I]
```

The type names here describe a proposed interface; they are not claims that declarations with those exact names already exist in Mathlib or the repository. A verified implementation should expose its universe parameters, exact imports, and axiom audit along with its theorem names.

15 A research agenda with precise targets

The following questions are divided between classification problems, sharpenings of the present results, and formalization tasks. Except where an examined published version is named, they are proposed research targets, not assertions that the entire literature has left them open. A literature and proof audit should precede any claim of priority.

15.1 Exponential and ω -map embeddings

Research question 15.1 (Characterize exponential value actions). Which ordered additive embeddings $\tau : \mathbf{No} \rightarrow \mathbf{No}$ occur as the natural value action of an exponential field self-embedding? Start with (11.4); add the logarithms of monomials and the multiplicative identities of the prospective field map. Cofinal value-displacement is necessary, but a sufficient compatibility criterion has not been supplied here.

Research question 15.2 (Remove or justify the cofinality hypothesis). Must two exponential self-embeddings with the same value action agree when both images are bounded? The proof of theorem 11.7 fails at one precise step: a bounded image cannot amplify a fixed nonzero difference arbitrarily. Resolve whether a different exponential argument repairs that step or whether bounded counterexamples exist.

Research question 15.3 (Proper exponential copies). Classify the possible ranges of proper exponential self-embeddings, including whether a bounded range can occur for the actual Gonshor exponential. The bounded field copies in this article are not examples of exponential copies. A candidate must be checked against the full exponential, not merely its infinitesimal power series.

Research question 15.4 (ω -map compatibility). Solve $\tau = H_\tau$ in the strong pure-monomial class, and then study Ω -compatible maps outside that class. Relate this to Questions 5.6 and 5.7 of the examined KKS version, concerning nontrivial Ω -automorphisms and extension of order automorphisms of generalized ε -numbers. Pointwise value fixation is already excluded by theorem 10.3; it is not the remaining case.

15.2 Combining tree structure with arithmetic

Research question 15.5 (Field and simplicity together). Classify field self-embeddings that preserve the sign-prefix relation. Neither the general Hahn construction nor the prefix construction solves this combined problem. An initial image is impossible for a proper map, but prefix preservation need not imply initiality of the image.

Research question 15.6 (Elementary embeddings of the sign hierarchy). Determine the proper elementary self-embeddings, if any, of $(\mathbf{No}, <, \prec_s)$ and of an explicitly two-sorted expansion by birthdays and ordinals. Such maps must fix the root and preserve labeled immediate successors. Identify which limit-stage properties become first-order visible and which remain external chain-completeness requirements.

Research question 15.7 (Weaker birthday controls). Replace the global inequality $b(jx) \leq b(x)$ by bounds on selected stages, a cofinal family, or a prescribed ordinal growth function. Determine when those weaker bounds force identity, surjectivity, or a specific substructure. The ordinal addition in prefix maps and ordinal multiplication in repetition maps provide useful sharp test cases.

15.3 Hahn images, fixed fields, and dynamics

Research question 15.8 (Beyond monomial images). Which strongly additive field embeddings admit a factorization into a pure exponent lift, a real character, and a controlled unit twist? The unit construction (10.2) proves that support control is essential. A general homomorphism into the unit group need not support a well-defined strongly additive substitution.

Research question 15.9 (Prescribe the fixed field). Which real-closed subfields of $\mathbf{No}$ arise as fixed fields of proper embeddings or of automorphisms? The formula $\mathbb{R}((\omega^{\text{Fix } \tau \cap \ker \chi}))$ solves a concrete subclass, and the pair J_-, J_g shows that identical fixed fields can occur on opposite sides of surjectivity. Seek invariants that distinguish the maps when the fixed field does not.

Research question 15.10 (Prescribe the reversible core). Which pairs of real-closed fields $L \subseteq K \subseteq \mathbf{No}$ can occur as $L = \text{Fix}(j)$ and $K = K_\infty(j)$? For a single injective map the image hierarchy already stabilizes at its countable intersection, by proposition 8.4; there is no further transfinite descent to discover. The remaining problem is to prescribe that stable core and its nontrivial automorphism, and to identify the restrictions imposed by a Hahn lift.

Research question 15.11 (The stabilizer structure). Study the composition and centralizers of the explicit families fixing $A \cup \mathbb{R} \cup \mathbf{On}$. The double lift is faithful, so order-embedding semigroup phenomena can be transported into the field category. Determine which of these phenomena survive stronger constraints such as fixed valuation, omnific floors, or a chosen derivation.

Research question 15.12 (Floor-preserving maps outside the monomial class). Characterize field embeddings preserving $\mathbf{Oz}$ and its floor without assuming monomial images. The unit twists above fail omnific preservation, whereas every real-weighted monomial lift succeeds. Identify the support conditions responsible for this boundary, and separately investigate pointwise fixation of $\mathbf{Oz}$ rather than just predicate preservation.

15.4 Topology, differentiation, and foundations

Research question 15.13 (Closed images beyond strong monomial maps). Which field self-embeddings have closed images in the full order topology? All real-weighted monomial images do. Determine what happens for non-strong embeddings and for immediate embeddings with unchanged value group and residue field. Do not infer density merely from cofinality.

Research question 15.14 (Differential centralizers). Classify embeddings commuting with the chosen Berarducci–Mantova derivation, with or without exponential compatibility. The necessary conditions of proposition 13.1 supply an inexpensive first filter. Any proposed construction by composition should specify its domain, chain rule, and summability theorem.

Research question 15.15 (A scale-sensitive calculus). Compare full order-topological differentiation, valuation-based asymptotics, and derivatives of formal transseries substitutions. Explain systematically which additional regularity conditions exclude the zero-derivative field automorphisms of proposition 9.6. Ordinary set sequences alone cannot supply the needed regularity test.

Research question 15.16 (Universe and cutoff absoluteness). For transitive universes or cutoff fields, determine which embedding properties are absolute: strong summability, cofinality, initiality, and being elementary in a fixed language. New supports and new cuts in a larger universe can change these answers. Coherent stage maps, rather than arbitrary stagewise existence statements, should be the basic data.

Research question 15.17 (Minimal foundational strength). Separate the axioms required for explicit Hahn lifts, global back-and-forth, general class-valued recursions, and a literal carrier of all class

embeddings. Give translations between formula schemes, GBC with specified recursion principles, and universe-indexed type theory. No one of these should be advertised as automatically equivalent to the others.

Research question 15.18 (Near-term formalization). First formalize the generic two-map comparison lemma and the cofinality/continuity dichotomy; both are finite algebra/order arguments. Next formalize support relabeling and the footprint theorem. Finally complete the actual surreal exponential/valuation bridge and derive the self-embedding rigidity corollary from the existing generic module. These tasks have clear interfaces and independently checkable endpoints.

16 Synthesis and an assumption checklist

The most informative answer is not a single yes or no to the existence of nontrivial self-embeddings. It is the following set of separations.

Flexibility with substantial arithmetic. There are proper elementary ordered-field self-embeddings that fix every real, every ordinal, and any additional prescribed set; preserve Hahn sums, corresponding normal-form truncations, omnific integrality and floors; and are continuous with cofinal closed nowhere-dense image. There are also bounded, nowhere-continuous elementary field copies fixing any prescribed set and all reals.

Rigidity from construction data. Nonincreasing birthdays force an order embedding to be the identity. Preserving ambient simplest cuts does the same. A simplicity-initial image of a full order copy must be the whole class. An order-and-simplicity automorphism is the identity.

Rigidity from exponential scale data. An exponential field self-embedding fixing natural-valuation values pointwise is the identity. Nontrivial exponential embeddings must move values cofinally in both directions. Among cofinal exponential embeddings, the value action determines the map uniquely.

For any proposed new theorem about a surreal self-embedding, check the following before treating it as established: the exact language; set versus class size; support admissibility; whether “initial” refers to signs or normal forms; whether valuation is fixed pointwise or only up to an induced map; whether continuity is topological or strong-Hahn; and whether an inverse or a uniform truth predicate has been used without being available. These checks are not peripheral bookkeeping. Several of the sharpest mathematical distinctions in this article occur exactly at those boundaries.

Part II

Source 02: Order, simplicity, algebra, analysis, and paradox-free foundations

17 Source 02 at a glance

Source 02 is the manuscript *Self-Embeddings of the Surreal Numbers — Order, simplicity, algebra, analysis, and paradox-free foundations*, “Prepared for Vladimir Reshetnikov”, dated 3 October 2026 (36 pages in its own typesetting). It pins the repository at snapshot `af04de3fa423f8f3253ee28db9afeb7fc6171604`, which is earlier than the base’s snapshot; the files it relies on are unchanged between the two. Like the base, it works with sign-sequence surreals

in GBC (with global choice only where a global back-and-forth is used), reads truth, elementarity and topology formula by formula, and treats set-sized birthday fragments $\mathbf{No}_{<\kappa}$ as honest sets. No elementary transfinite recursion is assumed anywhere. Its Section 13 (here section 24) is *conditional* on an amenable elementary embedding $j : V \rightarrow M$ with critical point κ ; every statement drawn from it says so.

Its organisation follows the same taxonomy as the base: sign tree, Hahn lifts, explicit field embeddings, constants and omnific integers, topology, exponential and differential constraints, model theory, cutoffs, large cardinals, lexicographic coding, formalization, and 15 research questions. This Part prints what the base does not already contain: ordinal sign blocks D_δ , localized tree embeddings fixing a set and all ordinals, the multiplicativity criterion for the first lift, the compression double lift J_c with exact fixed field $\mathbb{R}((\omega^{\mathbb{R}}))$, the fraction-field theorem $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$ (credited), the elementarity of ordered additive embeddings, a second proof of definable rigidity, cutoff cardinalities, the conditional large-cardinal embedding $\mathcal{J}_j$ with its first pointwise cut failure, the lexicographic word action, its formalization plan and its source audit.

[write] Source 02 inspected only the exponential-rigidity report and the README of the large-cardinal report among the collection’s reports. Much of its field half is therefore an independent re-proof of collection material it did not cite: the double Hahn lift and its functoriality (`opa:as:eq:innerlift`, `opa:as:lem:inner` and `opa:es:prop:lifts` of `docs/surreal/omnific-preserving-automorphisms`; `isc:thm:lift` of `docs/surreal/independent-surreal-copies`), the exact fixed-support formulas (`opa:es:thm:fixedsupport`), bounded copies built from the same compression (`opa:as:thm:parameters`), the continuity–cofinality criterion (`opa:as:thm:topology`) and the elementarity of field embeddings (`opa:as:thm:elementary`). Source 02 claims no priority for any of them (“the article asserts the formulas and proofs, not historical priority for the combination”); where its abstract says “we determine” or “we prove” for these items, read “we re-prove”.

Renames. The base’s notation is canonical. Source 02’s symbols were renamed as follows.

Here	Source 02	Reason
$\mathbf{b}(x)$	$\mathbf{bd}(x)$ (printed $\mathbf{b}$)	birthday, base notation
$x \preceq_s y, x \prec_s y$	$x \sqsubseteq y, x \sqsubset y$	sign prefix, base notation
$p \frown s$	$p \hat{\frown} s$	concatenation, base notation
$P_p[\mathbf{No}]$	cone C_p	image of the prefix map
D_δ (δ an ordinal)	B_δ	sign blocks; extends the base’s D_m ; base letter B is $\mathbb{R}((\omega^{\mathbf{No}_{\geq 0}}))$
Sw_ϵ	T	tree swap; base T is the Taylor automorphism
$\mathbb{R}((\omega^A))$	$\mathbb{R}((\omega^A))$ written $\text{Hahn}(A)$	base notation
A_f	$\mathcal{L}_f$	first (additive) lift
$J_f = A_{A_f} = H_{A_f}$	$E_f = \mathcal{L}_{\mathcal{L}_f}$	double lift; base $E(A)$ is the footprint
A_c, J_c	$\Phi = \mathcal{L}_c, J = E_c$	compression lifts
f_{sh}	f (Theorem 6.2)	the shift map $x \mapsto x - 1$ on $x < 0$
$H_{\tau, \chi}$	$T_{\varphi, \chi}$	weighted monomial embedding
S_q	D_q	exponent dilation; base D is the displacement
τ_σ	ρ_σ	value action
$\mathbf{v}(x)$	$\mathbf{v}(x)$	natural valuation
$\mathcal{J}_j$	J_j	large-cardinal map; base J_f is the double lift
$\text{+}_{\text{ord}}, \text{+}_{\text{ord}}$	$\text{+}_{\text{ord}}, \text{+}_{\text{ord}}$	ordinal operations (unchanged)

The tempting false reading to avoid: A_f is *not* a field embedding unless f is additive (proposition 19.1), and D_δ is not additive.

Correspondence table. Every numbered item of source 02, in source order, with its source numbering (theorem-like environments share one counter per section; the unnumbered “Reading and proof-status guide” precedes Section 1), followed by named unnumbered items and tagged displays. “Same proof” means the base’s proof is the source’s argument up to wording.

Source 02	Content	Where printed
Reading guide	proof-status guide	non-claims in section 29
§1 table, §1.2	taxonomy of embedding notions; first orientation	= table of section 2; distinct points in remark 18.1
§2	sign sequences, small cuts, GBC, truth and topology as schemes	= section 3; extra points in remark 18.2
Prop. 3.1	interval copy $c(x) = x/(1 + x)$	= proposition 4.1 (same proof)
§3.1 paragraph	set-sized partial order isomorphisms extend (global choice)	remark 18.3 (printed)
Prop. 3.2	prefix embeddings P_p	= proposition 4.2 (same proof); values of P_+ in remark 18.1
Prop. 3.3	repeated-sign embeddings B_δ , δ an ordinal	proposition 18.4 (printed; extends proposition 4.3 from integers $m \geq 2$ to ordinals; home)
Prop. 3.4	tree embedding fixing a set and all ordinals	proposition 18.5 (printed; home; 07 ord:localized points here)
§3.4	tree-swap automorphisms of the sign tree	remark 18.6 (explicit formula; base prose in section 4)
Lemma 4.1	no shortening for prefix embeddings	= first assertion of proposition 4.6 (same proof)
Thm. 4.2	birthday nonincrease is rigid	= theorem 4.5 (same proof)
Cor. 4.3	$\text{Aut}(\mathbf{No}, <, \preceq_s) = \{\text{id}\}$	= second assertion of proposition 4.6 (same proof); KKS Prop. 5.1
Thm. 4.4	canonical-cut rigidity	= proposition 4.7 (same proof); caveat in remark 18.7
Def. 4.5	initial subclass, initial image	= definition in section 5
Thm. 4.6	initial order copy is onto	= theorem 5.2 (same proof)
Prop. 4.7	convex subfield is the whole field	= proposition 5.3 (weaker; same proof)
Def. 5.1, (5.2)	first lift $\mathcal{L}_f$	= A_f , (6.1)
Thm. 5.2 (i)–(iv)	additive lifting theorem	= theorem 6.1 (same proof)
Thm. 5.2 (v)	A_f is a field embedding iff f is additive	proposition 19.1 (printed; the converse is not in the base)
Def. 5.3, (5.3)	double lift E_f	= J_f , (6.10)
Thm. 5.4	functorial field embeddings, faithfulness, onto iff onto	= theorem 6.3 (same proof)
Prop. 5.5	cofinality of the lifts	= cofinality clauses of theorems 6.1 and 6.3 (same proof)
§5.2 paragraph	ranges closed under truncation and summable families	remark 19.2
Lemma 5.6	reverse-well-order rigidity	= step in the proof of theorem 6.1
Thm. 5.7, (5.4), (5.5)	exact fixed-support formulas	= (6.4) with (6.9) (same proof); two-level form (19.1)
Prop. 5.8	fixed fields are real closed	= proposition 8.2 (same proof)
Thm. 6.1, (6.1)	bounded copy $J = E_c$; $\text{Fix} = \mathbb{R}((\omega^{\mathbb{R}}))$	theorem 19.3 (printed; bound sharpened in a note)
§6.1 paragraph	$\mathbb{R}((\omega^{\mathbb{R}}))$ is a set of size $2^{\aleph_0}$	remark 19.4
Thm. 6.2, (6.2)	proper cofinal embedding fixing all ordinals	case $A = \emptyset$ of theorem 7.4; the explicit map in example 19.5

Source 02	Content	Where printed
§6.2 paragraph	automorphism variant $2x$ on $x < 0$	= the map g in the paragraph after (8.3); noted in example 19.5
§6.3, (6.3)	exponent dilations D_q	= S_q of proposition 9.6; fixed field in remark 19.6
Thm. 7.1, (7.1), (7.2)	weighted monomial classification	= theorem 6.4, (6.13) (same proof); remark 19.7
§7.2	meanings of truncation preservation	= (10.4); remark 19.7
§8.1	what a field embedding fixes automatically; $r \mapsto r + \varepsilon$	remark 20.1; homogeneity: theorem 70.2 (home 07)
§8.2, (8.1)	value action ρ_σ	= τ_j , proposition 10.1, (6.8)
(8.2)	normal-form description of $\mathbf{Oz}$	= (8.1)
Prop. 8.1	weighted maps preserve $\mathbf{Oz}$ and its floor	= proposition 8.1 (same proof)
Thm. 8.2	$\text{Frac}(\mathbf{Oz}) = \mathbf{No}$; pointwise omnific rigidity	theorem 20.2 (printed; home; credits <code>odg:thm:fractions</code>)
Thm. 9.1	continuity iff cofinal image, any ordered field	theorem 67.1 (home 07) and theorem 9.3
Prop. 9.2	bounded image is uniformly separated	= last paragraph of the proof of theorem 9.3 (same proof)
Prop. 9.3	sets closed discrete; set nets eventually constant	= proposition 9.1 (same proof); remark 20.3
Lemma 10.1, (10.1)	unbounded displacement	= lemma 11.1 (same proof)
Lemma 10.2	$\ker(v \circ \exp) = \mathcal{O}$	= (11.3) (same argument)
Thm. 10.3, (10.2)	exponential–valuation rigidity for embeddings	= theorem 11.4 (same proof); remark 21.1
Prop. 10.4	omega-map rigidity	= theorem 10.3 (same proof)
Prop. 10.5	D_2 does not commute with $\exp$	proposition 21.2 (printed; special case of EAR <code>thm:dilation</code>)
Prop. 10.6	equal value action, cofinal reference map	= theorem 11.7 via lemma 11.6 (same proof); abstract form theorem 35.3 (home 04)
Prop. 10.7	necessary differential constraints	= proposition 13.1; D_2 clause in remark 21.3; translation parameter proposition 59.5 (home 06)
§11.1	elementarity in DLO, DOAG, RCF, exponential language	proposition 22.1 (DOAG clause printed; rest = proposition 12.1)
§11.2	D_2 not elementary in $(\mathbf{No}, <, \preccurlyeq_s)$; multisorted birthdays	remark 22.2 (with the base paragraph after proposition 12.1)
Thm. 11.1	o-minimal definable endomorphisms are trivial	second route after theorem 12.2 (section 22)
(12.1)	$ \mathbf{No}_{<\kappa} = 2^{<\kappa}$	proposition 23.1 (printed)
§12.1 claim	$\mathbf{No}_{<\lambda}$ a subfield exactly at epsilon numbers	remark 23.2 (source’s claim)
(12.2)	regular-cutoff Hahn description	= (14.1); internal lifts = proposition 14.1
§12.1 paragraph	prefix and block maps at cutoffs	proposition 23.1(ii)
§12.2	universes; transitive models $M \subseteq N$	remark 23.3
(13.1)	$J_j(s) = j(s)$	$\mathcal{J}_j$, theorem 24.2
Lemma 13.1	absoluteness of old cuts and arithmetic	lemma 24.1 (printed; credits <code>lce:lem:absolute</code>)
Thm. 13.2	the induced surreal embedding (conditional)	theorem 24.2 (printed; re-proves <code>lce:prop:J</code>)
§13.2	first pointwise cut failure at κ ; small cuts preserved	remark 24.3 (new, remark level)
(13.2), (13.3)	summation defect $\sum_{\alpha < \kappa} \omega^{-\alpha}$	remark 24.4 (= <code>lce:thm:witness</code> , instance of <code>lce:thm:defect</code>)
§13.4	why Kunen’s theorem is not evaded	remark 24.5
§14.1	lexicographic word action f_*	proposition 25.1

Source 02	Content	Where printed
§14.2	iterates $J_f^n = J_{f^n}$, $c^n(x) = x/(1 + n x)$	remark 25.2
§15	formalization plan, interfaces, verification levels	section 26
Q. 16.1	beyond the strong single-term class	research question 27.1
Q. 16.2	fixed fields of double lifts	research question 27.2
Q. 16.3	cofinal versus bounded images	research question 27.3
Q. 16.4	ordinal-fixing symmetries	research question 27.4
Q. 16.5	field and prefix simultaneously	research question 27.5
Q. 16.6	critical points recoverable from surreal structure	research question 27.6
Q. 16.7	summability thresholds	research question 27.7
Q. 16.8	noncofinal exponential self-copies	research question 27.8
Q. 16.9	faithfulness without a cofinal reference	research question 27.9
Q. 16.10	omega-map automorphisms	research question 27.10
Q. 16.11	differential–exponential centralizers	research question 27.11
Q. 16.12	minimal expressive expansions	research question 27.12
Q. 16.13	compatibility at singular cutoffs	research question 27.13
Q. 16.14	a verified embedding library	research question 27.14
Q. 16.15	class principles and parameterized homogeneity	research question 27.15
§17	conclusion	expository; its separations are the base’s overview in section 16
App. A	source audit and provenance	section 28
App. B	finite regression checks	section 28.3

18 Order and simplicity in source 02

Remark 18.1 (Orientation points of source 02). Three points of source 02’s taxonomy are not spelled out in the base. (a) *Pointwise fixation* of a named subcollection such as $\mathbb{R}$, **On** or **Oz** is a different condition from mapping that collection into itself; the two are listed as separate rows of its table, and [theorem 20.2](#) shows how far apart they are for **Oz**. (b) The omega-map is not the exponential: $\Omega(1) = \omega$ while $\exp(1) = e$, and one may not replace ω^x by $\exp(x \log \omega)$ for arbitrary surreal x . (c) The prefix maps are not arithmetic translations: for $p = +$,

$$P_+(0) = 1, \quad P_+(1) = 2, \quad P_+(-1) = \frac{1}{2},$$

so P_+ is not additive and does not fix 0.

Remark 18.2 (Foundational points [02]). Source 02’s foundations section agrees with [section 3](#) and adds two observations. First, *bounded does not mean small*: a bounded interval of **No** contains a proper class of elements, since order bounds do not bound birthdays; the bounded subfield of [theorem 19.3](#) makes this visible even for subfields. Second, no set-sized linear order without endpoints can fill every cut formed by its own subsets: taking all its elements as the left side already contradicts the proposed filling property. A universe-indexed construction must therefore distinguish its small cut-index types from the larger type containing all its elements.

Remark 18.3 (Order homogeneity under global choice [02]). Under global choice, a set-sized partial order isomorphism between subclasses of **No** extends to an order automorphism of **No**. Choose a set-like global enumeration of the sign sequences. At each ordinal stage the partial map has set-sized domain and range. The order requirements for the next element form a set cut; its interval of solutions is a proper class, so the cut can be filled while avoiding the set already used. Then perform the inverse

step. The union of the set stages is a class bijection. This is an extension statement for *set-sized* partial maps, not for arbitrary proper-class partial maps, and it avoids any appeal to a set-sized saturated model of the wrong cardinality.

Proposition 18.4 (Sign blocks of ordinal length [02]). *Let $\delta > 0$ be a set ordinal, and let $D_\delta(s)$ replace every sign of s by a block of δ copies of that sign. Then D_δ preserves and reflects both $<$ and $\preceq_s$, fixes the empty sequence, and*

$$\mathbf{b}(D_\delta(s)) = \delta \cdot_{\text{ord}} \mathbf{b}(s).$$

For $\delta > 1$ it is proper. It need not preserve addition.

Proof. The concatenation of $\mathbf{b}(s)$ blocks of length δ has length $\delta \cdot_{\text{ord}} \mathbf{b}(s)$. A first sign disagreement of two inputs remains a first disagreement of the corresponding blocks, with the same comparison. If one input terminates first, its image terminates at the start of the corresponding block. These observations prove preservation and reflection of both orders. When $\delta > 1$, every one-sign sequence is omitted. Finally $D_2(1) = 2$, whereas $D_2(1/2)$ has signs $++--$ and value $5/4$, not 1. $\square$

The order of the ordinal product matters: it is $\delta \cdot_{\text{ord}} \mathbf{b}(s)$, not $\mathbf{b}(s) \cdot_{\text{ord}} \delta$; neither is multiplication in the surreal field.

[write] This extends proposition 4.3 from integers $m \geq 2$ to ordinals $\delta \geq 2$, and the base's further conclusions extend with it. The proof of proposition 4.3 applies verbatim: an image point $z = D_\delta(x)$ is isolated by $(z \smallfrown (-), z \smallfrown (+))$, because any image point extending z begins its continuation with $\delta \geq 2$ copies of one sign; D_δ is cofinal and coinital since $D_\delta(\alpha)$ is the ordinal $\delta \cdot_{\text{ord}} \alpha \geq \alpha$; hence its image is discrete and it is nowhere continuous. For $\delta \geq \omega$ the map moves even small numbers far: $D_\omega(1) = \omega$ and $D_\omega(-1) = -\omega$.

Proposition 18.5 (Fixing any prescribed set [02]). *For every set $A \subseteq \mathbf{No}$ there is a nonidentity proper embedding of $(\mathbf{No}, <, \preceq_s)$ fixing A pointwise. It can also be chosen to fix every ordinal pointwise.*

Proof. Choose an ordinal α at least as large as every birthday in A , and a sign sequence p of length $\alpha \cdot_{\text{ord}} 1$ containing a minus sign. Inside the cone $P_p[\mathbf{No}]$ put

$$F(p \smallfrown s) = p \smallfrown D_2(s),$$

and outside it put $F(x) = x$. Propositions 4.2 and 18.4 handle pairs inside the cone. Convexity of the cone handles numerical comparisons across its boundary. A prefix of a member of the cone either extends p or is a prefix of p ; this proves preservation and reflection of simplicity across the boundary. The tails consisting of one sign are omitted, so F is proper, and $F(p \smallfrown (+)) = p \smallfrown (++)$ shows it is not the identity. No element of A extends p , by length, and no all-plus sequence extends a sequence containing a minus sign. $\square$

[write] Applying proposition 18.5 to the set $A \cup \mathbb{R}$ (every real has birthday at most ω) gives a proper nonidentity $(\mathbf{No}, <, \preceq_s)$ -embedding fixing $A \cup \mathbb{R} \cup \mathbf{On}$ pointwise: the tree-level counterpart of the field-level theorem 7.4. This is the home of that statement in the report; source 07's ord:localized points here. It does not contradict proposition 4.6: an embedding need not exhaust the predecessors of its image points.

Remark 18.6 (Tree-swap automorphisms [02]). If numerical order is forgotten, choose a sign $\epsilon(s) \in \{-1, 1\}$ for every node s of the sign tree and put

$$\text{Sw}_\epsilon(x)(\beta) = \epsilon(x \upharpoonright \beta) x(\beta) \quad (\beta < \mathbf{b}(x)).$$

The inverse is recovered recursively from the previously recovered input prefix. Sw_ϵ preserves birthdays and the prefix relation, but generally not numerical order; the constant choice $\epsilon \equiv -1$ is global sign reversal $x \mapsto -x$. A prefix-tree automorphism is thus a third notion, distinct from an order automorphism and from an automorphism of the doubly ordered structure (the base's prose in section 4).

Remark 18.7 (What canonical-cut preservation is not [02]). Merely sending separators to separators is automatic for order embeddings and is weaker than the hypothesis of [proposition 4.7](#). Likewise, replacing the pointwise option image $f[L]$ by a set-theoretic image object $j(L)$ changes the assertion; [remark 24.3](#) gives a precise example in which the two differ exactly at the critical point.

19 Hahn lifts and explicit field embeddings in source 02

Source 02's additive and double lifts are the base's A_f and $J_f = H_{A_f}$ (theorems 6.1 and 6.3); the following items go beyond the base.

Proposition 19.1 (Multiplicativity criterion for the first lift [02]). *For an order embedding $f : \mathbf{No} \rightarrow \mathbf{No}$, the additive lift A_f of (6.1) is a unital field embedding if and only if f is additive.*

Proof. If f is additive, then $f(0) = 0$ and $A_f = H_f$ is a field embedding by theorem 6.2. Conversely, if A_f is multiplicative, apply it to two monomials:

$$\omega^{f(a+b)} = A_f(\omega^{a+b}) = A_f(\omega^a)A_f(\omega^b) = \omega^{f(a)+f(b)}.$$

Uniqueness of the monomial exponent gives $f(a+b) = f(a) + f(b)$. $\square$

The first lift need not fix the real constants as elements: it sends r to $r\omega^{f(0)}$. This is consistent with $\mathbb{R}$ -linearity, which is scalar linearity rather than preservation of 1; the second lift repairs it automatically.

Remark 19.2 (Ranges of the lifts [02]). The image formula (6.2) gives more than truncation closure: a Hahn-summable family of elements of $\text{im } A_f$ has its sum in $\text{im } A_f$, by pulling the family back along A_f (summability is reflected). The same holds for J_f . None of this gives numerical convexity or simplicity-initiality; [proposition 5.3](#) and [theorem 5.2](#) preclude both for a proper self-copy. Combining (6.4) with (6.9), the fixed field of a double lift has the explicit two-level form

$$\text{Fix}(J_f) = \mathbb{R}((\omega^{\text{Fix}(A_f)})) = \mathbb{R}((\omega^{\mathbb{R}((\omega^{\text{Fix}(f)})})))), \quad (19.1)$$

where the inner $\mathbb{R}((\omega^{\text{Fix}(f)}))$ is an actual additive subgroup of $\mathbf{No}$ used as the exponent group of the outer Hahn field.

19.1 The compression double lift

Let $c(x) = x/(1+|x|)$ be the interval compression of [proposition 4.1](#), and write A_c for its additive lift and $J_c = H_{A_c}$ for its double lift.

Theorem 19.3 (A bounded strong field copy fixing $\mathbb{R}$ and ω [02]). *J_c is a proper, strongly $\mathbb{R}$ -linear field self-embedding of $\mathbf{No}$ with*

$$\begin{aligned} J_c(r) &= r \quad (r \in \mathbb{R}), & J_c(\omega) &= \omega, \\ J_c(\omega^\omega) &= \omega^{\omega^{1/2}}, & \text{Fix}(J_c) &= \mathbb{R}((\omega^{\mathbb{R}})). \end{aligned}$$

Its image is bounded; source 02 records the deliberately loose bound

$$|J_c(x)| < \omega^{\omega^2+1} \quad (x \in \mathbf{No}). \quad (19.2)$$

The image is a real-closed proper-class subfield, closed under Hahn summation and truncation, but neither initial nor convex.

Proof. Theorem 6.3 gives a strongly $\mathbb{R}$ -linear field embedding, proper because c is not onto. The first lift satisfies $A_c(1) = \omega^{c(0)} = 1$ and $A_c(\omega) = \omega^{c(1)} = \omega^{1/2}$, which gives the displayed values. Every exponent of a nonzero $A_c(x)$ lies in $(-1, 1)$, so $|A_c(x)| < \omega^2$; the leading exponent of a nonzero $J_c(y)$ has the form $A_c(a)$, which is below ω^2 , and (19.2) follows. The only solution of $c(a) = a$ is $a = 0$, since for $a \neq 0$ cancellation would give $1 + |a| = 1$; so (6.4) gives $\text{Fix}(A_c) = \mathbb{R}((\omega^{\{0\}})) = \mathbb{R}$, and (6.9) gives $\text{Fix}(J_c) = \mathbb{R}((\omega^{\mathbb{R}}))$. The image is isomorphic to the real-closed field $\mathbf{No}$; it is a proper class because an inverse map on a set-sized range would make the domain a set. Summation and truncation closure are remark 19.2; initiality or convexity would force surjectivity by theorem 5.2 and proposition 5.3. $\square$

[write] The bound sharpens: every exponent of a nonzero $A_c(x)$ is below 1, so already $|A_c(x)| < \omega$, every exponent of $J_c(y)$ is below ω , and $|J_c(y)| < \omega^\omega$ for all y . This is the bound of source 04 (theorem 32.1). Bounded field copies as such are the base’s theorem 7.6 (fixing a prescribed set as well) and already the collection’s `opa:as:thm:parameters`, built there from the same compression on an upper tail; what J_c adds is the exact fixed field $\mathbb{R}((\omega^{\mathbb{R}}))$, compared with the base’s maps $J_{f_\gamma}^+$, whose fixed field is $\mathbb{R}((\omega^{\mathbb{R}^{\text{No} \leq \gamma}}))$. By theorem 9.3 (and corollary 9.5) J_c is nowhere continuous: it preserves every Hahn-summable family yet has a uniformly discrete image, while intrinsically its image is a densely ordered real-closed field.

Remark 19.4 (A set-sized fixed field of a class map [02]). Every reverse-well-ordered subset of $\mathbb{R}$ is countable: in its decreasing enumeration, assign a distinct rational between consecutive members. Hence $\mathbb{R}((\omega^{\mathbb{R}})) = \text{Fix}(J_c)$ is a *set* of cardinality $2^{\aleph_0}$, although it is the fixed field of a proper-class map. The countability concerns real *exponents*; general surreal supports need not be countable.

Example 19.5 (An explicit ordinal-fixing map [02]). Put $f_{\text{sh}}(x) = x - 1$ for $x < 0$ and $f_{\text{sh}}(x) = x$ for $x \geq 0$. Both pieces are increasing and the negative piece takes values below -1 , so f_{sh} is a cofinal order embedding omitting $[-1, 0)$. By theorem 6.3, $J_{f_{\text{sh}}}$ is a proper cofinal strongly $\mathbb{R}$ -linear field embedding; it fixes every ordinal by the argument of theorem 7.1 (Cantor normal forms have ordinal exponents, fixed by f_{sh} and then by $A_{f_{\text{sh}}}$), and

$$J_{f_{\text{sh}}}(\omega^{\omega^{-1}}) = \omega^{\omega^{-2}}. \quad (19.3)$$

This is the case $A = \emptyset$ of theorem 7.4 with a different exponent map from the base’s f_- of (7.1). The construction is unconditional: no measurable-cardinal hypothesis lies behind an ordinal-fixing field embedding (contrast theorem 24.2). Replacing the negative branch by $2x$ gives the base’s automorphism g (paragraph after (8.3)), a nonidentity field *automorphism* fixing every ordinal; naming the ordinal subclass therefore does not encode the ambient set theory in the ordered-field language.

Remark 19.6 (Fixed fields of exponent dilations [02]). For a positive surreal q , the dilation $S_q(\sum r_a \omega^a) = \sum r_a \omega^{qa}$ of proposition 9.6 is a strongly $\mathbb{R}$ -linear field automorphism with inverse $S_{1/q}$. For $q \neq 1$ the exponent map $a \mapsto qa$ fixes only 0, so (6.9) gives $\text{Fix}(S_q) = \mathbb{R}$ exactly. For example $S_2(\omega) = \omega^2$. These are field symmetries that are neither exponential (proposition 21.2) nor differential (remark 21.3).

Remark 19.7 (On the weighted classification [02]). Source 02 states theorem 6.4 with “onto exactly when τ is onto”, which is the image formula there. It stresses that the classification concerns a carefully specified subclass: it does not assert that every field embedding preserves the chosen monomial section, fixes the real coefficient section, or preserves infinite normal-form sums, and an automorphism decomposition such as that of [5] cannot simply be read as an embedding decomposition, since factors requiring an inverse may cease to exist. The maps $H_{\tau, \chi}$ preserve the relation “is an initial normal-form truncation of” and transported cutoffs as in (10.4); preserving the *same* numerical cutoff is a different, much stronger requirement, and prefix truncation of signs is a third notion.

20 Named constants, omnific integers, and limits in source 02

Remark 20.1 (What a field embedding fixes automatically [02]). A unital field embedding $\sigma : \mathbf{No} \rightarrow \mathbf{No}$ fixes $\mathbb{Q}$, preserves and reflects positivity (positive elements of a real-closed field are nonzero squares), hence preserves order, and fixes every real algebraic number: the ordered real roots of a rational polynomial are individually characterized by their position in that finite ordered set (equivalently, they lie in the real-closed fixed field of [proposition 8.2](#)). It need not fix a real transcendental r : rational comparisons show only that $\sigma(r) - r$ is infinitesimal, which the residue map of [proposition 10.1](#) cannot see. If σ maps $\mathbb{R}$ into $\mathbb{R}$, it fixes $\mathbb{R}$ pointwise, because an increasing field embedding of $\mathbb{R}$ is determined by rational cuts. “Fixes the residue field” and “fixes the coefficient section” are different statements.

Source 02 illustrates the motion of reals by homogeneity: for a real transcendental r and a positive infinitesimal ε , the assignment $r \mapsto r + \varepsilon$ is an ordered isomorphism $\mathbb{Q}(r) \rightarrow \mathbb{Q}(r + \varepsilon)$, and ordered-field homogeneity of $\mathbf{No}$ (a set-sized back-and-forth through real closures, using choice) extends it to a field automorphism. That homogeneity theorem is printed by source 07 as [theorem 70.2](#) and is the collection’s `saut:thm:orderedhomogeneity`; the base’s Taylor automorphism (10.3) gives an explicit formula for an automorphism moving a real by an infinitesimal.

Theorem 20.2 (Pointwise omnific rigidity [02]; after `odg:thm:fractions`). *One has $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$. Consequently a field self-embedding fixing every omnific integer pointwise is the identity.*

Proof. For nonzero x choose a positive surreal β so large that $a + \beta > 0$ for every $a \in \text{supp}(x)$; this is possible because the support is a set: choose β above 0 and above the set of negatives of its members. Then ω^β and $x\omega^\beta$ are omnific integers, the latter with zero constant coefficient, by (8.1). Hence $x = (x\omega^\beta)/\omega^\beta$ is a quotient of omnific integers, and a field embedding fixing numerator and denominator fixes x . $\square$

[write] $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$ is `odg:thm:fractions` of `docs/surreal/omnific-diophantine-geometry` (“global monomial clearing”, proved there for every set S with one common monomial), with the Lean declarations `surreal_eq_omnific_fraction` and `omnificSurrealIsFractionRing` in `Surreal/Foundations/OmnificSupportBounds.lean` (lines 114 and 133 at HEAD); source 02 does not cite it. This theorem is the report’s home for the statement; sources 05 and 06 point here. It contrasts with [proposition 8.1](#): preserving $\mathbf{Oz}$ as a predicate is compatible with every proper map of [section 7](#), including the bounded J_c , whereas fixing $\mathbf{Oz}$ pointwise forces the identity.

Within the strong category a weaker hypothesis already suffices: a strongly $\mathbb{R}$ -linear embedding fixing every monomial fixes every normal form and is the identity. The word “strong” is what licenses the passage through an arbitrary set-sized normal-form sum.

Remark 20.3 (Three kinds of limit [02]). By [proposition 9.1](#), $1/n$ does not converge to 0 in the full surreal order topology: the neighborhood of radius $1/\omega$ contains no term. This does not make $\mathbf{No}$ discrete; it says that no *set* of other points approaches a point at all surreal scales. Class-indexed approximation is a separate formal notion, not a standard set-indexed net. Three kinds of “limit” must therefore be kept apart: unions of increasing prefix chains (as in the closure theory of [4] and [theorem 4.9](#)), Hahn sums of support-controlled families, and numerical order-topological limits. The pair J_c (bounded, summation preserving, nowhere continuous) and $J_{f_{\text{sh}}}$ (cofinal, hence continuous by [theorem 9.3](#)) shows that properness is not the obstruction to continuity; failure of cofinality is.

21 Exponential, omega-map and differential constraints in source 02

Source 02’s displacement lemma, kernel lemma, exponential–valuation rigidity, omega-map rigidity and cofinal two-map comparison are the base’s [lemma 11.1](#), (11.3), [theorems 10.3](#), 11.4 and 11.7 with

the same proofs; source 02 attributes the displacement argument to the repository’s exponential-rigidity report [13] and states that it “is not presented as a new result of this article”. It stresses that the bound B in lemma 11.1 is a field element, which is decisive in **No** because all finite surreals are bounded by the single surreal ω . The following items are additional.

Remark 21.1 (Nontrivial exponential automorphisms exist [02]; imported homogeneity). Theorem 11.4 does not say that every exponential automorphism is trivial. In the o-minimal exponential structure, ω and 2ω have the same type over $\mathbb{R}$: every definable one-variable set over $\mathbb{R}$ is eventually constant at positive infinity, so both realize the same cut over the definably closed set $\mathbb{R}$. The elementary map between their definable closures over $\mathbb{R}$ extends to an automorphism of $(\mathbf{No}, \exp)$ by the homogeneity of the surreal exponential field for set-sized elementary data, which source 02 imports from the literature [5] and does not prove. By theorem 11.4, such an automorphism necessarily moves some exact natural values. The argument uses elementary exponential data, not an arbitrary partial field map.

Proposition 21.2 (A value action that does not lift exponentially [02]; after EAR `thm:dilation`). *The field automorphism S_2 of remark 19.6 does not commute with $\exp$.*

Proof. S_2 induces multiplication by 2 on the value group. If it commuted with $\exp$, then with $v = v \circ \exp$,

$$v(S_2x) = v(S_2(\exp x)) = 2v(x) = v(2x),$$

so $S_2x - 2x$ would be finite for every x by (11.3). At $x = \omega$ this difference is $\omega^2 - 2\omega$, which is infinite. $\square$

[write] This is the case $r = 2$ of `thm:dilation` of `docs/surreal/exponential-automorphisms-rigidity`, which shows that no rational dilation $r \neq 1$ of the value group has *any* exponential lift; source 02 cites that report but not this theorem. It is printed here once for the report (home 02, as the earliest Part containing it); source 05 proves the obstruction for every positive rational dilation $q \neq 1$.

Even a cofinal, onto, strongly linear field symmetry can therefore fail the exponential condition: an arbitrary action on the value group is not an exponential lifting theorem.

Remark 21.3 (Dilations are not differential [02]). By proposition 13.1, a field embedding commuting with the Berarducci–Mantova derivation ∂ (normalized by $\partial\omega = 1$) fixes $\mathbb{R}$ and satisfies $\sigma(\omega) = \omega + r$ with $r \in \mathbb{R}$. In particular S_2 is not differential, since $\partial(S_2\omega) = \partial(\omega^2) = 2\omega \neq 1$. These are necessary conditions, not a classification; the derivation and its normalization must be named precisely, since changing either changes the centralizer. The additivity of the translation parameter r is source 06’s proposition 59.5.

22 Elementarity and definability in source 02

Proposition 22.1 (Elementarity in three reducts [02]). *Read formula by formula (section 3):*

- (i) *a numerical order embedding of **No** is elementary in the language of dense linear orders;*
- (ii) *an ordered additive embedding of **No** is elementary in the language of divisible ordered abelian groups;*
- (iii) *a field embedding of **No** is elementary in the language of ordered real-closed fields.*

Proof. Each theory has quantifier elimination in its language, and an embedding preserves and reflects atomic formulas. For (ii), $(\mathbf{No}, +, <)$ is a divisible ordered abelian group and an ordered additive embedding preserves and reflects the atomic order and linear relations. $\square$

Items (i) and (iii) are [proposition 12.1](#); (ii) is the additional clause. [write] In particular every additive lift A_f of [theorem 6.1](#), which is an ordered additive embedding whether or not f is additive, is elementary for divisible ordered abelian groups. The bounded J_c is an elementary ordered-field self-embedding; there is no contradiction, because the external bound in (19.2) is not in the image subfield and a first-order formula with parameters from the image cannot name it. For the exponential language one needs in addition model completeness of the real exponential field and the fact that $(\mathbf{No}, \exp)$ satisfies its theory [9, 3]; an exponential field embedding is then elementary in that language, but a pure field embedding need not preserve $\exp$ ([proposition 21.2](#)).

Remark 22.2 (Adding simplicity changes the formulas [02]). An embedding of $(\mathbf{No}, <, \preceq_s)$ need not be elementary. The root is definable as the unique prefix of every element, and P_p moves it for nonempty p . Even a root-fixing embedding need not be elementary: D_2 maps the immediate prefix-successor $+$ of the root to $++$, which is no longer an immediate prefix-successor, and the existence of an intermediate prefix is first-order in this language. (This is the base’s paragraph after [proposition 12.1](#), with the witness made explicit.) One must also distinguish preservation of a birthday function into an ordinal sort from an inequality between externally computed birthdays: an elementary embedding of a multisorted structure may move the ordinal sort, so it need not preserve birthdays as fixed external ordinals.

Source 02 proves the definability rigidity of [theorem 12.2](#) (stated there for field self-embeddings, equivalently field endomorphisms) by o-minimal calculus instead of by the fixed field.

Second route to theorem 12.2 [02]. Let h be a definable field endomorphism of the real-closed field K with o-minimal expansion. A definable one-variable function is continuously differentiable on some nonempty interval, after a finite cell decomposition [15]; so h is differentiable at some point a . Additivity gives

$$\frac{h(a+t) - h(a)}{t} = \frac{h(t)}{t},$$

so h is differentiable at 0, and by the same identity at every point, with one constant derivative d . The definable mean value theorem and $h(0) = 0$ give $h(x) = dx$ for every x . Since $h(1) = 1$, $d = 1$. $\square$

For $\mathbf{No}$ the argument is read in the formula-by-formula o-minimal structure, or in set-sized elementary exponential fragments; parameters do not evade the conclusion. The map c is semialgebraic, but J_c invokes the globally specified Conway normal-form operation, which is not definable in the pure ordered field or its o-minimal exponential expansion. “Definable in the ambient set theory” differs from “definable in the surreal field structure”, and a class relation with an elementary-embedding property differs again from a set-theoretic elementary embedding of V .

23 Birthday fragments and universe changes in source 02

Proposition 23.1 (Cutoff cardinality and sign maps at cutoffs [02]). *Let κ be an infinite cardinal and $\mathbf{No}_{<\kappa} = \{x : \mathbf{b}(x) < \kappa\}$.*

(i) $\mathbf{No}_{<\kappa}$ is a set of cardinality

$$|\mathbf{No}_{<\kappa}| = 2^{<\kappa} := \sup_{\lambda < \kappa} 2^{|\lambda|}. \quad (23.1)$$

(ii) For $p \in \mathbf{No}_{<\kappa}$ and $0 < \delta < \kappa$, the prefix map P_p of [proposition 4.2](#) and the block map D_δ of [proposition 18.4](#) send $\mathbf{No}_{<\kappa}$ into itself.

Proof. (i) There are $2^{|\alpha|}$ sign sequences of length α ; summing over $\alpha < \kappa$ gives (23.1). (ii) The birthday formulas $\mathbf{b}(p) +_{\text{ord}} \mathbf{b}(s)$ and $\delta \cdot_{\text{ord}} \mathbf{b}(s)$ stay below the infinite initial ordinal κ , by the cardinality bounds for ordinal sums and products of two ordinals below κ ; commutativity of ordinal arithmetic is not used. $\square$

The birthday cutoff κ and the carrier cardinality $2^{<\kappa}$ should never be silently identified.

Remark 23.2 (Which cutoffs are fields: the claim of source 02). Source 02 states: “For arbitrary ordinal cutoffs, closure under field operations is not automatic. The precise standard theorem is that $\mathbf{No}_{<\lambda}$ is a subfield exactly at epsilon-number cutoffs $\omega^\lambda = \lambda$; these fields are real closed and admit the inherited exponential structure”, citing van den Dries and Ehrlich together with their erratum [3]. It adds that every uncountable cardinal is such a cutoff. For an uncountable *regular* κ it records the description (14.1) of the base; at singular cutoffs, bounding the support cardinality is not enough, since the birthdays of the exponents can be cofinal in the cutoff, and the exact bounded-exponent formulation must be used instead.

[write] The claim was checked when this report was written. The “if” direction is van den Dries–Ehrlich with its erratum [3]; the collection states it in docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets and, as the corrected characterization, in the subsection found:sub:cutoffs of docs/foundations-and-computation/foundations. The converse has a short self-contained proof. Suppose $\mathbf{No}_{<\lambda}$ is a subfield. Since $\mathbf{No}_{<1} = \{0\}$ and $\mathbf{No}_{<2} = \{-1, 0, 1\}$ are not fields, $\lambda > 2$. (i) $\lambda = \omega^{\omega^\gamma}$ for some γ . An ordinal has itself as birthday, and the surreal product of two ordinals is their Hessenberg natural product, which is an ordinal at least as large as their ordinal product. So $\alpha, \beta < \lambda$ implies $\alpha\beta < \lambda$, and the ordinals above 2 closed under ordinal multiplication are exactly the ω^{ω^γ} . (ii) λ is an epsilon number. If $\gamma = 0$ then $\lambda = \omega$ and $\mathbf{No}_{<\omega}$ is the ring of dyadic rationals, which omits $1/3$. Let $\gamma \geq 1$ and suppose λ is not an epsilon number; then $\omega^\gamma < \lambda$ (if $\omega^\gamma = \lambda = \omega^{\omega^\gamma}$ then $\omega^\lambda = \lambda$), and γ is not an epsilon number either (otherwise $\lambda = \omega^\gamma = \gamma$ would be one), so $1 + \gamma < \omega^\gamma$. Gonshor’s sign expansion of ω^γ [2, Chapter 5] replaces each minus sign of y preceded by no plus sign with ω minus signs, after a leading plus. For $y = -\omega^\gamma$ this gives $\omega^{-\omega^\gamma}$ the signs (+) followed by $\omega \cdot \omega^\gamma$ minus signs, so its birthday is $\omega^{1+\gamma} < \omega^{\omega^\gamma} = \lambda$ (consistently, ω^{-1} has birthday ω and ω^{-2} has birthday $\omega \cdot 2$). Its inverse is the ordinal $\omega^{\omega^\gamma} = \lambda$, of birthday λ , so $\mathbf{No}_{<\lambda}$ is not closed under inverses, a contradiction. Hence $\mathbf{No}_{<\lambda}$ is a subfield exactly when λ is an epsilon number. This argument is the writer’s, unrefereed, and rests on the cited sign-expansion formula. The clause on the inherited exponential structure is the source’s, citing the same paper, and was not checked here.

Remark 23.3 (Internal sums, universes and transitive models [02]). The first and double lifts work internally on $\mathbf{No}_{<\kappa}$ for uncountable regular κ provided their exponent maps take that field into itself and only the families admitted by (14.1) are summed (proposition 14.1); an external family of size κ must not be passed through an internal “ $< \kappa$ -strong” map as though no size bound existed. In a proof assistant, $\mathbf{No}_u$ built from the ordinals small in a universe u lives in a higher universe; a map $\mathbf{No}_u \rightarrow \mathbf{No}_u$ is an ordinary function at that level, cut and sum index types remain u -small, and moving to $\mathbf{No}_v$ for $v > u$ is a change of universe, not an additional birthday inside $\mathbf{No}_u$. For transitive models $M \subseteq N$, a sign sequence coded in M belongs to both, but N may have additional sequences of old lengths; algebraic absoluteness (lemma 24.1) does not imply that the collection of all sequences is unchanged, nor that saturation persists in the outer model, so the self-embeddings internal to M and those available in N can be different collections. Set-sized fragments have honest endomorphism sets and automorphism groups with ordinary satisfaction and cardinal counting; the price is that fragment-level constructions are not automatically compatible, and a coherent system must satisfy $f_\lambda(x) = f_\mu(x)$ on overlaps (with output bounds) before a class union is taken.

24 Embeddings induced by large cardinals (conditional)

Hypothesis. Throughout this section assume an *amenable elementary embedding* $j : V \rightarrow M$, where M is a transitive inner model containing all ordinals and j has critical point κ . A normal-measure ultrapower provides this setting under the corresponding measurable-cardinal assumption; amenability ensures that restrictions of j to sets and the set-indexed image constructions below are legitimate in the

chosen class framework. For a surreal sign sequence s put

$$\mathcal{J}_j(s) = j(s) \in \mathbf{No}^M \subseteq \mathbf{No}^V. \quad (24.1)$$

This is an embedding induced from a universe embedding, not a claim that $\mathcal{J}_j$ is elementary for membership on V . The collection's docs/foundations-and-computation/large-cardinal-embeddings-and-normal-forms develops this map, its strong companion and their normal-form defects in full [18]; source 02 read only that report's README and isolates the short arguments needed for its taxonomy.

Lemma 24.1 (Absoluteness of old cuts and arithmetic [02]; compare `lce:lem:absolute`). *Let $M \subseteq V$ be a transitive inner model with all ordinals. Numerical order and prefix order on $\mathbf{No}^M$ are restrictions of those on $\mathbf{No}^V$. For sets $L, R \in M$ of old surreals with $L < R$, the simplest separator computed in M is the simplest separator in V . Consequently the field operations and the omega-map on old inputs are absolute.*

Proof. First disagreement and the prefix condition are absolute. Let $z \in M$ be M 's simplest separator of L, R . Every proper prefix of z is in M , being a restriction of the set function z to an ordinal of M . If an ambient separator y failed to extend z , then either y would be a proper prefix of z , or the first disagreement would exhibit a proper prefix of z numerically between y and z . Either way a proper prefix of z lies in the convex class of separators; it belongs to M , contradicting minimality of z there. Hence z is a prefix of every ambient separator. The genetic definitions of addition and multiplication use set-sized cuts of expressions in simpler inputs; well-founded induction on those inputs, followed by the cut assertion, proves absoluteness of the field operations, and the recursive cut definition of ω^x gives the omega-map in the same way. No claim about new sequences of old length is needed. $\square$

[write] `lce:lem:absolute` is the same statement with canonical normal forms and strongly summable M -families added; it assumes the same reals as well, which source 02's cut-by-cut argument on old inputs does not need. Source 02's lemma is not a conditional statement: it uses no large cardinal.

Theorem 24.2 (The induced surreal embedding [02]; conditional on an amenable elementary $j : V \rightarrow M$ with critical point κ ; after `lce:prop:J`). *Under the hypothesis of this section, $\mathcal{J}_j$ is a proper cofinal self-embedding of $\mathbf{No}$ preserving the ordered field operations, the prefix relation and the omega-map. It satisfies*

$$b(\mathcal{J}_j(s)) = j(b(s)), \quad \mathcal{J}_j(\alpha) = j(\alpha) \quad (\alpha \in \mathbf{On}),$$

and fixes every sign sequence of birthday less than κ . In particular it is not the identity and fixes $\mathbb{R}$ pointwise.

Proof. Elementarity transfers the defining sign-sequence relations and arithmetic identities into M , and lemma 24.1 identifies these operations with the ambient ones. The domain of $j(s)$ is $j(b(s))$, which is the birthday identity. The all-plus sequence of an ordinal goes to the all-plus sequence of the image ordinal. If $b(s) < \kappa$, that domain and every sign are fixed, so $j(s) = s$; since every finite ordinal and ω are fixed by j , $\kappa > \omega$, which bounds the birthdays of all reals.

No ordinal maps to κ : those below κ are fixed and the others map to ordinals at least $j(\kappa) > \kappa$. If a nonordinal sequence mapped to the all-plus sequence κ , elementarity of the all-plus property would make the input an ordinal. Thus κ is omitted and the map is proper. Finally $j(\alpha) \geq \alpha$ for ordinals α , so the image ordinals are cofinal in $\mathbf{No}$. $\square$

[write] This re-proves, under the same hypothesis, `lce:prop:J` of docs/foundations-and-computation/large-cardinal-embeddings-and-normal-forms (ordered-field embedding fixing $\mathbb{R}$, $J(\omega^a) = \omega^{J(a)}$, $b(Jx) = j(bx)$, prefix preservation and reflection, elementarity in the ordered-field

language); properness is `lce:cs:prop:proper` there. Source 02 adds nothing to those statements except the explicit omitted point κ and the fixation of all sign sequences of birthday below κ . It is the report's home for $\mathcal{J}_j$; source 06's measurable first-failure statement (its `found:prop:firstfailure`) concerns the same map.

$\mathcal{J}_j$ is a conditional nontrivial *field-and-simplicity* embedding, which none of the unconditional block maps of section 18 supplies. Being cofinal, it is continuous (theorem 9.3). It is elementary in the ordered-field reduct, but source 02 does not infer that the inclusion $\mathbf{No}^M \subseteq \mathbf{No}^V$ is elementary in expansions by birthdays, prefixes or set coding, and extending the conclusion to other recursively defined analytic operators requires their own absoluteness verification.

Remark 24.3 (The first pointwise cut failure [02]; conditional on j as above). As a surreal ordinal, $\kappa = \{\alpha : \alpha < \kappa \mid \emptyset\}$, and this is its canonical cut. Every option is fixed by $\mathcal{J}_j$, but $\mathcal{J}_j(\kappa) = j(\kappa) > \kappa$; so $\mathcal{J}_j$ does not preserve this cut pointwise. What elementarity preserves is the transformed *set* $j(\kappa)$ of options, not the pointwise image $j[\kappa] = \kappa$. On the other hand, if L and R each have cardinality less than κ , then $j(L) = j[L]$ and $j(R) = j[R]$ (enumerate them by ordinals below the critical point), and elementarity with lemma 24.1 gives

$$\mathcal{J}_j(\{L \mid R\}) = \{\mathcal{J}_j[L] \mid \mathcal{J}_j[R]\}.$$

So $\mathcal{J}_j$ preserves every cut with both sides of size $< \kappa$ and fails at a canonical cut of size κ : the least size of a pointwise cut failure is exactly κ . This is consistent with proposition 4.7, whose hypothesis concerns *all* set-sized (indeed all canonical) cuts.

[write] The collection has no statement about pointwise cut preservation of $\mathcal{J}_j$; this remark is new to it. It is the cut-theoretic analogue of the strong-summation threshold `lce:cor:firstfailure`.

Remark 24.4 (A transparent summation defect [02]; conditional on j as above). Consider the legitimate normal form

$$x = \sum_{\alpha < \kappa} \omega^{-\alpha}, \quad (24.2)$$

whose support is decreasing of reverse order type κ . Each term is fixed by $\mathcal{J}_j$, but applying j to the whole indexed normal-form object and using absoluteness gives

$$\mathcal{J}_j(x) = \sum_{\beta < j(\kappa)} \omega^{-\beta} \neq \sum_{\alpha < \kappa} \mathcal{J}_j(\omega^{-\alpha}) = x, \quad (24.3)$$

the extra tail having leading term $\omega^{-\kappa}$. Thus $\mathcal{J}_j$ is not strongly additive for all set-indexed Hahn sums, although it is an ordered-field and prefix embedding. For a summable family indexed by a set of cardinality less than κ , an enumeration below κ makes the transformed index set exactly the pointwise image, and $\mathcal{J}_j$ commutes with the sum. The failure is an exact index-size obstruction: no test of addition, multiplication, or even of every countable family can detect (24.3) when κ is measurable.

[write] (24.3) is `lce:thm:witness` ($J(h_\kappa) = h_{j(\kappa)}$) and the witness of `lce:cor:firstfailure` (the least size of a family on which J fails strong summation is exactly κ), an instance of the exact defect formula `lce:thm:defect`.

Remark 24.5 (Why Kunen's theorem is not being evaded [02]). Kunen-type inconsistency statements concern elementary embeddings $j : V \rightarrow V$ in a suitable class theory with choice and enough replacement [16]. Here the target is M , and the inclusion $M \subseteq V$ is not assumed elementary for membership, so the composite is not a forbidden elementary self-embedding of the universe. Conversely, a field self-embedding such as J_c does not secretly supply a $j : V \rightarrow V$: its language lacks membership, and even with a simplicity relation added one would have to prove an interpretation of the relevant set theory, its preservation by the map, and the class axioms before any set-theoretic inconsistency theorem applies.

[write] Added 4 October 2026, batch 89. The research-report collection’s `SetTheory/Cardinals/docs/reports/ordinals-and-order-types/naming-elementary-embeddings` (batch-89 manuscript 05; AI-assisted, unrefereed, not formalized) cites this report as a control example. In its urelement kernel models every injection of the atoms lifts to an amenable elementary self-embedding of the set-theoretic universe that fixes every pure set, so moves no ordinal (`nee:thm:lifts`), and naming such lifts can destroy Replacement in the expanded language (`nee:thm:profile`). Those embeddings act on a universe with atoms, not on the surreal field, and give no embedding of the pure-set presentations of **No**; nothing here changes.

25 Lexicographic coding and iteration in source 02

Proposition 25.1 (The word action [02]). *Let f be an order embedding of **No**, α a set ordinal, and $s : \alpha \rightarrow \mathbf{No}$ a word. Put $f_*s = f \circ s$. For words of the same length, f_* preserves and reflects the lexicographic order; it also preserves injectivity, and so acts on injective enumerations of set-sized subsets.*

Proof. The first differing coordinate of two words of the same length is unchanged by f_* , and its comparison is preserved and reflected by f . Injectivity of f preserves injectivity of words. $\square$

For variable-length words a termination convention must be chosen and preserved, as in the sign-sequence order itself (lexicographic with a central termination symbol). Pointwise transport is generally not a self-map of enumerations of the entire carrier: a proper embedding omits values, and no bijection $\alpha \rightarrow \mathbf{No}$ exists for a set ordinal α . If a well-order is represented as a relation rather than an enumeration, transport means pushing the relation forward to the image. Source 02 offers this as a bridge to the repository’s work on lexicographic well-order codes; these coding categories must be kept separate, for the same surjectivity and size reasons as surreal self-embeddings.

Remark 25.2 (Iteration without an inverse [02]). The composition law of theorem 6.3 gives $J_f^n = J_{f^n}$ for $n \in \mathbb{N}$. For the compression,

$$c^n(x) = \frac{x}{1 + n|x|},$$

by induction on n for $x \geq 0$ (indeed $c(x/(1 + (n-1)x)) = x/(1 + nx)$) and oddness. This is a controlled family of increasingly compressed order maps and field embeddings $J_{c^n} = J_c^n$; it asserts no convergence as $n \rightarrow \infty$ in the full order topology, which proposition 9.1 rules out for ordinary sequences. For prefix-chain-preserving maps a different iteration is available, taking unions at set limit stages when the iterates form an increasing prefix chain (theorem 4.9 and [4]); a proper-class union need not be a surreal, and this iteration is not interchangeable with Hahn sums of iterates or with numerical convergence. Compare the base’s iterates $f_-^n(a) = a/(1 - na)$ in section 8.

26 The formalization plan of source 02

Source 02 grounds its plan in the same universe-indexed `SignSequence` carrier displayed in section 3 [12]; that representation directly supports the prefix and block constructions. It read in full the simplicity module [17], which supplies `existsUnique_prefix_of_ordConnected`, `existsUnique_minimum_birthday_of_ordConnected` and `existsUnique_simplest_separator`. The last has an explicit separator-existence premise: its uniqueness conclusion must not be cited as an existence theorem for all cuts, and small-cut existence is a separate construction in the library. The header of `Surreal/Algebra/ExponentialProfile.lean` lists generic ordered-field results, including displacement amplification and profile rigidity, and names the actual surreal exponential instantiation as pending [14]; source 02 does not promote that inventory to a new kernel-verification claim.

Interfaces that prevent overstatement. Start from an order embedding and add separate predicates or structures rather than one opaque “surreal embedding” class; a single record forcing surjectivity would lose most examples, and a bare field-homomorphism proof would lose the distinctions needed for cuts, sums and analytic operators.

Interface	Essential field or obligation
PrefixEmbedding	$x \preccurlyeq_s y \leftrightarrow f(x) \preccurlyeq_s f(y)$.
BirthdayNonIncreasing	$b(f(x)) \leq b(x)$ for every input.
SmallCutMorphism	a quantified small index universe and equality of the two simplest-cut operations.
StrongMap	summability of the image family and equality of sums; index size stated in the type.
MonomialMap	an additive ordered exponent embedding and a positive real character, with a reconstruction theorem.
ExactValuePreserving	equality of natural values, not merely equivalence of comparison relations.
ExpCommuting	$f(\exp x) = \exp(fx)$ for the actual named exponential.
DerivationCommuting	equality for the specified derivation and normalization.
InitialRange	downward prefix closure of the image, separate from prefix preservation of the map.

A dependency-ordered sequence. (1) Pure sign combinatorics: ordinal concatenation, block replacement, first disagreement and prefix reflection; [theorem 4.5](#) then needs only canonical options, their birthday bounds and the simplest-separator theorem, and [theorem 5.2](#) one additional small-preimage argument, whose smallness should be proved rather than assumed in a universe-indexed implementation. (2) Set-sized Hahn series over an abstract ordered exponent type: transport, support equality, summability reflection, convolution compatibility, composition and the fixed-support lemma, and only then transport through the library’s identification with actual surreal normal forms. (3) The topology criterion, the displacement lemma and the equal-action comparison in a generic ordered-field module; the concrete exponential theorem as an instance whose obligations visibly include the actual natural valuation, total exponential, monotonicity, exponential law and the finite-kernel statement. (4) A large-cardinal module separating the ambient set-model embedding, absoluteness, transport of set codes and comparison with pointwise images, with two names such as `imageSet` and `pointwiseImage` rather than one overloaded notation, so that the critical-point discrepancy of [remark 24.3](#) is not discovered only after an invalid rewrite.

Verification levels. Source 02 distinguishes four deliverables: a written proof; a Lean statement with unproved assumptions; a kernel-checked generic theorem; and a kernel-checked instantiation for the actual constructed surreal operations, the last two to be reported with an axiom audit and exact dependency versions. It records that the repository README at its snapshot reports Lean/mathlib 4.32.0 and an axiom audit allowing `propext`, `Quot.sound` and `Classical.choice` [11], without having run that build.

27 Research questions of source 02

Source 02 formulates its questions so that a positive construction, a negative theorem, or a relative-consistency result would each be progress. They are not all claimed to be new open problems; the two published questions it cites are pinned to KKS version 3.

Research question 27.1 (Beyond the strong single-term class [02]). Which field self-embeddings admit a factorization into a coefficient-section change, a strong monomial transport, and an internal leading-term-preserving map? Which parts of an automorphism decomposition survive without inverses?

A first target is a necessary and sufficient condition for the image to admit a compatible chosen real section and monomial section; [theorem 6.4](#) solves reconstruction only once these strong hypotheses are available, and the non-surjective case must not be obtained by replacing “automorphism” by “embedding” in a group factorization. Compare the base question “Beyond monomial images” in [section 15](#).

Research question 27.2 (Fixed fields of double lifts [02]). Which additive subgroups and real-closed subfields occur as $\text{Fix}(A_f)$ and $\text{Fix}(J_f)$? What restrictions come from being the fixed-point class of a numerical order embedding?

[Equation \(19.1\)](#) reduces the problem to fixed sets one layer down. $\text{Fix}(J_c) = \mathbb{R}((\omega^{\mathbb{R}}))$ and $\text{Fix}(S_q) = \mathbb{R}$ show that set-sized and very small fixed fields arise; a next step is to prescribe $\text{Fix}(f)$ while controlling cofinality and the omitted intervals. Compare the base question “Prescribe the fixed field”.

Research question 27.3 (Cofinal versus bounded images [02]). Can image subfields of self-embeddings be classified by their cuts in **No**, intrinsic saturation, and ambient topological separation? Which bounded real-closed subclasses are isomorphic to all of **No**?

The continuity criterion ([theorem 9.3](#)) applies once an embedding exists; it does not characterize which subfields are images. The bounded double lift is one example, not a classification.

Research question 27.4 (Ordinal-fixing symmetries [02]). Describe the strong field embeddings fixing **On** pointwise, and distinguish those that preserve the omega-map from those that merely fix ordinal values.

[Example 19.5](#) shows a large gap between these conditions. A promising invariant is the action of the first lift on nonordinal exponents occurring inside the exponent of a monomial. Compare the base question “ ω -map compatibility” and [\(10.5\)](#).

Research question 27.5 (Field and prefix simultaneously [02]). Determine the existence and consistency strength of nonidentity proper embeddings of $(\mathbf{No}, +, \cdot, <, \preceq_s)$ under additional natural requirements, such as definability in set theory, fixation of all ordinals, or strong summability.

The conditional $\mathcal{J}_j$ of [theorem 24.2](#) gives existence without global strong summability; the block maps give unconditional existence without arithmetic. A satisfactory theorem should identify which extra condition closes the gap, rather than infer a Kunen contradiction from notation. Compare the base question “Field and simplicity together”.

Research question 27.6 (Critical points recoverable from surreal structure [02]). For an embedding preserving field operations and prefixes, can the least failure of pointwise cut preservation recover a critical point or a measure, under suitable definability and coding hypotheses?

For induced embeddings the failure is explicit at κ ([remark 24.3](#)). A converse requires an interpretation theorem and an actual construction of a measure; a birthday discontinuity by itself is not that construction.

[write] For the induced maps themselves the collection already recovers the measure, though from a summation defect rather than a cut failure: `lce:thm:measure` reads the normal measure derived from j off one defect coefficient of $J(h_A)$, $A \subseteq \kappa$, and `lce:thm:measurablecriterion` characterizes measurability of κ by a functional on these masks without assuming j . The question for abstract field-and-prefix embeddings and for cut failures remains open.

Research question 27.7 (Summability thresholds [02]). Under which hypotheses on an additive or field embedding does preservation of all sums below a cardinal force preservation of all set-sized sums? When can the first failure have a prescribed cardinality?

(24.3) is a measurable-cardinal conditional obstruction to an unrestricted upgrade. The repository has substantially broader claims about these thresholds [18]; source 02 asks for independent review and formalization of those claims before they are used as a classification theorem.

[write] For induced maps the threshold is exactly κ (`lce:cor:firstfailure`).

Research question 27.8 (Noncofinal exponential self-copies [02]). Does **(No, exp)** admit a noncofinal field self-embedding commuting with `exp` in the specified GBC framework? What additional support or definability assumptions forbid it?

The bounded Hahn example J_c does not answer this: it is not exponential. Any positive example would be discontinuous in the full order topology and would move exact natural values by theorem 11.4. Compare the base question “Proper exponential copies”.

[write] Source 04 answers the question positively under GBC with elementary transfinite recursion and a class satisfaction relation, which go beyond source 02’s GBC framework: theorem 34.2 gives bounded proper elementary self-embeddings of the exponential field. In bare GBC it remains open here.

Research question 27.9 (Faithfulness without a cofinal reference [02]). Can two distinct exponential field embeddings induce the same natural value-group embedding when neither image is cofinal?

Theorem 11.7 settles the case with a cofinal reference map; the proof shows why a bounded reference image cannot amplify a fixed difference. Removing the hypothesis needs a new argument or a counterexample. Compare the base question “Remove or justify the cofinality hypothesis”.

Research question 27.10 (Omega-map automorphisms [02]). Classify the nontrivial automorphisms commuting with the omega-map, and determine whether an order isomorphism of the generalized epsilon numbers extends to such an automorphism.

These are the directions of Questions 5.6 and 5.7 of [5] (v3). The conditional $\mathcal{J}_j$ is proper and does not settle the automorphism question; theorem 10.3 treats only the exact-value kernel. Compare the base question “ ω -map compatibility”.

Research question 27.11 (Differential–exponential centralizers [02]). For the actual Berarducci–Mantova derivation, which field embeddings commute with both ∂ and `exp`? Which real translations of ω extend to such maps, and with what summability properties?

The necessary condition $\sigma\omega = \omega + r$ (proposition 13.1) is not a sufficiency theorem and selects no unique extension. Compare the base question “Differential centralizers” and source 06’s proposition 59.5.

Research question 27.12 (Minimal expressive expansions [02]). Which natural expansions of the ordered field make a nontrivial Hahn lift internally definable? Which of them remain tame enough for useful quantifier elimination or decidability results?

Theorem 12.2 rules out o-minimal hosts. Naming support extraction, coefficient functions, a monomial section or a discrete integer predicate changes the problem substantially; the interpretation strength should be measured, not guessed.

Research question 27.13 (Compatibility at singular cutoffs [02]). Classify coherent self-embedding systems on epsilon-number birthday fragments, especially at singular limits, and determine which give class embeddings after union.

The support-cardinality condition alone fails at singular cutoffs; a criterion must control the birthdays of transported exponents and the agreement of the stage maps on overlaps. Compare the base question “Universe and cutoff absoluteness” and the club of closure ordinals in section 14.

Research question 27.14 (A verified embedding library [02]). Can the transport theorem, double lift, exact fixed-support theorem, continuity criterion, and concrete exponential rigidity instance be added to ProveIt with minimal extra assumptions and an explicit axiom audit?

These form a dependency-ordered project: generic Hahn algebra, sign-sequence rigidity, generic topology, then the actual surreal instantiation. The first milestone should be a kernel-checked statement distinguishing strong summation from continuity via one explicit proper double lift. Compare the base question “Near-term formalization”.

Research question 27.15 (Class principles and parameterized homogeneity [02]). What is the weakest class theory needed for each global extension theorem used in surreal embedding arguments? Can the results be recast as coherent set-level schemes with explicit choice and recursion requirements?

This separates termwise constructions using only set histories from back-and-forth procedures that choose arbitrary classes at stages, and would make comparisons between GBC, stronger class recursion and universe-polymorphic type theory precise. Compare the base question “Minimal foundational strength”.

28 Source audit and regression checks of source 02

28.1 Repository boundary

Source 02 pinned all repository references to snapshot `af04de3fa423f8f3253ee28db9afeb7fc6171604` and read, through connected GitHub tools: the surreal project `README.md` (partly truncated; no claim to have audited every entry or the formalization ledger); `Surreal/Foundations/SignSequence.lean` lines 1–110; `Surreal/Foundations/SignSequenceSimplicity.lean` in full; `docs/surreal/exponential-automorphism-rigidity/README.md` (theorem inventory and status; partly truncated, not an audit of the whole manuscript); `Surreal/Algebra/ExponentialProfile.lean` lines 1–160 (not kernel-checked); and lines 1–170 of the `README` of `docs/foundations-and-computation/large-cardinal-embeddings-and-normal-forms` (the 109-page article itself was not read, so its equalizer, independence and measurable-spectrum assertions were not imported). Its repository claims were checked when this report was written:

Claim of source 02	Verdict
<code>ExponentialProfile.lean</code> leaves the actual surreal instantiation pending	true (module header)
<code>existsUnique_simplest_separator</code> has an existence premise; cut filling is separate	true (<code>SignSequenceSimplicity.lean:72</code> ; existence in <code>SignSequenceCutOperation.lean</code> , <code>exists_separator_of_small_sets</code>)
<code>existsUnique_prefix_of_ordConnected</code> and <code>existsUnique_minimum_birthday_of_ordConnected</code> exist	true (lines 41 and 52, at the pin and at HEAD)
the sign carrier is universe indexed; bounded fragments small, carrier not small	true (<code>SignSequence.lean</code>)

Claim of source 02	Verdict
README reports Lean/mathlib 4.32.0 and the three-axiom audit	true (README lines 117 and 128 at the pin)
the exponential-rigidity report resolves KKS Question 5.4 (v3) and its generic formalizations do not yet instantiate the surreal exponential	true (EAR thm:no, cor:question54; header of the Lean module)
the large-cardinal report develops J and its normal-form defects	true (lce:prop:J, lce:thm:defect)

No claim of source 02 says the collection lacks something it has; its omissions are the uncited re-proofs listed in the note of section 17.

28.2 External literature boundary

Source 02 checked the normal-form and cutoff results against van den Dries–Ehrlich, in particular its set/class conventions, its Proposition 4.7 and its exponential-field discussion, and inspected the published erratum, which corrects a support-order bound while retaining the main results [3]. Bagayoko–van der Hoeven (read as arXiv:2305.02001v1) was used for the setting of surreal substructures, concatenation and prefix-chain closure, not as a claim that the field lifts preserve simplicity [4]. Kaplan–Krapp–Serra was read as arXiv:2509.22374v3 (submitted 23 April 2026; PDF dated 27 April 2026); printed page 10 was checked for Propositions 5.1, 5.2, 5.5 and Questions 5.4, 5.6, and the source does not depend on any proposed infinite-support example of a nonstrong automorphism from that paper [5]. Berarducci–Mantova was checked for the derivation’s constant field, exponential compatibility and $\partial\omega = 1$ [8]. The Kunen discussion follows Hamkins–Kirmayer–Perlmutter [16]. Model completeness [9] and o-minimal calculus [15] are imported, not reproved.

28.3 Finite regression checks

The delivered script ships as `code/02-paradox-free-validate.py` (Python 3.10+, standard library only, exact rational arithmetic, random seed 20261003), with its recorded output `data/02-paradox-free-validation_results.json`: status PASS, 198,220 assertions; a fresh rerun when this report was written reproduced the output byte for byte up to line endings. Its 29 groups test, on all sign words of length at most 6, first-disagreement comparison against dyadic values, and order and prefix preservation by P_+ , P_{+-} , D_2 , D_3 and the localized map of proposition 18.5 (with $p = + - +$), including its fixation of short words and of all-plus words; on a rational grid, the range, inverse, unique fixed point, order and iterates $c^n(x) = x/(1 + n|x|)$ of the compression; on random finite supports, additivity, composition and the fixed-support criterion of the compression lift, multiplicativity of the lift by $a \mapsto 2a$, and the counterexample showing that the compression lift is not multiplicative; on Laurent polynomials under the substitution $t \mapsto 2t$, the twisted displacement identity (11.2), a counterexample to its untwisted form, and explicit two-probe amplification witnesses; and the explicit values $D_2(1/2) = 5/4$ and $P_+(-1) = 1/2$. The script writes `validation_results.json` beside itself (under its original name), so it should be rerun on a copy. The shipped `code/02-paradox-free-build.sh` also compiles the source’s own `surreal_self_embeddings.tex`, which is not shipped; its data record `data/02-paradox-free-build_validation.json` carries three SHA-256 values of the delivered files (verified correct at intake) and is a provenance record, not a manifest.

A successful run checks finite algebra and the implemented formulas only. It cannot prove reverse-well-order preservation at all ordinal lengths, test a proper-class image, verify a measurable-cardinal hypothesis, or establish global strong summability from finite samples; the written proofs carry those conclusions.

29 Status and non-claims of source 02

Source 02 states, in its article, README and 02-paradox-free-PROOF_STATUS.md:

- “Proved here” means that a mathematical proof is supplied; it does not mean historically new, independently refereed, or Lean-checked. No priority claim is made for the deductions assembled; the new combinations and examples have not received independent specialist review.
- The large-cardinal construction is conditional on its explicitly stated embedding hypothesis (an amenable elementary $j : V \rightarrow M$); it does not assert an elementary membership embedding $V \rightarrow V$, and a field-embedding conclusion is kept distinct from an elementary-embedding conclusion in an expanded surreal language.
- The exponential–valuation argument is attributed to the repository’s exponential-rigidity research and “is not presented as a new result”. The repository’s generic algebraic formalizations do not, by themselves, provide a machine-checked instance for the actual surreal exponential; that bridge remains a formalization task.
- The taxonomy is an interpretation framework, not a classification of every expansion of the surreal field; theorem 6.4 classifies only a specified subclass.
- The research questions are not all claimed to be new open problems.
- The finite checks are regression tests, not a formal verification of the transfinite theorems; the double-lift theorem and its infinite-support consequences rest on the written proofs.
- No new Lean file is included and no Lean or Mathlib build was executed; module headers are reported as source inventory, not re-verified kernel results. The entire repository was not audited.
- The broader threshold claims of the large-cardinal report need independent review before use as a classification theorem.
- Standard structural results on the surreal field, its exponential, its derivation, model completeness and o-minimal calculus are imported from the literature, as is the homogeneity of the exponential field used in remark 21.1.
- Delivery: the PDF compiled without unresolved references or LaTeX warnings and all 36 pages were visually reviewed (provenance only; the PDF is not shipped).

Part III

Source 04: A Structural and Foundational Atlas

30 Source 04 at a glance

Source 04 is *Self-Embeddings of the Surreal Numbers: A structural and foundational atlas*, subtitled “Order, simplicity, Hahn series, exponentiation, model theory, and universe-sensitive formalization”, dated 3 October 2026 and headed “Research monograph”. Its title page reads “Research article prepared with ChatGPT” (also its PDF author field); it is the only one of the six sources that names the tool used. It pins VladimirReshetnikov/ProveIt at commit a006a77a0a8bacdfef089b70d3b963b17228a089 [22]. The delivered PDF has 35 pages; the source has 54 numbered environments (14 of them research questions), two appendices (a proof-obligation ledger and a compact audit of common invalid arguments),

a README, a source audit, and a finite regression script. Its foundational setting is GBC (Gödel–Bernays class theory with global choice) for class maps, a ZFC definable-class reading for the explicit support substitutions, and *GBC plus elementary transfinite recursion (ETR)*, with a fixed class satisfaction relation, for its model-theoretic section only (section 34); it treats regular-cardinal cutoffs and universe-polymorphic types separately.

Its distinctive contributions, all printed in this Part, are: the explicit bounded double lift J_c with bound ω^ω (theorem 32.1); a proper cofinal embedding with fixed field exactly $\mathbb{R}$ (example 32.2); the exact iterated-image formula $K_\infty(J_f) = \mathbb{R}((\omega^{\mathbb{R}((\omega^{\cap_n f^n(\mathbf{No}))}))})$ (theorem 32.3); bounded-image and proper cofinal *elementary* self-embeddings of every o-minimal class expansion of the surreal field, including the surreal exponential field, under GBC+ETR (theorems 34.2 and 34.5); the abstract exponential value-displacement and cofinal pair-rigidity theorems (theorems 35.1 and 35.3); the exact criterion for a monomial lift H_g to commute with $\exp$ (proposition 35.4); the failure of nontrivial rational exponent dilations to be differential (proposition 36.3); the converse classification of conjugation-compatible surcomplex extensions (proposition 36.6); and a five-stage Lean integration plan (section 38). Its conclusion section is expository: no single rigidity answer exists; support images control explicit field embeddings, cofinality controls ambient continuity, set-cut saturation controls elementary extensions, and the twisted Leibniz identity controls exponential growth rigidity.

30.1 Renames

Here	Source 04	Reason
$b(x)$	$\text{bd}(x)$	report birthday symbol
$\mathbb{R}((\omega^A))$	$\mathcal{H}(A)$	base notation for the class of series supported in A
H_g (g ordered additive)	A_g , “normalized Hahn-field embedding”	the report keeps A_f for the additive first lift only; source 04 writes A for both lifts
$J_f = H_{A_f}$	$\mathcal{J}_f = A_{A_f}$	base double lift; $\mathcal{J}_j$ is reserved for large-cardinal maps
D_δ (ordinal δ)	B_d	report sign-block symbol; B is the base’s $\mathbb{R}((\omega^{\mathbf{No}_{\geq 0}}))$
$K_\infty(j)$	$\text{Core}_\omega(j)$	base core symbol
T (base (10.3)) and T^{-1}	T_D, T_{-D}	the same operator; $T = \mathcal{T}_{\Delta,1}$ in the opa notation
Δ (derivation of $\mathbb{R}$), r_0 with $\Delta(r_0) = 1$	D, b	D is the report’s displacement; b is a bound in section 34
$D(x) = j(x) - x$; $d = j - k$	δ	report displacement; base lemma 11.6 writes d
τ_j	τ (in its exponential section)	report value-action symbol
$\mathfrak{M}, \mathfrak{M}_{\text{exp}}, \mathfrak{N}$	$\mathcal{M}, \mathcal{M}_{\text{exp}}, N$	$\mathcal{M}_\varphi$ is the report’s multiplicative conjugate
$\text{Th}(\mathfrak{M})$	T	T is the Taylor automorphism
$K_0; X, X'; Y; \pi$	$K; A, A'; B; f$	elementary submodels, image, partial map; K, A, B, f have base meanings
ℓ (target landmark)	c	c is the interval compression
$s > 1$ with $w(s) < 0$	T	as above
$\Gamma_c = \mathbb{R}((\omega^{(-1,1)}))$	Γ	Γ is a value group in section 35
$S_q = H_{a \mapsto qa}$	$A_g, g(x) = qx$	report exponent dilation
$\mathcal{M}_h = \exp \circ h \circ \log$	unnamed	report multiplicative conjugate

Here	Source 04	Reason
$\mathbf{No}(i)$, $\mathbf{No}_U$	$\mathbf{No}[i]$, $\mathbf{No}^u$	base notation

Kept, with disclosure: $I = \mathbb{R}((\omega^{\mathbf{No}_{>0}}))$ (purely infinite surreals, including 0; a subgroup of the base's $B = \mathbb{R}((\omega^{\mathbf{No}_{\geq 0}}))$ and not $\mathbf{Oz}$); $e_* : I \rightarrow \mathbf{No}$ (equation (35.3)); $\mathbf{m}$ for the infinitesimals and $\mathbf{O}$ for the finite surreals (as in the base). The tempting false reading to avoid: source 04's " A_g " for an *additive* g is a field embedding (here H_g), while A_f for an arbitrary increasing f is only additive.

30.2 Correspondence table

Numbers are those of the source PDF (counter shared within each section). "Same proof" means the source's proof is the cited one up to wording and notation.

Source 04 item	Content	Where printed
§§1–2 (unnumbered)	carrier, embedding specification, map of conclusions, foundations, $\eta_{\mathbf{On}}$ interpolation, three notions of definability	base sections 2 and 3; source-specific points in section 31
Prop. 3.1	interval compression	= proposition 4.1 (same proof)
§3.2 prefix maps	P_p , examples $P_+(0) = 1$ etc.	= proposition 4.2; examples in remark 31.1
§3.2 block maps	D_δ for ordinal δ , commutes with negation, $D_2(1) = 2$	home 02, proposition 18.4; integer case proposition 4.3; remark 31.1
§3.2 branch swaps	tree automorphisms without order	base prose after proposition 4.3
Thm. 3.2	birthday rigidity	= theorem 4.5 (same proof)
Cor. 3.3	birthday-preserving or ambient-cut-preserving $\Rightarrow$ id	= theorem 4.5 and proposition 4.7 (same proof); P_+ remark in remark 31.1
Thm. 3.4	$\text{Aut}(\mathbf{No}, <, \preceq_s) = \{\text{id}\}$	= proposition 4.6 (same proof)
Thm. 3.5	simplicity-initial image $\Rightarrow$ onto	= theorem 5.2 (same proof)
Cor. 3.6	field embedding with initial image is onto; identity if simplicity is preserved	= paragraph after theorem 5.2; compare <code>isg:cf:prop:coherence</code>
Prop. 3.7	convex subfield is everything	= proposition 5.3 (elementary variant of the same argument)
§4.1	support classes, strong summation	base Definition (strong summability)
Thm. 4.1	first lift A_f	= theorem 6.1 (same proof); "multiplicative iff f additive": home 02, proposition 19.1; "fixes $\mathbb{R}$ iff $f(0) = 0$ ": remark 31.1; credit <code>opa:as:lem:inner</code>
Rem. 4.2	A_c fixes 1 but is not multiplicative	remark 31.1 (printed)
Def. 4.3	$J_f = H_{A_f}$	= (6.10)
Thm. 4.4	double lift, image $\mathbb{R}((\omega^{\mathbb{R}((\omega^{f(\mathbf{No}}))}))$), faithful	= theorem 6.3 with (6.2), (6.6) (same proof, same witness (6.11)); credit <code>opa:es:prop:lifts</code>
Thm. 4.5	bounded J_c , bound ω^ω	theorem 32.1 (printed)
Thm. 4.6	J_- fixes all ordinals, proper cofinal	= theorem 7.1 (same proof)
Thm. 4.7	weighted monomial classification, composition law	= theorem 6.4 (same proof); credit <code>isc:prop:classification</code> , <code>saut:thm:monomial</code>
Lem. 5.1	rigidity of reverse well-orders	= step in the proof of theorem 6.1
Thm. 5.2	exact fixed-support formulas	= (6.4), (6.9), (6.13) (same proof); credit <code>opa:es:thm:fixedsupport</code>
Ex. 5.3	$\text{Fix}(J_c) = \mathbb{R}((\omega^{\mathbb{R}}))$	home 02, theorem 19.3; restated after theorem 32.1
Ex. 5.4	f_{gap} : $\text{Fix} = \mathbb{R}$ exactly	example 32.2 (printed)

Source 04 item	Content	Where printed
Prop. 5.5	j bijective on its core, maximality	= proposition 8.4 (same proof; source states it for any injective class self-map)
Thm. 5.6	$K_\infty(J_f) = \mathbb{R}((\omega^{\mathbb{R}((\omega \cap f^n(\text{No})))}); K_\infty(J_c)$	theorem 32.3 (printed)
Prop. 5.7	fixed field real closed; no finite cycles	= proposition 8.2 (same proof); credit <code>opa:es:lem:realclosed</code>
Prop. 6.1	field embeddings ordered, fix real algebraics, RCF-elementary	= propositions 8.2, 10.1 and 12.1; credit <code>opa:as:thm:elementary</code>
Prop. 6.2	$j^{-1}(O) = O$, residues fixed, $j(\mathbb{R}) \subseteq \mathbb{R} \Rightarrow$ fixed	= proposition 10.1 and the proof of proposition 13.1 (same proof)
Prop. 6.3	Taylor twist T	= base (10.3) paragraph (same proof); credit <code>saut:thm:coeffflow</code> , <code>opa:as:thm:explicit</code> ; Oz failure in remark 36.5
§6.4 (unnumbered)	$\tau_j, \tau_{H_g} = g, \tau_{J_f} = A_f, \tau_T = \text{id}$	remark 33.1 (printed)
Thm. 7.1	continuity $\Leftrightarrow$ cofinality, any ordered field; τ_j clause	home 07, theorem 67.1, and base theorem 9.3; τ_j clause and J_f corollary in proposition 33.2 (credit <code>opa:as:thm:topology</code>)
Prop. 7.2	convergent set nets eventually constant	= proposition 9.1 (same proof); strong sum versus limit in remark 33.3
§8.1	GBC+ETR class-model setting	section 34 (printed)
Lem. 8.1	saturation over every set	lemma 34.1 (printed; home)
Thm. 8.2	bounded elementary self-embeddings	theorem 34.2 (printed; home)
Rem. 8.3	what the theorem does not say	remark 34.3 (printed)
Lem. 8.4	omission-preserving extension	lemma 34.4 (printed; home)
Thm. 8.5	proper cofinal elementary self-embeddings	theorem 34.5 (printed; home)
§8.4 (unnumbered)	homogeneity; moving r to $r + \varepsilon$	remark 34.6 (printed); ordered-field version home 07, theorem 70.2
Thm. 8.6	o-minimal definability rigidity	second route after theorem 12.2 (printed in section 34)
Lem. 9.1	unbounded displacement, one map and two maps	= lemmas 11.1 and 11.6 (same twisted Leibniz proof); credit <code>EAR lem:amplify</code>
Thm. 9.2	abstract exponential value displacement	theorem 35.1 (printed; home)
Cor. 9.3	value-pointwise exponential embeddings are trivial	corollary 35.2 (printed; home; = theorem 11.4, route through 9.2)
Thm. 9.4	abstract cofinal pair rigidity	theorem 35.3 (printed; home)
Prop. 9.5	H_g commutes with exp iff $g \circ e_* = e_* \circ H_g$ on I	proposition 35.4 (printed)
Prop. 10.1	Ω -compatibility of H_g ; $\Omega +$ pointwise valuation $\Rightarrow$ id	proposition 36.1 (printed; = (10.5), theorem 10.3)
§10.2 (unnumbered)	additive, multiplicative, vector-space readings	remark 36.2 (printed)
Prop. 10.2	differential: j fixes $\mathbb{R}$, $j(\omega) = \omega + r$	= proposition 13.1 (same proof); translation parameter home 06, proposition 59.5
§10.3 (unnumbered)	rational dilations not differential	proposition 36.3 (printed)
§10.3 (unnumbered)	substitution domains, ADH universality, H -field homogeneity	remark 36.4 (printed)
§10.4 (unnumbered)	naming additional predicates	remark 36.2 (printed)
§11.1 (unnumbered)	Oz and omnific floor	base (8.1)
Prop. 11.1	H_g preserves Oz and floor	= proposition 8.1 (same proof)
§11.1 (unnumbered)	T does not preserve Oz	remark 36.5 (printed)

Source 04 item	Content	Where printed
§11.2 (unnumbered)	complexification, converse, modulus	proposition 36.6 (printed); credit <code>opa:as:thm:complexembed</code> , <code>saut:thm:axis</code> , <code>hset:thm:two-lifts</code>
§12.1 (unnumbered)	$\mathbf{No}_{<\lambda}$ a subfield iff λ is an ε -number	section 37 (printed as the source’s claim); see remark 23.2
§12.1 (unnumbered)	regular κ : interpolation, $2^{<\kappa}$, $\kappa^{<\kappa} = \kappa$ caveat	remark 37.1; explicit lifts = proposition 14.1
§12.2–12.3 (unnumbered)	type-theoretic version; local workspaces	section 37 (printed)
§13 (unnumbered)	repository integration, Stages A–E, pseudocode	section 38 (printed)
Questions 14.1–14.14	research questions	section 40 (all printed)
§15	conclusions	summarized in this section
App. A	proof-obligation ledger	section 39 (printed)
App. B	audit of invalid arguments	section 39.1 (printed)

31 Foundational setting and small examples of source 04

Source 04 reads concrete constructions in ZFC through defining formulas with set parameters, states assertions about arbitrary class maps in GBC, and uses GBC+ETR only in section 34; it calls that foundation “a sufficient foundation, not a minimality claim” and refers to the relation between ETR and class truth as a genuine foundational issue [20, 21]. Its name for the set-cut interpolation property of $\mathbf{No}$ is used below:

$$L, R \text{ sets, } L < R \implies \exists x (L < x < R) \quad (\text{the } \eta_{\mathbf{On}} \text{ interpolation property}). \quad (31.1)$$

It distinguishes three notions that the base treats more briefly (section 12): an *external* set-theoretically definable map may mention birthdays, normal forms and a global well-order; an *internally definable* map in $(\mathbf{No}, +, \cdot, <, \exp)$ is given by one first-order formula with finitely many parameters; a *class map* in GBC need not be definable without class parameters. “Elementary” always refers to a specified language, and a field self-embedding is not an elementary embedding of $(V, \in)$ (as in base section 3).

Remark 31.1 (Small examples [04]). (i) For $p = +$ the prefix map has $P_+(0) = 1$, $P_+(1) = 2$, $P_+(-1) = \frac{1}{2}$; it does not preserve 0, so there is no conflict with field rigidity. Its image of the empty ambient cut is 1, whereas the empty cut is 0 in $\mathbf{No}$: a prefix map preserves the *relative* simplest-between operation of its image, not the ambient one (compare the base remark on relative cuts after proposition 4.7). (ii) The block map D_δ (each sign repeated δ times, δ a nonzero ordinal; home 02, proposition 18.4) fixes 0 and commutes with negation; $D_2(1) = 2$, so it is not unital. Prefix and block maps refute the assertion that every order-and-simplicity self-embedding is the identity. (iii) The first lift A_f satisfies $A_f(r) = r \omega^{f(0)}$ for $r \in \mathbb{R}$, so it fixes $\mathbb{R}$ pointwise exactly when $f(0) = 0$. For the compression c the first lift A_c fixes 1 but is not multiplicative: $A_c(\omega^2) = \omega^{c(2)} = \omega^{2/3} \neq \omega = A_c(\omega)^2$, because $c(2) = 2/3 \neq 1 = 2c(1)$. A single exponent relocation by an order map is not yet a field embedding; the second lift is the crucial step.

32 Hahn lifts: a bounded copy, fixed fields, and cores

32.1 An explicit bounded field copy

The following is the same map as source 02’s bounded embedding (theorem 19.3), with an exact image formula and a smaller explicit bound; base theorem 7.6 with $A = \emptyset$ gives a different map with the bound $\omega^{\omega^{\gamma+1}}$, $\gamma > 0$.

Theorem 32.1 (A bounded-image field self-embedding [04]; after `opa:as:thm:parameters`). *Let $c(x) = x/(1 + |x|)$ and $J_c = H_{A_c}$. Then J_c is a proper strongly $\mathbb{R}$ -linear ordered-field self-embedding with*

$$\text{im}(J_c) = \mathbb{R}((\omega^{\Gamma_c})), \quad \Gamma_c = \mathbb{R}((\omega^{(-1,1)})), \quad |J_c(x)| < \omega^\omega \quad (x \in \mathbf{No}). \quad (32.1)$$

It fixes every real and fixes ω , but $J_c(\omega^\omega) = \omega^{\omega^{1/2}}$.

Proof. The image description is (6.2) followed by (6.6). If $0 \neq \gamma \in \Gamma_c$ has leading exponent $a < 1$, then $|\gamma| < \omega$: the ratio $\omega/\omega^a = \omega^{1-a}$ dominates every positive real coefficient. Thus every element of Γ_c lies between $-\omega$ and ω . If $0 \neq y \in \mathbb{R}((\omega^{\Gamma_c}))$ has leading exponent $\gamma < \omega$, the same comparison gives $|y| < \omega^\omega$. The image is therefore proper. Since $c(0) = 0$, $A_c(1) = 1$ and $J_c(\omega) = \omega$. On the other hand $A_c(\omega) = \omega^{c(1)} = \omega^{1/2}$, which gives the displayed moved value. $\square$

The image is a real-closed field, being isomorphic to $\mathbf{No}$, yet it is bounded in the ambient field. Applying the bound to $1/y$ shows that every nonzero image element satisfies $|y| > \omega^{-\omega}$: the image has a uniform ambient gap around 0, despite being densely ordered internally. (The repository's parameter theorem `opa:as:thm:parameters` has the same kind of two-sided bound $\Omega^{-1} < |y| < \Omega$ for its discrete copy.) Since the only fixed point of c is 0, (6.9) gives $\text{Fix}(J_c) = \mathbb{R}((\omega^{\mathbb{R}}))$ (source 04's Example 5.3; printed by source 02): the real-exponent Hahn subfield, not merely $\mathbb{R}$; ω^r is fixed for every real r , although ω^ω is not. By theorem 9.3, J_c is nowhere continuous, while it preserves all strong sums.

[write] The bound ω^ω is smaller than source 02's bound for the same map and than the base's bound for its (different) bounded map; it is not claimed to be optimal. For instance $\omega^{\omega-1}$ also bounds the image, since $\omega^a < \omega - 1$ for every exponent $a < 1$.

32.2 A proper cofinal embedding with fixed field exactly the reals

Example 32.2 (A proper embedding with fixed field exactly $\mathbb{R}$ [04]). Let

$$f_{\text{gap}}(x) = \begin{cases} x + 1, & x < 0, \\ x + 2, & x \geq 0. \end{cases}$$

This is an increasing injection onto $(-\infty, 1) \cup [2, \infty)$ with no fixed points. Hence $J_{f_{\text{gap}}}$ is proper (theorem 6.3) and, by (6.4) and (6.9),

$$\text{Fix}(J_{f_{\text{gap}}}) = \mathbb{R}((\omega^{\mathbb{R}((\omega^{\mathbb{Q}}))})) = \mathbb{R}((\omega^{\{0\}})) = \mathbb{R}.$$

Its image is cofinal: the inner exponent group contains monomials $\omega^{f_{\text{gap}}(a)}$ with arbitrarily large exponents, and the outer field then contains arbitrarily large monomials. By theorem 9.3 this is a proper *continuous* field embedding whose fixed field is precisely the designated real field, in contrast to the larger fixed fields $\mathbb{R}((\omega^{\mathbb{R}}))$ of J_c and $\mathbb{R}((\omega^B))$ of J_- ((8.3)).

32.3 The iterated-image core of a double lift

Base proposition 8.4 is stated in source 04 for any injective class self-map $j : X \rightarrow X$, with the same proof: j restricted to $K_\infty(j) = \bigcap_{n \in \mathbb{N}} j^n(X)$ is a bijection, and this is the largest subclass on which j restricts to a bijection. The quantifier over n is set-sized, so no collection of class images is formed.

Theorem 32.3 (Iterated-image formula [04]). *For every increasing injection $f : \mathbf{No} \rightarrow \mathbf{No}$,*

$$K_\infty(J_f) = \mathbb{R}((\omega^{\mathbb{R}((\omega^{\bigcap_{n \in \mathbb{N}} f^n(\mathbf{No}))}))}). \quad (32.2)$$

Proof. The composition law gives $J_f^n = J_{f^n}$ (theorem 6.3). For any set-indexed family of classes (A_i) , the support definition gives $\bigcap_i \mathbb{R}((\omega^{A_i})) = \mathbb{R}((\omega^{\bigcap_i A_i}))$. Apply this twice to the image formula $\text{im } J_{f^n} = \mathbb{R}((\omega^{\mathbb{R}((\omega^{f^n(\mathbf{No})}))}))$. $\square$

This contains base (8.5) (one level) and (8.6) (the case $f = f_-$) as instances. For the compression, direct algebra gives

$$c^n(x) = \frac{x}{1 + n|x|}, \quad c^n(\mathbf{No}) = (-1/n, 1/n) \quad (n \geq 1).$$

With $\mathfrak{m}$ the additive group of all infinitesimals (including 0),

$$K_\infty(J_c) = \mathbb{R}((\omega^{\mathbb{R}((\omega^{\mathfrak{m}}))})). \quad (32.3)$$

This is strictly larger than $\text{Fix}(J_c) = \mathbb{R}((\omega^{\mathbb{R}}))$: for $\varepsilon = \omega^{-1}$ and $z = \omega^{\omega^\varepsilon}$ the outer exponent ω^ε lies in $\mathbb{R}((\omega^{\mathfrak{m}}))$, so z is in the core, but $\omega^\varepsilon \notin \mathbb{R}$, so z is not fixed. Thus a globally non-surjective embedding can act as a nontrivial automorphism on the part surviving every finite inverse stage. No limit iterate j^λ is defined; a transfinite iteration would need an additional limit rule, and an intersection of finite images is not such a rule by itself (compare the base comment after proposition 8.4).

33 Automatic preservation, the Taylor twist, and value actions

Source 04's Propositions 6.1–6.3 are base results (correspondence table). Its comment on Proposition 6.2 is kept: it is incorrect to argue that the rational cut of a real forces its image to equal that real *inside* $\mathbf{No}$, since that cut also has infinitesimally perturbed realizations; the Taylor twist T of (10.3) is an explicit witness. The twist is a formal Taylor construction (no analytic convergence of a possibly discontinuous derivation of $\mathbb{R}$ is asserted), and it is not exponential, by theorem 35.1.

Remark 33.1 (Value actions of the explicit maps [04]). The value action of proposition 10.1 is $\tau_j(a) = v(j(\omega^{-a}))$. Equivalently, factor a normal form into its leading monomial, its real leading coefficient, and a factor $1+\varepsilon$; the last two and their images are valuation units, which gives $v(jx) = \tau_j(vx)$ directly. By (6.8), $\tau_{H_g} = g$ for every ordered additive g , hence

$$\tau_{J_f} = A_f, \quad (33.1)$$

and $\tau_T = \text{id}$ for the Taylor twist. “Valuation-preserving” is ambiguous: preservation of Archimedean comparisons is automatic, whereas $v(jx) = v(x)$ for all x is the much stronger pointwise assertion that produces exponential rigidity.

Proposition 33.2 (Value-group form of the continuity criterion [04]; after `opa:as:thm:topology`). *For a field self-embedding j of $\mathbf{No}$, the image $j(\mathbf{No})$ is cofinal if and only if $\tau_j(\mathbf{No})$ is cofinal in $\mathbf{No}$; so by theorem 9.3 both are equivalent to continuity of j . For an increasing injection f ,*

$$J_f \text{ is continuous} \iff f(\mathbf{No}) \text{ is cofinal in } \mathbf{No}.$$

Proof. The image $\tau_j(\mathbf{No})$ is an additive subgroup, hence symmetric. If it is cofinal and $y \neq 0$ is given, choose a with $\tau_j(a) < v(y)$ and put $x = \omega^{-a}$; then $v(jx) = \tau_j(a) < v(y)$, so the positive element jx exceeds $|y|$. Conversely, if $j(\mathbf{No})$ is cofinal and β is given, choose x with $jx > \omega^{\beta+1}$; then $v(jx) \leq -(\beta+1)$, and $\tau_j(-vx) = -v(jx) > \beta$. For the double lift, (33.1) gives $\tau_{J_f}(\mathbf{No}) = A_f(\mathbf{No}) = \mathbb{R}((\omega^{f(\mathbf{No})}))$, which is cofinal exactly when $f(\mathbf{No})$ is: if the exponents are bounded above by b , every element has absolute value below ω^{b+1} ; if they are cofinal, the monomials are. $\square$

Consequently J_c is strongly summation-preserving but discontinuous, whereas J_- and $J_{f_{\text{gap}}}$ are proper continuous topological field embeddings. The bounded image of theorem 32.1 is discrete as a subspace of $\mathbf{No}$ but has a nondiscrete, densely ordered *intrinsic* order topology (radii from the subfield only); these are different topologies. The equivalence of cofinality, coinitality of $j(\mathbf{No}_{>0})$ in $\mathbf{No}_{>0}$, continuity at 0, topological embedding, and uniform discreteness otherwise, for any ordered field, is source 04's Theorem 7.1, printed by source 07 (theorem 67.1) and for $\mathbf{No}$ by the base.

Remark 33.3 (Strong sums are not topological limits [04]). The Hahn sum $\sum_{n \geq 0} \omega^{-n}$ exists, but its finite partial sums do not converge to it in the full order topology: every nonzero tail dominates $\omega^{-\omega}$. Strong summation is a support-and-coefficient operation. Sequential continuity is an ineffective test on the full class: by proposition 9.1 every function preserves the (eventually constant) convergent set-indexed nets, including the discontinuous J_c . The statement is relative to the full class and to set-sized index sets; a larger-universe net need not be small relative to the original universe.

34 Elementary self-embeddings of o-minimal surreal structures (under GBC+ETR)

Hypothesis for this whole section (except the final second route). Work in GBC+ETR, with global choice included in GBC, and fix a class satisfaction relation for the class structure $\mathfrak{M}$ below. Every lemma, theorem and remark of this section is proved only under this hypothesis, which source 04 calls “a sufficient foundation, not a minimality claim”. Source 05 has parallel theorems and points here (theorem 49.5).

34.1 The class-model setting

Let $\mathfrak{M}$ be a class structure on $\mathbf{No}$ in a set-sized language, expanding an ordered field, whose complete first-order theory $\text{Th}(\mathfrak{M})$ is o-minimal. The hypotheses include the ordered surreal field and the natural surreal exponential field

$$\mathfrak{M}_{\text{exp}} = (\mathbf{No}, 0, 1, +, -, \cdot, <, \exp).$$

For the latter, the model-of- $\text{Th}(\mathbb{R}_{\text{exp}})$ assertion uses van den Dries–Ehrlich and Wilkie [3, 9]; it is not inferred from the existence of some unspecified exponential operation. Standard consequences of o-minimality for ordered-field expansions are used: unary definable sets are finite unions of points and intervals; endpoints over parameters A belong to $\text{dcl}(A)$; definable Skolem functions exist; definable closure has the exchange property [15]. The proofs spell out how these set-sized facts are used with the class order, rather than invoking an undefined compactness theorem for proper-class languages.

Lemma 34.1 (Saturation over every set [04]; under GBC+ETR with a class satisfaction relation). *Every complete one-variable type over a set of parameters in $\mathfrak{M}$ is realized in $\mathfrak{M}$. Consequently every set-sized partial elementary map from a set model of $\text{Th}(\mathfrak{M})$ into $\mathfrak{M}$ extends to an elementary embedding of that set model.*

Proof. For a set A , the class $\text{dcl}(A)$ is a set: formulas form a set and each uniquely specified value is obtained by replacement. An algebraic one-variable type is realized by a point of $\text{dcl}(A)$, since finite definable sets in a linear order have individually definable elements. A nonalgebraic type is determined by its strict cut in $\text{dcl}(A)$: every unary definable set over A has its finitely many endpoints there, so the cut determines membership in every such set. The two sides are sets and are strictly separated, so by (31.1) a surreal lies strictly between them, and it realizes the type.

To extend a partial elementary map, well-order the remaining domain of the set model. At a successor stage transport the complete type of the next element over the already mapped set and realize it by the

preceding argument. At a limit stage take the union. The domain at each stage is a set, and the final map is elementary because each formula uses only finitely many parameters. $\square$

Set-sized elementary submodels containing a prescribed set exist by the usual Skolem-hull construction (in the stated foundation). In an o-minimal ordered-field expansion, $\text{dcl}(X)$ of an elementary submodel together with extra parameters is an elementary submodel, because definable Skolem functions give the Tarski–Vaught witness property; in particular $\text{dcl}(X) = X$ when $X \prec \mathfrak{M}$.

34.2 Bounded elementary images

Theorem 34.2 (Bounded-image elementary self-embeddings [04]; under GBC+ETR with a class satisfaction relation). *Assume GBC+ETR and a class satisfaction relation for $\mathfrak{M}$. Let $K_0 \prec \mathfrak{M}$ be set-sized, and let b be a positive surreal with $k < b$ for every $k \in K_0$. There is an elementary class self-embedding $j : \mathfrak{M} \rightarrow \mathfrak{M}$ fixing K_0 pointwise such that*

$$|j(x)| < b \quad (x \in \mathbf{No}). \quad (34.1)$$

In particular, under the same hypothesis, the natural surreal exponential field has bounded-image proper elementary self-embeddings fixing every real, with any prescribed positive infinite b as a bound on the absolute values of the image.

Proof. Extension step. Suppose $X \prec \mathfrak{M}$ is a set-sized elementary submodel and $\pi : X \rightarrow \mathfrak{M}$ is elementary, with $Y = \pi(X)$ and every element of Y below b . Let $X' \prec \mathfrak{M}$ be any set-sized elementary submodel containing X and an additional source element to be processed. Use ordinary set-sized compactness on the elementary diagram of X' together with a new constant z and the inequalities $z > x'$ for all $x' \in X'$: every finite part is realized in X' by a sufficiently large element. Thus there is a set model $\mathfrak{N} \succeq X'$ with $z > X'$. Over the elementary submodel X , the element z realizes the unique type at positive infinity; o-minimality ensures uniqueness, since each unary definable set over X is eventually all or eventually empty, with finite endpoints in X . As $Y \prec \mathfrak{M}$ and $\text{dcl}(Y) = Y < b$, the element b has exactly the corresponding type over Y . Therefore $\pi \cup \{z \mapsto b\}$ is a partial elementary map from $\mathfrak{N}$ into $\mathfrak{M}$. By lemma 34.1, extend it to an elementary embedding of $\mathfrak{N}$. Its restriction π' to X' extends π , and $\pi'(x') < b$ for every $x' \in X'$ because $x' < z$ in $\mathfrak{N}$.

Recursion. Choose a set-like global enumeration of $\mathbf{No}$ by the ordinals. Start with $X_0 = K_0$ and $\pi_0 = \text{id}_{K_0}$. At each successor stage apply the extension step to include the next enumerated source element. At each set-ordinal limit take the union of the elementary submodels and the compatible maps; it is again a set-sized elementary submodel and an elementary map, and its image is still below b . The class recursion and its union are available in the stated foundation. The resulting class map j is total and elementary with $j(x) < b$ for all x ; applying this to $-x$ and using $j(-x) = -j(x)$ gives (34.1). The image omits b , so it is proper.

For the final assertion take $K_0 = \mathbb{R}_{\text{exp}}$, an elementary substructure of the natural surreal exponential field; every positive infinite b lies above all of K_0 . $\square$

Remark 34.3 (What the theorem does not say [04]). The constructed embedding is not claimed to preserve strong Hahn sums or the canonical monomial cross-section. It cannot be ambient-continuous, by theorem 9.3. It is not a first-order definable function in the exponential-field language, by theorem 12.2. Fixing a set-sized elementary subfield is compatible with boundedness; fixing a cofinal subclass such as all ordinals is not.

The protected point b is not part of the source structure being embedded at that step: it is the image of an auxiliary element z placed above a *set* model. Trying to place z above the entire proper class $\mathbf{No}$ would be invalid and is unnecessary.

[write] Under GBC+ETR this answers the base question “Proper exponential copies” (section 15) in its bounded-range clause: bounded ranges occur for the actual Gonshor exponential. The maps are not strongly additive and not monomial; the strong version is research question 40.2 below. The bounded *field* copies of theorems 7.6 and 32.1 are not exponential.

34.3 Proper cofinal elementary images

Lemma 34.4 (Omission-preserving extension [04]; under GBC+ETR with a class satisfaction relation). *Let $\pi : X \rightarrow Y \subseteq \mathfrak{M}$ be an elementary map between set-sized elementary submodels, and let $b \notin Y$. For every set-sized elementary $X' \prec \mathfrak{M}$ containing X , there is an elementary extension $\pi' : X' \rightarrow \mathfrak{M}$ whose image still omits b .*

Proof. The type $\text{tp}(b/Y)$ is nonalgebraic, because $\text{dcl}(Y) = Y$ and finite definable sets have individually definable elements. Transport it through π^{-1} to a nonalgebraic type p over X . The elementary diagram of X' together with $p(z)$ and $z \neq x'$ for every $x' \in X'$ is consistent: a finite conjunction from p defines an infinite set already in X , which cannot be covered by finitely many forbidden points of X' . Compactness produces a set elementary extension $\mathfrak{N} \succeq X'$ with $z \notin X'$ realizing p . The map $\pi \cup \{z \mapsto b\}$ is partial elementary; extend it into $\mathfrak{M}$ by lemma 34.1. Injectivity and $z \notin X'$ ensure that the restriction to X' does not hit b . $\square$

Theorem 34.5 (Proper cofinal elementary self-embeddings [04]; under GBC+ETR with a class satisfaction relation). *Assume GBC+ETR and a class satisfaction relation for $\mathfrak{M}$. For every set-sized $K_0 \prec \mathfrak{M}$ and every $b \in \mathbf{No} \setminus K_0$ there is a cofinal elementary class self-embedding of $\mathfrak{M}$ fixing K_0 pointwise and omitting b from its image.*

Proof. Maintain an elementary map $\pi : X \rightarrow Y$ between set-sized elementary submodels, extending id_{K_0} , with $b \notin Y$. Alternate two kinds of stages. A *forth stage* includes the next source element by lemma 34.4. A *cofinality stage* makes the image exceed a prescribed target landmark ℓ while preserving the omission: choose

$$y > \text{dcl}(Y \cup \{b\}) \quad \text{and} \quad y > \ell,$$

possible because the displayed definable closure is a set, and choose $x > X$ in the source. Both x over X and y over Y realize the unique positive-infinity type, so π extends to a partial elementary map taking x to y , and hence uniquely to the definable hulls $\text{dcl}(X, x) \rightarrow \text{dcl}(Y, y)$. These hulls are set-sized elementary submodels. The image omits b : otherwise $b \in \text{dcl}(Y, y) \setminus \text{dcl}(Y)$, and exchange would give $y \in \text{dcl}(Y, b)$, contrary to its choice. Process all source elements and all target landmarks in a set-like ordinal enumeration and take unions at limits. The union map is total, elementary, cofinal, and still omits b . $\square$

For $\mathfrak{M}_{\text{exp}}$ (with $K_0 = \mathbb{R}_{\text{exp}}$) this yields, under GBC+ETR, proper cofinal and therefore continuous (theorem 9.3) exponential self-embeddings fixing the designated reals. Unlike theorem 7.1, this construction does not fix every ordinal: its cofinality stages deliberately move large elements.

Remark 34.6 (Homogeneity, and moving a real infinitesimally [04]; under GBC+ETR with a class satisfaction relation). The same saturation lemma, with alternating forth and back stages, extends any set-sized partial elementary map of $\mathfrak{M}$ to a class automorphism; this is the standard strong-homogeneity principle of the surreal literature, and its construction again keeps set-sized maps at every ordinal stage. (The ordered-field version is printed by source 07, theorem 70.2, after `saut:thm:orderedhomogeneity`.) It shows why exponential automorphisms need not fix the actual copy of $\mathbb{R}$: in the countable language of exponential fields, $\text{dcl}_{\mathfrak{M}_{\text{exp}}}(\emptyset)$ is a countable subset of $\mathbb{R}$. Choose $r \in \mathbb{R}$ outside it and an infinitesimal $\varepsilon > 0$. Then r and $r + \varepsilon$ have the same cut over that definable closure, hence the same complete one-variable type, and a class automorphism of $\mathfrak{M}_{\text{exp}}$ can

send r to $r + \varepsilon$. It moves growth values somewhere else, as required by theorem 35.1; it is not an exponential 1-automorphism.

34.4 Internal definability, by a calculus argument

The base proves theorem 12.2 through the fixed field (an infinite definable subfield contains an interval). Source 04 states the same theorem, “in any o-minimal expansion of an ordered field, a definable unital field self-embedding is the identity; parameters are allowed”, and proves it by o-minimal calculus.

Second route to theorem 12.2 [04]. Let j be definable and additive. Definable differentiability on cells provides a point a at which j is differentiable. Additivity gives

$$\frac{j(a+h) - j(a)}{h} = \frac{j(h)}{h},$$

so j is differentiable at 0 with some derivative c , and the same identity gives derivative c at every point. The definable mean value theorem, valid over o-minimal ordered fields, implies that $j(x) - cx$ is constant on every interval; it vanishes at 0, so $j(x) = cx$ for every x . Finally $j(1) = 1$ gives $c = 1$. $\square$

[write] Source 04 places this theorem inside its GBC+ETR section, but it concerns one formula and uses only first-order consequences of o-minimality; the base’s proof shows that no class-theoretic hypothesis is needed. It rules out internal definability of the proper embeddings above, not their external definability from normal forms or a class well-order, and it shows that naming any nonidentity field self-embedding as a new function symbol destroys o-minimality of the expansion.

35 Exponential growth-scale rigidity for embeddings

Source 04’s Lemma 9.1 is base lemma 11.1 (with $k = \text{id}$) together with lemma 11.6: if $j, k : F \rightarrow F$ are field embeddings, $k(F)$ is cofinal and $j \neq k$, then $\{j(x) - k(x)\}$ is cofinal and coinital, by the twisted Leibniz rule $d(ax) = j(a)d(x) + d(a)k(x)$ for $d = j - k$. Source 04 says this isolates the part of the repository’s growth-scale argument that does not require surjectivity (EAR lem:amplify, for unital endomorphisms, in docs/surreal/exponential-automorphism-rigidity).

Setting. Let F be an ordered field, $E : (F, +) \rightarrow (F_{>0}, \cdot)$ an increasing group isomorphism with inverse log, and $w : F^\times \rightarrow \Gamma$ a surjective nontrivial convex valuation, so that $v = w \circ E$ is the base’s profile. Suppose an exponential field self-embedding j induces an ordered additive embedding $\tau_j : \Gamma \rightarrow \Gamma$ with $w(j(u)) = \tau_j(w(u))$. For the natural valuation of **No** this compatibility is automatic (proposition 10.1).

Theorem 35.1 (Exponential embedding displacement [04]; abstract form of theorem 11.5). *If $j \neq \text{id}$, then*

$$\{\tau_j(\gamma) - \gamma : \gamma \in \Gamma\} \tag{35.1}$$

is cofinal and coinital in Γ . In particular, if $\tau_j = \text{id}$, then $j = \text{id}$. Surjectivity of j is not required.

Proof. For $D(x) = j(x) - x$, exponential compatibility gives

$$w(E(D(x))) = w(E(jx)) - w(E(x)) = w(j(E(x))) - w(E(x)) = \tau_j(w(E(x))) - w(E(x)). \tag{35.2}$$

By lemma 11.1, $D(F)$ is cofinal and coinital in F . Since E is increasing and onto, $E(D(F))$ is cofinal and coinital among positive elements. Convexity of w means that $0 < u < v$ implies $w(u) \geq w(v)$.

Given $\gamma \in \Gamma$, choose a positive element of valuation strictly larger than γ and one of valuation strictly smaller; coinitiality and cofinality of $E(D(F))$ then give values of $w \circ E \circ D$ on both sides of γ . Thus the left side of (35.2) has cofinal and coinitial range, contained in (35.1). If $\tau_j = \text{id}$, that range is $\{0\}$, which cannot be cofinal in a nontrivial ordered abelian group. $\square$

[write] The base proves the surreal case (theorem 11.5, with $v = v \circ \exp$). The abstract statement is `EAR thm:cofinal` (docs/surreal/exponential-automorphism-rigidity), printed there for exponential automorphisms although its proof uses no surjectivity; and it is already the statement of the generic Lean theorem `cofinal_value_displacement` (Surreal/Algebra/ExponentialProfile.lean), which takes a unital endomorphism $\sigma : F \rightarrow_{+*} F$. Source 04 did not inspect that file. The instantiation with the actual surreal exponential is pending there.

Corollary 35.2 (No nontrivial exponential 1-self-embeddings [04]; = theorem 11.4). *A self-embedding of the natural surreal exponential field preserving the leading term of every nonzero element is the identity. More strongly, preservation of natural valuation values alone forces the identity.*

Proof. Leading-term preservation implies $v(jx) = v(x)$. The value action exists by proposition 10.1, and pointwise preservation means $\tau_j = \text{id}$. Apply theorem 35.1 with $E = \exp$, $w = v$. $\square$

No strong additivity or real-coefficient fixation is needed. For automorphisms the implication bears directly on the question about exponential 1-automorphisms printed as Question 5.4 in KKS version 3 [5]. Source 04 writes that the repository “already proposes a negative answer in its growth-scale report” and that “the proof here states and establishes the non-surjective extension, rather than treating that proposal as a machine-checked black box”.

[write] At source 04’s pin the growth-scale report already stated the non-surjective extension as its `cor:embedding` (“Self-embedding version”), and its answer to KKS Question 5.4 as `thm:no` and `cor:question54`; the generic Lean counterparts `eq_id_of_commute` and `eq_id_of_profile` also take endomorphisms. Source 04 inspected only the opening 235 lines of that article. Corollary 35.2 is therefore a re-proof, by a different route (through theorem 35.1), not a new extension.

There is no conflict with the bounded exponential embeddings of theorem 34.2 (under `GBC+ETR`): their value-group embeddings are not the identity. A map whose image is small in the ambient order can have very large displacement from the identity.

Theorem 35.3 (Faithfulness on cofinal exponential embeddings [04]; abstract form of theorem 11.7). *Let $j, k : F \rightarrow F$ be field embeddings commuting with E and inducing the same action on the nontrivial convex valuation w , that is $w(j(u)) = w(k(u))$ for all $u \neq 0$. If $k(F)$ is cofinal in F , then $j = k$. In particular, on cofinal exponential self-embeddings of **No** the natural value-group action determines the embedding uniquely.*

Proof. Set $d = j - k$. Equality of the value actions gives

$$w(E(d(x))) = w(j(E(x))) - w(k(E(x))) = 0 \quad (x \in F).$$

Choose $s > 1$ with $w(s) < 0$. Every positive valuation unit u satisfies $s^{-1} < u < s$; otherwise convexity would contradict $w(u) = 0$. Therefore $-\log s < d(x) < \log s$ for all $x \in F$, and the two-map form of the amplification lemma (lemma 11.6) forces $d = 0$. $\square$

[write] This strengthens `EAR cor:two-profiles` (“Comparing two automorphisms by profile”) and its Lean counterpart `eq_of_profile_eq` (ExponentialProfile.lean), both of which need one of the two maps to be surjective (the Lean statement takes $\tau : F \simeq_{+*} F$), to one map with merely *cofinal* image. The cofinal strengthening is not formalized. The base’s theorem 11.7 is the surreal case, proved

there through the finite kernel (11.3); source 05's two-profile theorem and source 07's two-embedding comparison point here.

The cofinality hypothesis is used in the twisted Leibniz identity. The proof does not establish that two arbitrary bounded-image exponential embeddings with the same value action coincide (research question 40.5). Comparing a single self-embedding with the identity has no such problem, because the identity is cofinal.

35.1 A criterion for monomial lifts to preserve exponentiation

Let $I = \mathbb{R}((\omega^{\mathbf{No}_{>0}}))$ be the additive class of purely infinite surreals, including 0. For the Gonshor exponential, $\exp(I)$ is exactly the class of monomials ω^y . Define the additive order isomorphism $e_* : I \rightarrow \mathbf{No}$ by

$$\exp(x) = \omega^{e_*(x)} \quad (x \in I). \quad (35.3)$$

This packages the usual normal-form exponential formula without confusing exponentiation with the omega-map [2].

Proposition 35.4 (Exponential criterion for monomial lifts [04]). *Let g be an ordered additive self-embedding of $\mathbf{No}$. The field lift H_g commutes with the Gonshor exponential if and only if*

$$g(e_*(x)) = e_*(H_g(x)) \quad (x \in I). \quad (35.4)$$

Proof. H_g maps I into I , since g preserves the sign of exponents. On I the two sides of exponential commutation are $H_g(\exp x) = \omega^{g(e_*(x))}$ and $\exp(H_g x) = \omega^{e_*(H_g(x))}$; the omega-map is injective, so this gives necessity and commutation on I . Every surreal splits uniquely as $x = x_\infty + r + \varepsilon$ with $x_\infty \in I$, $r \in \mathbb{R}$ and ε infinitesimal. The map H_g preserves this decomposition, fixes r , and preserves the strong power-series sum $\exp(\varepsilon) = \sum_{n \geq 0} \varepsilon^n / n!$. Therefore commutation on I implies commutation on all three components, and hence on x . $\square$

The condition is exact but still global. Without an additional strong-additivity property of g , testing it on a generating list of monomials does not justify its extension through arbitrary strong sums; this is one of the principal obstacles in classifying exponential Hahn lifts. For the double lifts J_- and J_{f_β} (that is, $g = A_{f_-}$ or $g = A_{f_\beta}$) the base shows failure directly (paragraph after theorem 11.5).

36 The omega-map, other languages, omnific integers, and complexification

Proposition 36.1 (Omega compatibility [04]; = (10.5) and theorem 10.3). *For a field lift H_g of an ordered additive embedding g , commutation with the omega-map, $H_g(\omega^x) = \omega^{H_g(x)}$ for all x , is equivalent to*

$$g(x) = H_g(x) \quad (x \in \mathbf{No}). \quad (36.1)$$

For any field self-embedding j , commutation with the omega-map together with pointwise preservation of the natural valuation forces $j = \text{id}$.

Proof. $H_g(\omega^x) = \omega^{g(x)}$, and the omega-map is injective. For the second assertion apply v to $j(\omega^x) = \omega^{j(x)}$ to get $-x = -j(x)$. $\square$

The second clause is the automorphism observation of the literature with surjectivity removed [5]. It is tempting to solve (36.1) by claiming that every exponent in a normal form is simpler than

the number. That induction is invalid: generalized epsilon numbers satisfy $x = \omega^x$, so the exponent can be the number itself; fixed-point structure and non-well-founded dependency patterns cannot be discarded. KKS's questions on nontrivial omega-preserving automorphisms and on extending order automorphisms of generalized epsilon numbers remain the reference points [5]. Source 04 gives the displayed reductions, not a complete classification of their solutions.

Remark 36.2 (Other readings of “self-embedding” [04]). Before field structure is required the problem is much looser: the first lift gives ordered additive embeddings with arbitrary order-relocation data, and $x \mapsto ax$ ($a > 0$) is an ordered additive automorphism, unital only for $a = 1$. An $\mathbb{R}$ -linearity requirement should say that it uses the designated copy of $\mathbb{R}$, not only the residue field. The total exponential transports additive to multiplicative problems, $h \mapsto \mathcal{M}_h = \exp \circ h \circ \log$: an ordered additive embedding h gives an ordered multiplicative embedding of $\mathbf{No}_{>0}$, generally not a field map. No unital field self-embedding of a real closed field can reverse its uniquely determined order. Preserving birthdays *without* order permits arbitrary permutations within birthday levels; preserving the prefix tree alone permits branch swaps; preserving an external well-order is again unrelated to $<$ or birthdays. Naming $\mathbb{R}$, the monomials, **Oz**, birthdays, simplicity or a coding of sets changes the elementary-embedding question: a field map fixing $\mathbb{R}$ pointwise is an embedding of the field with constants for all reals, but not automatically elementary in a language with a predicate for a larger designated subring. There is no finite list of all expansions; the useful exhaustive framework is an explicit embedding specification (carrier, language, size convention, definability convention, image requirements) with a theorem for each named combination.

36.1 Differential self-embeddings

For the Berarducci–Mantova derivation ∂ (constant field $\mathbb{R}$, $\partial\omega = 1$) source 04's Proposition 10.2 is base proposition 13.1. The translation parameter $j(\omega) = \omega + c_j$ is studied by source 06 (proposition 59.5). The explicit J_c fixes ω and $\mathbb{R}$, but that alone is not a proof of differential compatibility.

Proposition 36.3 (Rational exponent dilations are not differential [04]). *For a rational $q > 0$, $q \neq 1$, the exponent dilation $S_q = H_{a \mapsto qa}$ does not commute with ∂ .*

Proof. $S_q(\omega) = \omega^q$. Writing $q = m/n$ and differentiating the algebraic relation $(\omega^q)^n = \omega^m$ gives $\partial(\omega^q) = q\omega^{q-1}$. Commutation would give $\partial(S_q\omega) = S_q(\partial\omega) = S_q(1) = 1$, but $q\omega^{q-1}$ is infinite or infinitesimal for $q \neq 1$. $\square$

[write] It also follows at once from proposition 13.1, since $\omega^q - \omega \notin \mathbb{R}$. It is a different property from `EAR thm:dilation` (rational dilations do not lift *exponentially*).

Remark 36.4 (Known differential extension theorems [04]). Composition and substitution on transseries subfields are natural candidate constructions, but their domains must be checked: a theorem about omega-series or a specified hyperserial subfield does not by itself supply a total self-map of **No** commuting with the chosen derivation, a distinction already significant in [27]. A safe extension theorem states the source subfield, the admissible substituted argument, preservation of all relevant sums, and the chain rule. Strong positive global theorems are known in the ordered valued differential language: Aschenbrenner, van den Dries and van der Hoeven prove that the natural transseries embedding into the Berarducci–Mantova field is elementary and establish its universality for Hardy fields [26]; van den Dries, Ehrlich and Mildenberger prove that isomorphisms between set-sized substructures that are models of the relevant H -field model companion, with constant field $\mathbb{R}$, extend to automorphisms of the full surreal H -field [39]. These should not be presented as unsolved. Their language does not name arbitrary strong Hahn sums, the monomial cross-section, or the Gonshor exponential; simultaneous preservation of those structures remains a separate condition.

36.2 Omnific integers

Source 04's Proposition 11.1 is base proposition 8.1 (same proof), and applies to every double lift J_f . The bounded J_c therefore preserves the discrete omnific integer-part structure as well as the field while being proper and discontinuous; there is no analogue of the Archimedean assertion that fixing integers and reals forces all field points to be fixed.

Remark 36.5 (The Taylor twist does not preserve **Oz** [04]). For the real r_0 with $\Delta(r_0) = 1$, $\Delta^2(r_0) = 0$,

$$T(r_0\omega^{1/2}) = r_0\omega^{1/2} + \omega^{-1/2}.$$

The input is omnific and the output is not. Thus pointwise natural-value preservation and leading-term preservation (both hold for T) do not imply preservation of the omnific ring; compare the base's U_β (proposition 10.2). None of these assertions claims elementarity of the pair (**No**, **Oz**): preservation and reflection of the predicate, and commutation with floor, are the precise claims.

36.3 Complexification

Proposition 36.6 (Conjugation-compatible extensions [04]; after `opa:as:thm:complexembed`). *Let $\mathbf{No}(i)$ carry its standard conjugation.*

- (i) *A field self-embedding j of $\mathbf{No}$ has exactly the two extensions $j_\pm(a + ib) = j(a) \pm i j(b)$ fixing the real form; both commute with conjugation.*
- (ii) *Conversely, every field self-embedding of $\mathbf{No}(i)$ commuting with conjugation is j_+ or j_- for j its restriction to $\mathbf{No}$.*
- (iii) *$|j_\pm(z)| = j(|z|)$ with $|a + ib| = \sqrt{a^2 + b^2}$; if j is exponential, the logarithmic modulus is preserved: $\log |j_\pm(z)| = j(\log |z|)$ for $z \neq 0$.*

Naming i selects the positive extension.

Proof. (i) is the base paragraph on $\mathbf{No}(i)$ in section 12. (ii) The fixed field of conjugation is $\mathbf{No}$; for $a \in \mathbf{No}$, $\sigma(\overline{a}) = \sigma(\bar{a}) = \sigma(a)$, so σ maps $\mathbf{No}$ into itself, and $\sigma(i)$ is one of the two roots $\pm i$ of $X^2 + 1$. (iii) A field embedding of $\mathbf{No}$ preserves order and therefore positive square roots, so $j(\sqrt{a^2 + b^2}) = \sqrt{j(a)^2 + j(b)^2}$; apply log and exponential compatibility. $\square$

Without naming conjugation or a real form, arbitrary embeddings of the algebraically closed field $\mathbf{No}(i)$ need not reduce to embeddings of the designated real field; this is another language-dependent interpretation.

[write] Clause (ii) is the restriction step of the proof of `opa:as:thm:complexembed` (docs/surreal/omnific-preserving-automorphisms), which classifies the *strong* conjugation-compatible embeddings of complex Hahn fields; for automorphisms it is the centralizer statement `saut:thm:axis`, and for elementary embeddings of birthday cutoffs `hset:thm:two-lifts`. Source 04 cites none of these.

37 Cutoffs and universe-polymorphic formalization

For an ordinal λ , $\mathbf{No}_{<\lambda}$ is a set in ZFC and need not be a field. Source 04 states: “The precise classical criterion is that $\mathbf{No}_{<\lambda}$ is a subfield exactly when λ is an epsilon number; in that case it is also closed under the natural exponential and is an elementary extension of the real exponential field” [3], adding that the original paper has an erratum to a support-order-type lemma (Lemma 4.5) whose main field and

exponential conclusions remain the relevant results [19]; the paper must not be used without accounting for the erratum.

[write] Printed as source 04’s claim. Source 02 makes the same claim; the “only if” direction is proved in the note after [remark 23.2](#). The collection states the characterization as van den Dries–Ehrlich’s (the subsection `found:sub:cutoffs` of `docs/foundations-and-computation/foundations`), and `docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets` cites the “if” direction. The closure under the exponential is the source’s claim and was not checked here.

Remark 37.1 (Regular cutoffs and model-theoretic constructions [04]). For an uncountable regular cardinal κ , $\mathbf{No}_{<\kappa}$ is a κ -bounded Hahn field over its own exponent group (base (14.1)). It satisfies interpolation for cuts whose two sides have size less than κ , and, for a language of size less than κ , the cut argument of [lemma 34.1](#) gives saturation over parameter sets of size less than κ . Its cardinality is $2^{<\kappa}$, not automatically κ : a back-and-forth of length κ covering the whole carrier needs a size hypothesis such as $\kappa^{<\kappa} = \kappa$, and without one, saturation over fewer than κ parameters cannot simply be iterated through a larger carrier while keeping all domains of size $< \kappa$. The explicit Hahn constructions are less sensitive ([proposition 14.1](#)): if an increasing f maps $\mathbf{No}_{<\kappa}$ into itself, both lifts are defined on the cutoff, c and f_- are examples, and their range and fixed-support formulas keep their form. Birthday rigidity and the initial-image argument also restrict, since the canonical options of an element of `birthday <  $\kappa$` form sets of size $< \kappa$. A construction that starts by fixing every real uses a base set of size $|\mathbb{R}|$; it is not a $< \kappa$ -parameter construction when $|\mathbb{R}| \geq \kappa$.

The type-theoretic version. A universe-polymorphic sign-sequence type has the schematic form $\mathbf{No}_U : \text{Type}(u + 1)$, with lengths drawn from u -small ordinals (base [section 3](#)). It is a type but not u -small; its function space and powerset live at a larger level, and a proof claiming otherwise would undo the size barrier behind the Burali–Forti paradox. Normal forms must carry a proof that their support is u -small: a Hahn series on the large exponent type with an arbitrary large well-ordered support is not automatically a surreal in $\mathbf{No}_U$. Replacement becomes preservation of `Small.{u}` under images and explicit equivalences. An embedding $\mathbf{No}_U \rightarrow \mathbf{No}_V$ obtained by changing universes is not a self-embedding of one fixed carrier; elementarity, saturation and convergence claims depend on which parameter sets, supports and nets are small. Universe lifting is legitimate; collapsing the distinction is not.

Local workspaces. A reliable proof-engineering pattern keeps every individual algebraic calculation inside a common small Hahn workspace: collect the supports of finitely many surreals and of a set-sized summable family, take the additive subgroup or monoid they generate, record its smallness, compute there, and transport back through the normal-form equivalence. The class-theoretic model constructions of [section 34](#) have a different local invariant (at each stage the partial map, its elementary domain and image, and the compactness language are sets; only the final union is a class) and need a separate formalized satisfaction and recursion layer.

38 Integration with the ProveIt repository

38.1 What source 04 inspected

At the pin `a006a77a0a8bacdf089b70d3b963b17228a089` source 04 inspected the surreal README, the opening 235 lines of the growth-scale report `docs/surreal/exponential-automorphism-rigidity/article.tex` (blob `ffb9a14697e7`), and three files in `Algebra/SurrealNumbers/Surreal/Foundations/`: `NormalFormExponentAutomorphisms.lean` (full returned source, blob `a78aedc6ad3f`), `ExponentAutomorphismStrongSums.lean` (through line 120, blob `f1a81bf562a1`)

and `SignSequenceExpLog.lean` (opening 95 lines, blob `07fe74404032`). No Lean build or transitive axiom audit was run, and no new theorem is labeled machine checked on the strength of a filename or README claim. Its audit adds that a request for `Algebra/SurrealNumbers/docs/FORMALIZATION.md` “returned an empty content field”, so no claim rests on that ledger.

[write] The empty `FORMALIZATION.md` was a fetch artefact: the ledger exists in the repository. The four blobs named are unchanged between source 04’s pin and the write-time repository.

38.2 Existing interfaces

`NormalFormExponentAutomorphisms.lean` supplies a small-support exponent-transport construction whose input is an *ordered additive equivalence* of `SignSequence.{u}` (Lean $\approx +o$); visible declarations include `mapExponents`, `exponentRingEquiv` and `exponentAutomorphism`, transported through `SmallNormalForm.cutEvaluationRingEquiv`, with formulas on real constants and omega-powers in `exponentAutomorphism_ofReal` and `exponentAutomorphism_omegaPower`. `ExponentAutomorphismStrongSums.lean` supplies `StronglySummable.exponentAutomorphism`, `stronglySummable_exponentAutomorphism_iff`, `exponentAutomorphism_strongSum`, `valuation_exponentAutomorphism` and `exponentOrderAutomorphism`; its strong-sum equality has an explicit `Small.{u}` hypothesis on the index type, whereas a summability-transfer statement can be formulated for a larger index universe – exactly the distinction between a locally summable family and a total sum in a fixed small-support carrier. `SignSequenceExpLog.lean` describes itself as an *infinitesimal* exponential/logarithm development, defines `infExp` and `infLog`, and proves the strong power-series identities; it is a dependency for the infinitesimal part of [proposition 35.4](#), but not by itself for [theorems 34.2](#) and [35.1](#) specialized to the Gonshor exponential.

[write] Every declaration named here was verified to exist (write-time check): `mapExponents` (`NormalFormExponentAutomorphisms.lean:49`), `exponentRingEquiv` (`:85`), `exponentAutomorphism` (`:109`), `exponentAutomorphism_ofReal` (`:120`), `exponentAutomorphism_omegaPower` (`:128`); `SmallNormalForm.cutEvaluationRingEquiv` (`SmallNormalFormFieldEquiv.lean:51`); `StronglySummable.exponentAutomorphism` (`ExponentAutomorphismStrongSums.lean:27`), `stronglySummable_exponentAutomorphism_iff` (`:37`), `exponentAutomorphism_strongSum` (`:46`, with `[Small.{u} ι]`, as source 04 says), `valuation_exponentAutomorphism` (`:79`), `exponentOrderAutomorphism` (`:96`); `infExp`, `infLog` (`SignSequenceExpLog.lean:22, :26`). Source 04’s statement that the exponent input is an ordered additive *equivalence* is correct. In the repository the field lift exists only for equivalences; no proper $\mathbf{No} \rightarrow \mathbf{No}$ self-embedding, no first lift of a non-additive f , and no declaration about maps preserving `IsPrefix/Simpler` is formalized. The generic displacement lemmas `eq_id_of_commute`, `eq_id_of_profile`, `cofinal_value_displacement` (`Surreal/Algebra/ExponentialProfile.lean`) take endomorphisms; the cofinal two-map strengthening [theorem 35.3](#) is not formalized.

38.3 A staged implementation plan [04]

Stage A: order embeddings on supports. Generalize the exponent-transport layer from ordered additive equivalences to injective ordered additive maps. Inverse maps are unnecessary for preservation of multiplication or strong sums. Replace uses of surjectivity in coefficient-extensionality arguments with an explicit split: the queried exponent is in the image, or its coefficient is zero. Prove support image equality and range membership before packaging a field map.

Stage B: first and double Hahn lifts. Implement arbitrary increasing exponent relocation as an additive map, separate from multiplicative relocation. Establish the composition law (6.3) and the

exact multiplicativity criterion. Then obtain the double lift by composition of interfaces, not by a second independent arithmetic proof; the image formula should become a reusable support-membership equivalence.

Stage C: exact invariants. Formalize rigidity of order automorphisms of reverse well-ordered supports; this yields the fixed-support formulas without transfinite coefficient recursion. Add the finite-iteration composition law and the support-intersection lemma for [theorem 32.3](#). Formalize the concrete c and f_- examples with their omitted monomials, as regression theorems against accidental surjectivity assumptions.

Stage D: topology and exponential algebra. Prove the continuity–cofinality theorem first for an arbitrary ordered field. Prove the displacement lemma generically from the twisted Leibniz identity. Then add an abstract ordered exponential and a convex valuation to prove [theorems 35.1](#) and [35.3](#). Only the final specialization should depend on the global surreal exponential, keeping the hard surreal input out of the short algebraic rigidity proof.

Stage E: class and universe model theory. Formalize the saturation, compactness and ordinal-recursion proofs of [section 34](#) either in an explicit set/class theory or as universe-bounded theorems with correctly stated cardinal hypotheses. Do not encode an unrestricted class elementary embedding as an ordinary small Lean function without explaining the universe shift and the range of allowable types.

Suggested interface, not claimed executable code. The following specification pseudocode avoids pretending that exact Mathlib constructor names or the required proofs already exist:

```
FirstLift(f : order embedding No No) : additive order embedding No No
FieldLift(g : additive order embedding No No) : field embedding No No
DoubleLift(f) := FieldLift(FirstLift(f))

support_FirstLift(f, x) = image(f, support(x))
FirstLift(g) o FirstLift(f) = FirstLift(g o f)
range_DoubleLift(f) = H(H(range(f)))
fixed_DoubleLift(f) = H(H(fixed(f)))
continuous(j) <-> cofinal(range(j))

exp_compatible(j) and pointwise_value_preserving(j) -> j = identity
```

A checked contribution should include all support-smallness hypotheses, coefficient-zero-outside-image lemmas, and a dependency report.

38.4 The finite regression script

Source 04’s `finite_checks.py` is shipped as `code/04-atlas-finite_checks.py`, with its recorded output `data/04-atlas-finite_check_results.json`. It uses only the Python standard library and exact rational arithmetic (seed 20261003): prefix and doubling maps on all sign words of length at most five (63 words), the rational maps c , f_- , f_{gap} (monotonicity, range, inverse, exact iterates c^n , no fixed points of f_{gap}), first and double lifts on random finite Hahn polynomials with rational exponents (additivity, multiplicativity, order, composition, fixed-point criteria, ordinal fixation by J_-), and the formal Taylor twist on coefficient polynomials (inverse and multiplicativity). It reports 22,169 checks, all passing. It prints its JSON record to standard output only and writes no file. Its own scope statement:

exact finite rational/sign/Hahn-polynomial regression checks only; not verified are arbitrary ordinal signs, infinite Hahn summability, class recursion, elementary embeddings, and Lean compilation.

[write] A rerun on a fresh copy at write time passed with the same 22,169 checks; the output equals the recorded file except for the Python version string (3.13.5 recorded) and line endings.

39 Proof-obligation ledger and audit of invalid arguments of source 04

Component (source 04)	Status and dependency (as stated by source 04; locations here)
Normal forms, sign cuts, real closedness	Classical background [1, 2]; used with set-sized supports.
Double Hahn lift and exact image	Proved (theorems 6.1 and 6.3); no class compactness.
Bounded and ordinal-fixing explicit maps	Proved (theorems 7.1 and 32.1); no exponential claim.
Fixed fields and iterated-image core	Proved (theorem 32.3 and the base fixed-support formulas); finite iterate intersection only.
Continuity iff cofinality	Proved for ordered fields (theorem 9.3 and proposition 33.2).
Bounded and proper cofinal elementary maps	Proved (theorems 34.2 and 34.5); GBC+ETR, o-minimality, definable Skolem functions and exchange.
Surreal exponential model theory	Established external input [3, 9]; not inferred from local series alone.
Exponential pointwise-value rigidity	Proved for self-embeddings (theorem 35.1); develops the repository's unrefereed related automorphism argument.
Faithfulness for two cofinal exponential maps	Proved (theorem 35.3); cofinality explicitly used.
Complete omega/differential classification	Not claimed; necessary conditions and precise reductions supplied.
Lean coverage of this article	No new Lean certification claimed. Selected interfaces inspected; no project rebuild performed.
Finite checks in the package	Exact rational/sign computations, not proofs for arbitrary ordinals or classes.

39.1 A compact audit of common invalid arguments [04]

- (1) “*Every field embedding fixes $\mathbb{R}$ because it fixes $\mathbb{Q}$.*” It fixes real residues, but infinitesimally moved real representatives have the same rational cuts; the Taylor twist (10.3) is an explicit counterexample.
- (2) “*Every order-and-simplicity embedding is the identity.*” Surjectivity, birthday nonincrease, or ambient canonical-cut preservation is needed; prefix and block maps are proper counterexamples without them.
- (3) “*The image is a field, so it cannot be bounded.*” A bounded nonconvex field copy can omit all sufficiently small ambient positive scales and is then discrete as an ambient subspace, while remaining intrinsically densely ordered (theorem 32.1).
- (4) “*Strong sums are limits of finite partial sums.*” Not in the full order topology: set-indexed convergent nets are eventually constant, and strong sums are a different support operation (remark 33.3).

- (5) “*An order relocation of exponents is multiplicative.*” Only if the exponent map is additive; the second Hahn lift repairs this without pretending the first lift was multiplicative.
- (6) “*A pointwise valuation-preserving field map is rigid.*” It need not be; the Taylor twist is a nontrivial example. Exponential compatibility turns the additive error into a detectable change of multiplicative scale.
- (7) “*Every normal-form exponent has smaller birthday.*” Generalized epsilon numbers invalidate this; it cannot justify a recursion solving omega-preservation equations.
- (8) “*All automorphisms form a class group in GBC.*” Their graphs are proper classes, not class elements; factorization and composition assertions are statements about class variables, or move to a specified higher universe.
- (9) “*Compactness adds a number above all surreals.*” The bounded elementary construction adds an element above a set model only; at no point is a proper-class elementary diagram passed to ordinary compactness.
- (10) “*A file called ExpLog proves the global exponential.*” The inspected repository file proves only the infinitesimal power-series clauses; a global specialization needs a separate global construction and its laws.

40 Research questions of source 04

Source 04 states that its questions ask for additional classification or compatibility, not that every formulation is known to be globally open. The base’s questions (section 15) are unlabelled and are cited by name.

40.1 Classification and explicit constructions

Research question 40.1 (Beyond strong monomial embeddings [04]). Give useful necessary and sufficient conditions for an arbitrary field self-embedding to factor into coefficient motion, a monomial transport, and a residual strongly additive or non-strong component. Which parts of an automorphism decomposition survive without surjectivity?

The first target is a one-sided factorization theorem with an explicit image condition; arbitrarily transferring inverses from the automorphism case would conceal precisely the new obstruction. Compare the base question “Beyond monomial images”.

Research question 40.2 (Strong exponential compression [04]). Does there exist a bounded-image self-embedding of the natural surreal exponential field that is strongly $\mathbb{R}$ -linear? Can it additionally send monomials to monomials?

The bounded elementary theorem answers the question without the strong requirements (under GBC+ETR, theorem 34.2). In the monomial subclass, (35.4) is the exact equation to solve. A proof of impossibility would be equally informative and would separate first-order exponential structure from strong summation. Compare the base question “Proper exponential copies”.

Research question 40.3 (Ordinal fixation with exponentiation [04]). Is there a proper exponential self-embedding fixing every ordinal and every real? How does the answer change when strong summation or monomial preservation is imposed?

The field-only version is settled affirmatively by theorem 7.1; the exponential version is not supplied by that construction (the base shows J_- is not exponential, after theorem 11.5). Its image would necessarily be cofinal, so theorem 35.3 applies to any candidate's value action.

Research question 40.4 (Realizing value-group embeddings [04]). Characterize the ordered additive maps $\tau : \mathbf{No} \rightarrow \mathbf{No}$ that occur as natural value-group actions of cofinal exponential self-embeddings.

Uniqueness is proved (theorem 35.3); existence is the remaining lifting problem. Cofinal displacement of $\tau - \text{id}$ is necessary for every nonidentity candidate but should not be mistaken for a sufficient condition. Compare the base question “Characterize exponential value actions”.

Research question 40.5 (Two bounded embeddings with equal value action [04]). Can distinct bounded-image exponential self-embeddings induce the same natural value-group map? If not, replace the cofinality step in theorem 35.3 by an argument valid for bounded images.

This identifies the exact limit of the present comparison proof. Compare the base question “Remove or justify the cofinality hypothesis”.

40.2 Omega and differential constraints

Research question 40.6 (The omega self-consistency equation [04]). Classify the ordered additive embeddings g satisfying $g = H_g$, first in restricted normal-form subfields and then on the full class. Determine how prescribed motion of generalized epsilon numbers constrains an extension.

This is the strong monomial version of the omega-map classification questions in the literature; a viable proof must handle self-referential exponent dependencies, not use an invalid birthday induction. Compare the base question “ ω -map compatibility”.

Research question 40.7 (Differential extension with extra structure [04]). Which extensions supplied by H -field homogeneity can simultaneously preserve strong Hahn sums, the Gonshor exponential, or a specified action on monomials? Characterize proper cofinal differential self-embeddings extending prescribed maps of a fixed set-sized differential subfield.

The necessary normalization is proposition 13.1. The known homogeneous-universal theorem [39] already supplies a strong extension mechanism in its own language; the question concerns additional compatibility and proper-image requirements, not basic differential homogeneity. Compare the base question “Differential centralizers”.

40.3 Images, fixed fields, and dynamics

Research question 40.8 (Which fixed fields occur? [04]). Characterize the real closed subclasses that can be fixed fields of surreal self-embeddings under each of the field, exponential, and strong monomial specifications.

The formula $\text{Fix}(J_f) = \mathbb{R}((\omega^{\mathbb{R}((\omega^{\text{Fix } f}))}))$ supplies a substantial explicit family (including $\mathbb{R}$ itself, example 32.2) and reduces part of the problem to fixed sets of order embeddings; it does not show that all real closed subclasses occur. Compare the base question “Prescribe the fixed field”.

Research question 40.9 (Automorphic cores and conjugacy [04]). Classify the possible pairs $(K_\infty(j), j \upharpoonright K_\infty(j))$ and use them as invariants for conjugacy of proper embeddings. When does the fixed field determine the core, and when do the two differ essentially?

The compression example (32.3) proves that fixed fields and cores can differ; the exact support formulas give a tractable testing class. Compare the base question “Prescribe the reversible core”.

Research question 40.10 (Birthday distortion [04]). For a nonidentity field self-embedding j , quantify the ordinals on which $b(j(x)) > b(x)$ must occur. Can the distortion be bounded by a uniform ordinal function for the explicit double lifts?

Theorem 4.5 proves that some increase is unavoidable. Sharper bounds would connect the Hahn construction with the sign-sequence implementation and reveal which cutoff restrictions are optimal. Compare the base question “Weaker birthday controls”.

40.4 Foundations and verified implementation

Research question 40.11 (Minimal class assumptions [04]). Reduce the GBC+ETR hypotheses of the two elementary-embedding constructions (theorems 34.2 and 34.5) for the specific theories of real closed fields and real exponential fields. Can their uniform satisfaction relations and set-sized-stage recursions be constructed in a weaker theory without losing the parameter-fixing clauses?

This is a proof-theoretic optimization problem; the stronger foundation is sufficient and no necessity result is claimed. Compare the base question “Minimal foundational strength”.

Research question 40.12 (Universe coherence [04]). Construct compatible lifts on a tower of surreal universes and determine which embedding invariants are absolute under enlargement. In particular, separate preservation of old small sums from preservation of newly available large sums.

A commuting diagram of universe lifts is a more meaningful target than an unqualified “universal surreal type”. Compare the base question “Universe and cutoff absoluteness”.

Research question 40.13 (The pair (No, Oz) [04]). Which field embeddings preserving the omnific predicate are elementary in the expanded language? Does adding floor to the language produce further rigidity for bounded-image embeddings?

The explicit examples already preserve floor and the predicate (proposition 8.1); what remains is preservation of quantified formulas about the pair. Compare the base question “Floor-preserving maps outside the monomial class”.

Research question 40.14 (A verified embedding library [04]). Implement Stages A–D of section 38, with the bounded and ordinal-fixing examples as end-to-end tests. Then formalize the class or cardinal-bounded saturation lemma needed for Stage E.

This separates a concrete extension of already inspected exponent-transport code from the substantially different model-theoretic formalization task. Compare the base question “Near-term formalization”.

[write] For Stage D, the abstract displacement theorem (theorem 35.1) and the single-map rigidity are already formalized generically in `Surreal/Algebra/ExponentialProfile.lean`; the remaining generic item is the cofinal two-map comparison (theorem 35.3).

41 Status and non-claims of source 04

- *Status* (title page, README): an unrefereed research manuscript supplying conventional proofs and an implementation roadmap, not a Lean-certified development; established results, deductions proved there, repository proposals and research questions are distinguished. No exhaustive priority (or novelty) claim is made; the literature comparison is selective. The title page records “Research article prepared with ChatGPT”.

- “Proved here” identifies where a proof is supplied, not a certified claim of historical priority. The exponential embedding results develop an argument already pursued in the repository’s unrefereed growth-scale manuscript (see the correction after corollary 35.2: the self-embedding form was already stated there).
- GBC+ETR is “a sufficient foundation, not a minimality claim”, used only for the class-model constructions of section 34; explicit support substitutions have a weaker, direct definable-class reading.
- The bounded elementary embedding is not claimed to preserve strong Hahn sums or the canonical monomial cross-section (remark 34.3).
- No complete classification of omega-compatible or differential embeddings is claimed: “This article gives the displayed reductions, not a complete classification of their solutions”; necessary conditions only for the differential case.
- Differential universality and homogeneity are acknowledged as established positive results and are not recast as new or unsolved (remark 36.4).
- The article does not claim a complete classification for every possible expansion of the surreal structure.
- The repository was inspected selectively, not cloned, audited or rebuilt; no Lean build, kernel check, transitive axiom audit or independent verification of the repository’s foundation was performed; README-level assertions are not substitutes for those checks; files seen only as directory entries support no claim. “No new Lean certification claimed.”
- The pseudocode is “specification pseudocode”, not claimed executable code, and does not pretend that Mathlib constructor names or proofs exist.
- The 22,169 finite checks are regression tests only; they cannot validate arbitrary ordinals, infinite Hahn supports, class recursion, model-theoretic arguments, or Lean compilation.
- The research questions are not an assertion that every formulation is globally open.
- The open-question status of KKS Questions 5.4, 5.6, 5.7 (version 3, consulted 3 October 2026) is distinguished from the repository’s later unrefereed proposals.

Repository-audit claims of source 04, checked at write time. (a) The exponent transport takes an ordered additive equivalence: true. (b) The thirteen named declarations exist, and the strong-sum theorem carries a `Small.{u}` index hypothesis: true (section 38). (c) `SignSequenceExpLog.lean` is infinitesimal only: true. (d) `FORMALIZATION.md` “returned an empty content field”: a fetch artefact; the ledger exists. (e) The growth-scale report is unrefereed and does not claim proof-assistant certification of its main rigidity arguments: as stated by that report. (f) The claim that the self-embedding extension is established here rather than in the repository is stale (corollary 35.2 and the following note).

Part IV

Source 05: Order, simplicity, algebra, topology, exponential structure, and foundations

42 Source 05 at a glance

Source 05 is the manuscript *Self-Embeddings of the Surreal Numbers: Order, simplicity, algebra, topology, exponential structure, and foundations*, labelled “Research article / expanded study”, prepared for Vladimir Reshetnikov and dated 3 October 2026 (39 pages; repository pin 51b0a69d7385a786661e0169dc13ee7ab02b921c). It describes itself as incorporating and extending earlier working notes prepared for the same project. Its setting is Gödel–Bernays class theory with global choice (GBC), the sign-sequence model, and set-sized supports only; the model-theoretic section 49 additionally assumes uniform satisfaction and definable Skolem operations for an o-minimal expansion of the field **No**, with GBC plus elementary transfinite recursion (ETR) named as one sufficient framework. Its finite checks are shipped as `code/05-comprehensive-verification.py` with the recorded run data/`05-comprehensive-verification.json` and build record data/`05-comprehensive-BUILD_AUDIT.json`; its source audit is `05-comprehensive-SOURCE_AND_PROOF_AUDIT.md`.

Most of source 05 proves the common spine of this report in its own words. This Part prints only what is new relative to the base text and to the collection: its local padding description of all order-and-simplicity embeddings (home of that result for the whole report), an exact differentiability criterion for the strong weighted maps, a sharper bounded annulus, a uniform form of set discreteness, the cofinal profile theorem in a form not requiring exponential compatibility, a continuous non-strong automorphism, the commutant of the diagonal derivations, its versions of the conditional elementary embeddings, and its questions, formalization architecture and audit.

Renames. Source 05’s symbols are renamed to the canon of this report as follows; every other symbol is as in the source.

Here	Source 05	Reason
j (an embedding)	e	base letter for embeddings
$x \preceq_s y, x \prec_s y$	$\sqsubseteq, \sqsubset$	base canon for sign prefixes
$c(x) = x/(1 + x)$	$C(x)$	base proposition 4.1; C is the core class in section 8
τ (additive exponent embedding), H_τ	ϕ, H_ϕ	base theorem 6.2
$J_f = H_{A_f}$	E_f	base theorem 6.3; E is the exponential of exponential-automorphism-rigidity and the footprint $E(A)$ here
$H_{\tau, \chi}$, character χ $\mathbb{R}((\omega^C))$ (series supported in a class C)	$H_{\phi, c}, c$ Γ_C and $\mathbb{H}(G)$	base theorem 6.4; c is the compression base notation $\mathbb{R}((\omega^C))$ covers both: an additive group for a subclass, a Hahn field for a subgroup
$S_q = H_{a \mapsto qa}$	T_q	base proposition 9.6; T is the Taylor shift and $T_{>c}$ truncation
τ_j (value action) $\text{sh}_\tau(a) = \tau(a) - a$	ρ_e $\Delta(a)$	base proposition 10.1 Δ is the coefficient derivation of the base Taylor shift

Here	Source 05	Reason
∂_λ (diagonal derivation)	D_λ	D is the displacement and D_m the sign repetition
$\mathcal{M}_\varphi = \exp \circ \varphi \circ \log$	M_ϕ	report canon; M also names OPA's monomial maps
$\mathfrak{M}$ (o-minimal class model)	M	avoids $\mathcal{M}_\varphi$; source 04 writes $\mathcal{M}$
f_{ann} (annulus exponent map)	h	h is the increment in difference quotients
b (omitted point, external bound)	p	p is the prefix in P_p ; source 04 also writes b
$[a]_{-1}$ (coefficient of ω^{-1} in a)	$\lambda(a)$ in its Section 9.5	base notation
ordinal $+$, $\cdot$ (in words)	+_{ord} , 'ord	base convention

Two letters are kept and disclosed: K denotes a set-sized elementary submodel in section 49 (not the core $K_\infty(j)$ of proposition 8.4), and λ denotes an additive map $\mathbf{No} \rightarrow \mathbb{R}$ in section 50 (not a birthday stage).

Correspondence table. Every numbered item of source 05, in source order and with its own numbering, together with its unnumbered subsections of mathematical content. “Same proof” means the same argument up to wording.

Source 05	Where printed
§1.1 map of results	A summary table of the results below; dropped as a summary.
§1.2 inherited / developed	Section 43, with a correction (remark 43.1).
§2.1 presentations, weak reducts	Presentations: = last paragraph of section 12. Weak reducts: remark 43.2 (printed here).
§2.2 sign class, canonical cone	= section 3 (canonical options; the numbers between them are exactly the sign extensions).
§2.3 three frameworks	= section 3 and (14.1); the inaccessible-cutoff and type-theoretic remarks: remark 43.2.
§2.4 recursion, truth	= section 3; the extra framework of its Section 12 is stated in section 49.
§2.5 set supports	= strong summability in section 3; “a Hahn sum need not be a topological limit”: remark 46.3.
§3.1 compression, affine maps	= proposition 4.1 (same proof).
§3.2 prefix and block maps	= propositions 4.2 and 4.3.
Theorem 3.1 thm:local	Theorem 44.1 (printed here; home of this result for the report). Background [4]; new to the collection.
Corollary 3.2 cor:rank	Corollary 44.2 (printed here); its rank clause = first clause of proposition 4.6.
Proposition 3.3 prop:treefix	Proposition 44.3 (printed here); hypothesis weaker than theorem 4.9.
remark after 3.3	Remark 44.4 (printed here).
§3.5 tree without order	= “Simplicity without order” in section 4.
Theorem 4.1 thm:birthday	= theorem 4.5 (same proof).
Corollary 4.2 cor:firstmove	Corollary 44.5 (printed here).

Source 05	Where printed
Corollary 4.3 cor:initial	Second route after theorem 5.2 and proposition 4.6: corollary 44.6. Embedding part: case $E = \mathbf{No}$ of isg:cf:prop:coherence; automorphism part: [5, Proposition 5.1].
Proposition 4.4 prop:cutrigid	= proposition 4.7 (same proof). Its remarks on relative cuts = the remark after proposition 4.7; initial subfields: corollary 44.6.
Proposition 5.1 prop:Af	= theorem 6.1 (same proof); after opa:as:lem:inner. Its “multiplicative iff f additive”: remark 45.1 (printed here).
Theorem 5.2 thm:Hphi	= theorem 6.2 (same proof); after isc:thm:lift, exponential-automorphism-rigidity’s prop:hahn-lift (automorphisms) and opa:as:thm:classification.
Theorem 5.3 thm:double	= theorem 6.3 (same proof; its equation (6) = (6.11)); OPA’s double lifts M_{1,t_μ} (opa:as:lem:inner, opa:as:thm:four).
§5.4 algebraic distinction	= proposition 10.1 and the remark before it.
§5.5 one group operation	Remark 45.2 (printed here).
Theorem 6.1 thm:bounded	Theorem 45.3 (printed here: a sharper annulus than theorem 7.6); after opa:as:thm:parameters.
Theorem 6.2 thm:ordinalfix	= theorem 7.1 with proposition 8.1 (same map f_- , same witness, same proof).
§6.3 truncation vs. simplicity	Example 45.4 (printed here).
Theorem 7.1 thm:topology	= theorem 9.3 (same proof); after opa:as:thm:topology. Its remark “any ordered field into another”: theorem 67.1 (home 07).
Theorem 7.2 thm:small-discrete	Theorem 46.1 (printed here: uniform form of proposition 9.1); after saut:prop:discrete.
Corollary 7.3 cor:nets	Corollary 46.2 (printed here).
remarks after 7.3	Remark 46.3 (printed here).
Proposition 8.1 prop:derivative-rigid	Proposition 46.4 (printed here); embedding version of saut:thm:derivative.
Theorem 8.2 thm:dilation	Derivative clauses = proposition 9.6 (same proof; saut:thm:zeroderiv). Fixed field $\mathbb{R} = (6.9)$, since $\text{Fix}(a \mapsto qa) = \{0\}$ for $q \neq 1$ (source 05 argues directly by an increasing sequence in the support). Mean-value remark: = remark after proposition 9.6.
§9.1 three conditions, τ_j	= proposition 10.1 and the three constraints of section 10.
Theorem 9.1 thm:weighted	= theorem 6.4 (same proof); after saut:thm:monomial (automorphisms), isc:prop:classification, opa:as:thm:classification; Lean Surreal/HahnSeries/CharacterTwist.lean.
Lemma 9.2 lem:support-rigid	= (6.4) and (6.13) (same proof); its divisibility clause: remark 47.1.
Corollary 9.3 cor:double-fixed	Immediate from (6.4) and (6.9); recorded in remark 47.1.
equation (9) eq:weighted-compose	= composition formula of theorem 6.4; its remark that the image does not determine the map: remark 47.1.
Example 9.4 ex:bounded-fixed	Example 47.2 (printed here).
fixed field of the ordinal fixer	= (8.3).
Theorem 9.5 thm:derivative-criterion	Theorem 47.3 (printed here; new).
Corollary 9.6 cor:valuation-non-differentiable	Corollary 47.4 (printed here; new).
remark: bounded maps diverge	Remark 47.5 (printed here).

Source 05	Where printed
§9.5 pointwise valuation example	Example 47.6 (printed here; a variant of the example after proposition 10.1).
Lemma 10.1 lem:displacement	= lemma 11.1 and corollary 11.2 (same proof); exponential-automorphism-rigidity's lem:amplify, cor:bounded-displacement.
Theorem 10.2 thm:exp-rigid	= theorem 11.4 (same proof; its detector (10) = (11.3)); exponential-automorphism-rigidity's cor:embedding, thm:no; corollary 35.2.
Lemma 10.3 lem:two	Lemma 48.1 (printed: extends lemma 11.6 from $F \rightarrow F$ to $F \rightarrow K$; proof verbatim).
Theorem 10.4 thm:two-profile	Theorem 48.2 (printed: profile form, cf. theorem 11.7 and theorem 35.3 (home 04)); strengthens exponential-automorphism-rigidity's cor:two-profiles.
§10.3 rational dilations	Proposition 48.3 (printed: embedding version of saut:prop:rationalnonlift and exponential-automorphism-rigidity's thm:dilation).
Proposition 11.1 prop:omega	= theorem 10.3 and the remark after it (same proof); self-consistency $j = H_j =$ (10.5).
Theorem 11.2 thm:oz	Predicate part = proposition 8.1 (same proof). $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$ and pointwise $\mathbf{Oz}$ -rigidity = theorem 20.2 (home 02; same proof), after odg:thm:fractions.
Proposition 11.3 prop:fixedfield	Real closedness = proposition 8.2 (same proof); exp/log closure: remark 47.1.
§12.1 language	= proposition 12.1 and the exponential paragraph after it.
§12.2 framework	Section 49 (printed).
Lemma 12.1 lem:small-types	Lemma 49.1 (printed); realization cf. lemma 34.1 (home 04); homogeneity cf. saut:thm:orderedhomogeneity and theorem 70.2 (home 07).
Theorem 12.2 thm:omit	Second route after theorem 34.5 (home 04); theorem 49.2.
Corollary 12.3 cor:exp-proper	Corollary 49.3 (printed).
Lemma 12.4 lem: bounded-extension	Lemma 49.4 (printed).
Theorem 12.5 thm: bounded-elementary	Second route after theorem 34.2 (home 04); theorem 49.5.
Corollary 12.6 cor:exp-bounded	Corollary 49.6 (printed, with the $\log^{\text{on}} \omega$ bound).
remarks after 12.6	Remark 49.7 (printed).
Proposition 12.7 prop:nonstrong	Proposition 49.8 (printed; new).
Theorem 12.8 thm:definable	Second route after theorem 12.2: theorem 49.9.
§12.7 birthday languages	Remark 49.10 (printed).
§13.1 constant field $\mathbb{R}$	= first clause of proposition 13.1.
Criterion 13.1 crit:deriv	Criterion 50.1 (printed).
Proposition 13.2 prop: diagonal-deriv	Proposition 50.3 (printed; new).
§13.3 hyperserial substitution	Remark 50.4 (printed); cf. the substitution paragraph of section 13.
§13.4 surcomplex, quaternions	Surcomplex = “Algebraic variants” in section 12; quaternions: remark 50.5 (printed).
§14.1–14.4 formalization	Section 51 (printed).
Question 15.1	Research question 52.1.
Question 15.2	Research question 52.2.
Question 15.3	Research question 52.3.
Question 15.4	Research question 52.4.

Source 05	Where printed
Question 15.5	Research question 52.5; conditionally answered by <code>lce:prop:J</code> .
Question 15.6	Research question 52.6.
Question 15.7	Research question 52.7.
Question 15.8	Research question 52.8.
Question 15.9	Research question 52.9.
Question 15.10	Research question 52.10.
Question 15.11	Research question 52.11.
Question 15.12	Research question 52.12.
Question 15.13	Research question 52.13.
Question 15.14	Research question 52.14.
Question 15.15	Research question 52.15; converse partly by proposition 60.2 (home 06).
Question 15.16	Research question 52.16.
suggested sequence of work	Section 52 (printed).
§16 conclusion	Summary; dropped.
Appendix A ledger	Section 51 (printed).
Appendix B audit	Section 53 (printed).

43 Scope and foundations in source 05

Source 05 states its inheritance as follows. The elementary normal-form mechanisms are classical Hahn-field constructions; prefix concatenation and the distinction between rigid automorphisms and flexible embeddings belong to the established surreal-substructure theory [4]; the fraction-field identity for omnific integers and exponential valuation rigidity already occur in the repository [13, 24], and are proved again so that the dependence is mathematical rather than bibliographic. It then lists as “developments assembled here” the local padding description, the two-level lift with exact fixed fields and cofinality detection, the topological distinction between sets and the full class, the exponent criterion for differentiability, bounded elementary exponential copies, and the cofinal two-embedding profile theorem. It adds that some of these may be familiar to specialists in another formulation, that “developed here” describes the contents of the article and is not a priority claim, and that the classification of arbitrary exponential, omega-field or differential self-embeddings is not asserted to be complete.

Remark 43.1 (What the list means relative to the collection). [write] Source 05 inspected only the READMEs of exponential-automorphism-rigidity and omnific-preserving-automorphisms and none of docs/surcomplex/surcomplex-field-automorphisms, independent-surreal-copies or discrete-initial-subgroups-and-omnific-normalization. Item by item: the local padding description (theorem 44.1) is new to the collection, and sources 06 and 07 proved it independently. The two-level lift is OPA’s double lift M_{1,ι_μ} (opa:as:lem:inner, opa:as:thm:classification), and its cofinality detection is the content of opa:as:thm:four together with opa:as:thm:topology of docs/surreal/omnific-preserving-automorphisms; the exact fixed-field formulas are opa:es:thm:fixedsupport there for automorphisms, and their embedding form is also in the base ((6.4), (6.13)) and in sources 02, 04 and 07. The topological distinction between sets and the full class is saut:prop:discrete of docs/surcomplex/surcomplex-field-automorphisms (stated there for $\mathbf{No}(i) \supseteq \mathbf{No}$), and the continuity-cofinality dichotomy for field embeddings is opa:as:thm:topology. The ordinary derivative analysis of exponent dilations, and the dichotomy “a nonzero derivative forces the identity”, are saut:thm:zeroderiv and saut:thm:derivative there (for automorphisms); the exponent criterion theorem 47.3 is new. The bounded elementary exponential copies are new to the collection, conditional on the class framework of section 49, and were found independently by source 04 (theorem 34.2). The cofinal profile theorem is new relative to cor:two-profiles of docs/surreal/exponential-automorphism-rigidity and

its Lean form `eq_of_profile_eq`, which need a surjective comparison map; it is proved independently in the base (lemma 11.6 and theorem 11.7) and in sources 02, 04 and 07.

Remark 43.2 (Foundational distinctions specific to source 05). [05] The base’s foundations (section 3) are those of source 05. Source 05 adds the following distinctions.

- (a) *Weak redacts.* With equality alone a self-embedding is any injective class map. If only the birthday function is named, permutations within a birthday level preserve that datum and need not preserve order or simplicity; if only the comparison of birthdays is named, even the action on levels may move ordinals. Literal preservation of birthday values is a stronger demand.
- (b) *Domains.* An endomorphism has domain all of the chosen carrier. A substitution defined on a transseries subfield, an embedding of one birthday fragment into a later fragment, and a full self-embedding are different assertions; an embedding into an elementary extension of a set model is not a self-embedding of that model.
- (c) *Inaccessible cutoffs.* For a strongly inaccessible κ , $\mathbf{No}_{<\kappa} = \bigcup_{\alpha < \kappa} \{-, +\}^\alpha$ is a set of cardinality κ with the cut-filling property for cuts with fewer than κ options, and relativizing the surreal constructions to V_κ gives a set-sized surrogate. Quantifiers, supports and maps must stay relativized; a map of the surrogate need not extend to the external class $\mathbf{No}$. Strong inaccessibility is sufficient, not necessary: for uncountable regular κ the bounded normal-form theorem (14.1) applies [3, Proposition 4.7], and “ $< \kappa$ ” may not be replaced by “arbitrary set” inside the carrier. An arbitrary birthday cutoff is not a field: the finite-birthday surreals are the dyadic rationals, not closed under division by 3.
- (d) *Type theory.* Universe-polymorphic ordinals and sign sequences need no inaccessible cardinal postulated inside the development; the interpretation of a hierarchy of universes in a chosen set-theoretic model is a separate metatheoretic question. An implementation must never use `Type : Type`, nor treat a universe-relative ordinal type as containing every ordinal of its own larger universe.
- (e) *Recursion.* Recursion through ordinal stages with set-valued states (set recursion with class parameters) suffices for the sign constructions and for set-sized back-and-forth arguments; it is not class-valued elementary transfinite recursion [10], which source 05 invokes only for the uniform model theory of section 49.

44 Order and tree self-embeddings

Write $s^- = s \smallfrown (-)$ and $s^+ = s \smallfrown (+)$ for the immediate sign successors of a sign sequence s . The following theorem is the home of the local description of all order-and-simplicity embeddings in this report; sources 06 and 07 point to it.

Theorem 44.1 (Local padding description [05]). *A class assignment $s \mapsto t_s$ is an embedding of $(\mathbf{No}, <, \preceq_s)$ into itself exactly when the following conditions hold:*

- (i) t_0 is an arbitrary sign sequence;
- (ii) t_{s^-} and t_{s^+} properly extend t_s , and their first new signs, at coordinate $\mathbf{b}(t_s)$, are respectively $-$ and $+$;
- (iii) if $\mathbf{b}(s)$ is a nonzero limit ordinal, then t_s extends

$$\bigcup_{\beta < \mathbf{b}(s)} t_s \upharpoonright \beta.$$

Thus at a successor node one may append an arbitrary nonempty block beginning with the prescribed sign; at a limit node one may append an arbitrary further block, possibly empty, to the union of the earlier images.

Proof. For an embedding, prefix preservation gives the extension statements. Since $s^- < s < s^+$, the first new signs have the specified orientations. At a limit, every earlier image is a prefix of t_s , so their union is one as well.

Conversely, transfinite induction on length shows that an input prefix has an output prefix. Consider two inputs. If one extends the other, their order is decided by the first sign after the shorter one, which has the same orientation in the images by (ii). If neither is a prefix of the other, let u be their longest common prefix. Their images extend t_{u^-} and t_{u^+} in the appropriate order. These outputs first differ at coordinate $b(t_u)$, with opposite signs. Hence the output order is correct and the outputs are incomparable in simplicity. This proves strict order, injectivity, and reflection of simplicity. $\square$

The theorem shows again that strict order preservation together with one-way preservation of the prefix relation implies reflection, as in lemma 4.4; no separate reflection hypothesis is needed. The prefix maps P_p (append p at the root, no padding elsewhere) and the repetitions D_m (blocks of length m at successors, no padding at limits) of propositions 4.2 and 4.3 are the simplest instances.

Corollary 44.2 (Meet and rank [05]). *Every order-and-simplicity embedding j preserves the longest common prefix of two inputs, in the sense that the longest common prefix of $j(x)$ and $j(y)$ is $j(u)$ when u is that of incomparable x, y ; and $b(j(s)) \geq b(s)$ for every s .*

Proof. For incomparable inputs, the proof of theorem 44.1 locates their first output disagreement exactly after the image of their common prefix. The comparable case is immediate. Along the chain of prefixes of s , the birthdays of the images are strictly increasing, so their ordinal ranks cannot be smaller than the corresponding input ranks. $\square$

The rank clause is the first clause of proposition 4.6; the meet clause is new here.

Call a simplicity embedding *chain-continuous* if it preserves unions of nonempty set-sized prefix chains. In theorem 44.1 this is exactly the prohibition of extra padding at nonzero limit nodes: equality in (iii). This continuity is in the tree sense, not the order topology.

Proposition 44.3 (Least fixed points [05]). *A chain-continuous order-and-simplicity embedding j has a unique simplest fixed point,*

$$z = \bigcup_{n < \omega} j^n(0).$$

More generally, when $x \preceq_s j(x)$, the same construction started at x is the least fixed extension of x in the prefix order.

Proof. The empty sequence is a prefix of $j(0)$; applying prefix preservation repeatedly gives an increasing chain. Its union is a set sign sequence. Chain continuity gives $j(z) = \bigcup_n j^{n+1}(0) = z$. Any fixed point extends all the iterates of zero, and hence extends z . The second assertion uses the identical argument above x . $\square$

[write] Chain continuity is equivalent to the image-closure hypothesis of theorem 4.9: if j is chain-continuous and $(j(x_i))$ is a set-sized prefix chain in the image, reflection makes (x_i) a chain, and $j(\bigcup x_i) = \bigcup j(x_i)$ lies in the image; the converse is the first step of the proof of theorem 4.9. So proposition 44.3 is the least-fixed-point part of that theorem ($z = c_j(0)$), by the same iteration. It is printed because sources 06 and 07 cite the padding description together with this clause.

Remark 44.4 (Prefix maps with fixed points [05]). The map P_+ is proper but fixes ω , because ordinal addition gives $1 + \omega = \omega$. More generally, for nonempty p the ω -fold concatenation $p \smallfrown p \smallfrown \dots$ is the simplest fixed point of P_p , by [proposition 44.3](#). A claim that every nontrivial prefix map strictly increases every birthday would be false.

Corollary 44.5 (First movement [05]). *For any nonidentity order self-embedding j there is a least birthday α at which something moves. Every moved input of birthday α is sent to a proper sign extension of itself, hence to a strictly larger birthday.*

Proof. At the least such α , every canonical option of a moved x of birthday α is fixed, so $j(x)$ lies between L_x and R_x and extends x , as in the proof of [theorem 4.5](#); it is a proper extension since $j(x) \neq x$. $\square$

This does not say that every later birthday increases; a field automorphism may increase some birthdays and decrease others.

Corollary 44.6 (Initial images [05]). *An embedding of $(\mathbf{No}, <, \preceq_s)$ with prefix-initial image is the identity. In particular every automorphism of the doubly ordered structure is the identity.*

The base reaches this by [theorem 5.2](#) (initiality alone forces surjectivity) followed by [proposition 4.6](#). Source 05 argues directly through birthdays.

Second route to theorem 5.2 with proposition 4.6 [05]. For any x , every proper prefix of $j(x)$ is in the image, by initiality. Reflection of simplicity identifies the preimages of those prefixes with exactly the proper prefixes of x . The two predecessor chains have order types $b(j(x))$ and $b(x)$, so the birthdays are equal. Apply [theorem 4.5](#). $\square$

The automorphism clause is [5, Proposition 5.1]; the argument isolates why a proper, noninitial image escapes it. Source 05 adds that an *initial subfield* (a subfield that is prefix-initial in $\mathbf{No}$) is not thereby the image of a simplicity embedding: an arbitrary field isomorphism onto it need not preserve simplicity, and the corollary needs both hypotheses.

[write] The embedding clause is the case $E = \mathbf{No}$ of `isg:cf:prop:coherence` of `docs/surreal/discrete-initial-subgroups-and-omnific-normalization` (uniqueness of the isomorphism of the numerical and prefix orders from E onto an initial subclass); source 05 does not cite it. By [theorem 5.2](#) a self-embedding of the whole of $\mathbf{No}$ with initial image is onto, so the initial subfields in source 05's last remark are those that are not images of $\mathbf{No}$, for instance set-sized ones.

45 Hahn lifts, explicit copies, and the two truncations

Source 05's two lifts and their image, composition and cofinality statements are those of [theorems 6.1](#) to [6.3](#). It adds the following.

Remark 45.1 (One-level reindexing [05]). For an increasing f , the additive lift A_f preserves multiplication if and only if f is additive. Necessity follows by applying A_f to $\omega^a \omega^b = \omega^{a+b}$, which gives $\omega^{f(a+b)}$ on one side and $\omega^{f(a)+f(b)}$ on the other; sufficiency is the convolution argument of [theorem 6.2](#) (and then $f(0) = 0$, so $A_f = H_f$ is unital). This is why an arbitrary order map cannot be turned into a field map in one step. Source 05's finite checks record an explicit failure of multiplicativity for a one-level reindexing (an exponent map with $f(2) = 2/3$ but $2f(1) = 1$).

Remark 45.2 (Keeping only one group operation [05]). The additive and multiplicative reducts have their own embedding problems. For $c > 0$, $x \mapsto cx$ is an ordered additive automorphism but a unital

field map only for $c = 1$. The first lift supplies proper additive embeddings, and an entire additive copy of $\mathbf{No}$ can lie in the infinitesimals: for the map f_{ann} of theorem 45.3 below, every nonzero $A_{f_{\text{ann}}}(a)$ is infinitesimal. Using the Gonshor exponential and logarithm, an ordered additive embedding φ gives an ordered multiplicative embedding

$$\mathcal{M}_\varphi : \mathbf{No}_{>0} \rightarrow \mathbf{No}_{>0}, \quad \mathcal{M}_\varphi(x) = \exp(\varphi(\log x)).$$

Conjugation by $\exp$ and $\log$ is a bijective correspondence between these two kinds of group embeddings, preserving composition and surjectivity. It does not preserve field addition: $\varphi(x) = 2x$ gives $\mathcal{M}_\varphi(x) = x^2$, and $(1+1)^2 \neq 1^2 + 1^2$. Extending by $\mathcal{M}_\varphi(-x) = -\mathcal{M}_\varphi(x)$ embeds the group $\mathbf{No}^\times$; setting $\mathcal{M}_\varphi(0) = 0$ still gives no field embedding. Thus a multiplicative group embedding, an exponential field embedding and the field lift H_φ are three different constructions even from the same additive map φ .

Theorem 45.3 (Bounded field self-copy with a uniform ambient gap [05]). *Let*

$$f_{\text{ann}}(a) = -2 + \frac{1}{2} \cdot \frac{a}{1 + |a|},$$

an increasing map with image $(-5/2, -3/2)$, and let $j = H_{A_{f_{\text{ann}}}} = J_{f_{\text{ann}}}$. Then j is a proper, strongly $\mathbb{R}$ -linear ordered-field self-embedding, and every nonzero image element satisfies

$$\omega^{-1} < |j(x)| < \omega.$$

In particular the entire image lies in $(-\omega, \omega)$, and no nonzero image point lies in $[-\omega^{-1}, \omega^{-1}]$.

Proof. For nonzero a , the leading exponent of $A_{f_{\text{ann}}}(a)$ lies in $(-5/2, -3/2)$, so $A_{f_{\text{ann}}}(a)$ is infinitesimal; also $A_{f_{\text{ann}}}(0) = 0$. If $x \neq 0$ has leading exponent a , then $j(x)$ has leading exponent $A_{f_{\text{ann}}}(a)$, which lies strictly between -1 and 1 . Comparison of distinct leading exponents gives the displayed strict inequalities, whatever the nonzero real leading coefficient. The map is a field embedding by theorem 6.2, and is not surjective because it misses ω . $\square$

[write] Theorem 7.6 fixes a prescribed set and ω , with the weaker bound $|j(y)| < \omega^{\gamma+1}$; the present map fixes no prescribed set (it sends ω to $\omega^{\omega^{-2}}$) but places the image in the annulus $\omega^{-1} < |y| < \omega$. A parameter-fixing annulus $\Omega^{-1} < |y| < \Omega$ with an explicit Ω is `opa:as:thm:parameters` of `docs/surreal/omnific-preserving-automorphisms`, and the bounded case of `opa:as:thm:four` is the same phenomenon; explicit families of bounded independent copies are `isc:thm:explicit` of `docs/surreal/independent-surreal-copies`. Source 05 cites none of these.

There is no conflict with arbitrarily large elements *inside* this field copy: its order is internally unbounded but bounded in the ambient field, and its positive elements are internally cointial at zero but have the external lower bound ω^{-1} . The image is still a proper class. If its birthdays were bounded by an ordinal, it would lie in a set of sign sequences, and applying the inverse embedding to that set would make $\mathbf{No}$ a set by class replacement. Numerical boundedness and ambient discreteness therefore imply neither set-sizedness nor bounded birthdays.

Example 45.4 (Hahn truncation is not simplicity [05]). Every H_τ commutes with taking an initial segment of a normal form, with exponent labels reindexed by τ ((10.4)); this is Hahn truncation, not sign-prefix simplicity. The automorphism $S_2, \sum r_a \omega^a \mapsto \sum r_a \omega^{2a}$, does not preserve simplicity. Every positive infinite surreal has its first ω signs positive, and $\sqrt{\omega} \neq \omega$, so $\omega \prec_s \sqrt{\omega}$. But $S_2(\omega) = \omega^2$ and $S_2(\sqrt{\omega}) = \omega$, and $\omega^2 \not\prec_s \omega$. Preservation of one kind of truncation cannot be substituted for the other.

46 Topology and ordinary differentiability

Source 05's continuity theorem is theorem 9.3, with the same proof; it adds that the proof applies verbatim to an embedding of any ordered field into another, with cofinality in the target, which is theorem 67.1 (home 07).

[write] The dichotomy is opa:as:thm:topology of docs/surreal/omnific-preserving-automorphisms, stated there through the value group; source 05 does not cite it.

Theorem 46.1 (Sets are uniformly discrete [05]). *Every set $S \subseteq \mathbf{No}$ is uniformly discrete in the full surreal order topology: there is $\eta > 0$ with $|s - t| > \eta$ for all distinct $s, t \in S$. It is closed in the precise sense that every $x \notin S$ has an open interval about it disjoint from S . Every convergent net indexed by a set is eventually equal to its limit.*

Proof. The class

$$\{|s - t|^{-1} : s, t \in S, s \neq t\}$$

is a set. Choose a positive surreal B larger than every member of it, by a set cut, and put $\eta = 1/B$. Empty or singleton sets cause no difficulty: choose any positive B . For closedness apply the same argument to $S \cup \{x\}$; the interval $(x - \eta, x + \eta)$ misses S . If $(x_i)_{i \in I}$ is a set-indexed net converging to x , its range together with x is a set by class replacement; the separation gives a radius η whose neighbourhood of x contains no other value of the net, so $x_i = x$ eventually. $\square$

This strengthens proposition 9.1, which isolates each point with its own radius, to one radius for the whole set.

[write] Closedness, discreteness and eventual constancy of set-indexed nets are saut:prop:discrete of docs/surcomplex/surcomplex-field-automorphisms, proved there for $\mathbf{No}(i)$ and hence for $\mathbf{No}$; the uniform radius is source 05's addition.

Corollary 46.2 (Set-sized nets detect nothing [05]). *Every map $g : \mathbf{No} \rightarrow \mathbf{No}$, continuous or not, preserves all limits of set-indexed convergent nets. There is no set of positive radii coinital at zero, and no set-sized local base at any point.*

Proof. An eventually constant net remains eventually constant after applying any map. If R were a set of coinital positive radii, $\{d^{-1} : d \in R\}$ would have a surreal upper bound B , and $1/B$ would be smaller than every member of R , a contradiction. A set-sized local base would produce such a set of radii by choosing an interval inside each basic neighbourhood. $\square$

Remark 46.3 (Consequences [05]). The usual characterization of continuity by preservation of nets cannot be transferred while restricting nets to set-sized domains: its proof builds a net on a neighbourhood system of the relevant size. An **On**-indexed “net” behaves differently: $\omega^{-\alpha} \rightarrow 0$ as α runs cofinally through the ordinals, a class-indexed ε - δ statement, not a set sequence and not a Hahn sum over all ordinals. The full class is not discrete (every interval about x of radius ε contains $x + \varepsilon/2$); on a set-sized cutoff only the small-family statement is available, and applying the full-class theorem to the cutoff as its own ambient field would be an error. Two consequences: a continuous map from a connected set-sized topological space into the full surreal line is constant, since its image is a set with discrete subspace topology; in particular the topology induced on $\mathbb{R}$ by all surreal radii is discrete. And Hahn summation is not topological convergence: $\sum_{n \geq 1} \omega^{-n}$ exists, but its partial sums do not converge to it, since every nonzero tail exceeds $\omega^{-\omega}$. Likewise a computation along a set sequence of small increments cannot establish a surreal derivative; the derivative statements below quantify over all surreal radii and increments.

For a class function g , $g'(x) = L$ means $\forall \varepsilon > 0 \exists \delta > 0 \forall h (0 < |h| < \delta \Rightarrow |(g(x+h) - g(x))/h - L| < \varepsilon)$. A *finite derivative* is a limit belonging to the field, as opposed to a quotient tending to the external symbol $+\infty$; no Archimedean bound on the limit is assumed. This is the order-topological derivative, not the Berarducci–Mantova derivation.

Proposition 46.4 (A nonzero derivative forces the identity [05]). *If a field self-embedding j of $\mathbf{No}$ has a finite derivative at one point, it has the same derivative at every point. If that derivative is nonzero, then $j = \text{id}$ and the derivative is 1.*

Proof. Additivity makes the difference quotient equal to $j(h)/h$, independently of the base point. Suppose its limit is L . For fixed $a \neq 0$, replacing h by ah and using multiplicativity gives

$$\frac{j(ah)}{ah} = \frac{j(a)}{a} \cdot \frac{j(h)}{h}.$$

Multiplication by a fixed nonzero scalar is a homeomorphism near zero, so taking limits gives $L = (j(a)/a)L$. If $L \neq 0$, then $j(a) = a$ for every $a \neq 0$, and 0 is fixed too. $\square$

[write] This is `saut:thm:derivative` of `docs/surcomplex/surcomplex-field-automorphisms`, proved there for field automorphisms of $\mathbf{No}(i)$ by the same scaling identity; source 05's version is for self-embeddings of $\mathbf{No}$ and does not cite it. The zero-derivative dilations of source 05's Theorem 8.2 are `proposition 9.6` (`saut:thm:zeroderiv`).

47 Strong weighted maps: fixed fields and an exact differentiability criterion

Remark 47.1 (Fixed fields and images of the strong lifts [05]).

- (a) Combining (6.4) and (6.9), for every order self-embedding f ,

$$\text{Fix}(J_f) = \mathbb{R}((\omega^{\mathbb{R}((\omega^{\text{Fix}(f)})}))).$$

In particular a fixed-point-free f gives $\text{Fix}(J_f) = \mathbb{R}$ (source 05's Corollary 9.3).

- (b) The fixed-exponent subgroup $\text{Fix}(\tau) \cap \ker \chi$ in (6.13) is additive and divisible, by the two homomorphism laws and uniqueness of roots in $\mathbb{R}_{>0}$.
- (c) The image $\mathbb{R}((\omega^{\tau(\mathbf{No})}))$ of $H_{\tau, \chi}$ does not depend on χ . So two distinct weighted embeddings can have the same image and the same value action and different fixed fields: a description of the image is not a classification of the embedding.
- (d) If j is an exponential field self-embedding, its fixed field (real closed by `proposition 8.2`) is closed under $\exp$ and under the logarithm of positive elements: exponential closure is immediate, and logarithmic closure follows from uniqueness of logarithms and commutation with $\exp$ (source 05's Proposition 11.3).

Example 47.2 (A bounded copy fixing a familiar Hahn field [05]). Take the compression $c(a) = a/(1 + |a|)$ of `proposition 4.1`. Its only fixed point is 0, so by `remark 47.1(a)`

$$\text{Fix}(J_c) = \mathbb{R}((\omega^{\mathbb{R}})) = \mathbb{R}((\omega^{\mathbb{R}})).$$

Moreover A_c fixes real exponents, and J_c fixes $\mathbb{R}$ and ω . Every nonzero $A_c(a)$ has leading exponent in $(-1, 1)$, so $|A_c(a)| < \omega$; comparing leading exponents at the outer level gives

$$\omega^{-\omega} < |J_c(x)| < \omega^{\omega} \quad (x \neq 0).$$

This copy is bounded, proper and everywhere discontinuous. It complements the tighter annulus of theorem 45.3, whose map does not fix ω .

For the strong weighted maps of theorem 6.4 put

$$\text{sh}_\tau(a) = \tau(a) - a.$$

The phrase “ $\text{sh}_\tau(a) \rightarrow -\infty$ as $a \rightarrow -\infty$ ” means the class statement $\forall d \in \mathbf{No} \exists b \in \mathbf{No} \forall a < b \text{ sh}_\tau(a) < d$; the $+\infty$ statement reverses the last inequality.

Theorem 47.3 (Exponent criterion [05]). *For $j = H_{\tau, \chi}$, the ordinary derivative is zero everywhere if and only if*

$$\tau(a) - a \longrightarrow -\infty \quad (a \longrightarrow -\infty).$$

Its difference quotient tends to $+\infty$ at every point if and only if the same exponent difference tends to $+\infty$. The positive real character χ has no effect on either criterion. A finite ordinary derivative exists precisely when the first condition holds or $j = \text{id}$; in the latter case the derivative is 1.

Proof. If $h \neq 0$ has leading exponent a , then $j(h)/h$ is positive, with leading exponent $\text{sh}_\tau(a)$ and leading coefficient $\chi(a) \in \mathbb{R}_{>0}$.

To prove derivative zero, let $\varepsilon > 0$ have leading exponent d . Choose b such that $\text{sh}_\tau(a) < d$ whenever $a < b$, and use the increment bound $|h| < \omega^{b-1}$. The leading exponent of h then satisfies $a \leq b-1 < b$, so $j(h)/h < \varepsilon$. No uniform real bound on $\chi(a)$ is needed: a strict exponent inequality dominates every positive real coefficient.

Conversely, suppose $j(h)/h \rightarrow 0$ and fix $d \in \mathbf{No}$. Use the target $\varepsilon = \omega^{d-1}$. There is $\delta > 0$ such that the quotient is below this bound whenever $0 < |h| < \delta$. If b is the leading exponent of δ and $a < b$, the monomial $h = \omega^a$ is allowed. Thus $\chi(a)\omega^{\text{sh}_\tau(a)} < \omega^{d-1}$, which forces $\text{sh}_\tau(a) \leq d-1 < d$. This is the stated limit.

For divergence to $+\infty$, use the same argument with targets of leading exponent d in the sufficient direction and the monomial target ω^{d+1} in the necessary direction. A nonzero finite derivative forces $j = \text{id}$ by proposition 46.4, completing the classification. $\square$

For $\tau(a) = qa$ with a surreal $q > 1$, $\text{sh}_\tau(a) = (q-1)a \rightarrow -\infty$, and for $0 < q < 1$ it tends to $+\infty$; this recovers proposition 9.6.

[write] The criterion is new relative to the collection and to the other sources; it answers the derivative-classification question of source 05’s earlier edition, which source 05 removes from its agenda. It treats exactly the strong weighted sector of theorem 6.4; the unit twists proposition 10.2, the Taylor shift (10.3) and non-strong maps are outside it (research question 52.12).

Corollary 47.4 (Pointwise values and continuity do not give differentiability [05]). *A nonidentity weighted map $H_{\text{id}, \chi}$ has no finite ordinary derivative anywhere. In particular, fixing every natural-valuation value and being continuous do not imply ordinary differentiability.*

Proof. Its exponent difference is identically zero and so does not tend to $-\infty$; apply theorem 47.3. It is an automorphism, hence continuous by theorem 9.3. $\square$

Remark 47.5 (Bounded embeddings diverge [05]). Every *bounded* field self-embedding j , even one not given by a Hahn formula, has difference quotient tending to $+\infty$. By theorem 9.3 there is $\eta > 0$ with $|j(h)| > \eta$ for all $h \neq 0$; since j preserves sign, $0 < |h| < \eta/M$ implies $j(h)/h > M$. This is compatible with everywhere discontinuity, and is not a finite derivative.

Example 47.6 (Pointwise valuation does not alone force rigidity [05]). Put $\chi(a) = \exp_{\mathbb{R}}([a]_{-1})$, where $[a]_{-1}$ is the coefficient of ω^{-1} in a . Coefficient extraction is additive, so χ is a positive character, and $H_{\text{id},\chi}$ is a nonidentity field automorphism fixing every valuation value, every real and every ordinal (ordinal exponents have no ω^{-1} term), while

$$H_{\text{id},\chi}(\omega^{\omega^{-1}}) = \exp_{\mathbb{R}}(1) \omega^{\omega^{-1}}.$$

It preserves the omnific predicate (proposition 8.1), cannot preserve the surreal exponential (theorem 11.4), and has no finite ordinary derivative (corollary 47.4). This is the base example after proposition 10.1 with the base 2 replaced by $\exp_{\mathbb{R}}(1)$; the derivative clause is source 05’s addition.

48 Exponential comparison, the omega-map and omnific integers

Source 05’s bounded-displacement lemma and exponential valuation rigidity for self-embeddings are lemma 11.1, corollary 11.2, and theorem 11.4, with the same proofs; it credits them to the repository [13]. Its comparison lemma is stated for embeddings between two fields.

Lemma 48.1 (Cofinal bounded-difference rigidity [05]). *Let $\sigma, \tau : F \rightarrow K$ be embeddings of ordered fields, with $\tau[F]$ cofinal in K . If $|\sigma(x) - \tau(x)| \leq B$ for one $B > 0$ in K and every $x \in F$, then $\sigma = \tau$.*

This extends lemma 11.6 from self-embeddings $F \rightarrow F$ to embeddings $F \rightarrow K$; the proof of lemma 11.6, through the identity $d(ab) = \sigma(a)d(b) + \tau(b)d(a)$ and a b with $\tau(b) > B(1 + |\sigma(a)|)/|d(a)|$, applies verbatim, since it never compares values in F with values in K .

With the exponential profile $\nu(x) = v(\exp x)$ of section 11, additive with kernel the finite surreals $\mathcal{O}$ (11.3):

Theorem 48.2 (Cofinal profile uniqueness [05]). *Let $\sigma, \tau : \mathbf{No} \rightarrow \mathbf{No}$ be field self-embeddings with $\tau[\mathbf{No}]$ cofinal. If*

$$\nu(\sigma(x)) = \nu(\tau(x)) \quad (x \in \mathbf{No}),$$

then $\sigma = \tau$. In particular, on cofinal exponential field self-embeddings the induced action on the natural value group is faithful: $\tau_{\sigma} = \tau_{\tau}$ implies $\sigma = \tau$.

Proof. Since ν is additive, profile equality puts every difference $\sigma(x) - \tau(x)$ in $\mathcal{O}$, so all differences are bounded by ω ; apply lemma 48.1. If both maps commute with $\exp$ and have the same value action, then $\nu(\sigma x) = \tau_{\sigma}(\nu x) = \tau_{\tau}(\nu x) = \nu(\tau x)$. $\square$

The first clause assumes no exponential compatibility of either map, only equality of profiles; the second is theorem 11.7, and for an abstract ordered exponential field (F, E, w) it is theorem 35.3 (home 04). Source 05 notes that the inspected Lean file’s two-profile comparison assumes surjectivity of the second map, that lemma 48.1 isolates the weaker condition actually needed for this bounded-difference proof, cofinality, and that it supplies no new Lean proof of that strengthening. Removing cofinality by the same argument is invalid: a bounded image cannot provide an arbitrarily large $\tau(b)$ (compare the base question “Remove or justify the cofinality hypothesis” in section 15, and research question 52.6).

[write] The surjective case is `cor:two-profiles` of `docs/surreal/exponential-automorphism-rigidity` and its Lean form `eq_of_profile_eq` in `Surreal/Algebra/ExponentialProfile.lean`, whose first map is a ring homomorphism and whose second is a ring equivalence (`RingEquiv`); the automorphism case of the faithfulness clause is `saut:thm:expfaithful` of `docs/surcomplex/surcomplex-field-automorphisms`. The cofinal version strengthens both; the Lean generalization (replace the `RingEquiv` by a ring homomorphism with cofinal range) is source 05’s research question 52.1.

Proposition 48.3 (Rational dilations do not lift, for embeddings [05]). *Let j be an exponential field self-embedding of $\mathbf{No}$ whose value action is $\tau_j = q \text{ id}$ for some $q \in \mathbb{Q}_{>0}$. Then $q = 1$ and $j = \text{id}$.*

Proof. For every x , exponential compatibility and $\mathbb{Q}$ -linearity of the additive map v give $v(j(x)) = \tau_j(v(x)) = qv(x) = v(qx)$, so $v(j(x) - qx) = 0$ and $j(x) - qx$ is finite. At $x = \omega$ and $x = \omega^2$ write the finite errors as ϵ_1, ϵ_2 . Multiplicativity $j(\omega)^2 = j(\omega^2)$ gives

$$(q^2 - q)\omega^2 + 2q\omega\epsilon_1 + \epsilon_1^2 - \epsilon_2 = 0.$$

After division by ω^2 every term except $q^2 - q$ is infinitesimal, so the rational $q^2 - q$ vanishes and $q = 1$. Then $\tau_j = \text{id}$, and theorem 11.4 gives $j = \text{id}$. $\square$

Yet every positive rational dilation lifts to the plain-field automorphism S_q ; exponential compatibility is a real additional restriction. The value displacement theorem theorem 11.5 does not exclude $\tau_j = q \text{ id}$ by itself, since $(q - 1) \text{ id}$ has cofinal and cointial range for $q \neq 1$.

[write] For automorphisms this is `thm:dilation` of `docs/surreal/exponential-automorphism-rigidity` and `saut:prop:rationalnonlift` of `docs/surcomplex/surcomplex-field-automorphisms`, by the same X, X^2 computation; the Lean form is `surreal_not_lift_dilation` in `Surreal/HahnSeries/ValueGroupLifts.lean`, with an `OrderedExp` hypothesis. Source 05's version needs no surjectivity.

Remark 48.4 (Omega-map and omnific integers in source 05). [05] Source 05's omega-map proposition and the self-consistency equation are theorem 10.3 and (10.5); in this report every normal-form ω^x is the Conway map, which is not $\exp(x \log \omega)$ for arbitrary x . Its omnific theorem is proposition 8.1 (predicate preserved and reflected by every $H_{\tau, \chi}$) together with $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$ and pointwise $\mathbf{Oz}$ -rigidity, printed as theorem 20.2 (home 02; same proof, after `odg:thm:fractions` of `docs/surreal/omnific-diophantine-geometry`, which source 05 credits). Source 05 adds two cautions: the fraction-field identity uses the full field's ability to bound every *set* of exponents and cannot be transferred without qualification to small Hahn fields; and an automorphism of the pair $(\mathbf{No}, \mathbf{Oz})$ preserves the predicate rather than fixing it pointwise, with no unrestricted automatic-strongness theorem for pair embeddings assumed. Source 05 also notes that the general omega-field question [5, Question 5.6] is not solved.

49 Elementary embeddings and internal definability

By quantifier elimination every field self-embedding of $\mathbf{No}$ is elementary in the ordered-field language, formula by formula (proposition 12.1); so the bounded, cofinal and ordinal-fixing copies are already elementary there. For the ordered exponential field the background is that $\mathbf{No}$ is an elementary extension of the real exponential field [3], recorded as [7, Proposition 1.5]. Source 05 uses the standard o-minimal consequences: one-variable types over a definably closed model are determined by cuts, definable closure has exchange, and definable Skolem hulls are elementary. These are explicit model-theoretic inputs, not consequences of order saturation alone.

Standing hypothesis of this section [05]. $\mathfrak{M}$ is an o-minimal expansion of the ordered field $\mathbf{No}$ in a set-sized first-order language, and the ambient class theory provides global choice, a uniform satisfaction relation for $\mathfrak{M}$, and definable Skolem operations. Then the definable closure $\text{dcl}(A)$ of a set A is a set; it is an elementary submodel, because definable Skolem functions supply witnesses to existential formulas; and over a definably closed set a nonalgebraic one-variable type is determined by its cut, since a definable subset of the line is a finite union of intervals and points with endpoints in the definable closure. For the pure field language, quantifier elimination supplies the uniform description directly, and everything below can be read in GBC without a general truth predicate. For the exponential

language source 05 uses GBC plus elementary transfinite recursion, or explicitly supplied satisfaction and definable Skolem classes: a sufficient framework, not a minimality claim. Every statement of this section except proposition 49.8 and theorem 49.9 is conditional on this hypothesis.

[write] Source 04 proves the cofinal and bounded elementary embeddings in its own framework (theorems 34.2 and 34.5, home 04, with lemma 34.1); the versions below are source 05's independent proofs. Source 06, in its exponential section (remark 59.1, Part V), proposes handling the satisfaction and definable closures of the exponential structure through set-sized epsilon birthday stages: these are models of the theory of the real exponential field, model completeness makes their inclusions elementary, a set of parameters lies in one stage, and its definable closure can be computed there. If that route is carried out, it would interpret the class statements of corollaries 49.3 and 49.6 without a general satisfaction class and might weaken the GBC+ETR hypothesis. This is unchecked here.

Lemma 49.1 (Set types and elementary extensions [05]; under the standing hypothesis). *Every consistent nonalgebraic one-variable type over a set-sized elementary submodel of $\mathfrak{M}$ is realized by a proper class of elements of $\mathfrak{M}$. Any elementary isomorphism between set-sized elementary submodels extends across one new domain element. By alternately extending domain and range, it extends to an automorphism of $\mathfrak{M}$.*

Proof. Write a nonalgebraic type as its set cut $L < R$. Its class C of separators is proper: if C were a set, the cut $L \cup C < R$ would have a separator, which would belong to C while exceeding every member of C . All separators realize the type, by the interval description of definable sets. Hence a realization can be chosen outside any prescribed set.

Transfer the type of a new domain element through the elementary isomorphism, choose a realization in the target, and extend to the definable closures. Equality of complete types makes this extension well defined and elementary, and both closures are sets. For an automorphism, fix a set-like class enumeration of $\mathfrak{M}$ and alternately add the next missing domain element and the next missing range element, using the same construction for the inverse map at a back step. At a set limit take the union. Each stage is a set map, and the final class union has full domain and range. $\square$

[write] For the pure ordered field, the automorphism clause is `saut:thm:orderedhomogeneity` of `docs/surcomplex/surcomplex-field-automorphisms` (ordered-field isomorphisms between set-sized subfields extend to automorphisms of $\mathbf{No}$, under global choice); in this report ordered-field homogeneity is theorem 70.2 (home 07). The realization clause is lemma 34.1 (home 04); “by a proper class” is source 05's addition.

Theorem 49.2 (Omitting a point [05]; uniform satisfaction, definable Skolem operations, global choice). *Under the standing hypothesis, let $K \preccurlyeq \mathfrak{M}$ be a set-sized elementary submodel and $b \in \mathfrak{M} \setminus K$. There is a cofinal proper elementary self-embedding $j : \mathfrak{M} \rightarrow \mathfrak{M}$ fixing K pointwise and omitting b from its image.*

This is theorem 34.5 (home 04).

Second route to theorem 34.5 [05]. Fix a class enumeration $(a_\alpha)_{\alpha \in \mathbf{On}}$ of $\mathfrak{M}$. Construct increasing elementary isomorphisms $j_\alpha : K_\alpha \rightarrow K'_\alpha$ between set-sized definably closed elementary submodels, with $b \notin K'_\alpha$, starting from the identity on K .

At a forth step take the next enumerated $a \notin K_\alpha$, transfer its cut to K'_α , and choose a realization $y \notin \text{dcl}(K'_\alpha \cup \{b\})$, possible by lemma 49.1 since the excluded hull is a set. Extend the partial elementary map to the definable hulls of $K_\alpha \cup \{a\}$ and $K'_\alpha \cup \{y\}$. Exchange keeps b omitted: if $b \in \text{dcl}(K'_\alpha, y)$, then, as $b \notin \text{dcl}(K'_\alpha)$, exchange would put y in $\text{dcl}(K'_\alpha, b)$.

At a cofinality step choose a fresh source element above all of the current source model; its type is the cut above that model. Choose the target realization above all of the current target model and above a_α , again outside the definable hull of that target model together with b . The same extension and exchange argument applies.

At a set limit stage take unions; elementary-chain unions and finite character of definable closure preserve the induction hypotheses, and each stage remains set-sized. The final union has domain $\mathfrak{M}$, is elementary, fixes K , omits b , and has cofinal image because of the cofinality steps. $\square$

Corollary 49.3 (A cofinal proper exponential copy [05]; under GBC+ETR or the standing hypothesis). *The ordered exponential field of surreals has a cofinal proper elementary self-embedding fixing $\mathbb{R}$ pointwise and omitting ω .*

Proof. Take $K = \mathbb{R}$ with its exponential structure and $b = \omega$ in theorem 49.2; $\mathbb{R}_{\text{exp}} \preccurlyeq (\mathbf{No}, \text{exp})$ is the standard surreal exponential theorem. GBC plus elementary transfinite recursion is one sufficient class framework for the uniform satisfaction bookkeeping. $\square$

This is an existence construction. It supplies no strong Hahn formula, and no compatibility with simplicity, the omega-map, the omnific predicate or a derivation. Since the image is cofinal, the embedding is continuous by theorem 9.3; since it is not the identity, its value action is not the identity, by theorem 11.4.

Lemma 49.4 (Bounded extension [05]; under the standing hypothesis). *Let $f : A \rightarrow B$ be an elementary isomorphism between set-sized elementary submodels of $\mathfrak{M}$, and let b be greater than every element of B . For every $a \in \mathfrak{M}$ there is an elementary extension $f' : \text{dcl}(A, a) \rightarrow B'$ whose image B' is a set-sized elementary submodel containing B , all of whose elements are still less than b .*

Proof. The case $a \in A$ is immediate. Otherwise first ignore the bound and extend f across a , by lemma 49.1, to an isomorphism onto $C = \text{dcl}(B, y_0)$. Choose q greater than every element of the set C . Both q and b realize the same type over B , namely the cut above all of B ; this uniqueness of the type uses o-minimality, not merely an order-preserving map. Hence there is an elementary isomorphism

$$\theta : \text{dcl}(B, q) \longrightarrow \text{dcl}(B, b)$$

fixing B and taking q to b . Extend θ across y_0 by lemma 49.1; its extended domain contains C . For every $c \in C$ the inequality $c < q$ is preserved, so $\theta(c) < b$. Thus $B' = \theta[C]$ is the required bounded elementary submodel; compose the first extension of f with this restriction of θ . $\square$

Theorem 49.5 (A bounded elementary copy below any external bound [05]; under the standing hypothesis). *Let $K \preccurlyeq \mathfrak{M}$ be a set-sized elementary submodel and let b be greater than every element of K . There is an elementary self-embedding $j : \mathfrak{M} \rightarrow \mathfrak{M}$ fixing K pointwise with*

$$|j(x)| < b \quad (x \in \mathfrak{M}).$$

It is proper and everywhere discontinuous in the ambient order topology.

This is theorem 34.2 (home 04). Source 04 moves the bound by realizing the type at $+\infty$ in a set model obtained by compactness; source 05 transports an auxiliary upper bound q to b by the small isomorphism θ of lemma 49.4.

Second route to theorem 34.2 [05]. Start with the identity on K . Enumerate the source class and at each stage apply lemma 49.4 to add the next source element while keeping the target bound b . Take unions at set limit stages. Stage domains and ranges remain sets, elementary and definably closed. Every element of the final range belongs to some stage and so is below b ; since the range is a field, negatives are also below b , giving the absolute bound. The final map has full domain, is elementary and fixes K ; it omits b and is noncofinal, so theorem 9.3 gives everywhere discontinuity. $\square$

Corollary 49.6 (Bounded exponential copy [05]; under GBC+ETR or the standing hypothesis). *The ordered exponential field of surreals has a proper elementary self-embedding j fixing $\mathbb{R}$ pointwise with*

$$\omega^{-1} < |j(x)| < \omega \quad (x \neq 0).$$

It commutes with the total surreal exponential and logarithm, despite being everywhere discontinuous, and its difference quotient tends to $+\infty$ at every point. Moreover its range is bounded above by every positive infinite $\log^{on} \omega$, $n \in \mathbb{N}$.

Proof. Take $K = \mathbb{R}$ and $b = \omega$ in theorem 49.5; the exponential elementary-extension theorem gives $K \prec \mathfrak{M}$. Elementarity preserves exp; logarithmic compatibility follows from uniqueness and surjectivity of exponentiation. The lower bound follows by applying the upper bound to $j(x)^{-1} = j(x^{-1})$, and the quotient assertion is remark 47.5. For u in the range, $\exp^{on}(u)$ is in the range and so is less than ω ; applying n logarithms gives $u < \log^{on} \omega$. $\square$

[write] This (with theorem 34.2) answers, only under the stated class framework, the base question “Proper exponential copies” of section 15 as to whether a bounded range can occur, and the corresponding questions of sources 02 and 07. Source 05 records that it settles a question of its own earlier edition. The map is not explicit: no strong, monomial or omnific compatibility is claimed.

Remark 49.7 (What the elementary copies separate [05]). The range in corollary 49.6 is a proper class lying below every $\log^{on} \omega$; this is possible because the range is a proper class, not because a set escaped a cardinality restriction, and its birthdays remain unbounded. Together, theorems 49.2 and 49.5 separate exponential elementarity from both surjectivity and cofinality. They make no claim of strong summability, monomial control or simplicity preservation, and do not contradict exponential valuation rigidity: their induced value-group embeddings are not the identity. For plain real closed fields the same back-and-forth explains why a real need not be fixed: a transcendental real and an infinitesimal perturbation of it have the same cut over the real algebraic numbers, and a small ordered-field isomorphism between their real closures extends to an automorphism of **No** that does not preserve the named real subfield.

Proposition 49.8 (Lacunary-sum witness [05]; GBC). *There is a field automorphism σ of **No** fixing $\mathbb{R}$ and ω pointwise but not preserving a specified set-indexed Hahn sum. In particular continuity does not imply strong additivity.*

Proof. Put $t = \omega^{-1}$ and

$$z = \sum_{n \geq 1} t^{n!}, \quad z_N = \sum_{1 \leq n \leq N} t^{n!}.$$

This is a valid Hahn sum: its t -exponents strictly increase, that is, its Conway exponents strictly decrease. First, z is transcendental over $\mathbb{R}(t)$. Suppose $P(t, z) = 0$ for a nonzero $P(T, Y) \in \mathbb{R}[T, Y]$ with $\deg_T P = m$ and $\deg_Y P = d$. For all large N , $P(t, z_N) \neq 0$, since $P(t, Y)$ is a nonzero polynomial in Y (t being transcendental over $\mathbb{R}$) with at most d roots, and the z_N are distinct. As a polynomial in t of degree at most $m + dN!$, the nonzero element $P(t, z_N)$ has t -adic order at most $m + dN!$. But

$$P(t, z) - P(t, z_N) = (z - z_N) Q(t, z, z_N)$$

for a real polynomial Q ; every monomial of $Q(t, z, z_N)$ has nonnegative t -adic order, and $z - z_N$ has order $(N + 1)!$. Hence $P(t, z_N) = -(z - z_N)Q$ has order at least $(N + 1)!$. Choosing N with $(N + 1)! > m + dN!$ is a contradiction.

Let F be the real closure of $\mathbb{R}(t)$ in **No**, a set-sized real closed subfield; z is transcendental over F as well. The positive distances $|z - a|$, $a \in F$, form a set, so choose $\varepsilon > 0$ below all of them. Then z and $z + \varepsilon$ have the same cut over F , hence the same nonalgebraic one-variable type in the real closed

field language. The resulting elementary isomorphism between their definable closures extends to an automorphism σ of $\mathbf{No}$ by the pure-field case of lemma 49.1. It fixes $F \ni t, \omega$ and sends z to $z + \varepsilon \neq z$. All summands $t^{n!}$ are fixed, so

$$\sigma\left(\sum_{n \geq 1} t^{n!}\right) \neq \sum_{n \geq 1} \sigma(t^{n!}).$$

Every field automorphism is continuous by theorem 9.3. $\square$

The separation does not treat formal summation as a topological limit. It complements corollary 9.5, which shows that strong Hahn summability does not give continuity.

[write] The proposition uses only the pure ordered-field case of lemma 49.1, which by quantifier elimination needs only GBC with global choice; the extension step is `saut : thm : orderedhomogeneity` (theorem 70.2). The witness is new to the collection.

Theorem 49.9 (Internally definable field maps [05]). *Let M be an o-minimal expansion of a real closed field. Every unital field self-embedding whose graph is definable in M , with parameters, is the identity. This applies to the pure surreal field and to its standard o-minimal exponential structure, with the class interpretation fixed above.*

This is theorem 12.2, which the base proves by an interval in the definable fixed field. Source 05 argues through calculus.

Second route to theorem 12.2 [05]. The differentiability theorem for definable functions gives a nonempty interval on which the embedding is continuously differentiable. Choose $u < v$ there; strict increase gives $j(v) - j(u) > 0$, and the definable mean value theorem gives a point where the derivative is nonzero. Proposition 46.4, whose proof works over any ordered field with these limits, forces $j = \text{id}$. $\square$

Only ordinary o-minimal calculus is used [23]. The theorem shows why the zero-derivative automorphisms S_q cannot have internally definable graphs, and warns against conflating “explicit” (as S_q and J_f are, in set theory via normal forms) with “internally definable”.

Remark 49.10 (Birthday languages and set-theoretic embeddings [05]). Adding a birthday function, sign-prefix predicates or coding relations can destroy the elementarity of maps elementary in the pure field. The literal external equality $b(j(x)) = b(x)$ is already rigid by theorem 4.5; but preserving a birthday function as a symbol of an expanded structure can instead mean $j(b(x)) = b(j(x))$, where ordinal values may move, and the latter cannot be replaced by the former without proof. An elementary $j : V \rightarrow M$ from large-cardinal theory acts on a sign sequence as the set $j(s)$, with an external target, an altered ordinal domain, and possible extra coordinates in $j(\alpha) \setminus j[\alpha]$; a surreal field map requires absoluteness of the chosen arithmetic between the universes, and is not the coordinatewise prescription on image coordinates. Conversely none of the explicit surreal field embeddings determines an elementary map of (V, ϵ) .

[write] The large-cardinal map is carried out in `docs/foundations-and-computation/large-cardinal-embeddings-and-normal-forms`: its `lce : prop : J` gives, from an elementary class embedding $j : V \rightarrow M$ into a transitive M with a critical point (for instance a normal ultrapower by a measurable cardinal), an ordered-field embedding $\mathcal{J}_j$ of $\mathbf{No}$ fixing $\mathbb{R}$, preserving the omega-map and prefix simplicity, with $b(\mathcal{J}_j(x)) = j(b(x))$, that is, the second reading above. Source 05 does not cite it; source 02 treats it in Part II.

50 Differential, hyperserial and higher-dimensional variants

A differential self-embedding satisfies $j(\partial x) = \partial(j(x))$ for a named derivation ∂ . This is unrelated to whether the *function* j has an ordinary derivative: the zero-derivative phenomenon of [proposition 9.6](#) does not say that S_q commutes with any derivation. For a derivation with constant field exactly $\mathbb{R}$, as in the Berarducci–Mantova setting [\[8, 26\]](#), every differential self-embedding fixes $\mathbb{R}$ pointwise ([proposition 13.1](#)); this fails for a derivation with a larger constant field.

Criterion 50.1 (A monomial test [\[05\]](#)). *Suppose ∂ is strongly $\mathbb{R}$ -linear and annihilates $\mathbb{R}$, and j is a strongly $\mathbb{R}$ -linear field embedding, where strongness includes preservation of each summable family occurring in the compositions below. Then $j\partial = \partial j$ if and only if $j(\partial\omega^a) = \partial(j(\omega^a))$ for every Conway monomial ω^a .*

Proof. For a normal form $x = \sum r_a \omega^a$, strongness gives $j(\partial x) = \sum r_a j(\partial\omega^a)$ and $\partial(j(x)) = \sum r_a \partial(j(\omega^a))$, and the monomial equalities identify the sums. Necessity is immediate. $\square$

This reduces a verification problem; it constructs nothing, and it shows why an unjustified interchange of a class sum and an operator would be fatal.

Definition 50.2 (Diagonal derivations [\[05\]](#)). For an additive class map $\lambda : (\mathbf{No}, +) \rightarrow (\mathbb{R}, +)$, for example extraction of a fixed normal-form coefficient, put

$$\partial_\lambda \left(\sum r_a \omega^a \right) = \sum r_a \lambda(a) \omega^a.$$

The output support is contained in the input support. Coefficientwise addition and $\lambda(a + b) = \lambda(a) + \lambda(b)$ give the Leibniz rule through finite coefficient convolutions; ∂_λ is strongly $\mathbb{R}$ -linear and annihilates $\mathbb{R}$.

Proposition 50.3 (Commutant of a diagonal derivation [\[05\]](#)). *For a weighted embedding $j = H_{\tau, \chi}$,*

$$\partial_\lambda j = j \partial_\lambda \iff \lambda(\tau(a)) = \lambda(a) \text{ for all } a \in \mathbf{No}.$$

The constant field of ∂_λ is exactly $\mathbb{R}((\omega^{\ker \lambda}))$.

Proof. On ω^a the two compositions have the coefficients $\chi(a)\lambda(\tau(a))$ and $\chi(a)\lambda(a)$ at the exponent $\tau(a)$. Since $\chi(a) \neq 0$, equality is the displayed condition, and [criterion 50.1](#) extends it to every normal form. A series is killed by ∂_λ exactly when all its nonzero terms have exponents in $\ker \lambda$. $\square$

These are genuine differential self-embeddings for a named derivation, but not for the Berarducci–Mantova derivation, and their constant fields can be much larger than $\mathbb{R}$. Specifying the derivation is as important as specifying the exponential. For example, for $\lambda(a) = [a]_{-1}$ every ordinal-fixing J_{f_β} of [theorem 7.4](#) with $\beta < -1$ commutes with ∂_λ exactly when A_{f_β} preserves the coefficient of ω^{-1} , which it does, since f_β fixes every exponent $\geq \beta$. [\[write\]](#)

Remark 50.4 (Composition and hyperserial substitutions [\[05\]](#)). Substitution by another positive infinite surreal is a natural candidate source of embeddings compatible with exponential or hyperexponential operations. The hyperseries literature gives precise embedding results on specified hyperseries subfields [\[28\]](#); extension from a specified subfield to all of $\mathbf{No}$, with the desired compatibility and summability, must be verified separately, and a notation $x \mapsto x \circ g$ is not by itself a total self-embedding theorem (compare the substitution paragraph of [section 13](#)). The omega-map, the Gonshor exponential, hyperexponentials and a chosen derivation should each be an explicit symbol of the language; preserving one does not automatically preserve the others, and joint compatibility is a promising source of rigidity questions.

Remark 50.5 (Quaternionic extensions [05]). Besides the two extensions of a field embedding to $\mathbf{No}(i)$ (“Algebraic variants” in section 12), coordinatewise application of a field embedding j embeds the quaternion algebra over $\mathbf{No}$ when the three standard units are fixed, since their multiplication table has integer coefficients. Allowing the units themselves to move adds an independent algebraic symmetry problem. Neither extension supplies a contour theory or an analytic embedding merely from its field properties.

51 Formalization architecture and finite checks of source 05

Separate interfaces. Source 05 proposes separate layers for sign sequences, the field, normal forms, valuation, exponentiation and summability, noting that the repository’s generic displacement machinery and sign-sequence work do not mean that all bridges between these layers have been verified. The following are *interface descriptions*, not submitted or tested Lean code:

```

SignSequence.{u}      -- ordinal-length signs in a larger universe
birthday : SignSequence -> Ordinal.{u}
IsPrefix : SignSequence -> SignSequence -> Prop
OrderEmbedding        -- order preservation and reflection
TreeOrderEmbedding    -- OrderEmbedding + prefix compatibility
SmallCutData          -- explicitly small left and right option types
HahnSummable          -- set support + local finite occurrence
StrongMap              -- preserves HahnSummable and its sums
FieldEmbedding         -- unital ring homomorphism between fields
ValueAction           --  $v(e\ x) = \rho(v\ x),\ x \neq 0$ 
ExpCompatibility       --  $e(\exp\ x) = \exp(e\ x)$ 
CofinalRange          -- every target has a larger image element

```

The inspected repository declares `SignSequence : Type (u + 1)` with `birthday` in `Ordinal.{u}`; its `cut` constructor accepts `SmallCutData.{u, u + 1}` and combines the simplest-separator theorem with `small_cut_fillers`, so existence is supplied explicitly rather than hidden in a uniqueness statement [12, 25]. A universe-polymorphic surreal type is small in one universe only by being large relative to another and does not contain every ordinal of its own larger universe; cut filling must carry a smallness parameter, otherwise a type-level Burali–Forti error is hidden in the interface. In particular theorem 46.1 becomes a theorem about appropriately small families.

A short route to the new proof obligations.

1. Reuse the repository’s canonical-cut and prefix lemmas to formalize birthday-nonincreasing rigidity (sign model and order only).
2. Implement successor and limit padding as a recursion and prove theorem 44.1; derive rank noncontraction and the initial-image theorem.
3. Prove the bounded-difference identity for two ring homomorphisms and lemma 48.1 over arbitrary ordered fields, with cofinality as the exact hypothesis replacing surjectivity.
4. Formalize the topology dichotomy for ordinary ordered fields before instantiating it on a surreal type.
5. Formalize A_f and H_τ first on ordinary set-indexed Hahn groups and fields; a second module handles the universe-relative normal-form bridge.

The proof of the dilation derivatives should then be split into an algebraic leading-exponent lemma and an ε - δ theorem; a proof for rational exponents only, or monomial inputs only, is not the full assertion.

What source 05 inspected. At its pinned commit, through the GitHub connector: `Surreal/Algebra/ExponentialProfile.lean` (opening declarations and initial ordered-field proofs; the file marks the actual surreal exponential/valuation instantiations as pending); `docs/surreal/exponential-automorphism-rigidity/README.md` (opening 190 lines); `docs/surreal/omnific-preserving-automorphisms/README.md` (opening 100 lines) and its `13-omnific-isomorphisms-SOURCE_AUDIT.md` (opening 110 lines); `Surreal/Foundations/SignSequence.lean` (opening 115 lines); and `Surreal/Foundations/SignSequenceCutOperation.lean` (opening 130 lines). This was not a complete repository audit or a local rebuild, and an implementation comment was not treated as a substitute for checking assumptions; in particular the existence premise of a simplest-separator theorem must not disappear when quoted.

[write] The claims checked against the repository at its HEAD: the `SignSequence` carrier and its universe levels, the cut constructor on `SmallCutData.{u, u + 1}`, the pending surreal instantiation of the profile lemmas (recorded in the project's `FORMALIZATION.md`), the surjectivity hypothesis of `eq_of_profile_eq`, the repository's answer to [5, Question 5.4] (`exponential-automorphism-rigidity's cor:question54`), and the fraction-field identity (`odg:thm:fractions`) are all accurate. The files source 05 read are byte-identical at the placement commit.

Reproducible finite checks. Source 05's script, shipped as `code/05-comprehensive-verification.py` (standard library only, exact fractions and sparse symbolic polynomials, deterministic seed 20261003, `-seed` and `-output` options), checks first-difference comparisons and prefix/meet compatibility of finite sign words up to length 6, prefix embeddings, rational compression and the rational ordinal fixer, finite additive reindexing, nested Hahn convolution under the two-level lift, weighted characters, composition and omnific predicates in the finite model, explicit monomial witnesses, the lacunary degree gap, and the twisted displacement and two-map product identities; it records an explicit counterexample to multiplicativity of unrestricted one-level reindexing (remark 45.1). The recorded run, `data/05-comprehensive-verification.json`, passes 146,914 assertions in 28 groups. Finite calculations cannot certify set-sized summability, arbitrary ordinal padding, existence of class maps, the ε - δ derivative on all surreals, or the model-theoretic constructions; those rest on the written proofs and their stated inputs. Its `build.sh`, shipped as `code/05-comprehensive-build.sh`, runs the checks under their delivery names and rebuilds the delivered PDF.

[write] Rerun on a copy at placement: PASS, 146,914 assertions, output equal to the delivered record up to line endings. The earlier edition's suite (not shipped) had further groups (canonical cone, sign repetition, dilation derivative signs, two-probe amplification) that this rewrite does not keep under those names.

Proof-status ledger of source 05. Source 05's appendix ledger, with the location of each item in this report.

Result or input	Status in source 05	Here
Sign model, set-cut filling, normal forms	Classical inputs; no proper-class cut or support.	Section 3.
Local padding; birthday rigidity	Written sign-sequence proofs; related general theory credited to the surreal-substructure literature.	Theorem 44.1; theorem 4.5.

Result or input	Status in source 05	Here
Field lift; two-level lift	Written proofs from the Hahn support/convolution lemma and class replacement on set supports.	Theorems 6.2 and 6.3.
Bounded copy; ordinal-fixing copy	Explicit deductions from the two-level lift; no large-cardinal hypothesis.	Theorem 45.3; theorem 7.1.
Topology dichotomy	Written ordered-field proof, independent of Hahn theory.	Theorem 9.3.
Sets uniformly discrete	Smallness argument from set-cut bounds and class replacement; not a claim that the whole line is discrete.	Theorem 46.1.
Dilations; exponent criterion	Leading-exponent and ε - δ proofs; finite tests do not establish limits.	Proposition 9.6; theorem 47.3.
Support rigidity, fixed fields	Invariant-support proof, exact in the specified strong sector.	(6.13); remark 47.1.
Exponential valuation rigidity	Repository result, reproved; generic Lean coverage separate from the surreal instantiation.	Theorem 11.4.
Cofinal profile uniqueness	Written strengthening to cofinal comparison maps; not newly Lean-verified.	Theorem 48.2.
Weighted classification	Standard Hahn mechanism for embeddings in the specified sector; not a classification of all embeddings.	Theorem 6.4.
Omnific theorem	Elementary normal-form proof; fraction-field result already in the repository.	Proposition 8.1; theorem 20.2.
Omission; bounded elementary copies	Constructions conditional on explicit satisfaction, Skolem and o-minimal inputs; GBC+ETR sufficient.	Theorems 49.2 and 49.5.
Lacunary witness	Lacunary transcendence and small-model homogeneity; no infinite topological summation.	Proposition 49.8.
Diagonal derivations	Coefficientwise calculation, not about the Berarducci–Mantova derivation.	Proposition 50.3.
Internal definability	Deduction from o-minimal differentiability and the definable mean value theorem.	Theorem 49.9.
Research questions	Not declared solved; no assertion of novelty or exhaustive literature coverage.	Section 52.

52 Research questions of source 05

Source 05 presents these as scoped research targets, not claims that their precise formulations have never appeared. It records that two tempting questions are already settled in it, and removed from its agenda: the strong weighted differentiability question, by [theorem 47.3](#), and the existence of noncofinal elementary exponential copies, by [corollary 49.6](#) in its class framework (both were questions of its earlier edition). The base questions cited below are those of [section 15](#), named by title.

52.1 Near-term formalization projects

Research question 52.1 (Cofinal profile formalization [05]). Replace the surjective comparison-map hypothesis in the repository’s two-profile theorem by cofinality, using [lemma 48.1](#). Bundle the resulting faithful value-group action for cofinal exponential embeddings in a universe-relative setting.

This is a small generic ordered-field project; it should precede a surreal instantiation, so that missing

exponential or valuation bridges cannot be mistaken for a proof of the concrete application.

[write] Compare the base question “Near-term formalization”. The target theorem is `exponential-automorphism-rigidity’s cor:two-profiles`, Lean `eq_of_profile_eq`; open.

Research question 52.2 (A programmable tree interface [05]). Formalize the local padding description with explicit successor blocks and limit padding. Which restrictions on those data preserve a given birthday cutoff, are chain-continuous, or admit effective descriptions on ordinal notations?

The implementation should derive simplicity reflection and meet preservation rather than include them as redundant axioms, and should distinguish tree continuity from order-topological continuity.

Research question 52.3 (Reusable Hahn-lift and fixed-field libraries [05]). Formalize A_f , $H_{\tau, \chi}$, their composition law, support rigidity and the exact fixed-field formulas. Can the two-level functor be implemented without identifying the ambient surreal field with a same-universe Hahn series type?

The first target is an ordinary set-sized Hahn field; universe lifting and the surreal normal-form bridge should be separate theorems.

[write] Partial existing Lean material: `workspaceEmbedding` (Surreal/HahnSeries/WorkspaceEmbedding.lean, Hahn series along an injective additive exponent map), `hahnLift` (Surreal/HahnSeries/ValueGroupLifts.lean, for order automorphisms of the exponent group only) and `Surreal/HahnSeries/CharacterTwist.lean`. Not checked here whether they meet the question’s specification.

Research question 52.4 (Formal topology at the correct size [05]). Formalize the cofinality–continuity dichotomy and the collapse of small convergent nets. Construct a framework for class-indexed neighbourhood nets, or formulate continuity entirely by neighbourhood predicates, without positing a set of all open subclasses.

A useful test is to represent both a nowhere-continuous bounded embedding and its preservation of every small net limit, so that their coexistence is explicit rather than an apparent inconsistency in the library.

52.2 Structural classification projects

Research question 52.5 (Simultaneous field and simplicity embeddings [05]). Give explicit, assumption-audited constructions or obstructions for proper self-embeddings preserving both field operations and sign-prefix simplicity. Does adding fixation of $\mathbb{R}$, chain continuity, or preservation of selected simplest cuts force rigidity?

The padding theorem alone supplies no field laws, and the Hahn lift alone no simplicity laws (example 45.4); a result must bridge the two structures, not exchange their notions of truncation.

[write] Compare the base question “Field and simplicity together”. A conditional construction exists in the collection: `lce:prop:J` of `docs/foundations-and-computation/large-cardinal-embeddings-and-normal-forms` gives, from an elementary $j : V \rightarrow M$ with a critical point, a nonidentity ordered-field embedding fixing $\mathbb{R}$ and preserving and reflecting prefix simplicity, and it is proper (`lce:cs:prop:proper`). In ZFC or GBC alone the question remains open.

Research question 52.6 (Two noncofinal exponential embeddings [05]). If two exponential self-embeddings have the same induced natural value-group embedding but both have bounded images, must they agree? Which additional hypotheses recover the cofinal profile uniqueness theorem?

Theorem 49.5 shows that noncofinal exponential maps are a genuine case. The bounded-difference proof fails at a precisely identified step: the comparison map cannot provide arbitrarily large target values. For plain fields, two bounded images can have uniformly bounded difference without the maps being equal.

[write] Compare the base question “Remove or justify the cofinality hypothesis”; source 04 asks the same. Open.

Research question 52.7 (Strong exponential proper copies [05]). Can the bounded or cofinal elementary exponential embeddings constructed here be chosen strongly $\mathbb{R}$ -linear, monomial-preserving, or with a prescribed class of supports? Which ordered-group embeddings actually occur as τ_j ?

The rational-dilation obstruction (proposition 48.3) already excludes many valid ordered-group embeddings; an extension theorem must address all exponential equations and all relevant set-indexed summability conditions.

[write] Compare the base questions “Characterize exponential value actions” and “Proper exponential copies”, and exponential-automorphism-rigidity’s `q:image` (the second sentence restates it for embeddings). Source 04 gives a criterion for normalized Hahn lifts to preserve exponentiation (proposition 35.4, Part III), which bears on the first sentence; whether it yields a strong proper exponential copy is not checked here.

Research question 52.8 (Omega-field self-consistency [05]). Analyze proper solutions of $j = H_j$ and the role of choices at omega fixed points. Can one describe an invariant that decides when a partial monomial assignment extends to a total omega-compatible field embedding?

Recursion on normal-form exponents is not automatically well founded: an epsilon-type number can be its own leading exponent. This is a concrete obstruction to a superficial uniqueness proof.

[write] Compare the base question “ ω -map compatibility” and `opa:as:q:exp` of `docs/surreal/omnific-preserving-automorphisms`. Source 05 does not mention that `lce:prop:J` of `docs/foundations-and-computation/large-cardinal-embeddings-and-normal-forms` already supplies, conditionally on an elementary $j : V \rightarrow M$ with a critical point, proper (`lce:cs:prop:proper`) field embeddings $\mathcal{J}_j$ commuting with the omega-map ($\mathcal{J}_j(\omega^a) = \omega^{\mathcal{J}_j(a)}$); that report shows they are not strongly additive, so they do not solve $j = H_j$ in the strong monomial sector, which stays open.

Research question 52.9 (Ordinals and omnific integers [05]). Classify embeddings of the pair **(No, Oz)** under cofinality, and determine how pointwise fixation of all ordinals interacts with exponential, omega-map or differential compatibility. Which hypotheses force an abstract pair embedding to be strongly additive?

The proper ordinal-fixing plain-field copy is explicit (theorem 7.1); pointwise fixation of every omnific integer is rigid by $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$. Results about surjective pair automorphisms must not be transferred silently to proper embeddings.

[write] Compare the base question “Floor-preserving maps outside the monomial class”. Proper strong **Oz**-preserving embeddings with each combination of cofinality and coinitality of the value group are `opa:as:thm:four`, and parameter-fixed ones `opa:as:thm:parameters`; automatic strongness for the pair is the subject of `docs/surreal/omnific-preserving-automorphisms` Part III. The classification asked here is open.

Research question 52.10 (Fixed-field realization [05]). Which real closed subclasses arise as fixed fields of arbitrary surreal self-embeddings? Which arise in the strong monomial sector, the exponential sector, or the differential sector?

The strong weighted answer reduces to $\mathbb{R}((\omega^{\text{Fix } \tau \cap \ker \chi}))$ and the double lift gives $\mathbb{R}((\omega^{\mathbb{R}((\omega^{\text{Fix } f}))}))$; determine their limitations and the closure constraints imposed by named functions.

[write] Compare the base question “Prescribe the fixed field”. The common fixed field of all automorphisms is $\mathbb{Q}^{\text{fc}}$ (saut:prop:fixedfields of docs/surcomplex/surcomplex-field-aut omorphisms); open otherwise.

Research question 52.11 (Beyond monomial-to-term maps [05]). Classify strong embeddings whose monomial images have nontrivial unit factors rather than a single term. Which summability and separation conditions on the unit factors suffice for multiplicativity and injectivity?

The character $\chi(a)$ is real-valued; replacing it by a general positive unit introduces support unions and possible infinite coefficient interactions, and the finite convolution proof is no longer enough.

[write] Compare the base question “Beyond monomial images”. The unit twists U_β (proposition 10.2) and the Taylor shift (10.3) are examples with controlled unit factors; the coefficient flow saut:thm:coeffflow and OPA’s Taylor maps $\mathcal{T}_{d,\delta}$ are further ones. The classification is open.

Research question 52.12 (Ordinary derivatives beyond the strong sector [05]). For embeddings not known to preserve Hahn sums, what structural conditions force derivative zero, a finite derivative, or divergence? Can a non-strong field automorphism have ordinary derivative zero everywhere?

Theorem 47.3 settles the strong monomial-to-term sector only. A finite nonzero derivative is already rigid; the variable cases are zero derivative and failure of a finite limit.

[write] Compare the base question “A scale-sensitive calculus”. Open.

Research question 52.13 (Distinguished differential compatibility [05]). Find usable monomial assignments satisfying criterion 50.1 for the Berarducci–Mantova derivation. Study the common commutant of that derivation, the exponential and specified hyperserial operations among proper self-embeddings.

The diagonal derivations ∂_λ are a controlled test case but must not be confused with the distinguished derivation; constant fields and the domain of any composition must be part of every statement.

[write] Compare the base question “Differential centralizers”. A necessary condition is proposition 13.1, sharpened by the translation parameter $j(\omega) = \omega + c_j$ of proposition 59.5 (home 06). Open.

52.3 Foundational and computational projects

Research question 52.14 (Minimal class-theoretic assumptions [05]). Determine the weakest class theory supporting the uniform bounded-copy and omission constructions for the surreal exponential field. Can a fixed syntactic presentation remove the need for the full satisfaction apparatus used here?

GBC plus elementary transfinite recursion is a sufficient setting, not a lower bound; the generic algebraic and sign-sequence arguments use much less.

[write] Compare the base question “Minimal foundational strength”. Source 06’s epsilon-stage interpretation (remark 59.1) is a candidate answer to the second sentence; unchecked (see the note at the start of section 49).

Research question 52.15 (Coherent universe-relative maps [05]). Given embeddings on a cofinal system of birthday cutoffs, when can they be chosen compatibly and assembled into a full class embedding? Conversely, which full class embeddings restrict to self-maps of a prescribed unbounded family of cutoffs?

A separately existing embedding at each cutoff is not a coherent family; the target birthday bound may depend on both cutoff and map, and neither global choice nor a cardinality count supplies compatibility.

[write] Compare the base question “Universe and cutoff absoluteness”. For the converse, every class map restricts to self-maps of the stages $\mathbf{No}_{<\lambda}$ for a closed unbounded class of λ (“Coherent restrictions” in section 14; proposition 60.2, home 06); a *prescribed* family is not covered. Open.

Research question 52.16 (Effective descriptions and minimal rigid languages [05]). For computable notations for restricted surreal expressions, which of the explicit lifts admit effective evaluation and equality testing? Separately, what small expansions of the field language force every internally definable or every structure-preserving embedding to be the identity?

These are different questions: a finite expression procedure does not compute arbitrary ordinal-length signs, and set-theoretic definability is not definability inside an o-minimal field.

Suggested sequence of work [05]. Begin with the generic cofinal two-map lemma, small-cut birthday rigidity and the topology dichotomy; next implement the two Hahn lifts and their support and fixed-field statements; only then combine the normal-form, valuation and exponential interfaces. The last stage should address either a strong exponential construction or rigidity under simultaneous field and simplicity preservation, with a separate class-theoretic dependency audit.

53 Status and non-claims of source 05

- *Status.* Source 05 is an unrefereed research article with written proofs and explicitly scoped background results. It is not independently refereed, includes no new Lean certificate, and no repository-wide Lean build was run. No claim of historical priority is made; “developed here” describes its contents (remark 43.1).
- *Credited repository results.* Exponential valuation rigidity and the omnific fraction-field identity are credited to the repository and proved again; the generic Lean source was inspected separately from the pending concrete surreal exponential/valuation instantiations. The repository answer to [5, Question 5.4] is supported mathematically by the proof, but source 05 claims no independent publication, priority or community verification of it.
- *Scope of classifications.* No complete classification of arbitrary exponential, omega-field or differential self-embeddings is claimed. The weighted theorem classifies a specified sector only; the two-level lift is a representation, not a classification of all field embeddings. [5, Question 5.6] is not solved.
- *Conditional results.* The elementary-copy constructions (section 49) rest on uniform satisfaction, definable Skolem operations, o-minimality and global choice; GBC+ETR is a sufficient framework, not a minimal one. These copies carry no strong-sum, monomial, simplicity, omega-map, omnific or derivation compatibility.
- *Lean.* The cofinal strengthening of the two-profile comparison has no new Lean proof; the interface list of section 51 is design, not code.
- *Finite checks.* They test implementations and algebraic identities, not transfinite recursion, arbitrary Hahn summability, full-class ε – δ limits, proper-class existence, class truth, or the elementary-copy constructions.
- *Sources and audit.* Repository reads were targeted, through the GitHub connector at a pinned commit; no clone, rebuild or full proof build was completed, which limits any statement about

the repository’s overall coverage. The arXiv metadata of [5] was checked on 3 October 2026 and pages 4 and 10 of its v3 PDF were inspected for the valuation and leading-term conventions and Questions 5.4 and 5.6; the experimental HTML, with a different internal date, was not used as a separate version. The primary PDF of [3] was consulted, including Proposition 4.7 on printed page 183. Conway, Gonshor and Coste were not audited cover to cover; Coste’s archive access returned a bot challenge, so its o-minimal calculus inputs are stated rather than freshly audited. Exact-phrase literature searches are not evidence that a result has no antecedent.

- *Earlier drafts.* Earlier project drafts were read and used as working material, not as peer-reviewed authority.
- *Build record.* The delivered PDF had 39 pages, 16 research questions, no LaTeX errors, no undefined references, no overfull boxes and no replacement glyphs (data/05-comprehensive-BUILD_AUDIT.json); the repository was not modified.

[write] Source 05 is the expanded edition of an earlier manuscript of the same batch, which is not a separate source of this report (see the provenance note of the report). Its repository claims were checked at placement and none is false; its omission of `surcomplex-field-automorphisms`, `independent-surreal-copies` and `discrete-initial-subgroups-and-omnific-normalization` is corrected label by label in this Part.

Part V

Source 06: Structure, Rigidity, and Foundations

54 Source 06 at a glance

Source 06 is *Self-Embeddings of the Surreal Numbers: Structure, Rigidity, and Foundations. A systematic study with explicit constructions, proofs, and a research agenda*, subtitled “A study of the ProveIt development” and dated 3 October 2026 (47 pages; 10 sections, 22 bibliography entries, 11 numbered research questions plus unnumbered target lists; no code). Its repository reading is pinned to ProveIt commit 79049d58cc66341825ac5b1009ad7fd3b7bebf31. Its foundational setting is Gödel–Bernays class theory with Global Choice (GBC), with the Replacement and comprehension instances for the class maps involved stated explicitly; surreals are set-length sign sequences and Conway normal forms have set supports. Uniform truth predicates and class-valued recursions (ETR, Kelley–Morse) are named whenever used, and large-cardinal constructions are kept conditional.

About half of source 06 is a *credited synthesis* of collection material: Part `opa:part:as of docs/surreal/omnific-preserving-automorphisms`, `docs/surreal/exponential-automorphism-rigidity` (EAR), `docs/foundations-and-computation/large-cardinal-embeddings-and-normal-forms` (LCE) and `docs/surreal/omnific-diophantine-geometry` (ODG). Those statements are reprinted once below under their collection labels, as reprints and never as new results. Its genuinely new material is the exponential defect cocycle (proposition 59.6), the translation parameter (proposition 59.5), the club localization (proposition 60.2), the transcendence corollary corollary 57.2, the exponential automorphisms moving ω (proposition 59.2), the omega-series restriction (proposition 59.10) and a set of literature consequences in section 59; the Bagayoko–van der Hoeven fixed-point theorem is printed here as the home of that overlap (theorem 55.8).

Renames. Source 06’s symbols are renamed to the report’s canonical notation as follows.

Here	Source 06	Reason
$\mathbf{b}(x)$	$\ell(x)$	canonical birthday
$\preccurlyeq_s, \prec_s$	$\sqsubseteq, \sqsubset$	canonical simplicity
$P_p[\mathbf{No}]$	C_p	image of a prefix map; C is reserved
$D_{a-, a+}, D_\kappa$	D, D_κ	block maps; ordinal blocks are home 02
$c(x) = x/(1 + x)$	$H(x)$	compression; H is the field lift
Inc	I	Bagayoko–van der Hoeven’s map Inc ; I clashes
Σ_ε	T	tree automorphism; T is the Taylor map
dbl_m	A	doubling on infinitesimals; A is reserved
$m, O, \mathbb{J}$	I or $I; \backslash \text{Fin}; \backslash \text{Pinf}$ or J	infinitesimals, finite elements, purely infinite elements
A_h	$T_h, G, G_\pm$	additive (first) lift; T is reserved
H_τ	H_g	field lift of an additive embedding
$J_h, J_{P_\pm}$	$F, F_\pm$	double lift $J_h = H_{A_h}$
$H_{\tau, \chi}$	$J_{g, \chi}$	weighted monomial embedding
τ_f	g (in Prop. 3.2)	value action
$\mathcal{T}_{d,1}$	V_d	Taylor automorphism; base T is the case $d(r) = 1$
$\partial_{\text{ct}}, \mathcal{U}_s$	D, U_s	contracting derivation and its flow; D, U are reserved
H_τ^ρ	$J_{\tau, \rho}$	block map with coefficient section ρ ; J is the double lift
Θ_d	J (Section 5.4 of 06)	the explicit proper map moving a real; opa’s $J_{\mu, d}$
ϕ	J (Theorem 5.5 of 06)	a generic strong exact-monomial embedding
gap_τ	C_τ	bottom gap (as in opa); C is reserved
$\mathcal{A}$	A	multiplier ring of Π
$\mathcal{U}_1, \dots, \mathcal{U}_m$	$U_1, \dots, U_m$	ultrafilters
Ψ	T	countably strong map in theorem 60.8
Ω	$\backslash \text{Om}$	omega-map
$\mathbf{No}_{<\lambda}$	N_λ	birthday stages
j, C_F	J, C, C_F	a class map and its club, Section 7 of 06
$\mathcal{J}_j$	J	map induced by $j : V \rightarrow M$
∂_{BM}	∂_{BM}	unchanged
S_ω	S_ω	omega-series field (unchanged)

Two letters keep their source meaning and are disclosed: $c_j \in \mathbb{R}$ is the *translation parameter* of a differential embedding (not the compression c and not a real character), and α_j is the *exponential defect*. $K_\Gamma = \mathbb{R}((t^\Gamma))$ is a Hahn field with subscript, not the iterated-image core K_∞ . Section 5 of source 06 uses the small-monomial convention $t^\gamma = \omega^{-\gamma}$; it is announced where used.

Correspondence table. Every numbered item of source 06 (its counter is per section and shared by all theorem-like environments; its Section 1 and Section 8 have none), and every unnumbered result, example or construction with mathematical content, in source order.

06 item	Content	Where printed
§1	taxonomy of embedding notions; repository context	= base table in section 2; status in section 63
Prop. 2.1	prefix embeddings	= proposition 4.2 (same proof); extra examples in remark 55.1
§2.2 text	block maps $D_{a-, a+}, D_\kappa$	remark 55.1; D_m = proposition 4.3; ordinal blocks proposition 18.4 (home 02)
Prop. 2.2	local data (root word, successor words, limit tails)	theorem 44.1 (home 05; same statement and proof); chain-union clause in remark 55.2
eq. (2)	$\text{Inc}, \mathbf{b}(\text{Inc } x) = 2 \mathbf{b}(x) + 1$	example 55.3

06 item	Content	Where printed
Lem. 2.3	$b(x) \leq b(fx)$	= proposition 4.6, first clause (same proof)
Thm. 2.4	order + birthday nonincrease $\Rightarrow$ id	= theorem 4.5 (same proof)
Cor. 2.5	$\text{Aut}(\mathbf{No}, <, \preceq_s) = 1$; unique parametrization	= proposition 4.6 and the uniqueness paragraph before theorem 4.9 (same proof)
Thm. 2.6	initial image $\Rightarrow$ onto; with simplicity $\Rightarrow$ id	= theorem 5.2 with proposition 4.6 (same proof); ambient brackets = proposition 4.7; relative bracket remark 55.4
§2.5 text	all prefix-tree automorphisms; orientation	proposition 55.5
§2.6 text	order automorphism fixing $\mathbb{R} \cup \mathbf{On}$; compression	example 55.6; compression = proposition 4.1
eq. (4)	$\text{Fix}(P_P) = P_{P \sim \omega}[\mathbf{No}] = \bigcap_n P_P^n[\mathbf{No}]$	example 55.7
Thm. 2.7	$\text{Fix}(f) = \bigcap_n f^n[\mathbf{No}]$ (BvH)	theorem 55.8 (printed here; home of the overlap)
§2.7 text	simplest fixed point under chain continuity	= theorem 4.9 (base proves more); remark 55.9
§2.8 text	D_2 is not order continuous	= proposition 4.3
Prop. 2.8	sets closed discrete, set nets eventually constant	= proposition 9.1 (same proof); credit <code>saut:prop:discrete</code>
Prop. 3.1	$f(r) - r$ infinitesimal; $f(\mathbb{R}) \subseteq \mathbb{R} \Rightarrow f _{\mathbb{R}} = \text{id}$	= proposition 10.1 (residue clause) and the proof of proposition 13.1; credit <code>opa:as:lem:natural</code>
Prop. 3.2	induced value embedding	= proposition 10.1 (same proof); <code>opa:as:lem:natural</code> ; formula in remark 56.1
Def. 3.3	strong additivity	= base Definition (Strong summability), section 3
Thm. 3.4	order-to-additive lift	= theorem 6.1 (base stronger; same proof)
Thm. 3.5	additive-to-field lift	= theorem 6.2; ct clause in proposition 8.1; dilations = S_c of proposition 9.6
§3.4 text	$J_{P_{\pm}}, J_{P_{\pm}}(\omega) = \omega^{\omega}, J_{P_{-}}(\omega) = \omega^{1/\omega}$	proposition 56.2
Prop. 3.6	cofinal copy; bounded copy below every $\omega^{1/n}$	proposition 56.2 (printed: sharper bound than theorem 7.6, theorem 45.3)
eq. (12)	$J_{P_{+}}[\mathbf{No}] \cap (\omega, 2\omega) = \emptyset$	proposition 56.2
Thm. 3.7	weighted monomial classification	= theorem 6.4 (same proof); credit <code>saut:thm:monomial</code> ; example example 56.3; caution remark 56.4
§3.6 text	contracting flow $\mathcal{U}_s, \mathcal{U}_1(\omega) = \omega + 1$	example 56.5
Thm. 3.8	Taylor automorphism moving θ	= base T of (10.3) (same proof); credit <code>saut:thm:coeffflow</code> ; $\mathbf{Oz}$ failure in remark 56.6
§3.8 text	nonstrong automorphism fixing $\mathbb{R}, \omega$, all $\omega^{1/n}$	second route after proposition 49.8, at (56.4)
Thm. 3.9	continuity $\iff$ cofinality	= theorem 9.3 (same proof); closedness and table in remark 56.7; <code>opa:as:thm:topology</code> ; $F \rightarrow K$ form theorem 67.1 (home 07)
Prop. 4.1	strong automorphism fixing $\mathbb{R} \cup \mathbf{On}$, preserving $\mathbf{Oz}$, moving $\omega^{\omega^{1/2}}$	covered by theorem 7.4 (automorphism clause) with proposition 8.1; its witness is example 57.1
Cor. 4.2	$\omega^{\omega^{1/2}}$ transcendental over $\mathbb{Q}(\mathbb{R} \cup \mathbf{On})$	corollary 57.2 (printed, new)
Thm. 4.3	parameter-fixed discrete and continuous copies	theorem 57.3 (reprint of <code>opa:as:thm:parameters</code>); compare theorems 7.4 and 7.6 and corollary 9.5
Prop. 4.4	field embeddings are L_{or} -elementary	= proposition 12.1 (same proof); <code>opa:as:thm:elementary</code>
§4.3 text	back-and-forth for set parameter fields	theorem 70.2 (home 07); remark 60.6
Thm. 4.5	o-minimal definable endomorphism = id	= theorem 12.2 (same proof); remark 57.4
§4.5 text	computability and presentations	remark 57.5
Prop. 5.1	$\text{Frac } \mathbf{Oz} = \mathbf{No}$; $\text{Aut } \mathbf{Oz}$ extends	Frac part theorem 20.2 (home 02; <code>odg:thm:fractions</code>); extension and proof remark 58.1

06 item	Content	Where printed
eqs. (18)–(19)	quadratic test for Π ; multiplier ring	lemma 58.2 (odg: def: thm: ideal, odg: def: thm: multiplier)
Prop. 5.2	Oz -automorphisms fix $\mathbb{R}$ and ct	proposition 58.3 (reprint of opa: cor: allautos)
Lem. 5.3	scalar detection of summability	lemma 58.4 (reprint of opa: as: thm: detect)
Thm. 5.4	automatic strongness of Oz -automorphisms	theorem 58.5 (reprint of opa: as: thm: main, via opa: as: thm: ctiso)
§5.3 text	bottom gap	lemma 58.6 (reprint of opa: as: lem: gap)
Thm. 5.5	bottom-gap classification	theorem 58.7 (reprint of opa: as: thm: classification)
Cor. 5.6	surjectivity and coefficient rigidity	corollary 58.8 (reprint of opa: as: cor: onto, opa: as: cor: rigidity)
§5.4 text	Taylor section; explicit proper map moving a real; overlap ker d ; omega-map boundary; missing detector	proposition 58.9 (reprint of opa: as: lem: taylorsection, opa: as: thm: taylorcriterion, opa: as: thm: explicit, opa: as: prop: overlap, opa: as: cor: homogeneity, opa: as: cor: image)
§5.5 text	nonelementarity in (No , Oz) and of the ring restriction	theorem 58.10 (reprint of opa: as: thm: elementary)
§6.1 text	ε -stage interpretation of exponential homogeneity	remark 59.1
Prop. 6.1	exponential automorphism with $j(\omega) = a$, any $a > \mathbb{R}$	proposition 59.2 (printed; consequence of [5, Rem. 2.10])
Lem. 6.2	displacement amplification	= lemma 11.1 (same proof); EAR lem: amplify
Lem. 6.3	$v(\exp x) = 0 \iff x$ finite	= (11.3); converse via log, remark 59.3
Thm. 6.4	exact-value exponential rigidity for embeddings	= theorem 11.4 (same proof); EAR cor: embedding; KKS Q. 5.4 = EAR cor: question54
Cor. 6.5	faithfulness for automorphisms	weaker than theorem 11.7; = saut: thm: expfaithful, EAR cor: faithful
Prop. 6.6	omega-map + exact values $\Rightarrow$ id	= theorem 10.3 (same proof); remark 59.4
Prop. 6.7	translation parameter c_j , additive	proposition 59.5 (printed, home; strengthens proposition 13.1)
Prop. 6.8	exponential defect α_j	proposition 59.6 (printed, new)
§6.3 text	every $c \in \mathbb{R}$ is a translation parameter	proposition 59.7 (imported literature)
§6.3 text	$c_j = 0$ does not force rigidity	proposition 59.8 (imported literature)
§6.3 text	ω^2 example; transported derivations	remark 59.9
eq. (29)	strong exponential maps restrict to substitution on $\mathcal{S}_\omega$	proposition 59.10 (printed; compare theorem 68.5)
§6.4 text	chain rule; BM obstruction to global composition	remark 59.11
§6.5 text	hyperserial fields; five targets; model-theoretic boundary	remark 59.12
§7.1–7.2	frameworks; available recursions	remark 60.1; base section 3
Prop. 7.1	club of stages closed under a class map	proposition 60.2 (printed, home; base prose in section 14)
Cor. 7.2	bijective and exact-range stages	corollary 60.3
§7.3 text	cardinal stages of prescribed cofinality; stage theorem	remark 60.4, remark 60.5
§7.4 text	ordered-field back-and-forth	theorem 70.2 (home 07; saut: thm: orderedhomogeneity); remark 60.6
§7.5 text	elementarity is not $\in$ -elementarity; Kunen	= base section 3; remark 60.7
Prop. 7.3	least failure of strongness is measurable	theorem 60.8 (reprint of lce: mf: thm: spectrum)
§7.6 text	measurable construction $\mathcal{J}_j$	theorem 24.2 (home 02) and proposition 60.9 (lce: prop: J, lce: lem: exptransfer, lce: thm: Oz, lce: mf: thm: equivalence)

06 item	Content	Where printed
§7.7 text	formalization boundary	remark 60.10
§8	Lean status table; roadmap	section 61
Q. 9.1	elementarity of pair embeddings	research question 62.1 (= opa:as:q:elementary)
Q. 9.2	countable strongness, proper pair maps	research question 62.2
Q. 9.3	beyond exact monomials	research question 62.3
Q. 9.4	omega-field automorphisms	research question 62.4
Q. 9.5	existence of exponential lifts	research question 62.5
Q. 9.6	realization of defects	research question 62.6
Q. 9.7	joint exponential and differential maps	research question 62.7
Q. 9.8	simplicity of field lifts	research question 62.8
Q. 9.9	stabilizer of $\mathbb{R} \cup \mathbf{On}$	research question 62.9
Q. 9.10	closed and dense images	research question 62.10
Q. 9.11	compatible local data	research question 62.11
§9 text	lce:cs:q:derivations; formalization projects	section 62
§10	source audit, conventions, errata, limits	section 63

55 Order, simplicity, and birthdays (source 06, Section 2)

Source 06 calls an *ordered-simplicity embedding* a class map preserving and reflecting both $<$ and $\preccurlyeq_s$; its image is a surreal substructure in the sense of [4]. Injectivity together with preservation of the two relations already gives reflection (lemma 4.4). Its prefix maps, birthday lemma, birthday rigidity, automorphism rigidity and initial-image theorem are the base's results, with the same proofs (see the table). Source 06 adds two comments: for numerical order alone the inequality $b(x) \leq b(fx)$ fails (translation by 1 sends -1 to 0); and, by the proof of theorem 4.5, at the first birthday where an arbitrary order embedding moves a number it moves it to a strictly later birthday (later birthdays may still decrease; compare source 05's first-move corollary). The following material is additional.

Remark 55.1 (Prefix and block examples [06]). The image $P_p[\mathbf{No}] = \{y : p \preccurlyeq_s y\}$ is convex and has no endpoints, so P_p is an order homeomorphism onto an open convex class. Concatenation is not field translation: for the one-sign word $+$, $P_+(0) = 1$ and $P_+(-1) = 1/2$, whereas $-1 + 1 = 0$.

More generally, choose nonempty words a_-, a_+ whose first signs are $-$ and $+$ respectively, and let $D_{a_-, a_+}(x)$ replace every sign σ of x by a_σ , concatenated over the ordinal length of x . This is an ordered-simplicity embedding (prefixes concatenate compatibly, and the first new block begins with the sign deciding each comparison); it fixes 0 and preserves unions of nonempty set-sized prefix chains. These are established examples of surreal substructures [4, Example 4.2]. Repeating each sign a nonzero ordinal κ times gives D_κ with $b(D_\kappa x) = \kappa \cdot b(x)$ (ordinal product); for integers this is proposition 4.3, and the ordinal sign blocks are printed in proposition 18.4 (home 02). Even D_2 is not additive: it sends 1 to 2 and $1/2$ (signs $+-$) to $++-- = 5/4$.

Remark 55.2 (Chain continuity in the local description [06]). Source 06's Proposition 2.2 is the local padding description theorem 44.1 (home 05): an ordered-simplicity embedding F is determined by a root word $r = F(\emptyset)$, nonempty successor words $w_{s,\sigma}$ beginning with σ (so $F(s \smallfrown \sigma) = F(s) \smallfrown w_{s,\sigma}$), and possibly empty limit tails v_s (so $F(s) = (\bigcup_{\alpha < \lambda} F(s \upharpoonright \alpha)) \smallfrown v_s$ when $b(s) = \lambda$ is a nonzero limit); conversely any such data define one. Its proof is the same. Source 06 adds a proof of the clause that F preserves unions of nonempty set-sized prefix chains *exactly when every limit tail v_s is empty*: the clause holds for the chain of all proper prefixes of a word of limit length by definition of v_s , and every nonempty prefix chain without a maximum is cofinal among the proper prefixes of its union, so this

special case gives the general one.

Example 55.3 (A fixed-point-free substructure [06]; [4, Example 4.14]). Let Inc have root $+$, successor words $\sigma \frown +$ and limit tail $+$ at every nonzero limit. Ordinal induction gives

$$b(\text{Inc}(x)) = 2b(x) + 1 > b(x), \quad (55.1)$$

so Inc is an ordered-simplicity embedding with no fixed point. The nonempty limit tails are exactly why the iteration-and-union construction of a fixed point (theorem 4.9) fails for it.

Remark 55.4 (Relative brackets [06]). Base proposition 4.7 shows that preserving *ambient* canonical brackets forces the identity. A proper surreal substructure $S = f[\mathbf{No}]$ instead satisfies the *image-relative* bracket identity

$$f(\{L \mid R\}_{\mathbf{No}}) = \{f[L] \mid f[R]\}_S. \quad (55.2)$$

Indeed, if $x = \{L \mid R\}$ and $f(y)$ lies between $f[L]$ and $f[R]$, then $L < y < R$, so $x \preccurlyeq_s y$ and $f(x) \preccurlyeq_s f(y)$; hence $f(x)$ is the simplest point of S in the image cut. This proves the remark following proposition 4.7.

Proposition 55.5 (All prefix-tree automorphisms [06]). *For a class choice of signs $\varepsilon_s \in \{-1, 1\}$, one for every sign sequence s , the map*

$$\Sigma_\varepsilon(x)(\alpha) = \varepsilon_{x \upharpoonright \alpha} x(\alpha) \quad (\alpha < b(x))$$

is an automorphism of the prefix tree $(\mathbf{No}, \preccurlyeq_s)$, and every automorphism of $(\mathbf{No}, \preccurlyeq_s)$ has this form. Σ_ε preserves numerical order exactly when all $\varepsilon_s = 1$ (then it is the identity), and reverses it exactly when all $\varepsilon_s = -1$ (then it is $x \mapsto -x$).

Proof. The map independently keeps or interchanges the two children at each node; prefixes transform compatibly, and the inverse is recovered by recursion along the length of a target sequence. Conversely, a prefix-tree automorphism preserves ordinal height and permutes the two immediate children of each node, which defines ε . If it preserves order, then $s \frown - < s \frown +$ forces $\varepsilon_s = 1$ at every node; if it reverses order, $\varepsilon_s = -1$ at every node. Class-indexed choices are class parameters. $\square$

Thus “preserves the surreal tree” needs a statement about branch orientation; the base’s paragraph “Simplicity without order” gives the existence half.

Example 55.6 (Order automorphisms fixing every real and every ordinal [06]). Let $\mathfrak{m}$ be the convex class of infinitesimals and put $\text{dbl}_{\mathfrak{m}}(x) = 2x$ for $x \in \mathfrak{m}$, $\text{dbl}_{\mathfrak{m}}(x) = x$ otherwise. Multiplication by 2 is an increasing bijection of $\mathfrak{m}$, and convexity makes $\text{dbl}_{\mathfrak{m}}$ a global order automorphism. It fixes every real and every ordinal (0 is the only real infinitesimal and nonzero ordinals are not infinitesimal) but moves every nonzero infinitesimal; it is not additive, since $\text{dbl}_{\mathfrak{m}}(1 + \epsilon) = 1 + \epsilon \neq 1 + 2\epsilon$. So the order reduct is far from rigid even over $\mathbb{R} \cup \mathbf{On}$; this does not conflict with the algebraic rigidity theorems. Likewise the compression c of proposition 4.1 is a homeomorphism onto the convex class $(-1, 1)$ fixing 0 and sending 1 to $1/2$: properness, bounded image, continuity and fixed familiar points are separate restrictions.

Example 55.7 (Fixed points of a prefix map [06]). For nonempty p , let $p^{\frown \omega}$ be countably many concatenated copies of p . Then

$$\text{Fix}(P_p) = P_{p^{\frown \omega}}[\mathbf{No}] = \bigcap_{n < \omega} P_p^n[\mathbf{No}]. \quad (55.3)$$

A fixed point begins with every finite repetition of p , hence with their union; conversely $p \frown p^{\frown \omega} = p^{\frown \omega}$ (ordinal absorption $b(p) + b(p) \cdot \omega = b(p) \cdot \omega$), which fixes every extension of it. For instance P_+ fixes exactly the surreals whose signs begin with ω consecutive pluses. This is ordinal absorption in the sign string, not cancellation in the field.

The intersection formula (55.3) is a case of the following established theorem. This report prints it here as the home of the overlap with source 07 (section 65.4 points here).

Theorem 55.8 (Fixed points are the iterated-image core [06]; [4, Proposition 5.2]). *For every ordered-simplicity embedding $f : \mathbf{No} \rightarrow \mathbf{No}$,*

$$\text{Fix}(f) = \bigcap_{n < \omega} f^n[\mathbf{No}].$$

Proof [06], using ambient common prefixes. The inclusion $\subseteq$ is immediate. Write A for the intersection and suppose $A \setminus \text{Fix}(f) \neq \emptyset$. Choose x_0 in it of least birthday and let x_n be its unique n th inverse image. By injectivity every x_n lies in A , and none is fixed (else $x_0 = f^n(x_n) = x_n$ would be). By proposition 4.6, $b(x_n) \leq b(fx_n) = b(x_{n-1})$, and minimality gives equality, so all x_n have the same birthday. They are distinct and strictly monotone numerically (order reflection propagates $x_0 < x_1$ to $x_n < x_{n+1}$); assume $x_0 < x_1 < \dots$, the other case being symmetric.

Let u_n be the longest common prefix of x_n, x_{n+1} . Equal birthdays and distinctness give $x_n < u_n < x_{n+1}$ and $b(u_n) < b(x_0)$. Moreover $f(u_{n+1}) \preceq_s u_n$, being a common prefix of $f(x_{n+1}) = x_n$ and $f(x_{n+2}) = x_{n+1}$. Hence

$$b(u_{n+1}) \leq b(f(u_{n+1})) \leq b(u_n).$$

This nonincreasing sequence of ordinals stabilizes, after which $f(u_{n+1}) = u_n$. Each sufficiently late u_n therefore lies in every $f^k[\mathbf{No}]$, hence in A . But $u_n < u_{n+1}$ and $f(u_{n+1}) = u_n$ prevent u_n from being fixed, contradicting the minimal birthday of x_0 . $\square$

The intersection may be empty, as for Inc in (55.1).

[write] Compare the field category: for a field self-embedding the iterated-image core $K_\infty(j)$ can strictly contain $\text{Fix}(j)$ (proposition 8.4 and (8.6)). Simplicity preservation is what collapses the core to the fixed class here.

Remark 55.9 (Nonempty fixed classes [06]). If f also preserves unions of nonempty set-sized prefix chains, the iterates of 0 form a prefix chain and $x_\omega = \bigcup_n f^n(0)$ is the simplest fixed point; this is the case $u = 0$ of $c_j(u)$ in theorem 4.9, and proposition 44.3. The whole fixed class is then a surreal substructure closed under those unions [4, Corollary 5.14]; this is theorem 4.9.

56 Field embeddings and normal forms (source 06, Section 3)

Source 06 notes that a field self-embedding of $\mathbf{No}$ fixes $\mathbb{Q}$ and every real algebraic number (an algebraic number is pinned down by its polynomial and an isolating rational interval), and preserves order. Its lifting theorems are the base's theorems 6.1, 6.2 and 6.4; source 06 states that they organize standard Hahn-field arguments with the exponent and character constructions of [31, 5], “not a claim that the lifting method is new”.

Remark 56.1 (A formula for the value action [06]). The value action of proposition 10.1 is computed by $\tau_f(a) = v(f(\omega^{-a}))$, and for an automorphism τ_f is an automorphism. Preserving comparisons of values, or (10.1), is automatic, while $v(fx) = v(x)$ is a substantial further requirement [13, 5]. This is `opa:as:lem:natural`.

Two explicit proper copies

Let $P_\pm$ be the one-sign prefix maps, order isomorphisms onto $\mathbf{No}_{>0}$ and $\mathbf{No}_{<0}$ (proposition 4.2). Their additive lifts have images

$$A_{P_+}[\mathbf{No}] = \mathbb{J}, \quad A_{P_-}[\mathbf{No}] = \mathfrak{m},$$

the additive groups of purely infinite surreals (with 0) and of infinitesimals, by (6.2). Hence the double lifts $J_{P_{\pm}} = H_{A_{P_{\pm}}}$ are proper strongly $\mathbb{R}$ -linear field embeddings with images $\mathbb{R}((\omega^{\mathbb{J}}))$ and $\mathbb{R}((\omega^{\mathbb{m}}))$; both images are real closed, being isomorphic to **No**. Since $P_+(0) = 1$ and $P_-(0) = -1$,

$$J_{P_+}(\omega) = \omega^{\omega}, \quad J_{P_-}(\omega) = \omega^{1/\omega}. \quad (56.1)$$

Proposition 56.2 (Cofinal and bounded copies [06]). *The image of J_{P_+} is cofinal in **No** but not dense:*

$$J_{P_+}[\mathbf{No}] \cap (\omega, 2\omega) = \emptyset. \quad (56.2)$$

Every element of the image of J_{P_-} is Archimedean-smaller than $\omega^{1/n}$ for every positive integer n ; in particular its absolute value is below ω , and no positive image element is as small as ω^{-1} .

Proof. Above any exponent there is a larger purely infinite ordinal, and a monomial with that exponent dominates any prescribed normal form; this gives cofinality. An element of $(\omega, 2\omega)$ has leading exponent $1 \notin \mathbb{J}$, proving (56.2). For J_{P_-} the leading exponent a of a nonzero image element is infinitesimal, so $-1/n < a < 1/n$ and the leading monomial is Archimedean-smaller than $\omega^{1/n}$ and larger than $\omega^{-1/n}$; a nonzero real leading coefficient does not change these comparisons. $\square$

The bound sharpens, for this unparametrized copy, the bounds of [theorem 7.6](#) and [theorem 45.3](#). The bounded image is still a proper class, and boundedness is measured by the external element ω , which the image omits. Cofinality without density separates algebraic elementarity, cofinality, density and continuity (the image is closed and nowhere dense by [theorem 9.4](#)).

Example 56.3 (A character moving a leading coefficient [06]). With $\chi(a) = \exp_{\mathbb{R}}(\text{ct}(a))$, where ct is the normal-form constant term (additive), the weighted embedding $H_{\text{id}, \chi}$ of [theorem 6.4](#) fixes every real and every valuation value but sends ω to $e\omega$. (The base's example uses $\chi(a) = 2^{\lfloor a \rfloor - 1}$, which fixes ω .) Every $H_{\tau, \chi}$ preserves the constant coefficient and preserves and reflects **Oz** ([proposition 8.1](#)). Preserving and reflecting the canonical-monomial predicate means $\chi = 1$; it does not mean fixing monomials individually. Composition acts on characters through the exponent map, $(\tau_1, \chi_1)(\tau_2, \chi_2) = (\tau_1 \tau_2, \chi_2 \cdot (\chi_1 \circ \tau_2))$, so the two kinds of coordinates form a semidirect, not a direct, product.

Remark 56.4 (The omega-map equation is not a birthday induction [06]). $H_{\tau, \chi}(\omega^a) = \omega^{H_{\tau, \chi}(a)}$ for all a forces $\chi = 1$ and then $\tau = H_{\tau}$, which is (10.5). It is unsafe to conclude $\tau = \text{id}$ from this by induction on birthdays: a generalized epsilon number $a = \omega^a$ has a self-dependent normal form. The omega-structure questions do not reduce to the unconstrained lifting theorem.

Example 56.5 (A contracting flow moving ω to $\omega + 1$ [06]). Let $\partial_{\text{ct}}(\sum_a r_a \omega^a) = \sum_a r_a \text{ct}(a) \omega^{a-1}$. Additivity of ct gives the Leibniz rule on monomials, so ∂_{ct} is a strongly $\mathbb{R}$ -linear derivation which strictly lowers exponents. Its exponential is

$$\mathcal{U}_s\left(\sum_a r_a \omega^a\right) = \sum_{a, n \geq 0} r_a \binom{\text{ct}(a)}{n} s^n \omega^{a-n} \quad (s \in \mathbb{R}), \quad (56.3)$$

that is, $\mathcal{U}_s(\omega^a) = \omega^a (1 + s\omega^{-1})^{\text{ct}(a)}$. The support lies in $\text{supp}(x) - \mathbb{N}$, so the Hahn support lemma gives summability; Leibniz's formula gives multiplicativity, and the exponential identity, justified coefficientwise, gives $\mathcal{U}_s \mathcal{U}_{s'} = \mathcal{U}_{s+s'}$. So $\mathcal{U}_s$ is a strongly $\mathbb{R}$ -linear automorphism with inverse $\mathcal{U}_{-s}$, fixing every real and every leading term, yet

$$\mathcal{U}_1(\omega) = \omega + 1, \quad \mathcal{U}_1(\omega^{\omega}) = \omega^{\omega}.$$

It does not preserve the canonical-monomial predicate. This illustrates the contracting-derivation mechanism of [5]; in the convention $v(t^a) = a$ a contracting shift is $a \mapsto a + 1$.

[write] $\mathcal{U}_s$ differs from the base's unit twist U_β of (10.2): U_β uses the coefficient $[a]_\beta$ with $\beta < 0$ and fixes $t = \omega^{-1}$, whereas $\mathcal{U}_s$ uses $[a]_0$ and moves t (to $t(1 + st)^{-1}$); its inverse comes from the flow property, not from fixing the binomial units. Neither preserves $\mathbf{Oz}$ in general: $\mathcal{U}_1(\omega) = \omega + 1 \in \mathbf{Oz}$, but $\mathcal{U}_1(\omega^{1/2}) = \omega^{1/2} + \frac{1}{2}\omega^{-1/2} + \dots \notin \mathbf{Oz}$ although $\omega^{1/2} \in \mathbf{Oz}$.

Remark 56.6 (The Taylor automorphism [06]). Source 06's Theorem 3.8 is the base's Taylor automorphism T of (10.3), written $\mathcal{T}_{d,1}$ here: for a derivation $d : \mathbb{R} \rightarrow \mathbb{R}$ with $d(\theta) = 1$ at a transcendental θ , $\mathcal{T}_{d,1}(\sum r_a \omega^a) = \sum_{a,n} \frac{d^n(r_a)}{n!} \omega^{a-n}$ is a strongly additive field automorphism fixing every canonical monomial, every real algebraic number and every leading term, with $\mathcal{T}_{d,1}(\theta) = \theta + \omega^{-1}$; it is not $\mathbb{R}$ -linear. Its proof is the base's (the composite with $\mathcal{T}_{-d,1}$ has the coefficient $\sum_{n=0}^k (-1)^n / (n!(k-n)!) = [k=0]$ at total iterate k). It separates four requirements: fixing the residue field does not fix the real representatives; strong additivity does not imply strong $\mathbb{R}$ -linearity; fixing all monomials does not fix all normal forms; and preserving each leading term does not force triviality. It is not asserted to preserve $\mathbf{Oz}$: for an infinitesimal exponent $a > 0$, the omnific integer $\theta \omega^a$ acquires the term ω^{a-1} with negative exponent.

[write] This is `saut:thm:coeffflow` of `docs/surcomplex/surcomplex-field-automorphisms` (there Φ_d , over $\mathbb{C}$ and with a general shift δ) and the Taylor flow of Part `opa:part:as` of `docs/surreal/omnific-preserving-automorphisms`; source 06 does not cite the `surcomplex` report. The general maps $\mathcal{T}_{d,\delta}$ with a gap are in proposition 58.9.

Ordinary field automorphisms need not respect Hahn sums. Source 06 proves the following separation, which is proposition 49.8 (home 05: an automorphism fixing $\mathbb{R}$ and ω but not a specified Hahn sum) by a different witness, the pseudolimit argument behind [5, Example 4.4]. Its automorphism also fixes every $\omega^{1/n}$.

Second route to proposition 49.8 [06]. Consider the valid decreasing support

$$a = \sum_{n \geq 1} \omega^{1/n}, \quad \varepsilon = \omega^{-\omega}. \quad (56.4)$$

Let $K_{\mathbb{P}}$ be the real Puiseux-series field in ω^{-1} , a set-sized real closed subfield. Since the support of a has unbounded denominators, $a \notin K_{\mathbb{P}}$; for every $b \in K_{\mathbb{P}}$ the nonzero difference $a - b$ has a rational leading exponent, hence $|a - b| > \varepsilon$. So a and $a + \varepsilon$ have the same cut over $K_{\mathbb{P}}$, and real closedness of $K_{\mathbb{P}}$ gives an ordered $K_{\mathbb{P}}$ -isomorphism between the real closures of $K_{\mathbb{P}}(a)$ and $K_{\mathbb{P}}(a + \varepsilon)$ taking a to $a + \varepsilon$. Set-sized homogeneity of the ordered field $\mathbf{No}$ ([5, Remark 2.10]; theorem 70.2, `saut:thm:orderedhomogeneity`) extends it to a field automorphism σ ; only a set-sized partial map is extended. Then σ fixes $\mathbb{R}$, ω and every $\omega^{1/n}$, but $\sigma(\sum_n \omega^{1/n}) = a + \varepsilon \neq a = \sum_n \sigma(\omega^{1/n})$. $\square$

Specifying the images of the monomials in one expansion does not determine the image of its infinite sum under an arbitrary field automorphism. The same example separates strong automorphisms from all automorphisms in the finite pointwise-convergence topology *on maps*: the neighbourhood $\{\tau : \tau(\omega) = \omega, \tau(a) = a + \varepsilon\}$ contains no strongly additive automorphism. (That topology on maps is not the order topology on values.) Replacing $1/n$ by $-1/n$ in (56.4) would give an increasing support with no greatest element, for which the Hahn sum is undefined.

Remark 56.7 (Closed discrete images, and four test maps [06]). Source 06's Theorem 3.9 is theorem 9.3 with the extra equivalent condition that $f[\mathbf{No}_{>0}]$ is coinital in $\mathbf{No}_{>0}$, and the extra conclusion that a noncofinal image is *closed* and uniformly discrete (as in `opa:as:thm:topology`, which source 06 credits). Proof of closedness: if $\varepsilon > 0$ lies below every positive image element, distinct image points are at distance $\geq \varepsilon$; an interval of radius $\varepsilon/3$ about a point outside the image contains at most one image point, and shrinking the radius avoids it. The test maps show why the field hypothesis matters:

Map	Structure preserved	Cofinal image	Continuous
A_{P_+}	ordered additive group	yes	no
A_{P_-}	ordered additive group	no	yes
J_{P_+}	ordered field	yes	yes
J_{P_-}	ordered field	no	no

For additive maps the relevant condition is positive coinitality and inversion is unavailable: nonzero purely infinite elements cannot approach 0, while the infinitesimals are coinital among positive surreals. Passing to the field lifts exchanges which feature of the additive image controls small positive values. By (56.2) a continuous cofinal field self-embedding need not have a dense image. The dichotomy for embeddings between arbitrary ordered fields is theorem 67.1 (home 07).

57 Fixed parameters, ordinals, and logic (source 06, Section 4)

Fixing $\mathbb{R}$, fixing ω , fixing every ordinal, and being definable in a specified language are different restrictions; source 06 separates them by explicit tests.

Example 57.1 (An ordinal-fixing automorphism moving $\omega^{\omega^{1/2}}$ [06]). Let $h(x) = 2x/(1+x)$ for $0 < x < 1$ and $h(x) = x$ otherwise. It is an increasing bijection of $\mathbf{No}$ (its inverse on $(0, 1)$ is $y \mapsto y/(2-y)$) fixing every ordinal, and it is not additive. Its double lift $J_h = H_{A_h}$ is a strongly $\mathbb{R}$ -linear field automorphism (theorem 6.3). It fixes $\mathbb{R}$ and every ordinal pointwise (every ordinal has a Cantor normal form with ordinal exponents and positive integer coefficients, and h fixes those exponents at both levels), and it preserves and reflects **Oz** (proposition 8.1). It is not the identity:

$$J_h(\omega^{\omega^{1/2}}) = \omega^{\omega^{2/3}}, \quad J_h(\omega^{1/2}) = \omega^{1/2},$$

since $h(1/2) = 2/3$ gives $A_h(\omega^{1/2}) = \omega^{2/3}$, while $A_h(1/2) = 1/2$. The two equalities are incompatible with $J_h(\omega^x) = \omega^{J_h(x)}$ at $x = \omega^{1/2}$, so J_h does not commute with the omega-map; and it is not simplicity-preserving, since an order-and-simplicity automorphism is the identity (proposition 4.6).

[write] The qualitative content of source 06's Proposition 4.1 (a nonidentity strong automorphism fixing $\mathbb{R} \cup \mathbf{On}$ and preserving **Oz**) is the automorphism clause of theorem 7.4 with $A = \emptyset$, together with proposition 8.1, since $J_{g_\beta} = H_{A_{g_\beta,1}}$. What the base does not give is this witness: J_{g_β} only moves exponents below $\beta < 0$ and fixes every x with $E(\{x\}) \subseteq [\beta, \infty)$, in particular $\omega^{\omega^{1/2}}$ (whose footprint is $\{1/2\}$). The witness is what corollary 57.2 needs. Source 06's comparison sentence (its Proposition 4.1 "handles one particular proper class of parameters by a separate explicit construction") should be read beside theorem 7.4, which fixes $A \cup \mathbb{R} \cup \mathbf{On}$ for every set A .

Corollary 57.2 (Ordinals and reals do not generate [06]). *Let K be the subfield of $\mathbf{No}$ generated by $\mathbb{R} \cup \mathbf{On}$. Then $\omega^{\omega^{1/2}}$ is transcendental over K . In particular, adjoining all ordinals to the reals and taking relative real closure does not produce $\mathbf{No}$.*

Proof. J_h fixes K . An increasing field automorphism fixing K fixes every element algebraic over K : it permutes the finite ordered list of roots in $\mathbf{No}$ of a polynomial over K , and an increasing permutation of a finite linear order is the identity (as in proposition 8.2). Since J_h moves $\omega^{\omega^{1/2}}$, that element is not algebraic over K . $\square$

The field generated by a class is defined by finite expressions, so no set of all generators is postulated. Source 06 adds that this is a direct application of the lifting method, "not a claim that the method itself is new".

Theorem 57.3 (Parameter-fixed copies [06]; reprint of `opa:as:thm:parameters`). *For every set $A \subseteq \mathbf{No}$ there are proper strongly $\mathbb{R}$ -linear exact-monomial field self-embeddings fixing $A \cup \mathbb{R}$ pointwise and preserving and reflecting $\mathbf{Oz}$. They can be chosen with a discrete bounded image, or with a continuous nondiscrete image, in the full ambient order topology.*

Mechanism [06]. Collect the normal-form exponents of all members of A , and then the exponents of those exponents; this two-level footprint is a set $((7.2))$. Choose an order embedding of the exponent order fixing that set and compressing, or inserting a gap, beyond it. The first lift fixes all exponents occurring in A and the second lift fixes A . A bounded compression gives the discrete example, a cofinal gap insertion the continuous one; the cited report proves the order constructions and the closure of the images. $\square$

[write] In this report the same conclusions are theorem 7.4, theorem 7.6 and corollary 9.5 (which also fix $\mathbf{On}$ in the cofinal case), with $\mathbf{Oz}$ from proposition 8.1; the collection statement `opa:as:thm:parameters` adds that no field automorphism of $\mathbf{No}$ carries one image onto the other and an explicit $\Omega > 1$ with $\Omega^{-1} < |y| < \Omega$ on the nonzero discrete image. The theorem is about a *set* of parameters; it does not permit replacing A by $\mathbf{No}$.

Remark 57.4 (Elementary is not definable [06]). Source 06's Proposition 4.4 and Theorem 4.5 are proposition 12.1 and theorem 12.2. In particular $J_{P_-}[\mathbf{No}]$ and $J_{P_+}[\mathbf{No}]$ are elementary subfields in $L_{\text{or}} = \{0, 1, +, -, \cdot, <\}$; this asserts nothing about the ambient topology, the named real section, birthdays, set-indexed sums, or the expanded language with $\mathbf{Oz}$ (theorem 58.10). The exponent-doubling automorphism S_2 is explicit in set theory, yet no formula of L_{or} with finitely many parameters defines it; likewise the nontrivial exponential automorphisms obtained from homogeneity have no parameter-definable graph in the exponential field. O-minimality must not be transferred to enrichments: simplicity, an ordinal predicate, $\mathbf{Oz}$, a normal-form operation or ∂_{BM} change the definable sets substantially. A class-sized back-and-forth for set parameter fields is needed for the abundance of automorphisms (remark 60.6).

Remark 57.5 (Computability and changes of presentation [06]). Computable self-embeddings require a specified presentation. Finite sign strings admit effective prefix and block substitution, and finitely supported normal forms admit exact symbolic reindexing when the exponent map is computable in the chosen notation; neither gives an algorithm whose finite inputs range over all ordinals or all surreals. A computable map on a notation system must respect the identification of codes for the same surreal and be injective after the quotient; on canonical sign sequences no such ambiguity arises, but the proper-class size remains. Computability questions are best posed on a selected set of notations, a set-sized subfield, or a provably closed symbolic fragment. Maps of raw game forms, of their value classes and of sign sequences are different objects until a compatibility theorem connects them, and changing the set universe adds new signs and supports (compare the base's last paragraph of section 12).

58 The omnific predicate (source 06, Section 5)

This section of source 06 is a credited synthesis of Part `opa:part:as of docs/surreal/omnific-preserving-automorphisms` [31] and of `docs/surreal/omnific-diophantine-geometry`; every statement below is a reprint under its collection label, with source 06's proof or proof sketch, and none is new. Adding $\mathbf{Oz}$ reconstructs the coefficient field and forces automorphisms to respect every Hahn sum, while proper embeddings can preserve and reflect $\mathbf{Oz}$ and still move an ordinary real.

Throughout this section use the small-monomial convention $t^\gamma = \omega^{-\gamma}$, $x = \sum_{\gamma \in S} r_\gamma t^\gamma$ with S a well-ordered set, $v(x) = \min S$, and

$$\Pi = \{x : \text{supp}(x) \subseteq \mathbf{No}^{<0}\}, \quad \mathbf{Oz} = \mathbb{Z} + \Pi,$$

so $x \in \mathbf{Oz}$ iff its support is nonpositive and $\text{ct}(x) = [t^0]x \in \mathbb{Z}$ (this is (8.1)). Π is the purely infinite ideal of $\mathbf{Oz}$; its elements need not be positive.

Remark 58.1 (Fractions [06]). $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$ is `odg:thm:fractions`, printed in this report as theorem 20.2 (home 02). Source 06's proof: for $x = \sum_{\gamma \in S} r_\gamma t^\gamma$ choose $\eta > 0$ above the set S ; then $b = t^{-\eta}$ and $a = xt^{-\eta}$ lie in Π and $x = a/b$. Consequently every automorphism σ of $\mathbf{Oz}$ extends uniquely to a field automorphism of $\mathbf{No}$, by $\tilde{\sigma}(a/b) = \sigma(a)/\sigma(b)$ (cross multiplication preserves equality of fractions, and the extension of σ^{-1} is the inverse). The support bound matters: it does not apply to a Hahn field on a fixed set-sized value group, where a set support can be cofinal.

Lemma 58.2 (Reconstruction [06]; after `odg:def:thm:ideal`, `odg:def:thm:multiplicator`). For $z \in \mathbf{Oz}$,

$$z \in \Pi \iff \exists y \in \mathbf{Oz} \ z^2 = 2y^2. \quad (58.1)$$

The multiplier ring $\mathcal{A} = \{a \in \mathbf{No} : a\Pi \subseteq \Pi\}$ satisfies

$$\mathcal{A} = \mathbb{R} + \Pi, \quad \mathcal{A}^\times = \mathbb{R}^\times. \quad (58.2)$$

Proof. Forward: $y = z/\sqrt{2} \in \Pi$. Reverse: with integer constant coefficients m, n of z, y , constant extraction is multiplicative on series with nonpositive support, so $m^2 = 2n^2$ and $m = n = 0$. For the multiplier ring, nonpositive support suffices; conversely a nonzero coefficient of a at $\delta > 0$ produces, after multiplication by $t^{-\delta} \in \Pi$, a nonzero constant term. A nonconstant element of $\mathcal{A}$ has negative valuation, so its inverse has positive valuation and is not in $\mathcal{A}$. $\square$

Proposition 58.3 (Omnific automorphisms fix the real coefficients [06]; reprint of `opa:cor:allautos`). Every field automorphism F of $\mathbf{No}$ with $F(\mathbf{Oz}) = \mathbf{Oz}$ fixes $\mathbb{R}$ pointwise and satisfies $\text{ct}(Fx) = \text{ct}(x)$ for all $x \in \mathbf{No}$.

Proof. By (58.1)–(58.2), F preserves Π , $\mathcal{A}$ and $\mathbb{R}$ setwise; its restriction to $\mathbb{R}$ is an ordered field automorphism, hence the identity. Order is preserved (positives are nonzero squares), so integer inequalities show that F preserves the infinitesimals $\mathfrak{m}$. In $\mathbf{No} = \Pi \oplus \mathbb{R} \oplus \mathfrak{m}$, F preserves each summand and fixes the middle one, which is the ct coordinate. $\square$

The multiplier condition quantifies over *all* purely infinite elements; an embedding into a proper subfield need not transport it to all target witnesses, which is the obstruction to applying this proof to proper maps.

Lemma 58.4 (Scalar detection of summability [06]; reprint of `opa:as:thm:detect`). Put $\langle x, y \rangle_0 = \text{ct}(xy)$ (nondegenerate: multiplying by $t^{-\gamma}$ tests the coefficient at γ). A set-indexed family (x_i) is Hahn summable iff for every $y \in \mathbf{No}$ only finitely many $\langle x_i, y \rangle_0$ are nonzero; it suffices to test y with countable support and coefficients in $\{0, 1\}$. For a summable family,

$$\left\langle \sum_i x_i, y \right\rangle_0 = \sum_i \langle x_i, y \rangle_0 \quad (58.3)$$

with a finite sum on the right.

Proof. For well-ordered S, T , $S \cap (-T)$ is finite (an infinite subset would give an increasing sequence in S whose negatives decrease in T). Applied to $S = \bigcup_i \text{supp } x_i$, $T = \text{supp } y$ this gives the forward implication and (58.3). Conversely, monomial tests $y = t^{-\gamma}$ force pointwise finiteness. If the union of supports were not well ordered, pass to a decreasing sequence $\gamma_0 > \gamma_1 > \dots$ in it and distinct indices i_n with $[t^{\gamma_n}]x_{i_n} \neq 0$ and $[t^{\gamma_m}]x_{i_n} = 0$ for $m > n$. Choose $c_n \in \{0, 1\}$ successively with $\sum_{m \leq n} c_m [t^{\gamma_m}]x_{i_n} \neq 0$ (at most one choice is forbidden, and later choices do not alter this row). Then $y = \sum_n c_n t^{-\gamma_n}$ is a countably supported surreal pairing nontrivially with every x_{i_n} , a contradiction. $\square$

The lemma detects the sum as well as its existence, by nondegeneracy; it does not claim first-order definability of indexed summation.

Theorem 58.5 (Automatic strongness of the omnific stabilizer [06]; reprint of `opa:as:thm:main`). *Every automorphism of $\mathbf{Oz}$ extends uniquely to an automorphism of $\mathbf{No}$ which is strongly $\mathbb{R}$ -linear with strongly $\mathbb{R}$ -linear inverse.*

Proof. Let F be the extension (remark 58.1) and (x_i) summable. Every target test is $y = F(z)$ by surjectivity, and proposition 58.3 gives $\langle F(x_i), F(z) \rangle_0 = \langle x_i, z \rangle_0$; so $(F(x_i))$ is summable by lemma 58.4. With $s = \sum_i x_i$ and $w = \sum_i F(x_i)$, $\langle F(s), F(z) \rangle_0 = \langle s, z \rangle_0 = \sum_i \langle x_i, z \rangle_0 = \langle w, F(z) \rangle_0$ for all z , so $F(s) = w$. Apply the same to F^{-1} ; fixation of $\mathbb{R}$ gives linearity. $\square$

The collection obtains it from the constant-term-covariant isomorphism theorem `opa:as:thm:ctiso`. Surjectivity supplies every target detector before an image sum is invoked. The theorem concerns the omnific stabilizer, not every field automorphism of $\mathbf{No}$.

Lemma 58.6 (The bottom gap [06]; reprint of `opa:as:lem:gap`). *Let Γ, Δ be nonzero divisible ordered abelian groups (sets, or the surreal additive class with set supports), $K_\Gamma = \mathbb{R}((t^\Gamma))$, $\Pi_\Gamma = \{x : \text{supp } x \subseteq \Gamma^{<0}\}$ and $I_\Gamma = \mathbb{Z} + \Pi_\Gamma$. For an ordered additive embedding $\tau : \Gamma \hookrightarrow \Delta$ put*

$$\text{gap}_\tau = \{c \in \Delta : n|c| < h \text{ for every } n \geq 1 \text{ and } h \in \tau(\Gamma)^{>0}\}. \quad (58.4)$$

Then gap_τ is the largest convex subgroup of Δ meeting $\tau(\Gamma)$ only in 0; it is divisible; $\text{gap}_\tau = 0$ iff $\tau(\Gamma)^{>0}$ is coinital in $\Delta^{>0}$; and the sum $\tau(\Gamma) + \text{gap}_\tau$ is direct, with $\tau(\Gamma)$ the outer lexicographic coordinate.

Proof [06]. Downward closure in absolute value gives convexity; for c, d in the gap, comparing each with $h/(2n)$ gives $n|c| + n|d| < h$, and division by positive integers preserves the condition. A nonzero image element is not smaller in absolute value than itself, so the gap meets the image only in 0. If a convex subgroup D disjoint from the image away from 0 contained c with $n|c| \geq h > 0$ for an image element h , convexity would put $h/n \in D$; this proves maximality and the block ordering (a difference of distinct outer coordinates dominates every gap correction). If positive coinitality fails, a positive c below every positive image element lies in the gap by divisibility. No completeness of either group is used. $\square$

Theorem 58.7 (Bottom-gap classification [06]; reprint of `opa:as:thm:classification`). *Strongly additive field embeddings $\phi : K_\Gamma \hookrightarrow K_\Delta$ with $\phi(t^\gamma) = t^{\tau(\gamma)}$ and $\phi(I_\Gamma) \subseteq I_\Delta$ correspond exactly to ordered field embeddings $\rho : \mathbb{R} \hookrightarrow \mathbb{R}((t^{\text{gap}_\tau}))$, via*

$$H_\tau^\rho\left(\sum_\gamma r_\gamma t^\gamma\right) = \sum_\gamma \rho(r_\gamma) t^{\tau(\gamma)}, \quad \rho = \phi|_{\mathbb{R}}. \quad (58.5)$$

Every such embedding satisfies $v(\phi x) = \tau(v x)$, $\text{lc}(\phi x) = \text{lc}(x)$ for $x \neq 0$, and $\text{ct}(\phi x) = \text{ct}(x)$, and preserves and reflects both I and Π .

Proof [06]. An ordered embedding of $\mathbb{R}$ into a real Hahn field has $\phi(r) = r + \text{an infinitesimal}$. For $\gamma > 0$, $rt^{-\gamma} \in \Pi_\Gamma$, and the existential test (58.1) (valid in these integer parts) puts $\phi(r)t^{-\tau(\gamma)}$ in Π_Δ ; so every exponent c of $\phi(r)$ is nonnegative and below every positive element of $\tau(\Gamma)$, that is, $c \in \text{gap}_\tau$. No strongness is used for this; strongness then forces (58.5). Conversely, for an ordered ρ and a well-ordered source support S , the blocks $\tau(\gamma) + \text{supp } \rho(r_\gamma)$ lie in distinct cosets of gap_τ ordered as S , so their union is well ordered; for a summable family only finitely many members contribute at each outer exponent. This proves admissibility and strong additivity; multiplicativity follows from Hahn convolution, $\rho(rs) = \rho(r)\rho(s)$ and additivity of τ . The first block starts at $\tau(v x)$ with the original leading coefficient; negative and positive outer blocks keep their sign; the zero block has constant coefficient r_0 and equals r_0 when $r_0 \in \mathbb{Z}$; distinct blocks cannot cancel. This gives the remaining assertions. $\square$

It is a classification within the strong exact-monomial category; character factors, unit corrections and nonstrong embeddings are outside it, and ct is not multiplicative on the whole Hahn field.

Corollary 58.8 (Surjectivity and coefficient rigidity [06]; reprint of `opa:as:cor:onto`, `opa:as:cor:rigidity`). H_τ^ρ is onto exactly when τ is onto, and then $\text{gap}_\tau = 0$ and $\rho = \text{id}_\mathbb{R}$. For fixed τ , every embedding of theorem 58.7 fixes $\mathbb{R}$ pointwise exactly when $\tau(\Gamma)^{>0}$ is coinital in $\Delta^{>0}$.

Proof. Surjectivity forces every target value into $\tau(\Gamma)$; an onto τ has zero gap, an ordered embedding $\mathbb{R} \rightarrow \mathbb{R}$ is the identity, and reindexing by τ^{-1} gives the inverse. Zero gap forces coefficient rigidity without surjectivity; for a nonzero gap the Taylor sections of proposition 58.9 are nontrivial. $\square$

Positive coinitality is the condition here; cofinality of the value image, used in the continuity criterion, is a different requirement.

Proposition 58.9 (Taylor sections and a proper embedding moving a real [06]; reprint of `opa:as:lem:taylorsection`, `opa:as:thm:taylorcriterion`, `opa:as:thm:explicit`, `opa:as:prop:overlap`, `opa:as:cor:image`). Let $b \in \mathbb{R}$ be transcendental and $d : \mathbb{R} \rightarrow \mathbb{R}$ a $\mathbb{Q}$ -derivation with $d(b) = 1$ (choice on the set $\mathbb{R}$ only).

(a) For $\delta > 0$,

$$\rho_{d,\delta}(r) = \sum_{n \geq 0} \frac{d^n(r)}{n!} t^{n\delta} \quad (58.6)$$

is an ordered field embedding with constant term r and $\rho_{d,\delta}(b) = b + t^\delta$; for $\delta \in \text{gap}_\tau$ it is an admissible coefficient section in theorem 58.7.

(b) Let $\mu(a) = a + \sqrt{a^2 + 1}$, an order isomorphism $\mathbf{No} \rightarrow \mathbf{No}^{>0}$ with $\mu^{-1}(y) = (y - y^{-1})/2$, and let $\tau = A_\mu$, i.e.

$$\tau\left(\sum_a s_a \omega^a\right) = \sum_a s_a \omega^{\mu(a)}. \quad (58.7)$$

Then τ is an ordered additive embedding with image $\mathbb{J}$, and gap_τ is exactly the class of finite surreals; in particular $1 \in \text{gap}_\tau$.

(c) The map $\Theta_d = H_\tau^{\rho_{d,1}}$,

$$\Theta_d\left(\sum_\gamma r_\gamma t^\gamma\right) = \sum_\gamma \sum_{n \geq 0} \frac{d^n(r_\gamma)}{n!} t^{\tau(\gamma)+n}, \quad (58.8)$$

is a proper strong ordered-field self-embedding with $\Theta_d^{-1}(\mathbf{Oz}) = \mathbf{Oz}$, $\text{ct} \circ \Theta_d = \text{ct}$, $\Theta_d(b) = b + \omega^{-1}$, $\Theta_d(\omega) = \omega^\omega$, fixing every real algebraic number.

(d) The ordinary reals in the image are

$$\Theta_d[\mathbf{No}] \cap \mathbb{R} = \ker d, \quad (58.9)$$

a proper real closed subfield of $\mathbb{R}$; the image still contains the copy $\rho_{d,1}(\mathbb{R})$ of $\mathbb{R}$, different from the embedded $\mathbb{R}$.

(e) $\Theta_d(\omega^1) = \omega^\omega$ while $\omega^{\Theta_d(1)} = \omega$: carrying monomials to monomials is not commuting with Ω .

(f) $\text{ct}(t \Theta_d(x)) = 0$ for every $x \in \mathbf{No}$, although $t \neq 0$.

Proof [06]. (a) The iterated Leibniz rule gives multiplicativity coefficientwise; the constant term r gives injectivity and order. (b) μ is onto $\mathbf{No}^{>0}$, so τ has image the purely infinite group and $\tau(1) = \omega$ (τ is not a field map). Every positive purely infinite element dominates every finite one; an infinite c with leading exponent $a > 0$ dominates the positive image element $\omega^{a/2}$, so it is not in the gap. (c)

is theorem 58.7; properness holds since $1 \notin \tau[\mathbf{No}]$, so $t \notin \Theta_d[\mathbf{No}]$; $\Theta_d(\omega) = t^{\tau(-1)} = \omega^\omega$. (d) If $\Theta_d(x)$ is real, every nonzero outer block vanishes, so $x = r$ is real, and $\rho_{d,1}(r)$ is constant iff $d(r) = 0$. $\ker d$ is real closed: an element of $\mathbb{R}$ algebraic over it is killed by d (differentiate its separable minimal polynomial). (e) is immediate. (f) Every support exponent of $\Theta_d(x)$ is $\tau(\gamma) + n$ with $n \geq 0$, never -1 , since a nonzero purely infinite value is not an integer. $\square$

This is the collection's construction `opa:as:thm:explicit` with its choice μ_{13} ; (d) is `opa:as:prop:overlap` and (d), (f) are `opa:as:cor:image`; the boundary (e) is recorded after `opa:as:cor:homogeneity`; the exact admissibility criterion for Taylor sections is `opa:as:thm:taylorcriterion`. Item (f) shows that tests from the image alone cannot reproduce the nondegeneracy argument used for automorphisms; and $\text{ct} \circ \Theta_d = \text{ct}$ does not imply $\Theta_d \circ \text{ct} = \text{ct}$ (at b it would assert $b + t = b$).

Theorem 58.10 (Predicate preservation is not elementarity [06]; reprint of `opa:as:thm:elementary`). *The embedding Θ_d preserves and reflects $\mathbf{Oz}$ but is not elementary as an embedding of $(\mathbf{No}, \mathbf{Oz})$, and its restriction to $\mathbf{Oz}$ is not an elementary ring embedding.*

Proof [06]. $\mathbb{R}$ is first-order definable in $(\mathbf{No}, \mathbf{Oz})$: (58.1) defines Π , the multiplier condition defines $\mathcal{A}$, and $\{0\}$ with the units of $\mathcal{A}$ is $\mathbb{R}$. An elementary embedding of the pair preserves this definition, but $\Theta_d(b) = b + t \notin \mathbb{R}$; the new target witness is $t^{-1} \in \Pi$ with $(b + t)t^{-1} = bt^{-1} + 1 \notin \Pi$. For the ring, the fraction field is uniformly interpretable by pairs (a, b) , $b \neq 0$, modulo $ad = bc$, with $\mathbf{Oz}$ as the fractions $(z, 1)$; an elementary ring embedding would induce an elementary embedding of the interpreted pair. The real b is represented internally by $(b\omega)/\omega$. $\square$

These distinctions explain why rigidity of omnific automorphisms coexists with abundant proper self-embeddings.

59 Exponential, differential, and hyperserial constraints (source 06, Section 6)

An exponential-field embedding satisfies $j(\exp x) = \exp(jx)$, an omega-field embedding $j(\omega^x) = \omega^{jx}$. They differ already at $x = 1$ ($\exp 1 = e \in \mathbb{R}$, $\omega^1 = \omega$), so an omega-field embedding fixes ω while an exponential one need not. Preserving the real powers $x^r = \exp(r \log x)$ ($x > 0$, $r \in \mathbb{R}$) is a third requirement; it follows from preserving $\exp$ and fixing $\mathbb{R}$, but does not recover $\exp$.

Remark 59.1 (Exponential homogeneity and its interpretation [06]). $(\mathbf{No}, \exp)$ is an elementary extension of $\mathbb{R}_{\exp}$ [3]. In the class framework of Kaplan, Krapp and Serra, every elementary embedding of a set-sized elementary exponential subfield into $\mathbf{No}$ extends to an exponential automorphism of $\mathbf{No}$ [5, Remark 2.10 and Section 5]; the hypothesis is elementarity in the exponential language, which an arbitrary ordered-field embedding of such a subfield need not satisfy. Source 06 proposes a coherent interpretation of the class statements without a truth predicate for arbitrary class structures: the set-sized birthday stages $\mathbf{No}_{<\lambda}$ at epsilon numbers λ are models of the real exponential theory, model completeness makes their inclusions elementary, a set of parameters lies in one stage and its definable closure can be computed there [3, 26].

[write] This route bears on the conditional hypotheses (GBC+ETR or a class satisfaction relation) of the bounded and cofinal elementary exponential copies of Part III and Part IV (theorem 34.2, theorem 34.5): it is a proposal recorded by source 06, not a proof that those hypotheses can be dropped. Unchecked here.

Proposition 59.2 (Moving the distinguished infinite argument [06]; imported literature, in the class framework of [5]). *For every $a > \mathbb{R}$ there is an exponential-field automorphism j of $\mathbf{No}$ fixing $\mathbb{R}$ pointwise with $j(\omega) = a$.*

Proof. By o-minimality, every subset of the line definable over the elementary submodel $\mathbb{R}$ is a finite union of points and intervals with endpoints in $\mathbb{R} \cup \{\pm\infty\}$, so any two elements above every real have the same complete type over $\mathbb{R}$. Hence the map fixing $\mathbb{R}$ and sending ω to a is partial elementary. Definable Skolem functions of an o-minimal expansion of an ordered field extend it to an elementary isomorphism between the definable closures of $\mathbb{R} \cup \{\omega\}$ and $\mathbb{R} \cup \{a\}$, which are sets. Apply the homogeneity theorem of remark 59.1. $\square$

Taking $a = \omega^2$ gives an exponential automorphism not preserving the omega-map. The construction provides no preservation of Conway summation or simplicity; those are additional requirements, not consequences of being elementary in a finitary language. This is a consequence of [5, Remark 2.10], not stated as such in the collection.

Remark 59.3 (Exact values detect exponential maps [06]). In contrast, the action on exact values detects every exponential self-embedding: source 06's Lemma 6.2, Lemma 6.3 and Theorem 6.4 are lemma 11.1, (11.3) and theorem 11.4 (EAR lem:amplify, cor:embedding), without surjectivity, a prior fixing of $\mathbb{R}$, or strong additivity. Its proof of the converse half of (11.3) is: $v(\exp x) = 0$ makes $\exp x$ a positive unit, so $n^{-1} < \exp x < n$ for an integer $n > 1$, and the increasing logarithm gives $-\log n < x < \log n$. Source 06 records, correctly, that Kaplan–Krapp–Serra's April 2026 version proves rigidity for strongly $\mathbb{R}$ -linear exponential 1-automorphisms and asks whether strong linearity can be removed [5, Proposition 5.2 and Question 5.4]; a 1-automorphism fixes every exact value, so the repository argument (EAR cor:question54) gives the negative answer and a stronger self-embedding statement, and that question should not be listed as open relative to ProveIt. Its Corollary 6.5 (faithfulness of the value action for exponential automorphisms) is weaker than theorem 11.7; it is saut:thm:expfaithful of docs/surcomplex/surcomplex-field-automorphisms and EAR cor:faithful, neither cited by source 06. Uniqueness of lifts does not give existence; existence is examined in [13].

Remark 59.4 (Omega-map rigidity [06]). Source 06's Proposition 6.6 is theorem 10.3; it extends the automorphism statement [5, Proposition 5.5] without further argument. Omega-compatibility forces $\tau_j(x) = j(x)$, and for a strong Hahn lift H_τ it is the self-consistency equation $\tau = H_\tau$ ((10.5)). Any omega-field embedding preserves the equation $x = \omega^x$, hence maps generalized epsilon numbers to generalized epsilon numbers; this is a necessary condition, not an extension theorem. Neither this calculation nor ordered-field homogeneity answers [5, Questions 5.6–5.7].

Fix the Berarducci–Mantova derivation ∂_{BM} , with $\ker \partial_{\text{BM}} = \mathbb{R}$, $\partial_{\text{BM}}\omega = 1$, $\partial_{\text{BM}}\exp x = (\exp x)\partial_{\text{BM}}x$; it is strongly additive and makes **No** a Liouville closed H -field [8]. A differential-field embedding satisfies $\partial_{\text{BM}}j = j\partial_{\text{BM}}$.

Proposition 59.5 (The translation parameter [06]). *Every differential-field self-embedding j of $(\mathbf{No}, \partial_{\text{BM}})$ fixes $\mathbb{R}$ pointwise and satisfies*

$$j(\omega) = \omega + c_j \quad \text{for a unique } c_j \in \mathbb{R}.$$

For compositions, $c_{j \circ k} = c_j + c_k$.

Proof. The kernel equation sends each real into $\mathbb{R}$; the restriction to $\mathbb{R}$ is an ordered field embedding $\mathbb{R} \rightarrow \mathbb{R}$, hence the identity by rational cuts. Next $\partial_{\text{BM}}(j\omega - \omega) = j(1) - 1 = 0$, so the difference is real. Finally apply j to $k\omega = \omega + c_k$ and use $j(c_k) = c_k$. $\square$

This strengthens proposition 13.1 (which has the first two assertions for any derivation with constant field $\mathbb{R}$ and $\partial\omega = 1$) by the additivity of c ; it is the home of the overlap with sources 02, 04, 05 and 07, and was not found as a statement in the collection.

Proposition 59.6 (The exponential defect of a differential embedding [06]). *For a differential-field self-embedding j of $(\mathbf{No}, \partial_{\mathbf{BM}})$ put*

$$\alpha_j(x) = \log(j(\exp x)) - j(x).$$

Then $\alpha_j : \mathbf{No} \rightarrow \mathbb{R}$ is additive and vanishes on every finite surreal, and

$$j(\exp x) = \exp(jx + \alpha_j(x)). \quad (59.1)$$

j preserves $\exp$ iff $\alpha_j = 0$; j preserves every real power of every positive surreal iff α_j is $\mathbb{R}$ -linear. For two differential self-embeddings,

$$\alpha_{j \circ k} = \alpha_j \circ k + \alpha_k.$$

Proof. $j(\exp x) > 0$ since j is increasing, so α_j is defined. Using the logarithmic derivative and $\partial_{\mathbf{BM}} j = j \partial_{\mathbf{BM}}$,

$$\partial_{\mathbf{BM}} \log(j(\exp x)) = \frac{j(\partial_{\mathbf{BM}} \exp x)}{j(\exp x)} = j(\partial_{\mathbf{BM}} x) = \partial_{\mathbf{BM}} j(x),$$

so $\alpha_j(x) \in \ker \partial_{\mathbf{BM}} = \mathbb{R}$. Additivity follows from $\exp(x + y) = \exp x \exp y$ and the logarithm of a product. For finite x with standard part r , both $j(x)$ and $\log(j(\exp x))$ have standard part r , because j fixes $\mathbb{R}$ (proposition 59.5) and preserves infinitesimals; their difference is real and infinitesimal, hence 0. Identity (59.1) is the definition and characterizes exponential preservation. For $r \in \mathbb{R}$,

$$j((\exp x)^r) = \exp(rj(x) + \alpha_j(rx)), \quad j(\exp x)^r = \exp(rj(x) + r\alpha_j(x)),$$

and every positive surreal is an exponential, so preservation of all real powers is $\mathbb{R}$ -linearity of α_j . Applying (59.1) twice, with j fixing reals and α_j vanishing on them, gives the composition law. $\square$

The proposition constrains embeddings that exist. It does not assert that every additive functional vanishing on finite elements is realized by a self-embedding of $\mathbf{No}$ (research question 62.6); proposition 59.8 realizes such data on a particular subfield.

The next two statements are consequences of imported literature, recorded by source 06 and not proved in this report beyond the cited theorems. Let $\mathbb{T}$ be the field of logarithmic-exponential transseries, identified with its canonical image in $\mathbf{No}$ with variable ω ; it is an elementary differential subfield of $(\mathbf{No}, \partial_{\mathbf{BM}})$ [26]. The homogeneity theorem of [39] extends an isomorphism between set-sized submodels of the complete theory of ω -free, newtonian, Liouville closed H -fields with small derivation and constant field $\mathbb{R}$ to an automorphism of the surreal differential field (it refers to models of that theory, not to all differential subfields).

Proposition 59.7 (Every translation parameter occurs [06]; imported literature: [26, 39], in the class framework of [39]). *For every $c \in \mathbb{R}$ there is a differential-field automorphism j of $(\mathbf{No}, \partial_{\mathbf{BM}})$ with $c_j = c$.*

Proof. Transseries substitution $f \mapsto f \circ (\omega + c)$ is an automorphism of $\mathbb{T}$ (inverse: substitution by $\omega - c$) commuting with differentiation by the chain rule. The homogeneity theorem extends it to a differential automorphism of $\mathbf{No}$. $\square$

Choosing an extension for each c does not by itself give a coherent action of $(\mathbb{R}, +)$ on $\mathbf{No}$; that would require an additional construction.

Proposition 59.8 (A zero translation parameter does not force rigidity [06]; imported literature: [29, Fact 5.7] with [39]). *There is a differential-field automorphism j of $(\mathbf{No}, \partial_{\mathbf{BM}})$ with $j(\omega) = \omega$ (so $c_j = 0$) and $j(\exp \omega) = e \exp \omega$; in particular j does not preserve $\exp$ and $\alpha_j(\omega) = 1$.*

Proof. By [29, Fact 5.7], for every additive $\alpha : \mathbb{T} \rightarrow \mathbb{R}$ vanishing on finite transseries there is a strongly $\mathbb{R}$ -linear differential automorphism s_α of $\mathbb{T}$ with $s_\alpha(\omega) = \omega$ and $s_\alpha(\exp f) = \exp(s_\alpha(f) + \alpha(f))$. Choose an $\mathbb{R}$ -linear such α with $\alpha(\omega) = 1$ (extend a vector-space basis; ω is not finite). Then $s_\alpha(\exp \omega) = e \exp \omega$. Differential homogeneity extends s_α to $\mathbf{No}$; the failure of exponential compatibility is witnessed inside $\mathbb{T}$. $\square$

The extension theorem does not promise global preservation of strong sums or real powers.

Remark 59.9 (Exponential and differential constraints are distinct [06]). The exponential automorphism of proposition 59.2 with $j\omega = \omega^2$ cannot be differential, since $\partial_{\text{BM}}(\omega^2) = 2\omega \neq 1$. If both structures are imposed, proposition 59.5 still applies, as a necessary condition. A field automorphism j transports ∂ to $j\partial j^{-1}$ and is then an isomorphism between two differential structures on one field, which is weaker than being an automorphism of one specified differential structure; the uniqueness-up-to-isomorphism theorem of [39] must be read with this distinction.

Let $\mathcal{S}_\omega = \mathbb{R}\langle\omega\rangle$ be the field of omega-series: the smallest subfield of $\mathbf{No}$ containing $\mathbb{R}$ and ω and closed under $\exp$, $\log$ and all existing set-indexed Conway sums; it is a proper class and a proper subfield of $\mathbf{No}$. Berarducci and Mantova construct a composition $\mathcal{S}_\omega \times \mathbf{No}^{>\mathbb{R}} \rightarrow \mathbf{No}$, $(f, g) \mapsto f \circ g$, whose right substitutions are strongly $\mathbb{R}$ -linear exponential-field embeddings [27, Theorem 6.3]; it restricts to omega-series and, on $\mathbb{T}$, to transseries composition.

Proposition 59.10 (Restriction to omega-series [06]; uses [27, Theorem 6.3]). *If $j : \mathbf{No} \rightarrow \mathbf{No}$ fixes $\mathbb{R}$, preserves $\exp$, and commutes with all summable families, then*

$$j(f) = f \circ j(\omega) \quad (f \in \mathcal{S}_\omega). \quad (59.2)$$

If j also commutes with ∂_{BM} , its restriction to $\mathcal{S}_\omega$ is substitution by $\omega + c_j$; in particular, under these combined hypotheses, fixing ω forces pointwise fixation of $\mathcal{S}_\omega$.

Proof. $j(\omega) > \mathbb{R}$ because j fixes $\mathbb{R}$ and is increasing. The maps j and $f \mapsto f \circ j(\omega)$ agree on $\mathbb{R}$ and on ω , and both preserve the field operations, $\exp$, $\log$ (inverse of $\exp$ on positives) and summation. The class where they agree is therefore a subfield containing $\mathbb{R} \cup \{\omega\}$ and closed under these operations, so it contains $\mathcal{S}_\omega$; equivalently, use uniqueness of the omega-series substitution. The differential case is proposition 59.5. $\square$

It does not give fixation of $\mathbf{No}$, because $\mathcal{S}_\omega \neq \mathbf{No}$.

[write] Source 07's theorem 68.5 classifies strong exponential self-embeddings of $\mathcal{S}_\omega$ (as C_g , all automorphisms; the differential ones are $C_{\omega+c}$). The proposition here concerns restrictions of global self-embeddings of $\mathbf{No}$; the two proofs use the same generation argument.

Remark 59.11 (Substitution and the global composition obstruction [06]). For $g \in \mathcal{S}_\omega^{>\mathbb{R}}$ the chain rule is $\partial_{\text{BM}}(f \circ g) = ((\partial_{\text{BM}}f) \circ g) \partial_{\text{BM}}g$, so substitution commutes with ∂_{BM} when $\partial_{\text{BM}}g = 1$; the translates $\omega + c$ are the immediate examples (compare C_g in section 13). On all of $\mathbf{No}$, [27, Theorem 8.4] excludes a global composition compatible with ∂_{BM} in the following sense: derivative-zero elements give constant functions, positive derivatives give increasing functions, and the chain rule holds. The reason is a proper class of log-atomic solutions of a growth differential equation, which such a composition would convert injectively into elements of derivative 1, whereas these all lie in the set $\omega + \mathbb{R}$. The theorem must not be expanded into a prohibition of every binary operation called composition, or of every individual embedding; its force lies in the simultaneous functional requirements.

Remark 59.12 (Hyperserial fields and remaining targets [06]). Write $\mathbf{No}_\lambda^{\text{hyp}}$ for Bagayoko's field of hyperserial strength bounded by an infinite additively indecomposable ordinal λ (a proper-class restriction on hyperserial representations, not a birthday stage). Its embeddings preserve real coefficients, sums, products, monomials and the hyperlogarithms and hyperexponentials of that strength; if a is λ -atomic

(each $L_\gamma(a)$, $\gamma < \lambda$, a single monomial) there is a unique such embedding $\mathbf{No}_\lambda^{\text{hyp}} \rightarrow \mathbf{No}$ sending ω to a and preserving the indices distinguishing nested expansions, and right substitutions by $\log \omega$ and $\exp \omega$ are inverse automorphisms of $\mathbf{No}_\omega^{\text{hyp}}$ [28, Theorem 2, Section 7.5, Corollary 3]. These are constructive results on an intermediate proper-class field, not a global right substitution on $\mathbf{No}$. Ehrlich–Kaplan show that the canonical transseries image, the exponential-logarithmic series field and the omega-series field are initial exponential subfields [7]; initiality of an image subtree must not be strengthened to simplicity preservation by every embedding with that image. Source 06’s list of targets:

- (1) characterize the value-group actions admitting global exponential lifts (lifts are unique by theorem 11.7 and source 06’s Corollary 6.5; existence is the substantial part);
- (2) classify omega-field self-embeddings, distinguishing automorphisms, strong Hahn lifts, and maps preserving generalized epsilon parameters;
- (3) determine which translations of omega-series extend to embeddings of $\mathbf{No}$ preserving both $\exp$ and ∂_{BM} , with which further summability; differential homogeneity alone does not answer this;
- (4) describe the kernel $c_j = 0$ for differential automorphisms and its intersections with exponential, monomial and strong-summation constraints (proposition 59.8 shows the unrestricted kernel is nontrivial);
- (5) extend the constructive hyperserial embedding theorems to more general arguments, treating nested indices and compatibility between strength bounds.

The canonical embedding $\mathbb{T} \rightarrow \mathbf{No}$ is elementary separately as an ordered exponential field and as an ordered differential field [26], and Kaplan’s September 2026 version establishes elementarity after adjoining the uniform real-power map and e [29, Section 6.2]. These do not give elementarity in the full differential-exponential language; that question and extension theorems for the combined structure are separate from the homogeneity theorems for the reducts.

60 Foundations, localization, and large cardinals (source 06, Section 7)

Remark 60.1 (Frameworks and recursions [06]). Each surreal is a set and each stage $\mathbf{No}_{<\alpha}$ is a set; $\mathbf{No}$ is a proper class (Replacement on birthdays). An injective class map cannot have a set-sized image, so no self-embedding lands in one $\mathbf{No}_{<\alpha}$, while a numerically bounded interval can contain a proper class. Source 06 lists four precise formulations, complementing the three registers of section 3: (1) ZFC with definable-class notation, general assertions being metatheoretic schemes; (2) GBC, quantifying over individual class maps, with Global Choice supplying enumerations (some references call it NBG; the choice convention is stated explicitly); (3) a set-stage or universe-relative formulation, where the surrounding universe remains part of the meaning of “all subsets” and “all summable families”; (4) an added recursion principle such as ETR, or Kelley–Morse, when a construction needs a uniform truth predicate or a genuine class-valued recursion. In none of these are proper classes members of classes, so “the class of all class self-embeddings” is not an object; a family is coded by one relation $F(i, x, y)$. Class quantifiers are model-relative: in a two-sorted model $(M, \mathcal{X})$ they range over $\mathcal{X}$, which need not contain every externally available subclass [33]. An $\mathbf{On}$ -long recursion that produces a set from a set-sized history is ordinary transfinite recursion, not an application of ETR (canonical sign prefixes, the dependency closure of one arithmetic computation, and reindexing a set support are of this form); a uniform truth recursion for an arbitrary class structure consults a proper class of earlier instances, and GBC does not in general prove that such recursions have solutions, even of height ω [21, 10].

The base’s section 14 gives the following argument as prose; source 06 states and proves it, and this report prints it here as the home of the overlap (source 07’s theorem 73.1 points here).

Proposition 60.2 (Localization to birthday stages [06]; in GB with the Replacement and comprehension instances for j). *Let $j : \mathbf{No} \rightarrow \mathbf{No}$ be a class function. There is a closed unbounded class C of nonzero limit ordinals such that*

$$j[\mathbf{No}_{<\lambda}] \subseteq \mathbf{No}_{<\lambda} \quad (\lambda \in C),$$

and C can be chosen to consist of epsilon numbers. Restrictions inherit injectivity and all preserved finitary operations and relations. A fixed witness to nonidentity or to global nonsurjectivity persists at every sufficiently large selected stage.

Proof. Class Replacement makes $F(\alpha) = \sup\{b(j(x)) + 1 : x \in \mathbf{No}_{<\alpha}\}$ an ordinal for every α ; F is nondecreasing and $x \in \mathbf{No}_{<\alpha}$ implies $b(j(x)) < F(\alpha)$. Let

$$C_F = \{\lambda : \lambda \text{ a nonzero limit ordinal and } F(\alpha) < \lambda \text{ for all } \alpha < \lambda\}.$$

Unbounded: above γ put $\delta_0 = \gamma + 1$, $\delta_{n+1} = \max\{\delta_n, F(\delta_n)\} + 1$, $\lambda = \sup_n \delta_n$ (a set-length recursion); for $\alpha < \lambda$ choose $\alpha < \delta_n$, so $F(\alpha) \leq F(\delta_n) < \delta_{n+1} < \lambda$. Closed: if λ is a nonzero limit of members of C_F and $\alpha < \lambda$, choose $\eta \in C_F$ with $\alpha < \eta < \lambda$; then $F(\alpha) < \eta < \lambda$. For $x \in \mathbf{No}_{<\lambda}$ choose $b(x) < \alpha < \lambda$; then $b(j(x)) < F(\alpha) < \lambda$, and Replacement gives the set graph of the restriction.

The epsilon numbers ($\omega^\lambda = \lambda$, ordinal exponentiation) form a closed unbounded class; for a direct proof of simultaneous unboundedness take δ_0 an epsilon number above γ and δ_{n+1} the least epsilon number above δ_n and $F(\delta_n)$; continuity of ordinal exponentiation makes the supremum an epsilon number, and the same estimate puts it in C_F . Closedness follows from the two closedness assertions. Injectivity and finitary identities restrict immediately. If $j(a) \neq a$, every selected $\lambda > b(a)$ contains a moved point; if $b \notin j[\mathbf{No}]$, every selected $\lambda > b(b)$ contains a point omitted by the restriction. $\square$

It localizes a map that already exists; it does not construct a coherent global map from unrelated local witnesses.

Corollary 60.3 (Automorphisms and exact local ranges [06]). *If j is bijective, there is a closed unbounded class of epsilon stages on which j restricts to a bijection. If j is merely injective, one can instead arrange $j[\mathbf{No}_{<\lambda}] = j[\mathbf{No}] \cap \mathbf{No}_{<\lambda}$.*

Proof. Simultaneous closure under finitely many maps follows by taking the maximum of their bounds F . In the bijective case use j and j^{-1} . In the injective case let $R(y)$ be the unique preimage of y if $y \in j[\mathbf{No}]$ and 0 otherwise; closure under j and R gives the equality. $\square$

Remark 60.4 (Families and cardinal stages [06]). A set-indexed family coded by one class relation is handled by taking the supremum over the product of its index set and $\mathbf{No}_{<\alpha}$; no class of maps is formed. There are also arbitrarily large closed cardinal stages of any prescribed infinite regular cofinality μ : build an increasing continuous μ -sequence of infinite cardinals, each next one above the current cardinal and its F -value, and take the supremum κ ; regularity of μ gives $\text{cf}(\kappa) = \mu$. The resulting cardinal need not be regular or inaccessible; localization neither assumes nor produces inaccessibles.

Remark 60.5 (What a bounded stage preserves [06]). By the classical stage theorem, $\mathbf{No}_{<\lambda}$ is a subfield precisely when λ is an epsilon number, and then it is real closed and closed under $\exp$ and $\log$ [3]. So a class field embedding restricts, on the epsilon stages of proposition 60.2, to set-sized field self-embeddings, elementary in the ordered-field language by quantifier elimination. Closure under j alone does not make a stage elementary in an expanded language, and $\forall \lambda \exists j_\lambda$ does not give one global map (the stage maps may disagree on overlaps). For an uncountable cardinal κ , the order of $\mathbf{No}_{<\kappa}$ fills cuts whose two sides have total size less than $\text{cf}(\kappa)$ (the birthdays are bounded below κ , so a simplest separator stays inside), while a cofinal set of ordinals below κ gives an omitted end cut [39]: a bounded-stage argument must keep its cardinal bound.

Remark 60.6 (Global extension needs an extension proof [06]). For the ordered-field reduct, source 06 gives the back-and-forth that extends an ordered isomorphism between set-sized real closed subfields of $\mathbf{No}$ to an automorphism, using Global Choice and set-sized stages (a transcendental a is sent to a simplest separator b of the transferred cut; polynomial signs over a real closed field are determined by root positions; unions of chains of real closed fields are real closed). This is the ordered-field homogeneity theorem printed as theorem 70.2 (home 07), which is `saut:thm:orderedhomogeneity` of `docs/surcomplex/surcomplex-field-automorphisms`; source 06 does not cite the latter. For richer structures a separate extension theorem is required: the homogeneous universal H -field theorem of [39] applies to its specified theory and common constant field $\mathbb{R}$, and the saturation of the underlying order cannot alone give exponential or differential homogeneity.

Remark 60.7 (Elementarity is not set-theoretic elementarity [06]). For a fixed formula ϕ , preservation of $\phi^{\mathbf{No}}(\bar{a}) \leftrightarrow \phi^{\mathbf{No}}(j\bar{a})$ is meaningful formula by formula; a uniform truth relation is additional information, and “one j satisfies every instance” must not be replaced by “for every instance some j exists”. As in section 3, an elementary embedding of the ordered field $\mathbf{No}$ is not an elementary embedding of $(V, \in)$, and a least moved birthday has no automatic large-cardinal meaning. Transporting $\in$ from V to $\mathbf{No}$ along a class bijection gives a coding-dependent structure, for which elementarity is membership-elementarity under conjugation and is subject to Kunen’s inconsistency under the usual choice assumptions [16]; literal membership between sign-code sets is representation-dependent and is not that transported copy.

Ordinary self-embeddings need no large cardinals, but some infinitary preservation profiles have exactly that strength [32].

Theorem 60.8 (Measurable first failure [06]; reprint of `lce:mf:thm:spectrum`). *Let $\Psi : \mathbf{No} \rightarrow \mathbf{No}$ be an $\mathbb{R}$ -linear class map preserving every countable Hahn-summable family and its sum. If Ψ is not strong for all set-indexed Hahn sums, the least cardinality of a family on which sum preservation fails is an uncountable measurable cardinal.*

Mechanism [06]. Nonsummability has a countable witness (infinitely many terms at one exponent, or an increasing sequence of exponents), so countable strength implies that every summable family has a summable image. Choose a first failing cardinal κ , a witnessing family and a target coefficient with nonzero sum defect; the combined input support is a reverse well-ordered set X , and restriction plus coefficient extraction give a functional $\psi : \mathbb{R}^X \rightarrow \mathbb{R}$ preserving point-finite sums of fewer than κ vectors, each point-finite family having only finitely many nonzero scalar images. Let $\mathcal{I}_\psi$ consist of the subsets of X on which every supported vector is annihilated by ψ ; it is closed under unions of fewer than κ sets, and $\mathcal{P}(X)/\mathcal{I}_\psi$ is finite (countably many disjoint nonzero elements would give a point-finite family with infinitely many nonzero scalar images). Its atoms give κ -complete ultrafilters $\mathcal{U}_1, \dots, \mathcal{U}_m$; a real function is constant on a set in any countably complete ultrafilter (successive bisection), so $\psi = \sum_{r=1}^m c_r \text{ev}_{\mathcal{U}_r}$. If all $\mathcal{U}_r$ were κ^+ -complete, ψ would preserve the witnessing κ -indexed sum; so one is not, its least partition into small pieces has size κ , and pushing it forward gives a nonprincipal κ -complete ultrafilter on κ . Every restriction and partition is a set. This is the complete-ultrafilter factorization of [32]; compare [30]. $\square$

Proposition 60.9 (The measurable construction: exponential and omnific clauses [06]; reprint of `lce:prop:J`, `lce:lem:exptransfer`, `lce:thm:Oz`; under a measurable cardinal). *Let $j : V \rightarrow M \subseteq V$ be a normal ultrapower by a measure on a measurable cardinal κ , and let $\mathcal{J}_j : \mathbf{No} \rightarrow \mathbf{No}$ apply j to sign codes and include $\mathbf{No}^M$ into $\mathbf{No}^V$. Besides the properties of theorem 24.2 (ordered field, $\mathbb{R}$ fixed, prefix simplicity preserved and reflected, $b(\mathcal{J}_j x) = j(bx)$, $\mathcal{J}_j(\Omega(x)) = \Omega(\mathcal{J}_j x)$), it satisfies*

$$\mathcal{J}_j(\exp x) = \exp(\mathcal{J}_j x), \quad \mathcal{J}_j(\log y) = \log(\mathcal{J}_j y) \ (y > 0), \quad \text{ct}(\mathcal{J}_j x) = \text{ct}(x),$$

*and preserves and reflects **Oz**. It fixes every surreal of birthday below κ , moves κ to $j(\kappa)$ and omits κ from its image; it preserves every Hahn sum with fewer than κ terms but not the sum of $(\omega^{-\alpha})_{\alpha < \kappa}$.*

Absoluteness argument [06]. All prefixes of an old sign sequence are old; the externally simplest separator of an old set cut is a prefix of its internally simplest separator, so lies in M , and absolute sign inequalities identify the two. Set-like recursion on canonical options then identifies the field operations and the omega-map in M and V . The same applies to Gonshor’s exponential, whose recursive cuts use simpler option exponentials, finite Taylor polynomials, field operations and inequalities with the same natural indices and real constants in both universes; elementary transfer followed by this identification gives the exponential identity, and uniqueness of the inverse gives log. The canonical-normal-form absoluteness lemma gives **Oz** and constant-coefficient preservation. Since κ is a prefix of $\mathcal{J}_j(\kappa)$ but is not in the image, the image is not initial despite prefix preservation and reflection. For the family $(\omega^{-\alpha})_{\alpha < \kappa}$ the image of the sum has terms at all indices below $j(\kappa)$, while the sum of the images has indices below κ only; the coefficient at exponent $-\kappa$ detects the discrepancy. $\square$

This is a joint preservation example with a precisely located first failure, not an automorphism or a strong embedding; it is a class map defined from a set measure, not a membership-elementary $V \rightarrow V$, and $M \subseteq V$ is not assumed elementary. Compatibility with ∂_{BM} is unproved; the collection asks for the required naturality or absoluteness theorem in `lce:cs:q:derivations`. Together with theorem 60.8 it gives the collection’s exact equivalence `lce:mf:thm:equivalence`: there is no measurable cardinal iff every countably strong coefficient-fixing, **Oz**-preserving field self-embedding is strong. The countable hypothesis is essential: an embedding failing a countable Hahn sum lies outside the theorem, and whether such a countable failure can coexist with coefficient fixing and omnific preservation is a separate question (research question 62.2). Neither order continuity nor finite algebraic elementarity may be substituted for countable Hahn strength.

Remark 60.10 (A formalization boundary [06]). A natural implementation separates set-small sign and normal-form constructions; bounded-stage embeddings with explicit coherence; and class-theoretic metatheorems with declared choice, recursion or large-cardinal hypotheses. Smallness belongs in the statement of every option and summation construction, and the collection’s foundational interfaces are the place to record those hypotheses [34]. A universe-polymorphic theorem is a family of level-specific statements, not a single carrier containing all universes.

61 Formalization status recorded by source 06 (its Section 8)

Source 06 compares manuscript scopes with the Lean development theorem by theorem, using inspected definitions and proof bodies together with the mapping ledger [35, 36]. “Actual” means the repository’s sign-sequence surreal carrier at a fixed universe. The table reports source-level scope, not a fresh compilation, kernel recheck, or transitive axiom audit.

Topic	Construction and precise boundary (source 06)
Actual normal forms	<code>SmallNormalFormFieldEquiv.lean</code> defines <code>cutEvaluationOrderRingIso</code> , an ordered ring equivalence between lower-universe-small formal normal forms and the actual <code>SignSequence</code> field, built from separately supplied addition, multiplication, extraction and comparison theorems. It is therefore incorrect to describe every subsequent exponent construction as merely a surrogate Hahn-field calculation.
Actual exponent automorphisms	<code>NormalFormExponentAutomorphisms.lean</code> and <code>OmnificExponentAutomorphisms.lean</code> construct actual field and omnific-ring automorphisms from an ordered additive <i>automorphism</i> of the exponent carrier, proving real-coefficient fixation, exact monomial and support transport, inverse laws and positive exponent dilations. Their input is bijective; they do not formalize the proper-embedding bottom-gap classification.

Topic	Construction and precise boundary (source 06)
Strong sums and valuation	<code>ExponentAutomorphismStrongSums.lean</code> proves preservation and reflection of strong summability for arbitrary index universes; the actual strong-sum identity additionally assumes the index type small in the birthday universe, exactly the domain of the existing sum. Leading exponents, the additive valuation with infinity, and the order are transported.
Omnific reconstruction	<code>OmnificAutomorphisms.lean</code> constructs the unique fraction-field extension of an actual omnific-ring automorphism and proves order preservation, fixation of embedded reals, finiteness and infinitesimality equivalences and standard-part preservation; these are automorphism results, consistent with the coefficient-moving proper embeddings of proposition 58.9 .
Character twists	<code>HahnSeries/CharacterTwist.lean</code> constructs a Hahn-field automorphism from a multiplicative character, preserving support and fixing constants; <code>Surcomplex/CharacterAutomorphisms.lean</code> transports it to the actual surcomplex field and Gaussian omnific ring. A general complex character need not preserve the real axis.
Exponential rigidity	<code>Algebra/ExponentialProfile.lean</code> contains generic displacement, visible-scale, self-embedding, value-action and coarsening rigidity proofs. In <code>HahnSeries/ValueGroupLifts.lean</code> the actual-surreal rational-dilation nonlifting theorem accepts $E : \text{OrderedExp SignSequence}$ as a hypothesis; constructing the canonical global surreal exponential in that interface is a separate obligation.

[write] Every claim of this table, and of the two paragraphs below, was checked true at the dossier commit and re-checked at the placement commit e1395c66c: the eight modules exist under `Algebra/SurrealNumbers/Surreal/`; `cutEvaluationOrderRingIso` is an order-ring equivalence `SmallNormalForm.{u}` to `SignSequence.{u}`; `mapExponents` takes an ordered additive *equivalence* `SignSequence.{u}` to itself; `exponentAutomorphism_strongSum` carries a `Small.{u}` hypothesis on the index type while the summability transport does not; `omnificAutomorphismExtension` takes an omnific-ring equivalence; `eq_id_of_commute` and `eq_id_of_visible_scale` take a ring endomorphism of F with E only a monotone homomorphism; and `surreal_not_lift_dilation` takes $E : \text{OrderedExp SignSequence}.{u}$ as a hypothesis. The automatic-strongness theorem and the proper-embedding classification of `Part opa:part:as` appear in `docs/FORMALIZATION.md` only as inventory rows.

The generic `eq_id_of_commute` concerns unital endomorphisms and `eq_id_of_visible_scale` does not need surjectivity of the exponential, so no bijectivity or strong-summation hypothesis should be silently added to the mathematical self-embedding rigidity statement. The canonical-surreal conclusions, including the answer to KKS Question 5.4, still need the canonical exponential for an unconditional Lean specialization [13, 35, 36]; the existing finite and infinitesimal exponentials do not supply that global structure. The omnific automatic strongness theorem and proper-embedding classification were not located as mapped Lean proofs; appearance in the ledger’s manuscript inventory is not implementation, and source-level inspection does not certify every imported dependency.

Source 06’s suggested order of continuation:

- (1) generalize exponent transport from ordered additive equivalences to ordered additive embeddings, proving support transport and preservation of lower-universe-small support before bundling the ring embedding and its valuation formula; surjectivity should appear only in the inverse theorem;
- (2) formalize gap_τ and its separated cosets, giving the admissibility proof for coefficient sections before any iterated sum, then integer-part preservation and reflection from block signs;
- (3) implement the scalar detector ([lemma 58.4](#)) with its countable triangular witness; constant-term covariance plus surjectivity then gives automatic strongness by finite scalar equations, while the

proper-embedding version must keep a separate target-test hypothesis;

- (4) build the canonical global exponential, related exactly to the existing finite and infinitesimal operations, before specializing the generic rigidity API (the omega-map is not a substitute), with universe-lift compatibility and small-support statements at each step.

A compatible family of typed constructions is a foundation for class-level claims, not permission to form a set containing all surreal self-maps.

62 Research questions of source 06

Source 06 notes that several broadly phrased questions are already answered in the collection or the literature: proper field self-embeddings exist, ordered-field elementarity is automatic, omnific automorphisms are strongly $\mathbb{R}$ -linear, and exact-value exponential self-embeddings are rigid. Its statuses describe the sources it examined, not a worldwide priority or exhaustive-literature claim; each target has a first step.

Research question 62.1 (Elementarity of coefficient-fixing pair embeddings [06]). Which strong exact-monomial, $\mathbb{R}$ -fixing self-embeddings are elementary in $(\mathbf{No}, +, \cdot, <, \mathbf{Oz})$?

This is the collection's `opa:as:q:elementary`. Moving a real coefficient is an obstruction (theorem 58.10), but fixing $\mathbb{R}$ is not known to suffice; $J_{P_{\pm}}$ are test cases. First step: a formula with a finite parameter tuple separating source and image, or a relative quantifier elimination or extension theorem; real-closed-field quantifier elimination does not decide the expanded language.

Research question 62.2 (Countable strongness for proper pair embeddings [06]). Can a proper field self-embedding fixing $\mathbb{R}$ and preserving and reflecting $\mathbf{Oz}$ fail to preserve a countable Hahn sum?

Automorphisms are covered by theorem 58.5; proper embeddings may lack target detectors, and the measurable first-failure theorem assumes countable strength, so it does not settle this, with or without measurable cardinals. First step: characterize when the image supplies enough constant-term pairings to detect a countable nonsummability witness.

Research question 62.3 (Removing the exact-monomial assumption [06]). Classify strongly additive omnific-preserving proper embeddings whose images of monomials may have lower terms.

Theorem 58.7 answers this in its exact-monomial category; the contracting-derivation examples (example 56.5) show that the category misses automorphisms. Intermediate problem: impose a uniform triangular form on monomial images and prove a summability criterion, checking both well-ordered support and finite coefficient incidence. Compare the base question “Beyond monomial images” in section 15.

Research question 62.4 (Omega-field automorphisms and strong proper maps [06]). Do nonidentity automorphisms of the field $\mathbf{No}$ commute with the omega-map? Which proper such embeddings can be constructed without large-cardinal assumptions, and which can be strong?

The automorphism question is [5, Questions 5.6–5.7]. Exact-value preservation forces identity (theorem 10.3) but does not classify the value actions. The measurable construction (proposition 60.9, theorem 24.2) already gives a *proper, nonstrong* omega-field embedding under its assumption, even preserving exp and simplicity; it gives no automorphism or strong example. Within strong monomial lifts the equation $\tau = H_{\tau}$ is a starting point, but generalized epsilon numbers prevent a naive birthday induction. Compare the base question “ ω -map compatibility”.

Research question 62.5 (Existence of exponential lifts [06]). Which ordered additive automorphisms of the natural value group arise from exponential-field automorphisms of **No**?

Uniqueness follows from exact-value rigidity; existence is the classification problem. Source 06 records that the repository rules out positive rational dilations other than the identity and changes confined to bounded value layers [13]. First step: a criterion via the logarithmic structure of monomials, then a test of whether compatible data on elementary birthday stages lift coherently; for proper embeddings, images and uniqueness separately. Compare the base question “Characterize exponential value actions” and EAR q:image.

Research question 62.6 (Realization of differential exponential defects [06]). Which real-valued additive maps vanishing on $\mathcal{O}$ occur as $\alpha_j(x) = \log(j(\exp x)) - j(x)$ for a global differential self-embedding?

Proposition 59.6 gives necessary identities and a composition law and characterizes zero defect and real-power compatibility; Kaplan’s transseries constructions realize substantial local families [29], but their extension by differential homogeneity need not preserve global strongness or powers. First step: an extension theorem for a specified defect on a set-sized differential submodel, then a coherence analysis.

Research question 62.7 (Joint exponential and differential embeddings [06]). Classify maps preserving both $\exp$ and the fixed ∂_{BM} , with optional strong summation and omnific preservation.

$j(\omega) = \omega + c$ is necessary for differential embeddings and the defect must vanish. Transseries translations are compatible with both structures on their domain; extending them globally with both properties is stronger than the separate homogeneity theorems. Focused target: a coherent real-parameter family of global extensions; the obstruction to global composition does not by itself exclude such a restricted family. A related collection question, lce:cs:q:derivations, asks whether ∂_{BM} is absolute under the inner-model comparison of the measurable construction; that naturality needs an analysis of its defining recursion and log-atomic data and does not follow from field or exponential elementarity.

Research question 62.8 (When does a field lift preserve simplicity? [06]). Give necessary and sufficient conditions on an ordered additive embedding τ for H_τ to preserve prefix simplicity.

The lifting theorem alone does not answer this; simplicity preservation by the inner parametrization A_{P_+} must not be passed to $H_{A_{P_+}}$ without proof. Approach: translate the sign-expansion comparison theorem for normal forms into local conditions on τ , first on two-term forms. Source 06 calls this a targeted program, not an assertion that no special cases are known. Compare the base question “Field and simplicity together” and source 05’s “Simultaneous field and simplicity embeddings”.

Research question 62.9 (Stabilizers of large parameter classes [06]). Compute the common fixed field of the automorphisms fixing every real and every ordinal, with or without omnific or strong-sum constraints.

Example 57.1 shows the stabilizer is nontrivial and corollary 57.2 gives a moved transcendental element. Source 06 records that the collection computes several fixed fields for specified shear subgroups [31] (unchecked here); those differ from the fixed field of the whole stabilizer. First target: order reindexings supported in one interval between ordinal parameters. Compare the base question “The stabilizer structure”.

[write] Partial information in this report: by theorem 7.4 the common fixed field of the strong automorphisms fixing $\mathbb{R} \cup \mathbf{On}$ contains no element moved by some $J_{g\beta}$, and by corollary 57.2 it omits $\omega^{\omega^{1/2}}$; it contains the relative real closure of the field generated by $\mathbb{R} \cup \mathbf{On}$ (proposition 8.2). Not answered.

Research question 62.10 (Images beyond the full normal-form categories [06]). Which cofinal field self-embedding images are closed in the ambient order topology? Which proper images can be dense?

Noncofinal images are uniformly discrete and closed (remark 56.7); full normal-form subfields and strong exact-monomial images are closed (opa:as:thm:closed, theorem 9.4). These do not classify cofinal, possibly nonstrong images. Ordinary sequences cannot detect the distinction in No (proposition 9.1); neighbourhood criteria or justified class-indexed approximations are required. Compare the base question “Closed images beyond strong monomial maps”.

Research question 62.11 (Compatible local data [06]). Characterize the set-stage embedding systems whose union is an available global class embedding preserving the chosen operations.

Proposition 60.2 supplies compatible restrictions of a given map, not a selection of compatible maps from unrelated existence claims. A useful formulation fixes a class-coded cofinal chain of stages, specifies restriction maps, and proves the extension property at successors and the small-support property at limits. Compare the base question “Universe and cutoff absoluteness”.

Formalization projects (source 06). The most direct are the general proper normal-form lift, the prefix-derived $J_{P_{\pm}}$ with their topology criterion, the finite algebraic proof of definability rigidity, and the explicit ordinal-fixing automorphism J_h . The defect identities of proposition 59.6 can be proved abstractly for an ordered differential-exponential field with real constant field before any surreal instantiation. The canonical exponential interface, automatic omnific strongness, and the measurable first-failure theorem need distinct further work and should not be bundled into one completion claim.

63 Status and non-claims of source 06

Source 06’s own audit, README and the batch dossier record the following; their substance is kept here.

- *Repository pin and access.* Repository material was pinned to 79049d58c (full identifier in section 54) and external sources checked on 3 October 2026. All collection labels it cites exist (verified for this report at e1395c66c, by which every report and Lean file it read is unchanged).
- *Synthesis, not discovery.* Results already in the repository are attributed as such; exact-value exponential rigidity and the proper omnific embeddings are existing repository results, treated as a synthesis rather than a second claim of discovery. A consequence proved in source 06 is not thereby claimed new in the literature. The normal-form lifting method is not claimed new.
- *Lean.* No new Lean theorem is claimed. The implementation assessment is a source audit (a proof term distinguished from an inventory entry, a generic result from a canonical-surreal instantiation, an automorphism API from a proper-embedding theorem); no fresh repository build, kernel recheck or transitive axiom audit was run.
- *What was reviewed.* The proofs of the two-level lifts, the ordinal-fixing automorphism, the topology criterion, scalar detection, the defect identities and birthday localization were independently reviewed during preparation; the final source compiled with pdfLaTeX with no unresolved references, warnings, or overfull/underfull boxes, and the 47-page PDF was rendered and visually reviewed; the archive was checked against its SHA-256 sums. (Not reproduced for this report; no PDF of source 06 is staged.)
- *Conventions.* In the decreasing ω -exponent convention $\sum_{n \geq 1} \omega^{1/n}$ is an admissible normal form and $\sum_{n \geq 1} \omega^{-1/n}$ is not; in the increasing t -exponent convention $\gamma \mapsto \gamma + 1$ is the contracting

valuation shift; coefficient families are reindexed under composition, so their composition with exponent permutations is generally semidirect, not direct. Source 06 says its explicit formulas avoid the opposite signs, and an unqualified direct-product reading, “in the corresponding examples and factorization display” of [5]; these are observations about concrete formulas, not a claim that the structural program of [5] fails. (Not checked for this report; compare the sign remarks of source 07.)

- *Errata.* The 2001 van den Dries–Ehrlich paper has an erratum correcting a quantitative support bound, leaving the closure results used here unchanged [3]; the erratum to the 2021 Ehrlich–Kaplan paper corrects reference numbering [7].
- *Open, not resolved.* The unrestricted omega-field automorphism problem, the expanded omnific elementarity question, countable strongness for proper pair embeddings, and the realization of global differential defects.
- *Conditionality.* Existing large-cardinal constructions are stated with their assumptions (proposition 60.9); the exponential and differential homogeneity consequences (proposition 59.2, proposition 59.7, proposition 59.8) rest on the cited class frameworks and theorems; local or intermediate-field composition results are not promoted to a global composition law on **No**.
- *Limited assertions.* The Taylor automorphism is “not asserted to preserve **Oz**” (it does not, remark 56.6); a research question is a proposed direction, not a claim that it is open in the whole literature; no repository file was modified by source 06.

[write] Corrections and qualifications made in this Part: credits added for re-proofs of `saut:prop:discrete`, `saut:thm:monomial`, `saut:thm:coeffflow`, `saut:thm:expfaithful` and `saut:thm:orderedhomogeneity`, which source 06 does not cite; its Proposition 4.1 recorded as covered by theorem 7.4 and proposition 8.1; its Corollary 6.5 recorded as weaker than theorem 11.7; the comparison sentence after its Theorem 4.3 read beside theorem 7.4. No mathematical error was found in source 06.

Part VI

Source 07: Order, simplicity, normal forms, analytic structure, and foundations without paradoxes

64 Source 07 at a glance

Source 07 is the manuscript *Self-Embeddings of the Surreal Numbers: Order, simplicity, normal forms, analytic structure, and foundations without paradoxes*, a “research article prepared for Vladimir Reshetnikov” dated 3 October 2026 (42 pages; one standalone 2,944-line LaTeX source with 22 bibliography entries and 18 research questions; no code, no data, no checksum file). It inspected the repository at the same commit as the base, `b0d4af4f30b24529fec877e2ad4a5952888146e9`. Its foundational setting is Gödel–Bernays class theory with Global Choice (GBC) for the back-and-forth and homogeneity arguments, ZFC formula schemes for the explicit maps, set-length sign sequences, and set-supported normal forms written both with large monomials ω^a and with small monomials $t^g = \omega^{-g}$. Its main contributions beyond the base are an exact criterion for convex images of order-and-simplicity embeddings (proposition 65.1), lifts of arbitrary increasing ordinal maps (corollary 65.3), the continuity

criterion for an embedding between arbitrary ordered fields (theorem 67.1), a truncation characterization of strong exact-monomial maps (proposition 66.4), a conditional classification of the strong exponential self-embeddings of the omega-series field (theorem 68.5), and, in the pure field language under GBC, a homogeneity theorem with an omitting-types refinement that determines exactly which elements are unavoidable (theorem 70.2, proposition 70.3, and corollary 70.4). It also found the false sentence in docs/surreal/omnific-preserving-automorphisms that was corrected before this report was placed (section 74).

Renames. Source 07's symbols are renamed to the report's notation as follows.

Here	Source 07	Reason
$b(x)$	$\ell(x)$	Report notation for the birthday (length).
$x \preceq_s y, x \prec_s y$	$x \preceq_s y, x \prec_s y$	Report macros; same relation.
$P_p[\mathbf{No}]$	C_p (prefix cone)	Report writes the image of P_p ; C is the base's class $C = \bigcap_n (-1/n, \infty)$ and 07's C_g is substitution.
j (a double-order embedding)	F	F is an ordered field in theorem 67.1; 07's F_τ is renamed below.
D_w (block map with words $w = (w_-, w_+)$)	B	$B = \mathbb{R}((\omega^{\mathbf{No}_{\geq 0}}))$ in the base ((8.2)); D_m, D_δ are the repetition special cases.
σ_η (tree automorphism)	T_η	T is the base's Taylor automorphism.
A_f, f	A_h, h	Report notation for the additive lift.
H_τ	$F_\tau (= A_\tau)$	Report notation for the field lift; A_f is not a field embedding unless f is additive.
$H_{\tau, \chi}$	$F_{\tau, \chi}$	Report notation (base theorem 6.4).
S_c	D_c	Exponent dilation; D_m is sign repetition and $D(a)$ displacement.
$\mathcal{T}_{d, \delta}$	$T_{d, \delta}$	As in omnific-preserving-automorphisms; the base's T is the case $\delta = 1$.
$[a]_0$	$\text{ct}(a)$	Constant coefficient, as in base (8.1).
gap_τ	G_τ	G is the base's group $\mathbb{R}((\omega^C))$; the collection writes gap_τ .
H_A (field lift of an additive A)	J_A	J is reserved for double lifts $J_f = H_{A_f}$.
$J_{P(-)} = H_{A_{P(-)}}$	J, A, h of 07 Thm. 5.2	07's $h(s) = (-) \frown s$ is the prefix map $P(-)$; its double lift is a J_f .
$\Omega(x) = \omega^x$	$W(x)$	Report notation for the omega-map.
$\mathcal{A} = \mathbb{R}\langle\langle\omega\rangle\rangle$	$\mathcal{A}$	Kept; not the additive lift A_f (Part V uses its own macro for it).
$\partial_{\mathcal{A}}, \partial_{\mathcal{A}, g}$	D, D_g	D is the displacement in the base.
j_c (differential automorphism)	J_c	J is reserved for double lifts.
$\mathbb{Q}(A)^{\text{rc}}$	K_A	K_∞ is the iterated-image core.
$x _U$	$x _U$	Truncation to a final segment U ; base $T_{>c}x = x _{(c, \infty)}$.
$\mathcal{J}_j$	$J(x) = j(x)$	Large-cardinal map from $j : V \rightarrow M$.

Correspondence. Every numbered item of source 07, in its own numbering, and its named unnumbered items, with the place where it is printed in this report.

07 item	Content	Where printed
§1	Table of interpretations by preserved structure	Base table in section 2 (same content).

07 item	Content	Where printed
§1.1	Three readings of “preserving a canonical feature”; four readings of preserving a subclass	Remark 64.1 (printed here); valuation part = base section 10.
§1.2, §2	Imported versus proved; sign, normal-form and size conventions	Base section 3 (same conventions); 07’s status sentences in section 76.
Prop. 3.1	Prefix maps P_p	= proposition 4.2 (same proof).
Prop. 3.2	Block maps $D_w, b(D_w s) = \mu \cdot b(s)$	Constant-word case of theorem 44.1 (home 05); repetition case = proposition 4.3, proposition 18.4 (home 02). Example $D_2(1/2) = 5/4 = \text{base}$.
Prop. 3.3	Convex images: no set-sized cofinal/coinitial subset	Proposition 65.1 (printed here; after [4, Prop. 4.18]).
Thm. 3.4	Coding of all double-order embeddings by root, successor and limit words	= theorem 44.1 (home 05); 07’s chain-suprema remark in remark 65.2.
Cor. 3.5	Every increasing $h : \mathbf{On} \rightarrow \mathbf{On}$ lifts with $b(js) = h(b s)$	Corollary 65.3 (printed here).
Thm. 3.6	Proper double-order embedding fixing a set $A, \mathbb{R}$ and $\mathbf{On}$	= proposition 18.5 (home 02) applied to the set $A \cup \mathbb{R}$; same method; 07’s example in remark 65.4.
Thm. 3.7	Birthday rigidity	= theorem 4.5 (same proof).
Cor. 3.8	$\text{Aut}(\mathbf{No}, <, \preceq_s) = 1$	= proposition 4.6 (same proof).
Cor. 3.9	Initial image of a double-order embedding $\Rightarrow$ identity	Base theorem 5.2 with proposition 4.6; second route in section 65.
§3 remarks	$x \preceq_s j(x)$ for all x forces identity; tree automorphisms σ_η ; relative brackets	Remarks 65.5 to 65.7 (printed here); ambient brackets = proposition 4.7.
Thm. 3.10	$\text{Fix}(j) = \bigcap_n j^n[\mathbf{No}]$ [4, Prop. 5.2]	= theorem 55.8 (home 06); 07’s shorter proof as a second route in section 65.
§3 examples	$\text{Fix}(P_p) = P_{p^\omega}[\mathbf{No}]$; a fixed-point-free double-order embedding	Example 65.8 (printed here).
Prop. 4.1	Field embeddings fix real algebraics, $f(r) = r + \varepsilon_r$, value action τ_f	= proposition 10.1 and proposition 8.2; collection <code>opa:as:lem:natural</code> , <code>saut:lem:order</code> .
Def. 4.2	Summable families, strong additivity	= base definition of strong summability (section 3).
Thm. 4.3	Additive lift, image, composition; field embedding iff f additive	= theorems 6.1 and 6.2; the converse clause in remark 66.1.
Ex. 4.4	Exponent dilation $S_c, S_c S_d = S_{cd}$, not Ω -compatible	= base S_c (proposition 9.6); $\text{Fix}(S_c) = \mathbb{R}$ by (6.9).
§4.2 text	Weighted lifts $H_{\tau, \chi}$ and their converse	= theorem 6.4.
Prop. 4.5	Fixed points of Hahn lifts	= (6.4), (6.9), (6.13) (same proof).
§4.2 example	Character $\chi(a) = 2^{ a _0}, \omega \mapsto 2\omega$	Example 66.2 (printed here; compare base example after proposition 10.1).
§4.3 text	$H_{\tau, \chi}$ preserves and reflects $\mathbf{Oz}$	= proposition 8.1.
Ex. 4.6	Proper double lift from $\mu(a) = a + \sqrt{a^2 + 1}$; $\text{Fix} = \mathbb{R}$	Collection display <code>opa:as:eq:mus</code> (“ μ_{13} ”) and <code>opa:as:lem:inner</code> ; (58.8) (home 06).
Thm. 4.7	Nonidentity strong automorphism fixing a set, all ordinals, preserving $\mathbf{Oz}$	Covered by theorem 7.4 (automorphism clause) with proposition 8.1; see remark 66.3.
Prop. 4.8	Strongly additive $\iff f = H_\tau \iff$ truncation compatible	Proposition 66.4 (printed here).
Prop. 4.9	Taylor flow $\mathcal{T}_{d, \delta}$, strong automorphism moving reals	Collection <code>saut:thm:coeffflow</code> , display <code>opa:as:eq:taylor</code> ; base T is $\delta = 1$ ((10.3)); Part V’s remark 56.6. Failure of $\mathbf{Oz}$ in remark 66.5.
Thm. 4.10	Coefficient-gap classification of strong exact-monomial $\mathbf{Oz}$ -embeddings	= <code>opa:as:thm:classification</code> (with <code>opa:as:def:gap</code> , <code>opa:as:lem:gap</code>); theorem 58.7 (home 06). Sharpness example in remark 66.6.
§4.8	Non-strong automorphism moving $\sum_n \omega^{1/n}$	(56.4) (home 06); 07’s topological comment in remark 66.7.
Thm. 5.1	Continuity $\iff$ cofinality for $j : F \rightarrow K$	Theorem 67.1 (printed here; HOME).

07 item	Content	Where printed
§5.1 text	$\mathbb{R} \hookrightarrow \mathbf{No}$ discontinuous; exponent criterion	Remark 67.2, (67.1).
Thm. 5.2	Strong bounded discontinuous copy, $\omega^{-1} < J(x) < \omega$	= theorem 45.3 (home 05); 07's map $J_{P(-)}$ as a second route in section 67.
Prop. 5.3	Set-sized sets closed discrete; set nets eventually constant	= proposition 9.1 (same proof); collection <code>saut:prop:discrete</code> . Class-indexed net in remark 67.3.
§6.1 text	$\omega^{[x]}$ versus $\Omega(x)$; exponential maps preserve powers	Remark 68.1 (printed here).
Prop. 6.1	Ω -field embeddings: $v(jx) = j(vx)$; with value fixation, identity	= theorem 10.3 and the remark after it (same proof; also source 05's Prop. 11.1). Induction caveat in remark 68.2.
Lem. 6.2	Unbounded displacement	= lemma 11.1, corollary 11.2; EAR <code>lem:amplify</code> .
Thm. 6.3	Exponential valuation rigidity for self-embeddings	= theorem 11.4 (same proof); EAR <code>cor:embedding</code> .
§6.2 text	Exponential automorphism with $\omega \mapsto \omega + 1$; abstract (F, E, w) mechanism	Proposition 59.2 (home 06; conditional); value-action witness in remark 68.3; abstract form = theorem 11.3, EAR <code>cor:embedding</code> .
Prop. 6.4	Value displacement of a nonidentity exponential embedding is cofinal and coinital	= theorem 11.5 (same proof); theorem 35.1; EAR <code>thm:cofinal</code> (automorphisms).
Prop. 6.5	Two exponential embeddings of (F, E, w) with equal w -action, one cofinal, are equal	= theorem 35.3 (home 04); base lemma 11.6 and theorem 11.7; improves EAR <code>cor:two-profiles</code> .
Prop. 6.6	∂_{BM} -embeddings: $j(\omega) = \omega + c$; Ω -map forces $c = 0$	First part = proposition 13.1, proposition 59.5 (home 06); the Ω clause is proposition 68.4 (printed here).
§6.4 text	Every real translation occurs in a differential automorphism j_c	Proposition 59.5 and the text after it (home 06; conditional on [26, 39]).
Thm. 6.7	Strong exponential self-embeddings of $\mathcal{A}$ are the substitutions C_g	Theorem 68.5 (printed here; conditional on imported results).
§6.5 text	Order reversal, $C_{2\omega}$, covariance law, domain caveat	Remark 68.6 (printed here).
Prop. 6.8	No exponential derivation with a uniform valuation-gain bound	= EAR <code>thm:derivation-valued</code> (more general there) and <code>cor:no-contraction</code> ; same amplification.
§6.6–6.7	Operator exponentials; intersection problems; [27, Thm. 8.4] boundary	Remark 68.7 (printed here); the BM obstruction is also discussed in Part V.
§7	Classes, coded families, ZFC/GBC/ETR	Section 69 (printed here, condensed; overlaps base section 3).
§8.1	Elementarity depends on the language; ε -stage exhaustion	= proposition 12.1; 07's exhaustion argument in remark 70.1.
Thm. 8.1	GBC: isomorphisms of set subfields extend to automorphisms	Theorem 70.2 (printed here; HOME); collection <code>saut:thm:orderedhomogeneity</code> .
Prop. 8.2	Embedding fixing a real-closed set subfield and omitting b	Proposition 70.3 (printed here).
Cor. 8.3	b unavoidable $\iff b \in \mathbb{Q}(A)^{\text{rc}}$	Corollary 70.4 (printed here; generalizes <code>saut:prop:fixedfields</code>).
§9	Set-sized fragments, saturation, universes	Section 71 (printed here, condensed).
§10	Set-theoretic elementary embeddings	Section 72 (printed here); collection <code>lce:prop:J</code> , <code>lce:thm:defect</code> .
Thm. 11.1	Club of invariant birthday stages for finitely many maps and operations	= proposition 60.2 with corollary 60.3 (home 06); 07's caveats in remark 73.1.
§11.2	Lean interface table and six formalization targets	Section 73 (printed here).
§12.1	Correction concerning Taylor supports	Section 74.1 (printed here, with the collection's correction).

07 item	Content	Where printed
§12.2	Two alleged sign slips in KKS v3 Examples 4.3–4.4; status of KKS Q5.4	Section 74.2 (source’s claim, not checked).
§12.3	What the verification establishes	Section 76.
Q13.1– 13.18	Research questions	Section 75 (all printed).

Remark 64.1 (Conventions of source 07). An *automorphism* is a bijective self-embedding whose inverse preserves the same structure, and for relational structures an embedding also reflects the named relations; one-way preservation is always stated separately. Preserving a subclass A may mean $j(A) \subseteq A$, $j^{-1}(A) = A$, $j(A) = A$, or pointwise fixation; these differ for a proper embedding. For $A = \mathbb{R}$ and a field embedding, setwise preservation $j(\mathbb{R}) \subseteq \mathbb{R}$ already gives pointwise fixation, by rational cuts inside $\mathbb{R}$; merely inducing the identity on the residue field gives neither. Likewise “preserving canonical monomials” ($j(\omega^a)$ is a monomial) is weaker than $j(\omega^a) = \omega^{j(a)}$, which links two levels of a normal form [07].

65 Order, simplicity and the sign tree [07]

Source 07 calls an embedding of $(\mathbf{No}, <, \preceq_s)$ a *double-order embedding*; its images are the surreal substructures of Bagayoko–van der Hoeven [4]. Its prefix maps, repetition maps, birthday rigidity and automorphism rigidity coincide with the base results of section 4. This section prints what it adds. Throughout, j denotes a double-order self-embedding of $\mathbf{No}$.

65.1 Convex images

Every set of surreals has strict upper and lower bounds, given by Conway cuts; in particular $\mathbf{No}$ has no set-sized cofinal or coinital subset. The following characterization is attributed by source 07 to [4, Proposition 4.18]; source 07 includes the order-theoretic argument to make the size qualification explicit.

Proposition 65.1 (Convex images of double-order embeddings [07]; after [4, Prop. 4.18]). *A convex subclass $C \subseteq \mathbf{No}$ is the image of a double-order embedding $\mathbf{No} \rightarrow \mathbf{No}$ if and only if C has no set-sized cofinal subset and no set-sized coinital subset.*

Proof. Necessity: transport a set cofinal or coinital in C back to $\mathbf{No}$, where every set has strict upper and lower bounds.

Sufficiency. First, every nonempty convex class has a unique simplest member. Choose $c \in C$ of minimal birthday. If c were not a prefix of some $d \in C$, their longest common prefix would either equal d or lie strictly between c and d ; in both cases it is a member of C (by convexity in the second case) with smaller birthday than c , a contradiction. Thus c is a prefix of every member, which also gives uniqueness.

Next, for sets $L < R$ contained in C there is a member of C strictly between them. If both are nonempty, their ambient Conway bracket lies in C by convexity. If only R is nonempty, the absence of a set-sized coinital subset supplies $c \in C$ below R , and $\{c \mid R\}$ is a witness in C . The case of nonempty L alone is symmetric, and the empty cut has a witness because $C \neq \emptyset$. The witnesses form a nonempty convex class and hence have a simplest member.

Recursively send each x to the simplest element of C between the already constructed images of L_x and R_x . The prefix and fork comparisons prove that this is a double-order embedding. For surjectivity, suppose $y \in C$ has minimal birthday among omitted elements. Its proper prefixes in C are all in the

image. Take the Conway cut of their preimages, separated according to their position below or above y . Each canonical option of this input is bounded on the appropriate side by a defining option, since otherwise it would be a simpler witness for that input cut. Hence y satisfies the inequalities that define the image of the input, and that image is a prefix of y . If it were a proper prefix, it would itself be one of the boundary elements of the defining cut, which is impossible. So the image is y , a contradiction. $\square$

In particular every set-defined open cut $\{y : L < y < R\}$ with $L < R$ sets is such an image: adjoining internal points to the appropriate boundary gives strict internal bounds. An open-looking description with class-sized boundaries does not supply this. For example, the class of finite surreals has $\mathbb{N}$ as a cofinal subset and so is not the image of any order self-embedding of all of **No** [07]. The prefix cones $P_p[\mathbf{No}]$ of proposition 4.2 are the simplest examples.

[write] The proposition is new to the collection (no report or other source manuscript states it). The attribution to [4, Prop. 4.18] is source 07's and was not checked against that paper.

65.2 Local coding data and ordinal lifts

Source 07's Theorem 3.4 describes every double-order embedding by a root word $p = j(0)$, nonempty successor words $w_{s,\varepsilon}$ beginning with ε , and possibly empty limit words v_s ; this is theorem 44.1, printed in Part IV. Its Proposition 3.2 (constant words w_-, w_+ of a common length μ , with $b(D_w s) = \mu \cdot b(s)$) is the constant-word case, and includes the concatenation products of [4, Section 3].

Remark 65.2 (Two continuities [07]). In the coding of theorem 44.1, j preserves the longest common prefix $s \wedge t$: incomparable inputs have images diverging immediately after $j(s \wedge t)$. Moreover j preserves ambient unions of nonempty set-sized prefix chains exactly when every limit word v_s is empty; preserving the union of the empty chain additionally requires $p = 0$. This prefix-tree continuity is unrelated to continuity in the numerical order topology: the sign doubling D_2 preserves chain unions, but every positive input maps above 1 and every negative input below -1 , so the inverse image of $(-1, 1)$ is the nonopen singleton $\{0\}$ (compare proposition 4.3). Prefix insertion P_p , whose image is convex, is an order-topological embedding.

Corollary 65.3 (Ordinal lifts [07]). *Every strictly increasing class function $h : \mathbf{On} \rightarrow \mathbf{On}$ admits a double-order self-embedding j of **No** with*

$$b(j(s)) = h(b(s)) \quad (s \in \mathbf{No}), \quad j(\alpha) = h(\alpha) \quad (\alpha \in \mathbf{On}).$$

Proof. Use the coding of theorem 44.1. Take the root word $+^{h(0)}$. At a node of length α , let $\mu_\alpha > 0$ be the unique ordinal with $h(\alpha) + \mu_\alpha = h(\alpha + 1)$ and replace the input sign ε by the word ε^{μ_α} of μ_α copies of ε ; it is nonempty because h is strictly increasing. At a node of nonzero limit length λ , append the all-plus word of length ν_λ , where $\sup_{\beta < \lambda} h(\beta) + \nu_\lambda = h(\lambda)$; this tail exists, possibly empty, because $\sup_{\beta < \lambda} h(\beta) \leq h(\lambda)$. Induction on $b(s)$ gives the birthday formula. All-plus inputs have all-plus outputs, which gives $j(\alpha) = h(\alpha)$. $\square$

Thus arbitrary strictly increasing ordinal behavior occurs under the two order constraints alone. No preservation of membership and no set-theoretic elementarity follows from such a lift [07].

Remark 65.4 (Invisible modifications [07]). Source 07's Theorem 3.6 is proposition 18.5 applied to the set $A \cup \mathbb{R}$, proved by the same method: a modification confined to one prefix cone. Source 07 takes an infinite limit ordinal γ above every birthday in A , puts $q = +^\gamma \smallfrown (-)$, and lets j insert one plus sign after q inside $P_q[\mathbf{No}]$ and be the identity outside; q is omitted, the cone misses A , every real, and every ordinal (its members have a minus sign at position γ). For fixing just $\mathbb{R}$ and **On** one can take $q = +^\omega \smallfrown (-)$, the sign sequence of $\omega - 1$. Hence no set of test values, even a whole birthday stage, certifies that a class double-order embedding is the identity. The statement asserts no compatibility with field operations.

65.3 Rigidity: initial images and prefix growth

The base proves that an order embedding with simplicity-initial image is onto (theorem 5.2), and that a double-order automorphism is the identity (proposition 4.6). Source 07’s Corollary 3.9 reaches the identity directly, without passing through surjectivity.

Second route to the identity conclusion after theorem 5.2 [07]. Let j be a double-order embedding with simplicity-initial image. Every proper prefix of $j(x)$ belongs to the image, and by reflection its preimage is a proper prefix of x . Thus j maps the chain of proper prefixes of x isomorphically onto that of $j(x)$, so $b(j(x)) = b(x)$ for every x . Apply theorem 4.5. $\square$

An initial *image* alone does not make an order embedding rigid: $x \mapsto x + 1$ is onto $\mathbf{No}$. Likewise “initial” in the literature on ordered-field embeddings must not be read as saying that the embedding preserves simplicity [07].

Remark 65.5 (Prefix growth forces identity [07]). An increasing map $j : \mathbf{No} \rightarrow \mathbf{No}$ with $x \preccurlyeq_s j(x)$ for every x is the identity. If $j(x) > x$, put $y = j(x) \smallfrown (-)$. Then y extends x by the same first new sign as $j(x)$, so $x < y$ and hence $j(x) < j(y)$; but $y \preccurlyeq_s j(y)$ forces $j(y) < j(x)$. If $j(x) < x$, use $y = j(x) \smallfrown (+)$.

Remark 65.6 (Tree automorphisms [07]). For any class assignment η of a sign $\eta(s) \in \{-1, 1\}$ to every node s , identifying signs with ± 1 ,

$$\sigma_\eta(x)(\xi) = \eta(x \upharpoonright \xi) x(\xi) \quad (\xi < b(x))$$

is a birthday-preserving automorphism of $(\mathbf{No}, \preccurlyeq_s)$; transfinite decoding recovers the input signs. Conversely every automorphism of $(\mathbf{No}, \preccurlyeq_s)$ has this form: it fixes the root, preserves or swaps the two children at each node, and preserves unions of prefix chains. It preserves numerical order exactly when every $\eta(s) = +1$, and reverses it exactly when every $\eta(s) = -1$, giving $x \mapsto -x$. This makes precise the base’s branch-swap discussion in section 4.

Remark 65.7 (Relative brackets [07]). For a double-order embedding with image I the correct naturality law is $j(\{L \mid R\}) = \{j[L] \mid j[R]\}_I$, the simplest witness *inside* I : a simplest witness in $\mathbf{No}$ is a prefix of every other witness, and the two relations transport this. The relative bracket can differ from the ambient one: sign doubling sends $1 = \{0 \mid \emptyset\}$ to 2, while the ambient $\{0 \mid \emptyset\}$ is still 1. Ambient bracket preservation forces the identity (proposition 4.7).

65.4 Fixed points

The identity $\text{Fix}(j) = \bigcap_{n < \omega} j^n[\mathbf{No}]$ for every double-order self-embedding is [4, Proposition 5.2], printed with a proof as theorem 55.8 in Part V. Source 07’s proof is shorter: it uses two predecessors instead of an infinite backward orbit.

Second route to theorem 55.8 [07]. Write K for the intersection. Fixed points lie in K , and injectivity gives $j[K] = K$. Suppose $x \in K$ is a nonfixed element of minimal birthday, and let $y = j^{-1}(x)$ and $z = j^{-1}(y)$, both in K . Simplicity embeddings do not decrease birthdays (proposition 4.6), so $b(z) \leq b(y) \leq b(x)$; a smaller birthday would make y or z fixed by minimality, and then x would be fixed. So the three birthdays are equal, the elements are distinct, and since j is increasing they form a monotone chain, say $z < y < x$ (the other case is symmetric). The common prefixes $m = x \wedge y$ and $n = y \wedge z$ have smaller birthday, lie in K (every $j^k[\mathbf{No}]$ is closed under common prefixes), and are therefore fixed. Preservation of common prefixes gives $j(n) = j(y) \wedge j(z) = m$, so $n = m$. But n lies strictly between z and y , and m strictly between y and x , which is impossible. $\square$

The intersection is defined by quantification over the set ω , so no collection of proper classes is formed; it may be empty [07].

Example 65.8 (Prefix maps and a fixed-point-free embedding [07]). For $p \neq 0$ let p^ω be the concatenation of ω copies of p . Then

$$\text{Fix}(P_p) = P_{p^\omega}[\mathbf{No}].$$

If $P_p(x) = x$, induction gives $p^n \prec_s x$ for every n , hence $p^\omega \prec_s x$; conversely $p \frown p^\omega = p^\omega$, so every extension of p^ω is fixed. This refines the base's least fixed extension c_j of theorem 4.9 for $j = P_p$.

On the other hand a double-order embedding can have no fixed point at all. In the coding of theorem 44.1 take root word $+$, every successor word equal to the single input sign, and every limit word $+$. Birthday induction gives $b(j(x)) = b(x) + 1$ (at a limit the lengths before the extra sign have supremum $b(x)$), so j has no fixed points; in fact $j(x) > x$ for every x , an ordinal α going to $\alpha + 1$. The obstruction is the extra sign at limit branches: internal suprema exist in the image but are not the ambient unions. This separates double-order embedding, prefix-chain continuity and existence of fixed points.

66 Normal-form lifts [07]

Source 07's two Hahn lifts, weighted lifts, fixed-field formula and omnific compatibility coincide with base theorems 6.1, 6.2 and 6.4 and proposition 8.1. It organizes, and cites, the embedding results of docs/surreal/omnific-preserving-automorphisms. This section prints its additional statements and examples.

Remark 66.1 (When the additive lift is a field map [07]). For an increasing $f : \mathbf{No} \rightarrow \mathbf{No}$, the additive lift A_f is a unital field embedding *exactly* when f is additive. Sufficiency is theorem 6.2. Conversely, a unital multiplicative A_f applied to $\omega^a \omega^b = \omega^{a+b}$ gives $\omega^{f(a)+f(b)} = \omega^{f(a+b)}$, so $f(a+b) = f(a) + f(b)$ by uniqueness of normal forms. Thus an arbitrary order embedding lifts once to an additive embedding and *twice* to a field embedding.

Example 66.2 (A constant-coefficient character [07]). The map $\chi(a) = 2^{[a]_0}$, with $[a]_0$ the constant coefficient of a , is a positive real character because coefficient extraction is additive. With $\tau = \text{id}$, $H_{\text{id},\chi}$ is a strongly $\mathbb{R}$ -linear automorphism fixing every real and every natural value, but $H_{\text{id},\chi}(\omega) = 2\omega$. By (6.13) its fixed field consists exactly of the normal forms whose exponents have constant coefficient 0. Fixing the value map and the coefficient field therefore still permits multiplicative changes of the chosen representatives of Archimedean classes. (The base's example after proposition 10.1 uses $[a]_{-1}$ instead and fixes ω .)

Remark 66.3 (Ordinal-fixing automorphisms fixing a set [07]). Source 07's Theorem 4.7 states: *for every set $B \subseteq \mathbf{No}$ there is a nonidentity strongly $\mathbb{R}$ -linear field automorphism of $\mathbf{No}$ fixing B and every ordinal pointwise and preserving $\mathbf{Oz}$* . Its construction modifies the exponent order only in a bounded window: with $c < 0$ below the two-level footprint $E(B)$ of (7.2), let μ_c be the identity outside $(c-1, c)$ and $\mu_c(c-1+u) = c-1+2u/(1+u)$ for $0 < u < 1$ (inverse $w \mapsto w/(2-w)$ on $(0, 1)$), and take J_{μ_c} . It moves $\omega^{\omega^{c-1/2}}$ to $\omega^{\omega^{c-1/3}}$. A parameter-free instance uses $\mu(x) = x/(x+2)$ on $(-1, 0)$ and the identity elsewhere: its double lift fixes every real and every ordinal and sends $\omega^{\omega^{-1/2}}$ to $\omega^{\omega^{-1/3}}$. The argument for ordinals does not need exponents smaller than the ordinal, so it covers epsilon numbers.

[write] Verified: this is covered by the base. The automorphism clause of theorem 7.4 gives, for every set A , a nonidentity strongly $\mathbb{R}$ -linear automorphism J_{g_B} fixing $A \cup \mathbb{R} \cup \mathbf{On}$; since $J_{g_B} = H_{A_{g_B}}$ is a weighted monomial embedding with $\chi = 1$, proposition 8.1 gives preservation and reflection of $\mathbf{Oz}$ and of the omnific floor. Source 07's window map μ_c differs from g_B only in modifying a bounded interval instead of a lower half-line. Part V's example 57.1 is the case $B = \emptyset$.

66.1 Truncation compatibility

For a final segment U of the exponent order write $x|_U = \sum_{a \in \text{supp}(x) \cap U} r_a \omega^a$; thus the base's $T_{>c}x$ is $x|_{(c, \infty)}$.

Proposition 66.4 (Strongness is truncation compatibility [07]). *Let $f : \mathbf{No} \rightarrow \mathbf{No}$ be an $\mathbb{R}$ -linear field embedding with $f(\omega^a) = \omega^{\tau(a)}$ for every a . The following are equivalent:*

- (i) f is strongly additive;
- (ii) $f = H_\tau$;
- (iii) $f(x|_{\tau^{-1}(U)}) = f(x)|_U$ for every x and every final segment U .

In (iii) it suffices to take $U = (b, \infty)$ and $U = [b, \infty)$ for all surreal b .

Proof. Here τ is an ordered additive embedding, by the converse part of theorem 6.4. Strong additivity applied to the family of normal-form terms of x gives (ii), and (ii) gives (iii) termwise. Conversely, subtract the identities (iii) for $U = [b, \infty)$ and $U = (b, \infty)$. The left side is 0 if $b \notin \tau(\text{supp } x)$, and is $r_a \omega^{\tau(a)}$ if $b = \tau(a)$ with $a \in \text{supp } x$; the right side is exactly the term of $f(x)$ at exponent b . This determines every coefficient of $f(x)$ and proves $f = H_\tau$, which is strongly additive by theorem 6.2. $\square$

The condition concerns commutation with truncations at *every* cut, including cuts in the gaps of $\tau(\mathbf{No})$; at $b = \tau(c)$ it is the base's covariance (10.4) with $\chi = 1$. Merely asking for a truncation-closed image is much weaker: the image of every automorphism is $\mathbf{No}$ [07].

[write] New to the collection. Compare `opa:as:cor:monodet` of `docs/surreal/omnific-preserving-automorphisms`: an **Oz**-preserving field automorphism fixing every monomial is the identity, because such automorphisms are automatically strong (`opa:as:thm:main`). The proposition replaces strongness, for embeddings sending monomials to monomials, by an equivalent truncation condition.

66.2 The Taylor flow and omnific integers

Source 07's Proposition 4.9 is the coefficient Taylor flow

$$\mathcal{T}_{d,\delta} \left(\sum_{g \in S} r_g t^g \right) = \sum_{g \in S} \sum_{n \geq 0} \frac{d^n(r_g)}{n!} t^{g+n\delta} \quad (\delta > 0),$$

for a derivation $d : \mathbb{R} \rightarrow \mathbb{R}$ with $d(b) = 1$ at a chosen transcendental b : a strongly additive field automorphism with inverse $\mathcal{T}_{-d,\delta}$ that fixes every monomial, every natural value and every leading coefficient, and sends b to $b + t^\delta$. This is `saut:thm:coefffflow` and display `opa:as:eq:taylor`; the base's T of (10.3) is the case $\delta = 1$, and Part V treats it in remark 56.6. Source 07's proof is the same support argument ($S + \mathbb{N}\delta$ is well ordered with finite fibers), with the binomial cancellation $\sum_{i=0}^k (-1)^i / (i!(k-i)!) = 0$ for $k > 0$ for the inverse.

Remark 66.5 (The flow is defined everywhere but does not preserve **Oz** [07]). The construction works on arbitrary normal forms: overlap between shifted supports causes finite coefficient addition, not failure of definition. What fails is omnific preservation. For the chosen b ,

$$\mathcal{T}_{d,\delta} \left(\frac{b}{2} \omega^\delta \right) = \frac{b}{2} \omega^\delta + \frac{1}{2} \notin \mathbf{Oz}, \quad \frac{b}{2} \omega^\delta \in \mathbf{Oz}.$$

Source 07 records that this corrects a sentence of `docs/surreal/omnific-preserving-automorphisms`; see section 74.1.

Source 07's Theorem 4.10 (the coefficient-gap classification of strongly additive field embeddings J with $J(t^g) = t^{\tau(g)}$ and $J(\mathbf{Oz}) \subseteq \mathbf{Oz}$, as the maps $J_{\tau,\rho}(\sum r_g t^g) = \sum \rho(r_g) t^{\tau(g)}$ with $\rho : \mathbb{R} \rightarrow \mathbb{R}((t^{\text{gap}_\tau}))$) is `opa:as:thm:classification` of `docs/surreal/omnific-preserving-automorphisms`, including its clause $J(\mathbb{R}) \subseteq \mathbb{R}((t^{\text{gap}_\tau}))$ without strongness, with the same proof (`opa:as:def:gap`, `opa:as:lem:gap`); Part V prints it as theorem 58.7. Source 07 states it for $\mathbf{No} \rightarrow \mathbf{No}$ and credits the repository.

Remark 66.6 (Sharpness of the gap criterion [07]). If $\text{gap}_\tau \neq 0$, choose $0 < \delta \in \text{gap}_\tau$ and $\rho(r) = \sum_{n \geq 0} d^n(r) t^{n\delta}/n!$; then $J_{\tau,\rho}$ moves b to $b + t^\delta$ while preserving and reflecting $\mathbf{Oz}$. For $\mu(a) = a + \sqrt{a^2 + 1}$, an order isomorphism $\mathbf{No} \rightarrow \mathbf{No}_{>0}$ with inverse $y \mapsto (y - y^{-1})/2$, the additive embedding $\tau = A_\mu$ has the purely infinite normal forms as image, so its fixed points are 0 only and $\text{Fix}(H_{A_\mu}) = \mathbb{R}$; every positive element of $\tau(\mathbf{No})$ is positive infinite, so $1 \in \text{gap}_\tau$. Hence $\mathcal{T}_{d,1} \circ H_{A_\mu} = J_{A_\mu,\rho}$ is an explicit proper $\mathbf{Oz}$ -preserving field embedding with $b \mapsto b + \omega^{-1}$. If instead $\tau(\mathbf{No})^{>0}$ is coinital, then $\text{gap}_\tau = 0$ and every embedding in the classification fixes the canonical reals. The purely infinite group is a subgroup of the exponent group, not a convex subgroup of the field: it contains ω and 0 but not 1.

[write] $\mu = \mu_{13}$ of display `opa:as:eq:mus` and its inner lift `opa:as:lem:inner`; the Taylor section and its admissibility criterion are `opa:as:lem:taylorsection` and `opa:as:thm:taylorcriterion`. Part V prints the same explicit map as (58.8).

Remark 66.7 (Strongness, continuity and Hahn sums [07]). Source 07's Section 4.8 uses the class homogeneity of the surreal ordered field to build a field automorphism fixing ω but moving $a = \sum_{n \geq 1} \omega^{1/n}$ to $a + \omega^{-\omega}$; this is the example (56.4) of Part V (pure field language; compare theorem 70.2). Source 07 adds that there is no conflict with order continuity of an automorphism: the finite partial sums of a do not converge to a in the order or valuation topology, because the first omitted monomial $\omega^{1/(m+1)}$ exceeds 1. A Hahn sum is specified by support and coefficients; its existence asserts no topological convergence of partial sums. Field operations, ambient order continuity and commutation with all Hahn sums are three separate requirements; a field automorphism has the first two and can fail the third. Normal-form prescriptions do not determine an arbitrary field embedding from its values on selected generators or from its value action.

67 Continuity and set nets [07]

Continuity of a class map f at a is the ordinary formula $\forall \varepsilon > 0 \exists \delta > 0 \forall x (|x - a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon)$ with class parameter f ; no object of all open subclasses is needed, and the same language applies to ordered fields of set or class size.

Theorem 67.1 (Continuity and cofinal image [07]; after `opa:as:thm:topology`). *Let F and K be ordered fields and $j : F \rightarrow K$ an order-preserving field embedding. The following are equivalent:*

- (i) j is continuous at 0 for the order topologies of F and K ;
- (ii) j is continuous everywhere;
- (iii) $j(F_{>0})$ is coinital in $K_{>0}$;
- (iv) $j(F)$ is cofinal in K .

If these conditions fail, $j(F)$ is a discrete subspace of K , although its intrinsic ordered-field topology is not discrete.

Proof. Suppose (i), and let $0 < \varepsilon \in K$. Choose $\delta > 0$ in F for this neighborhood; then $0 < j(\delta/2) < \varepsilon$, proving (iii). Conversely, given (iii) and ε , choose $d > 0$ in F with $j(d) < \varepsilon$; for $|x| < d$, order

preservation gives $|j(x)| = j(|x|) < j(d) < \varepsilon$, which is (i). Additivity transports continuity at 0 to every point, and continuity at any point back to 0, so (i) $\iff$ (ii). Conditions (iii) and (iv) are equivalent by reciprocation, $j(1/x) = 1/j(x)$.

If the image is not cofinal, choose $B > 0$ in K above every element of $j(F)$. By symmetry B bounds all absolute values in the image, and applying this to $j(1/x)$ gives $|j(x)| > B^{-1}$ for $x \neq 0$. So an ambient interval of radius B^{-1} about an image point contains no other image point. By contrast every intrinsic interval of radius $d > 0$ about 0 contains $d/2$. $\square$

[write] This is the home statement of the report for continuity versus cofinality. The case $F = K = \mathbf{No}$ is base [theorem 9.3](#) and `opa:as:thm:topology of docs/surreal/omnific-preserving-automorphisms`, where it is phrased through the value-group image $\tau_f(\mathbf{No})$; source 07 does not cite that theorem. The same statement is reproved in Part II (source 02's [Theorem 9.1](#)), Part III ([proposition 33.2](#)) and Part IV (source 05's [Theorem 7.1](#)). When the conditions fail, j is nowhere continuous, since continuity at one point gives (i).

Remark 67.2 (Subspace versus intrinsic topology [07]). An ordered-field isomorphism onto its image is always a homeomorphism for the image's intrinsic order topology; the theorem concerns the subspace topology from K . An automorphism of $\mathbf{No}$ is onto and so continuous; continuity does not distinguish the identity among automorphisms. Even the inclusion $\mathbb{R} \hookrightarrow \mathbf{No}$ is discontinuous: its image is bounded by ω , and the interval of radius $1/\omega$ about a real contains no other real. For a Hahn lift H_A of an ordered additive embedding A ,

$$H_A \text{ is order-continuous} \iff A(\mathbf{No}) \text{ is cofinal in } \mathbf{No}, \quad (67.1)$$

because cofinal $A(\mathbf{No})$ makes the monomials of the image dominate every surreal, while $A(s) < b$ for all s gives $|H_A(x)| < \omega^b$. Only leading exponents are used, not an exchange of limits with infinite sums (compare [theorem 6.2](#) and [corollary 9.5](#)).

Source 07's [Theorem 5.2](#) (a strongly $\mathbb{R}$ -linear proper field self-embedding fixing $\mathbb{R}$ with $\omega^{-1} < |J(x)| < \omega$ for $x \neq 0$, hence discrete image and nowhere continuous) is [theorem 45.3](#) of Part IV. Its map is different: the double lift of the one-sign prefix map $P_{(-)}$.

Second route to theorem 45.3 [07]. The prefix map $P_{(-)}(s) = (-) \smallfrown s$ is an order isomorphism of $\mathbf{No}$ onto $\mathbf{No}_{<0}$: every negative surreal begins with a minus sign, and deleting it (reindexing the remaining final segment by its order type) gives the unique preimage. By [theorem 6.1](#), $A = A_{P_{(-)}}$ is an ordered, strongly $\mathbb{R}$ -linear additive embedding whose image is the class of normal forms with only negative exponents, that is, the infinitesimals $\mathfrak{m}$; so A is an ordered additive isomorphism $\mathbf{No} \cong \mathfrak{m}$ and not a field map ($A(1) = \omega^{-1}$). By [theorem 6.2](#) the double lift $J = J_{P_{(-)}} = H_A$ is a strongly $\mathbb{R}$ -linear field embedding fixing $\mathbb{R}$, with image the normal forms whose exponents lie in $\mathfrak{m}$. For $x \neq 0$ the leading exponent of $J(x)$ is infinitesimal, so $|J(x)| < \omega$, and applying this to $J(1/x) = 1/J(x)$ gives $|J(x)| > \omega^{-1}$. The surreal ω is not in the image. The discreteness and discontinuity follow from [theorem 67.1](#). $\square$

Here the image is a bounded real-closed subfield of proper-class size: boundedness in the surreal order does not imply set size. Preservation of all Conway-summable families does not imply order continuity [07].

Source 07's [Proposition 5.3](#) is [proposition 9.1](#) (via the cut $\{0 \mid D\}$ below the set D of nonzero deviations; collection `saut:prop:discrete`).

Remark 67.3 (A class-indexed witness to discontinuity [07]). By [proposition 9.1](#), $1/n$ does not converge to 0 (the neighborhood of radius $1/\omega$ excludes every term), and every map $\mathbf{No} \rightarrow \mathbf{No}$ preserves the limits of all convergent set-indexed nets, since eventual constancy is preserved; such nets give no

continuity test. The full class has no set-sized local base at 0. Class-indexed nets do give tests without forming a collection of classes: $u_\alpha = \omega^{-\alpha}$ ($\alpha \in \mathbf{On}$) converges to 0 in the sense that for every $\varepsilon > 0$ there is α_0 with $0 < u_\alpha < \varepsilon$ for all $\alpha \geq \alpha_0$ (take $\alpha_0 > -b$ for b the leading exponent of ε ; ordinals are cofinal). For the map $J = J_{P(-)}$ above, $J(u_\alpha) = \omega^{-A(\alpha)} > \omega^{-1}$ because $A(\alpha)$ is a positive infinitesimal, so the image net never enters the ω^{-1} -neighborhood of 0. All of this uses only the class parameters J and u , set quantifiers and ordinal tail conditions.

68 Exponentiation, the omega-map and the derivation [07]

Source 07's displacement lemma, exponential valuation rigidity, value-displacement proposition and two-embedding uniqueness coincide with base lemma 11.1 and theorems 11.4, 11.5 and 11.7 and with Part III's theorem 35.3 (source 07's Proposition 6.5 is stated, as there, for an abstract ordered field F with a total increasing exponential E and a nontrivial convex valuation w). Its Proposition 6.1 is theorem 10.3 with the remark after it. Source 07 presents the rigidity theorem as a repository theorem (docs/surreal/exponential-automorphism-rigidity, cor:embedding), “not a new result claimed by the present article”. Throughout, strong additivity includes summability of the image family, and a strongly $\mathbb{R}$ -linear map fixes the distinguished coefficient field pointwise.

Remark 68.1 (Two exponential operations [07]). The omega-map $\Omega(x) = \omega^x$ parametrizes canonical monomials; Gonshor's $\exp$ is an increasing isomorphism $(\mathbf{No}, +) \rightarrow (\mathbf{No}_{>0}, \cdot)$. Write $a^{[x]} = \exp(x \log a)$ for the exponential-field power of a positive base. Then $\omega^{[x]}$ and $\Omega(x)$ are different functions: from $\exp(\omega) = \Omega(\omega)$ and $\log \omega > 1$, $\omega^{[\omega]} = \exp(\omega \log \omega) > \exp(\omega) = \Omega(\omega)$ [3, 26]. An exponential-field embedding preserves the powers, $j(a^{[x]}) = j(a)^{[j(x)]}$, since commuting with $\exp$ gives commuting with $\log$ on positive elements; this does not give $j(\Omega(x)) = \Omega(j(x))$. Conversely, the omega-map law forces $j(\omega) = \omega$ (as $\Omega(1) = \omega$ and $j(1) = 1$) but says nothing by itself about $\exp$.

Remark 68.2 (A tempting induction fails [07]). If j is strongly $\mathbb{R}$ -linear and commutes with Ω , then $j(\sum_i r_i \Omega(a_i)) = \sum_i r_i \Omega(j(a_i))$. This is a constraint, not a proof that $j = \text{id}$: the exponents a_i need not have smaller birthday than the original surreal, and at a generalized epsilon number $e = \Omega(e)$ the identity refers back to e itself. An induction needs a genuinely well-founded rank or a separate treatment of such fixed points. The existence and description of nontrivial omega-field automorphisms, and the extension of order automorphisms of the generalized epsilon numbers, are [5, Questions 5.6–5.7]; the displayed identity resolves neither. Source 07 notes that the pointwise-valuation clause is the self-embedding version of [5, Proposition 5.5].

Remark 68.3 (An exponential automorphism moving values [07]; conditional). Source 07, like Part V's proposition 59.2, obtains a nonidentity exponential-field automorphism J of $\mathbf{No}$ fixing $\mathbb{R}$ with $J(\omega) = \omega + 1$ (indeed $J(\omega) = g$ for any prescribed $g > \mathbb{R}$): by o-minimality, ω and $\omega + 1$ have the same type over $\mathbb{R}$; definable Skolem functions give an elementary isomorphism between the set-sized definable closures of $\mathbb{R} \cup \{\omega\}$ and $\mathbb{R} \cup \{\omega + 1\}$; and the strong homogeneity of the surreal exponential field extends it [3, 5]. This rests on that imported homogeneity theorem in its class framework and asserts no preservation of infinite sums. Source 07 adds an explicit witness that its value action is nontrivial: for $y = \exp(\exp \omega)$,

$$\frac{J(y)}{y} = \exp(\exp(\omega + 1) - \exp \omega) = \exp((e - 1) \exp \omega), \quad v(J(y)) - v(y) < 0,$$

because the exponent is positive infinite. So an automorphism can fix the value of ω while changing the value of a larger exponential expression, as theorem 11.5 requires. The mechanism is not specifically surreal: for (F, E, w) , the class $\{x : w(E(x)) = 0\}$ is a proper convex additive subgroup, bounded in F , and an exponential self-embedding fixing every w -value has its displacement in it (base theorem 11.3); the bound must live in the same field as the endomorphism.

Source 07's Proposition 6.6 states, for a differential field self-embedding j of $(\mathbf{No}, \partial_{\text{BM}})$ (using only $\ker \partial_{\text{BM}} = \mathbb{R}$ and $\partial_{\text{BM}}\omega = 1$ [8]), that j fixes $\mathbb{R}$ pointwise and $j(\omega) = \omega + c$ for a unique $c \in \mathbb{R}$: this is proposition 13.1 and Part V's proposition 59.5 (which adds $c_{j \circ k} = c_j + c_k$). Its additional clause is the following.

Proposition 68.4 (The omega-map removes the translation [07]). *Let j be a differential field self-embedding of $(\mathbf{No}, \partial_{\text{BM}})$, so that $j(\omega) = \omega + c$ with $c \in \mathbb{R}$. If j commutes with the omega-map, or merely sends every canonical monomial to a canonical monomial, then $c = 0$, that is, $j(\omega) = \omega$.*

Proof. With the omega-map, $j(\omega) = j(\Omega(1)) = \Omega(j(1)) = \Omega(1) = \omega$. Under the weaker hypothesis, $j(\omega) = \omega + c$ must be a monomial; for $c \neq 0$ its normal form $\omega + c\omega^0$ has two terms. $\square$

This is a necessary constraint, not global rigidity. Every real translation of ω occurs in a global differential automorphism j_c : substitution of $\omega + c$ is an exponential and differential automorphism of the field $\mathbb{T}$ of logarithmic-exponential transseries embedded in $\mathbf{No}$ with variable ω (inverse: substitution of $\omega - c$; chain rule since $(\omega + c)' = 1$; [27]), $\mathbb{T}$ is an elementary differential subfield [26, Theorem 1], and the homogeneity theorem for Liouville-closed, ω -free, newtonian H -fields with small derivation and constants $\mathbb{R}$ extends it [39]. Part V prints this argument after proposition 59.5. The extension is obtained in the differential language only: it need not preserve exp, Conway sums or the omega-map, and $c \neq 0$ already excludes the last. This is why differential automorphisms must be distinguished from the intersection of the differential and exponential categories [07].

68.1 The omega-series field

Let $\mathcal{A} = \mathbb{R}\langle\omega\rangle$ be the field of omega-series: the smallest subfield of $\mathbf{No}$ containing $\mathbb{R}$ and ω and closed under exp, log and all defined set-indexed sums. It is a proper class and a proper subfield of $\mathbf{No}$. Source 07 uses three imported results, which it distinguishes:

- (a) [27, Theorem 6.3]: for $g > \mathbb{R}$ there is a unique substitution $C_g : \mathcal{A} \rightarrow \mathbf{No}$, $C_g(f) = f \circ g$, strongly $\mathbb{R}$ -linear, preserving exp, with $C_g(\omega) = g$; composition is associative where defined.
- (b) [37, Corollary 3.20]: $(\mathcal{A}^{>\mathbb{R}}, \circ, \omega)$ is a group, so compositional inverses exist inside $\mathcal{A}$; this is stronger than the existence of substitutions alone.
- (c) [38, Theorem A]: an account that uses this inversion result and gives monotonicity in the substituted argument.

Theorem 68.5 (Strong exponential self-embeddings of omega-series [07]; conditional on (a)–(c)). *Assume the imported results (a)–(c). Every strongly $\mathbb{R}$ -linear exponential-field self-embedding $j : \mathcal{A} \rightarrow \mathcal{A}$ is C_g for the unique $g = j(\omega) \in \mathcal{A}^{>\mathbb{R}}$, and every such self-embedding is an automorphism of $\mathcal{A}$. Those commuting with ∂_{BM} are exactly the translations $C_{\omega+c}$, $c \in \mathbb{R}$.*

Proof. Order preservation and fixation of the reals give $g > \mathbb{R}$. Both j and C_g fix the real coefficients, send ω to g , preserve the field operations and exp, hence log, and commute with all the specified sums. The elements on which they agree form a subfield containing the generators and closed under these operations; minimality of $\mathcal{A}$ gives $j = C_g$. Let $g^{\text{inv}} \in \mathcal{A}^{>\mathbb{R}}$ be the inverse from (b). Associativity and the identity substitution give $C_{g^{\text{inv}}} \circ C_g = \text{id} = C_g \circ C_{g^{\text{inv}}}$, so C_g is onto, and evaluation at ω gives uniqueness of g . If C_g commutes with ∂_{BM} , evaluation at ω gives $\partial_{\text{BM}}g = 1$, so $g - \omega \in \ker \partial_{\text{BM}} = \mathbb{R}$. Conversely the chain rule on $\mathcal{A}$ gives $\partial_{\text{BM}}(f \circ (\omega + c)) = (\partial_{\text{BM}}f) \circ (\omega + c)$. $\square$

[write] New to the collection as a statement. It is conditional on the three cited results, which were not re-proved or checked here. Part V proves the restriction identity $j(f) = f \circ j(\omega)$ on $\mathcal{A}$ for global strong

exponential maps of **No** (proposition 59.10); source 07 adds surjectivity from Bagayoko’s inverses and the differential classification.

Remark 68.6 (Composition, covariance and the ambient domain [07]). Parameters compose in reverse order, $C_g \circ C_h = C_{h \circ g}$, so the strong exponential automorphisms of $\mathcal{A}$ form the opposite of the group $\mathcal{A}^{>\mathbb{R}}$ under composition, and the differential ones form the additive group of real translations. For example $C_{2\omega}$ is a strong exponential automorphism with inverse $C_{\omega/2}$, but it is not differential, since $\partial_{\text{BM}}(2\omega) = 2$. Requiring $j(\omega) = \omega$ makes every embedding in the theorem the identity. With $\partial_{\mathcal{A}} = \partial_{\text{BM}}|_{\mathcal{A}}$ the chain rule reads $\partial_{\mathcal{A}} C_g = (\partial_{\mathcal{A}} g) C_g \partial_{\mathcal{A}}$; as $\partial_{\mathcal{A}} g > 0$, the operator $\partial_{\mathcal{A},g} = (\partial_{\mathcal{A}} g)^{-1} \partial_{\mathcal{A}}$ is a derivation and C_g is a differential isomorphism $(\mathcal{A}, \partial_{\mathcal{A}}) \rightarrow (\mathcal{A}, \partial_{\mathcal{A},g})$. For real $\lambda > 0$, $\partial_{\mathcal{A}} C_g = \lambda C_g \partial_{\mathcal{A}}$ holds exactly when $g = \lambda\omega + c$ with $c \in \mathbb{R}$ (evaluate at ω , then use the constant field; the chain rule gives the converse). This changes the differential structure; it adds no automorphisms of the original differential field. The theorem classifies embeddings of $\mathcal{A}$, not of **No**: a global strong exponential map fixing $\mathbb{R}$ and ω fixes $\mathcal{A}$ by the generation argument but leaves the rest of **No** untouched by the proof, and $C_g : \mathcal{A} \rightarrow \mathbf{No}$ for general $g > \mathbb{R}$ need not be an endomorphism of $\mathcal{A}$, much less of **No**.

Source 07’s Proposition 6.8 (an exponential derivation D of **No** with a uniform bound $v(D(y)) \geq v(y) + \gamma$ is zero) is the case of `thm:derivation-valued` of `docs/surreal/exponential-automorphism-rigidity`, which needs neither the order nor convexity, proved by the same amplification; in particular ∂_{BM} is not globally contracting (`cor:no-contraction` there), while its smallness on the valuation ring is a local condition.

Remark 68.7 (Operator exponentials and the remaining intersections [07]). An operator exponential $\sum_n D^n(x)/n!$ of a derivation is not a general definition: summability and multiplicativity need hypotheses, and the previous paragraph excludes a derivation that respects `exp` and contracts every value uniformly. A valid flow must specify its domain and prove summability there (compare the controlled shift in (10.3)). The established intersections are: value fixation with exponential compatibility forces the identity (theorem 11.4); value fixation with the omega-map forces the identity (theorem 10.3); derivation preservation forces real fixation and $j(\omega) \in \omega + \mathbb{R}$; adding the omega-map fixes ω ; adding `exp` and strong additivity fixes all of $\mathcal{A}$. None of the last deductions, without value fixation, is a classification on all of **No**. The global composition programme also has a precise boundary: a well-behaved composition $\mathbf{No} \times \mathbf{No}^{>\mathbb{R}} \rightarrow \mathbf{No}$ is conjectured in [5, Remark 4.1], while [27, Definition 8.1 and Theorem 8.4] show that ∂_{BM} cannot support a global composition with their compatibility package (chain rule, monotonicity from the derivative sign); this excludes that package, not every composition and not individual embeddings. Source 07 lists as unresolved directions: the strong exponential self-embeddings of all of **No** and their action beyond $\mathcal{A}$; which differential automorphisms preserve `exp` globally; whether the omega-map, `exp`, ∂_{BM} and summation jointly force the identity without a value hypothesis; and cofinal versus noncofinal exponential embeddings with prescribed value action. The homogeneity, substitution and valuation theorems settle, respectively, an extension problem in one language, a generated-subfield problem and an identity-kernel problem; none substitutes for the others.

69 Foundations: objects and quantifiers [07]

Source 07 works with $\mathbf{No} = \bigcup_{\alpha \in \mathbf{On}} \{-, +\}^\alpha$, a proper class of set-length sign sequences with set-supported normal forms; there is no sign sequence of length **On**, and $\{L \mid R\}$ requires sets (take $L = \mathbf{No}$, $R = \emptyset$). Every set of surreals is bounded above and below by an ordinal exceeding the lengths of its members, which is why “saturated over every set of parameters” does not mean saturated over every subclass [3]. This agrees with base section 3; source 07 adds the following points.

Individual maps, coded families and collections. In GBC a map $J : \mathbf{No} \rightarrow \mathbf{No}$ is a class of ordered pairs; by class Replacement its restriction to a set has a set graph, which permits $J[S]$, transported cuts

and transported supports, and compositions, ranges and inverses of bijections are again classes. There is no class of *all* class maps; notation such as $\text{Aut}(\mathbf{No})$ is a predicate on class variables with its composition laws, the convention also used in docs/foundations-and-computation/surreal-fields-across-universes. A *specified* family is different: a class $F \subseteq I \times \mathbf{No} \times \mathbf{No}$, with I a set or a class of sets, codes maps F_i by its slices, so a set group acting on $\mathbf{No}$ or an ordinal-indexed family of embeddings needs no collection of classes. Quantifying over every class map uses second-order quantifiers; treating the whole collection as an object needs a further type level or a theory of hyperclasses. The same discipline applies to topology: a subclass U is open when each point lies in an interval inside U , and continuity is the local neighborhood condition.

ZFC, GBC and stronger recursion. Explicit formulas for embeddings can be handled in ZFC: for a fixed formula $\Phi(x, y, p)$ defining a unique output, preservation of an operation is a first-order assertion, and statements about all definable maps are schemes. Coding formulas by natural numbers does not remove this: a uniform evaluator for arbitrary set-theoretic formulas would be a forbidden definable truth predicate for the universe. One may instead use a fixed definable family, bounded formula complexity, or a stronger theory with the required satisfaction relation; the explicit constructions need none of these. For noncanonical global extension arguments source 07 uses GBC including Global Choice (the common convention for NBG in the surreal literature), which supplies an **On**-indexed enumeration of $\mathbf{No}$ and proves first-order *set-valued* transfinite recursion with class parameters: the history below any ordinal is a set, local recursions agree, and their graphs form a class by comprehension. This is not the principle ETR for class-valued stages, whose additional strength is analyzed by Gitman–Hamkins [20] (compare [10]); truth predicates for arbitrary class structures are a familiar source of that requirement. Kelley–Morse theory gives more than is needed, and a principle choosing a class witness for every set parameter is a further issue not assumed merely by naming KM [41, Section 2]. The extension proofs below choose individual sets and keep set-sized partial maps at every stage.

70 Elementarity and homogeneity in the field language [07]

Remark 70.1 (The language determines elementarity [07]). For $L_{\text{of}} = \{0, 1, +, -, \cdot, <\}$, quantifier elimination for real closed fields makes every unital field embedding $J : \mathbf{No} \rightarrow \mathbf{No}$ elementary (base proposition 12.1); evaluating the quantifier-free equivalent gives both a formula-by-formula reading and a uniform satisfaction definition for this particular structure, without a truth predicate for $(V, \in)$. It says nothing about birthdays, simplicity, the canonical reals, strong sums, the omega-map, exp or a derivation. For the exponential language, $(\mathbf{No}, \exp)$ is a model of the theory of the real exponential field [3] and Wilkie’s model completeness [9] makes exponential-field embeddings elementary. For the class carrier source 07 offers, besides the scheme reading, a set-model route: the set-sized exponential fields $\mathbf{No}_{<\varepsilon}$, ε an epsilon number, are elementary and form a cofinal exhaustion, so a tuple and the image of any set stage lie in a later stage and ordinary set-model elementarity gives the class statement, with no unrestricted class satisfaction theorem. (Part V makes the same observation for exponential homogeneity.)

Theorem 70.2 (Ordered-field homogeneity [07]; under GBC; after `saut:thm:orderedhomogeneity`). *Assume GBC with Global Choice. Every ordered-field isomorphism $f : A \rightarrow B$ between set-sized subfields of $\mathbf{No}$ extends to a class ordered-field automorphism of $\mathbf{No}$.*

Proof. First extend f to the relative real closures of A and B in $\mathbf{No}$. These are sets (set-many polynomials, each with finitely many roots), and the order-preserving isomorphism of real closures extending f is unique. So assume A, B real closed.

Extension step. Let $p : D \rightarrow E$ be an isomorphism of set-sized real closed subfields and $x \in \mathbf{No} \setminus D$. Then x is transcendental over D , since a real closed field has no proper ordered algebraic extension. Put

$L = \{p(d) : d \in D, d < x\}$ and $R = \{p(d) : d \in D, x < d\}$; these are sets, $L < R$, and $L \cup R = E$. The simplest separator $y = \{L \mid R\}$ lies outside E , hence is transcendental over E , and $x \mapsto y$ extends p to an ordered isomorphism $D(x) \rightarrow E(y)$: factor a polynomial over the real closed field D into linear factors and quadratic factors of constant sign; the signs of the linear factors are determined by the cut of x , which was transported exactly. Extend again to relative real closures.

Back-and-forth. Fix an **On**-indexed enumeration (a_α) of **No** (Global Choice). Starting from f , at stage α apply the extension step to put a_α into the domain if necessary, then apply it to the inverse partial isomorphism to put a_α into the range; take unions at limits. A union of an increasing chain of real closed fields is real closed, since each polynomial occurs at an earlier stage. For each α the partial fields, their isomorphism and the history below α are sets, so this is the set-valued recursion available in GBC. The class union is total, onto by the back steps, and preserves the ordered-field operations, because each finite computation occurs at one stage. $\square$

[write] This is `saut:thm:orderedhomogeneity` of `docs/surcomplex/surcomplex-field-aut` `omorphisms`, proved there by the same back-and-forth; source 07 does not cite it (it compares Ehrlich's absolutely saturated models [40] and the universes report). It is printed here as the report's home statement because source 07 makes the size bookkeeping and the separator choice explicit.

For example, the identity on $\mathbb{R}$ extends to an ordered isomorphism between the relative real closures of $\mathbb{R}(\omega)$ and $\mathbb{R}(\omega + 1)$ sending ω to $\omega + 1$, since both realize the same cut over $\mathbb{R}$; the theorem gives a nonidentity automorphism of **No** fixing every real and imposing none of the other structures studied in this report [07].

Proposition 70.3 (Omitting a prescribed transcendental element [07]; under GBC). *Assume GBC with Global Choice. Let $A \subseteq \mathbf{No}$ be a set-sized real closed subfield and $b \in \mathbf{No} \setminus A$. There is a field self-embedding of **No** fixing A pointwise whose image omits b .*

Proof. Run only the forth steps of the preceding proof, starting from id_A , and maintain that b is transcendental over the current range E . In an extension step at x , let S be the relative real closure of $E(b)$ in **No**, a set; with L, R as before put $C = \{s \in S : L < s < R\}$ and $y = \{L \cup C \mid R\}$, which exists because $L \cup C < R$. Then y realizes the cut (L, R) , and $y \notin S$, since $y \in S$ would give $y \in C$ and hence $y < y$. So y is transcendental over $E(b)$, and therefore b remains transcendental over the relative real closure of $E(y)$. At limit stages any algebraic relation would already occur earlier. The final domain is all of **No**, and b is transcendental over, hence absent from, the image. $\square$

[write] This is the pure-field, GBC-only analogue of the omitting theorems for o-minimal class models in Part III (theorem 34.2 and its omission lemma) and Part IV (theorem 49.2), which for the exponential language need GBC+ETR with a class satisfaction relation.

Corollary 70.4 (Exactly which elements are unavoidable [07]; under GBC). *Assume GBC with Global Choice. Let A be a set of surreals and $\mathbb{Q}(A)^{\text{rc}}$ the relative real closure of $\mathbb{Q}(A)$ in **No**. For a surreal b the following are equivalent:*

- (i) $b \in \mathbb{Q}(A)^{\text{rc}}$;
- (ii) every field self-embedding of **No** fixing A fixes b ;
- (iii) every field automorphism of **No** fixing A fixes b ;
- (iv) b lies in the image of every field self-embedding of **No** fixing A .

The quantification over maps is class-theoretic quantification, not an intersection over a set of all maps.

Proof. An embedding fixing A fixes $\mathbb{Q}(A)$. The finitely many roots in $\mathbf{No}$ of a polynomial over $\mathbb{Q}(A)$ form a finite ordered set mapped injectively and increasingly into itself, so each root is fixed; hence (i) implies (ii), (iii) and (iv). If $b \notin \mathbb{Q}(A)^{\text{rc}}$, proposition 70.3 applied to the set-sized real closed field $\mathbb{Q}(A)^{\text{rc}}$ gives an embedding fixing it whose image omits b , refuting (iv) and (ii). For (iii), let (L, R) be the cut of b over $\mathbb{Q}(A)^{\text{rc}}$ and $c = \{L \cup \{b\} \mid R\}$. Then $c > b$ and c realizes the same cut, so the ordered isomorphism over $\mathbb{Q}(A)^{\text{rc}}$ sending b to c extends through real closures and then, by theorem 70.2, to an automorphism of $\mathbf{No}$ moving b . $\square$

[write] New to the collection. The case $A = \emptyset$ of (i) $\iff$ (iii) is $\text{Fix}(\text{Aut}(\mathbf{No})) = \mathbb{Q}^{\text{rc}}$ in `saut : prop:fixedfields` of `docs/surcomplex/surcomplex-field-automorphisms`; the corollary extends it to an arbitrary set of parameters and to self-embeddings and their images. Compare base proposition 8.2.

This answers a parameter-constrained version of the self-embedding problem in the pure field language. Naming simplicity, **Oz**, `exp` or a derivation can change the equivalences, because the extension step no longer applies in the expanded language [07].

Remark 70.5 (Limits of the homogeneity method [07]). Theorem 70.2 does not start from a proper-class partial map: its extension step realizes a *set* cut, and fixing a proper subclass or meeting proper-class many constraints needs another argument. Nor does real closedness alone give the theorem for arbitrary proper-class target fields; the separator property is essential. The proof cannot simply be repeated after adding `exp`, a derivation or simplicity: polynomial factorization is what reduces the ordered-field extension step to a cut. Model completeness gives elementarity of maps that already preserve the structure; it neither constructs them nor realizes their expanded-language types from order saturation alone.

71 Set-sized fragments and universes [07]

For a set ordinal λ , $\mathbf{No}_{<\lambda} = \{x : \text{b}(x) < \lambda\}$ is a set, so its self-maps, automorphism group and order topology are ordinary set objects. It is a subfield of $\mathbf{No}$ exactly when $\omega^\lambda = \lambda$ (ordinal exponentiation), and for these epsilon cutoffs it is real closed and elementary in the surreal exponential field [3]. A regular uncountable κ is an epsilon number, and $\mathbf{No}_{<\kappa}$ realizes every cut with fewer than κ parameters (regularity bounds their lengths, and the simplest separator has length below κ); hence it is κ -saturated as a real closed field, since a nonalgebraic one-variable type over a real closed subfield of size below κ is determined by its cut. Its cardinality is $2^{<\kappa}$. If $2^{<\kappa} = \kappa$, ordinary back-and-forth of length κ extends every partial elementary map with domain of size below κ to an automorphism; for $\kappa = \omega_1$ the continuum hypothesis suffices, and an inaccessible κ is another sufficient choice. These are hypotheses for this saturation-to-homogeneity argument only, not necessary conditions for explicit self-embeddings; when $|\mathbf{No}_{<\kappa}| > \kappa$, κ -saturation alone does not justify a back-and-forth through the whole carrier [07]. Compare base (14.1) and proposition 14.1.

A global map need not preserve a preselected cutoff; Replacement only puts the image of a set stage inside a later stage. The club of invariant stages (remark 73.1) supplies cofinally many invariant cutoffs for specified maps and finitary operations; these need not be regular cardinals, preserving a carrier does not preserve its exponential, and closure under finitary operations does not give closure under unrestricted set-indexed Hahn sums.

For a transitive set universe U , $\mathbf{No}^U$ consists of the sign sequences and lengths in U : a proper class from inside U , a set from outside. For $U = V_\kappa$ with κ inaccessible it is $\mathbf{No}_{<\kappa}$; a general inner model may lack sign sequences of the outer universe, so its ordinal height does not determine its surreal carrier. Saturation does not transfer across universes: if a sign sequence s is new in an outer universe but all its proper prefixes are old (take a new one of least length), its left and right proper prefixes form an outer-set cut of old

surreals that no old surreal separates, since every separator has s as a prefix and restriction would recover s inside. The old field may therefore cease to be saturated over the outer universe's parameter sets; the universes report docs/foundations-and-computation/surreal-fields-across-universes investigates this. In type theory, a universe-polymorphic carrier lives one level up from its small option families; a theorem that every small cut has a separator must keep its smallness hypothesis and cannot be reapplied to the whole carrier. Source 07 recommends two formalization routes, explicit set cutoffs with closure and transport hypotheses, or an encoded background model of set or class theory, and warns that no large-cardinal hypothesis should be inferred from a universe annotation without stated semantics.

72 Set-theoretic elementary embeddings [07]

Under explicitly stated large-cardinal and amenability hypotheses, suppose $j : (V, \epsilon) \rightarrow (M, \epsilon)$ is elementary with $M \subseteq V$ a transitive inner class containing all ordinals. Applying j to sign-sequence codes gives $\mathcal{J}_j(x) = j(x)$, an ordered-field embedding of **No** by absoluteness of the set-based surreal arithmetic; here elementarity refers to the membership language, much stronger data than ordered-field elementarity. If $x : \alpha \rightarrow \{-, +\}$ then $j(x)$ has domain $j(\alpha)$ and $j(x)(j(\beta)) = x(\beta)$ for $\beta < \alpha$, which says nothing about the coordinates in $j(\alpha) \setminus j[\alpha]$; likewise for an infinite support S , $j(S)$ can strictly contain $j[S]$. So the map obtained by applying j to a whole normal-form code must be distinguished from a Hahn lift transporting each original term over the original index set. Conversely, an elementary self-embedding of the ordered field **No** yields no elementary embedding of (V, ϵ) ; Kunen-type inconsistency theorems concern the latter and do not prohibit the former [16]. Relating the two would need an explicit interpretation of sets and membership in an expanded surreal structure and a proof that the map preserves it [07].

[write] This is the subject of docs/foundations-and-computation/large-cardinal-embeddings-and-normal-forms: lce:prop:J proves that $\mathcal{J}_j$ is an ordered-field embedding fixing $\mathbb{R}$, commuting with the omega-map, preserving and reflecting prefix simplicity and elementary in the field language, and lce:thm:defect computes the exact defect between ambient and supportwise transport (the “critical-point defect analysis” that source 07 cites). All of it is conditional on the elementary embedding, not an existence theorem for large cardinals. Base section 3 makes the same separation.

73 A staged formalization programme [07]

Source 07 proposes proving local algebra first for arbitrary fields and ordered groups, and letting the surreal representation enter only where it supplies a cut, a normal form or an analytic operation, so that no global class claim is hidden in a type signature. Its Theorem 11.1 (a closed unbounded class of limit κ with **No**_{< κ} closed under finitely many named class functions and total finitary operations; for a field self-embedding, adding 0, 1, +, −, · and the inverse totalized by $0^{-1} = 0$ makes the stage a subfield invariant under the embedding; for an automorphism, adding its inverse makes the restrictions onto) is proposition 60.2 with corollary 60.3 of Part V, by the same rank-closure argument; the base gives the argument as prose in section 14.

Remark 73.1 (What the club theorem does not say [07]). It is a set-valued rank-closure argument. It does not assert that one cutoff is invariant under every class map, and it does not assume that all class maps form a class. A set-indexed family is accommodated only when a single class evaluation relation for the family is supplied; a family of arbitrary class functions cannot be inserted as a set parameter by notation. Closure under finitary operations does not imply closure under externally set-indexed Hahn sums, since a support can be cofinal in the cutoff; summation needs its own size and support

hypotheses. This is where a proof-assistant interface should separate an algebraic field embedding from an embedding of a specified summability structure.

Existing repository interfaces. Source 07 read the following declarations at the pinned commit, without a fresh Lean build. All of them exist at the current repository head, in `Surreal/HahnSeries/WorkspaceEmbedding.lean` and `Surreal/Foundations/SignSequenceHahnEmbedding.lean` (paths relative to `Algebra/SurrealNumbers/`).

Declaration	Role in the programme (source 07)
<code>workspaceEmbedding</code>	Transports a Hahn workspace along a strictly increasing additive exponent map; the starting point for the second normal-form lift.
<code>support_workspaceEmbedding,</code> <code>workspaceEmbedding_injective</code>	Identify the transported support and prove injectivity, needed for image descriptions and properness witnesses.
<code>mapExponentsFamily,</code> <code>hsum_mapExponentsFamily</code>	Preserve the support and finite-fiber conditions of summable families and prove commutation with Hahn sums.
<code>addVal_workspaceEmbedding</code>	Expresses the induced valuation map; to be kept separate from any theorem that this map is the identity.
<code>hahnEmbedding,</code> <code>hahnOrderRingHom</code>	Connect a small real Hahn workspace with the sign-sequence surreal field; a bridge from abstract series arguments to the repository's carrier.

Six targets. Source 07's sequence of new formalization targets:

1. the generic displacement identity and bounded-displacement rigidity for an ordered field endomorphism (finite algebra and inequalities only);
2. the exponential detector with an explicit convex valuation and one positive exponential scale outside its valuation ring, after which the surreal case needs only the interface for the canonical exponential and valuation;
3. birthday-nonincrease rigidity on ordinal-length sign sequences, separately from arithmetic, with the key lemma identifying the canonical prefix cut with a cone of extensions;
4. reuse of the Hahn workspace maps for double lifts, support images, composition, properness witnesses and the set-parameter stabilizer (the first lift needs a strictly increasing exponent map, the second also additivity);
5. continuity versus cofinality for arbitrary ordered-field embeddings (theorem 67.1), then its instance for double lifts, with continuity encoded locally or on a set carrier and no type of all open subclasses;
6. the cutoff closure theorem in a universe-relative development, recording exactly which global statements are represented by coherent families of smaller theorems.

Each target has a finite specification; none needs a datatype of all externally existing surreals or of all class functions. Conversely, checking a bounded model should not be described as verifying an unrestricted proper-class statement without a separate transfer argument [07]. Compare the base's layers in section 14.

74 Source audit and corrections of source 07

Source 07's repository inspection was targeted, at the pinned commit. It read, under `Algebra/SurrealNumbers/`: `docs/surreal/omnific-preserving-automorphisms` (normal-form

lifts, proper embeddings, coefficient drift; the portions used were checked and the main examples reproved); docs/surreal/exponential-automorphism-rigidity (the bounded-displacement argument and its exponential application, independently checked, and noted to include self-embeddings already, as `cor:embedding` there does); docs/foundations-and-computation/surreal-fields-across-universes (as background, not as a certificate for every global assertion); docs/foundations-and-computation/large-cardinal-embeddings-and-normal-forms (ambient versus supportwise transport; large-cardinal hypotheses never inferred from field embeddings); and the two Lean files of section 73 (signatures and proof bodies; no fresh full build). It did not read the whole repository, every preserved manuscript or every imported proof dependency.

74.1 A correction concerning Taylor supports [07]

The omnific report contains an accurate theorem that the coefficient Taylor map is a strong automorphism. A later sentence, after its explicit proper-embedding examples, said that the ordinary Taylor section cannot be applied to arbitrary normal forms because its integer shifts can meet. Taken as a claim about the termwise Taylor extension, this is incorrect. If S is a well-ordered small-monomial support and $\delta > 0$, then $S + \mathbb{N}\delta$ is well ordered and each target exponent has only finitely many decompositions $s + n\delta$; intersections are allowed, and finite multiplicity is the requirement. In the large-monomial convention the same statement concerns $S - \mathbb{N}\delta$. The proper exponent embedding in the coefficient-gap construction is needed for preservation of the omnific integers and for the block geometry, not for the existence of the Taylor automorphism (remarks 66.5 and 66.6).

[write] Confirmed and already corrected in the collection. The sentence was in Section 27.2 of docs/surreal/omnific-preserving-automorphisms, after its Theorem 27.3: “The ordinary Taylor section alone is not a map on arbitrary normal forms, since supports could meet its integer shifts”. It contradicted the report’s own display `opa:as:eq:taylor`, the flow $\mathcal{T}_{d,\delta}$ that `saut:thm:coeffflow` proves to be a strong automorphism defined on every normal form. Commit `f06e67d10` (3 October 2026), made before this report was placed, replaced it by: the ordinary Taylor flow alone does not give those maps, since $\mathcal{T}_{d,\delta}$ is a strong automorphism but never preserves **Oz**, and the proper exponent embedding separates the scales; followed by a dated correction note quoting the old wording and crediting this manuscript. No statement, label or numbering of that report changed, and no result was retracted. Source 07’s own wording (“no repository file was modified as part of preparing the article”) refers to its preparation and remains true.

74.2 Two sign conventions in the KKS preprint [07]

Source 07 reports that in the author PDF of [5] it inspected, the series displayed in Example 4.4 has exponents $-1/n$, an increasing sequence which is not summable under the stated Conway normal-form convention, and that $+1/n$ gives the decreasing-exponent series of the intended Puiseux argument (used in remark 66.7). It also reports that a displayed derivation in Example 4.3 shifts a small-monomial exponent by -1 while describing contraction for the valuation $v(t^g) = g$, whereas a $+1$ shift has the asserted direction. Source 07 calls these local formula issues and uses neither displayed expression.

[write] These are source 07’s claims about an external preprint. They were *not* checked against [5] for this report. Part V independently uses the decreasing support $\omega^{1/n}$ in its version of the Example 4.4 argument.

Source 07 treats [5, Question 5.4] (nontrivial exponential 1-automorphisms) differently from a notation issue: the repository gives a direct proof excluding them under the stated meaning of fixing leading terms (`cor:question54` of docs/surreal/exponential-automorphism-rigidity; base theorem 11.4), source 07 checked that argument, and it therefore does not list that question as open.

This is a mathematical check of the argument, not a claim of external peer review of the repository manuscript or that no other author has proved the same statement.

75 Research questions of source 07

Source 07 separates questions left open in the literature it checked from refinements suggested by its constructions; a proposed question is not a claim that its answer is unknown throughout the literature, and the first step is always to fix a language, a support convention and a foundational setting. Its immediate priorities are the generic rigidity formalization, the explicit double-lift interfaces, and a precise exponential lifting criterion. Base questions are those of section 15, cited by title.

75.1 The omega-map and analytic intersections

Research question 75.1 (Omega-compatible field self-embeddings [07]). Classify the unital field self-embeddings with $j(\omega^x) = \omega^{j(x)}$. Which are proper, and which are strongly additive? The automorphism problem is [5, Questions 5.6–5.7]. A tractable first target is an extension criterion for an order map on generalized epsilon numbers, with all the compatibility conditions imposed by the remaining normal forms.

[write] Compare the base question “ ω -map compatibility” and `opa:as:q:exp`. Conditionally on a nontrivial elementary $j : V \rightarrow M$ as in `docs/foundations-and-computation/large-cardinal-embeddings-and-normal-forms`, `lce:prop:J` supplies field self-embeddings commuting with the omega-map (section 72); this gives examples, not a classification.

Research question 75.2 (The image of the exponential value action [07]). Which ordered additive self-embeddings τ of the value group lift to an exponential field self-embedding? Rigidity controls the identity fiber and cofinality controls uniqueness between two lifts, but neither gives existence. Find necessary and sufficient conditions, first for a precisely specified family of τ .

[write] Compare the base question “Characterize exponential value actions” and `q:image of docs/surreal/exponential-automorphism-rigidity` (stated there for automorphisms).

Research question 75.3 (Removing the cofinality assumption [07]). If two exponential self-embeddings induce the same value map, must they coincide without a cofinal-image hypothesis? The missing step is that a value in one image must amplify a nonzero difference beyond a fixed ambient bound. Replace that step or construct an explicit pair for which it fails.

[write] Same question as the base’s “Remove or justify the cofinality hypothesis” and Part IV’s question on two noncofinal exponential embeddings.

Research question 75.4 (Cofinality of exponential images [07]). Which order-cofinality types occur for the image of an exponential self-embedding of all of **No**? In particular, does a bounded-image example exist? The bounded strong field embedding of section 67 does not commute with `exp` and does not answer this.

[write] Answered *conditionally* in this report: under GBC together with ETR and a class satisfaction relation for the exponential language, Part III’s theorem 34.2 and Part IV’s corollary 49.6 give proper elementary exponential self-embeddings with bounded image (in Part IV, $\omega^{-1} < |e(x)| < \omega$ for $x \neq 0$). Without those hypotheses (in ZFC, or in plain GBC) this report gives no answer. Compare the base question “Proper exponential copies”.

Research question 75.5 (Simultaneous exponential and differential structure [07]). Classify the self-embeddings preserving both $\exp$ and ∂_{BM} , optionally fixing ω . An extension theorem in either language alone does not give homogeneity in the combined language. A first target is extending a prescribed omega-series substitution to the whole combined structure.

[write] Compare the base question “Differential centralizers”. On $\mathcal{A}$ the strong ones are classified, conditionally, by [theorem 68.5](#).

Research question 75.6 (Differentiation together with the omega-map [07]). A differential embedding sends ω to $\omega + c$ with $c \in \mathbb{R}$, and the omega-map removes the translation ([proposition 68.4](#)). Does fixing both operations force all higher scales, or can a nontrivial map remain? The equation $j(\omega) = \omega$ is only a starting constraint.

75.2 Series structure and integer parts

Research question 75.7 (Beyond exact-monomial lifts [07]). Develop a classification allowing $j(t^g) = c_g t^{\tau(g)}(1 + u_g)$ and a deformed coefficient section. State separately the algebraic cocycle conditions and the support conditions keeping every required family summable; the exact-monomial case should be a proved special case.

[write] Compare the base question “Beyond monomial images”.

Research question 75.8 (When is strong additivity automatic? [07]). Find structural hypotheses on a proper self-embedding preserving and reflecting $\mathbf{Oz}$ that force strong additivity. An automorphism argument using all multiplier tests need not work in a proper image, because the missing test elements may carry exactly the lost information. Seek a sharp condition on the supply of multipliers in the image.

[write] For automorphisms this is `opa:as:thm:main` (automatic strongness of omnific automorphisms); the question concerns proper embeddings. For exact-monomial maps, [proposition 66.4](#) reduces strongness to truncation compatibility.

Research question 75.9 (Initiality of algebraic images [07]). Which additive or field self-embeddings have an image that is initial in the sign tree? Distinguish this from requiring the map to preserve simplicity: an order-and-simplicity embedding with initial image is rigid, while a field isomorphism onto an initial subfield is a different problem. Find effective normal-form criteria in a restricted family.

[write] For *order-preserving* self-maps of all of $\mathbf{No}$, which includes every field self-embedding and every ordered additive one, [base theorem 5.2](#) already answers this: the image is simplicity-initial exactly when the map is onto. The question keeps content for maps between cutoffs, for embeddings of set-sized fields into $\mathbf{No}$ (initial embeddings in the sense of [\[7\]](#)), and for additive maps that are not order preserving.

Research question 75.10 (Conjugacy of proper copies [07]). Classify the images of double lifts up to ambient field automorphism, and then up to automorphisms preserving $\mathbf{Oz}$, $\exp$ or the derivation. Value-image geometry and the coefficient section are candidate invariants. Do not transfer a classification in one language to a richer one without proof.

Research question 75.11 (Fixed points, intersections and iteration [07]). For an explicit proper embedding j , determine $\text{Fix}(j)$, $\bigcap_{n < \omega} j^n(\mathbf{No})$ and the corresponding value subgroups. For double lifts, begin with the iterated images of the underlying order map. Describe when these intersections keep real closedness, strong summation, or the structure of a surreal substructure.

[write] Partly answered in the base: [proposition 8.4](#) (the core $K_\infty(j)$ is real closed and the hierarchy stabilizes at the countable intersection), (8.5) for monomial lifts, and the computation (8.6) for J_- ; compare the base question “Prescribe the reversible core”. For double-order embeddings, [theorem 55.8](#) identifies the intersection with $\text{Fix}(j)$.

75.3 Simplicity and controlled deformation

Research question 75.12 (Sharp birthday-growth bounds [07]). For natural embedding families, determine the least useful ordinal function B with $b(jx) \leq B(b(x))$. Equality with the identity bound is rigid, so the first nontrivial bounds should say exactly where a sign-tree or normal-form construction increases complexity.

[write] Compare the base question “Weaker birthday controls”. For double-order embeddings, [corollary 65.3](#) shows that every strictly increasing $h : \mathbf{On} \rightarrow \mathbf{On}$ occurs as an exact birthday function.

Research question 75.13 (Rigidity from selected cut operations [07]). The full canonical-cut operation rigidifies the structure ([proposition 4.7](#)). Which restricted collection of cut constructors suffices? Separate finite cuts, bounded-birthday cuts, sign-chain limits and cuts subject to cardinality bounds. A good answer includes a rigidity theorem and a proper embedding showing that a nearby weaker condition fails.

Research question 75.14 (Order-and-simplicity versus arithmetic [07]). Characterize the order-and-simplicity self-embeddings that also preserve addition, multiplication or a prescribed set of arithmetic operations. The prefix and localized-cone examples generally fail arithmetic and the normal-form lifts generally fail simplicity, so the intersection is a substantive classification problem.

[write] Near-duplicate of the base question “Field and simplicity together”. Conditionally on a nontrivial elementary $j : V \rightarrow M$, $\text{lce} : \text{prop} : J$ gives nonidentity field self-embeddings preserving and reflecting prefix simplicity ([section 72](#)).

75.4 Foundations and proof-assistant targets

Research question 75.15 (Coherent cutoff embeddings [07]). Give necessary and sufficient coherence conditions on a family $j_\kappa : \mathbf{No}_{<\kappa} \rightarrow \mathbf{No}_{<\kappa}$ that produce a global class self-embedding. The club theorem gives restrictions of an existing map; it does not glue arbitrary local choices. A first target is a construction over a specified club sequence of invariant stages.

[write] Compare the base question “Universe and cutoff absoluteness” and Part V’s [corollary 60.3](#).

Research question 75.16 (Axiom strength of homogeneity statements [07]). Locate the exact principles needed for each global extension theorem: set recursion with class parameters, Global Choice, or a stronger class recursion principle. Distinguish homogeneity of a fixed class structure from an assertion needing a truth predicate for the whole universe.

[write] Compare the base question “Minimal foundational strength”.

Research question 75.17 (A reusable formal embedding library [07]). Implement interfaces for ordered maps, sign-tree maps, strong Hahn maps, valued-field maps and exponential or differential maps, with explicit forgetful constructions. Make failed implications into formal counterexamples where a suitable set-sized model is available. The generic amplification theorem and the continuity criterion are small, high-value first milestones.

[write] Compare the base question “Near-term formalization”.

Research question 75.18 (Ambient transport and supportwise transport [07]). For elementary maps between ambient set-theoretic models, compare the map on a surreal code with the external map on its original normal-form terms. Determine exact support thresholds for equality and for landing inside the target model, keeping all hypotheses on models, class definability and support size.

[write] Partly addressed, conditionally, by `lce:thm:defect of docs/foundations-and-computation/large-cardinal-embeddings-and-normal-forms`, which computes the exact defect $\mathcal{J}_j(x)$ minus the supportwise transport.

76 Status and non-claims of source 07

The following limitations of source 07 (article and `README.txt`) are kept as stated.

- It is a mathematical research exposition with complete proofs of its principal constructions and deductions. Repository claims were checked selectively; no new Lean certification, exhaustive classification or claim of literature priority is asserted. The set-parameter stabilizer and the cofinal uniqueness statement are presented as deductions, not priority claims.
- No fresh Lean build was run; the Lean interfaces were read at the pinned commit. No repository file was modified.
- The repository inspection was targeted, not a reading of the whole repository, every preserved manuscript or every imported dependency.
- Deep background theorems (differential homogeneity, composition, substitution and inversion of omega-series, exponential homogeneity) are cited with their roles and not reproved; results resting on them are conditional on them.
- A research question is a proposed direction, not a claim that its status has been exhaustively settled across the literature.
- Compilation and visual inspection certify presentation, not theorems; the mathematical support is the proofs and the cited results. No finite numerical experiment is offered as evidence for a class-sized assertion.
- The KKS remarks are local formula issues, not a claim that the programme of [5] fails; the exponential-rigidity check is not a claim of external review or priority.
- The club theorem gives no cutoff invariant under every class map, forms no class of all class maps, and closure under finitary operations does not give closure under set-indexed sums.
- Checking a bounded model is not verification of a proper-class statement without a transfer argument.
- Homogeneity in each separate language does not give homogeneity in the combined exponential and differential language.
- The verification checked the constructions at their logical bottlenecks: set size of supports, reverse well-ordering, finite coefficient multiplicity, convolution, positivity, properness, and forward preservation versus reflection; the rigidity arguments use explicit finite identities.

[write] Collection status: source 07 cites `omnific-preserving-automorphisms` and `exponential-automorphism-rigidity` but not `docs/surcomplex/surcomplex-field-automorphisms`,

whose `saut:prop:discrete`, `saut:thm:coeffflow`, `saut:thm:orderedhomogeneity` and `saut:prop:fixedfields` it reproves or extends; those credits are given above. Its one finding about the collection (section 74.1) was confirmed and corrected.

A Result and verification ledger

Result family	Mathematical basis	Verification status in this delivery
Sign sequences, normal forms, real closedness	Classical surreal theory; repository carrier and field development	Imported standard foundations; selected source inspection, not an independent build.
Order/simplicity examples and rigidity	First-disagreement and canonical cut arguments	Complete proofs in this article; finite sign tests only as regression checks.
Hahn lifts and weighted classification	Support transport and finite coefficient convolution	Complete proofs in this article; no Lean certification claimed.
Arbitrary-set stabilizers and bounded copies	Two-level support footprint plus rational maps	Complete proofs, explicit bounds and omissions; rational regression tests included.
Closed images and topology dichotomy	Ordered-field arguments and first missing exponent	Complete proofs in this article.
Fixed fields and reversible core	Ordered finite-root argument and support membership	Complete proofs in this article.
Unit twists and Taylor shifts	Neumann support control and formal binomial identities	Proofs given; finite binomial checks are not an infinite-support proof.
Displacement/profile rigidity	Repository finite engine and generic profile framework	Conventional proof reproduced; generic source inspected; actual surreal bridge not machine-checked here.
Cofinal comparison of two exponential maps	Two-map product identity and finite-kernel bridge	Complete proof in this article; suggested new generic formalization task.
Model-theoretic and differential background	Quantifier elimination, Wilkie, surreal exponential and derivation literature	Imported theorems are cited; consequences are proved or explicitly qualified.
Research agenda	Questions and proposed interfaces	No assertion that all listed questions are certified open in the literature.

The accompanying `verify.py` uses exact rational arithmetic. It checks finite sign-order and prefix identities, the rational branch formulas and iterates, finite Hahn relabeling identities, and binomial coefficients. These tests are designed to catch transcription mistakes. They do not prove statements about arbitrary ordinal-length signs, proper classes, real coefficients, or all well-ordered supports. The mathematical proofs are the argument for those generalizations.

[write] In the collection this program is `code/03-hahn-lifts-verify.py`, byte-identical to the delivered `verify.py`; it writes `verification.json` beside itself, and its delivered record is

`data/03-hahn-lifts-verification.json`. Run on a fresh copy at placement it passed 195,579 checks and reproduced the record up to line endings. The delivered `verification.txt`, a byte copy of the record, is not shipped. The finite checks of sources 02, 04 and 05 are described in their Parts; sources 06 and 07 shipped none.

B Source audit and reproducibility

The repository inspection was pinned to commit `b0d4af4f30b24529fec877e2ad4a5952888146e9`. The principal recent paper was read in arXiv version `2509.22374v3`, whose submission history dates that version to 23 April 2026. Version pinning is intentional: question numbers and foundational conventions should not be silently transferred between versions. Rendered HTML dates need not be publication or submission dates. The relevant PDF page containing Questions 5.4 and 5.6 was also visually inspected during research.

The report uses no bundled external font files and no external figures. The source is a single `article.tex`, with its bibliography included. A standard TeX Live installation with `newtx`, `amsmath`, `tcolorbox`, `hyperref`, and `cleveref` can build it with `latexmk -pdf article.tex`. The included build script and README record the same procedure. This delivery contains no uncompiled Lean file represented as a verified result; the API sketch in section 14 is explicitly a design sketch.

[write] This paragraph describes source 03’s delivery. The present `article.tex` is the merged report; it builds with `latexmk -pdf` under pdf $\LaTeX$ (MiKTeX was used), and its bibliography is the union of the six sources’. The delivered README is replaced by the report README; the delivered build script is `code/03-hahn-lifts-build.sh` and still names the delivered file names. The pins of the other sources are in Appendix C.

C Provenance of the merged report

[write] This appendix was written when the report joined the collection.

C.1 Archives, pins and placement

Seven archives arrived together in commit `06395052a` of ProveIt (“New research reports”, 3 October 2026, 16:54 PDT) under `docs/incoming/`. They are batch 84 of the collection’s intake, and the manuscript numbers below are their batch numbers, in archive-time order. Each names the ProveIt commit it read (its pin); all pins are ancestors of the placement commit. Between the pins and the placement, of the files the sources read only two changed: `omnific-preserving-automorphisms` (the correction `f06e67d10` described below) and `surreal-fields-across-universes` (a reciprocal note of another batch, `d68b65ea0`); no Lean file changed. The placement commit `e1395c66c` staged 20 delivered files of sources 02–05 byte for byte (the README lists them) and removed the seven archives from `docs/incoming`; each archive remains retrievable with `git show 06395052a:"docs/incoming/<archive>.zip"`.

No.	Archive	Pages	Pin (commit time, PDT)	Role here
01	<code>Surreal_Self_Embedding.s.zip</code>	30	<code>20aa9b9a5</code> (15:08)	earlier edition of 05; not a source
02	<code>surreal_self_embedding.s(1).zip</code>	36	<code>af04de3fa</code> (15:16)	Part II

No.	Archive	Pages	Pin (commit time, PDT)	Role here
03	Surreal_Self_Embedding_s(2).zip	33	b0d4af4f3 (15:20)	base: Part I, Appendices A, B
04	Surreal_Self_Embedding_s_Atlas.zip	35	a006a77a0 (16:03)	Part III
05	surreal_self_embedding_s_comprehensive.zip	39	51b0a69d7 (15:56)	Part IV
06	surreal_self_embedding_s(3).zip	47	79049d58c (15:37)	Part V
07	surreal_self_embedding_s(4).zip	42	b0d4af4f3 (15:20)	Part VI

Author lines as delivered: sources 02, 03, 05 and 07 (and 01) say “Prepared for Vladimir Reshetnikov”; source 06 says “A study of the ProveIt development”; source 04’s title page and PDF metadata say “Research article prepared with ChatGPT”, the only one of the seven to name the tool. All seven are dated 3 October 2026.

C.2 The superseded edition 01

Manuscript 01 (*Self-Embeddings of the Surreal Numbers: Order, simplicity, field structure, exponential rigidity, and foundations*) is an earlier edition of 05: 89% of its distinct eight-word sequences occur in 05, all 52 of its labels reappear there unchanged, the package layout, preamble and title coincide, 01’s pin and PDF precede 05’s, and 05 says that it “incorporates and extends earlier working notes prepared for the same project”. 05 also settles two of 01’s research questions (an exact criterion for the vanishing derivative, Part IV, and the existence of noncofinal elementary exponential copies, under the class framework of Part III). Nothing of 01’s mathematics is missing from 05. Nothing of 01 is shipped; it survives in its archive. Three things of 01 do not reappear in 05:

- one audit bullet, saying that the repository’s simplest-separator theorem has an explicit existence premise and that filling a cut is a separate statement (true: `existsUnique_simplest_separator` in `Surreal/Foundations/SignSequenceSimplicity.lean` and `exists_separator_of_small_sets` in `SignSequenceCutOperation.lean`);
- the first paragraph of its proof of the omission theorem, which 05 moved into a lemma on small types;
- its finite-check program, a different program from 05’s (61,042 against 146,914 assertions). Its groups for prefix cones (`canonical_cone`, `sign_prefix_*`), repetition by two and three (`sign_repeat_2_*`, `sign_repeat_3_*`), explicit dilation difference quotients (`derivative_q_gt_1`, `derivative_q_lt_1`) and numerical two-probe displacement (`two_map_displacement`, `two_probe_amplification`) have no counterpart group in 05’s program, which tests a random padding map and symbolic product identities instead. 01’s program passes on a fresh copy and reproduces its record up to line endings.

C.3 What each source contributed

- **03 (base).** The text of Part I and Appendices A and B, with every label prefixed `sse:` and every reference updated; its section n is Section $n + 1$ here, and its numbered statements keep their positions within their sections. Its finite checks are shipped.

- **02.** The local tree embeddings fixing a set and all ordinals, blocks of ordinal length, the bounded map J_c fixing ω , elementarity in the order and group languages, cutoff cardinalities, the conditional large-cardinal section, the lexicographic word action and iterates, and its questions and plan (Part II). Its finite checks are shipped.
- **04.** The sharpest bound for the bounded double lift, an embedding with fixed field exactly $\mathbb{R}$, the core formula, the conditional bounded and cofinal elementary embeddings, abstract exponential value displacement and cofinal pair rigidity, the exponent-lift criterion for commuting with $\exp$, a Lean integration plan and an audit of invalid arguments (Part III). Its finite checks are shipped.
- **05.** The local-padding description of all order-and-simplicity embeddings, the derivative criterion, a continuous non-strong automorphism, diagonal derivations, its own route to the elementary embeddings with a logarithmic range bound, and its questions (Part IV). Its finite checks and build record are shipped.
- **06.** A credited synthesis of the omnific-pair results of the collection, the exponential defect cocycle, the translation parameter, club localization, the fixed-point theorem of Bagayoko–van der Hoeven, a Lean status table and its agenda (Part V). It shipped no code.
- **07.** Convex tree images, the omega-series classification (conditional on imported literature), unavoidable points as a relative real closure, homogeneity, the correction of a sentence of omnific-preserving-automorphisms, two alleged sign slips in KKS, a staged formalization program and its questions (Part VI). It shipped no code.

C.4 Where the merge had to choose

- **Base 03 rather than 05** (section 1.1): 03 has the weaker hypotheses on the shared theorems and no ETR, satisfaction class or large cardinal; 05’s additions enter as Part IV.
- **One home per shared result.** A result proved by several members but not by the base is printed by one home source and pointed to by the others: local padding (05), local tree embeddings and ordinal blocks (02), elementary embeddings and abstract exponential rigidity (04), continuity for embeddings between arbitrary ordered fields and homogeneity (07), club localization, the Bagayoko–van der Hoeven fixed points and the translation parameter (06), $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$ (02). Different proofs are kept as second routes; each Part’s correspondence table accounts for every numbered item of its source.
- **Notation.** The base’s symbols are used throughout; the renames are listed in section 1.2 and in each Part.
- **Collection results.** Every re-proof of a result already in the collection is printed once with the existing label (section 1.3 and the Parts), never as new; sentences of the sources claiming such results as developments of their own are corrected in [write] notes.
- **A correction elsewhere.** Source 07 (Part VI) showed that a sentence of docs/surreal/omnific-preserving-automorphisms (Section 27.2) about the Taylor section was false as stated; it was corrected in commit f06e67d10, before the placement.
- **Checked claims.** The repository claims of the six sources (declaration names, file contents, labels of other reports) were checked at the placement commit; none was found false, and the Lean status statements are repeated as checked. The claim of sources 02 and 04 that the birthday cutoff $\mathbf{No}_{<\lambda}$ is a subfield exactly when λ is an epsilon number was checked, and its converse is proved in a [write] note (remark 23.2). Source 07’s two alleged sign slips in KKS were not checked.

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